

PHYSICS

QUIZ # 01

SECTION-I

Single Correct Answer Type

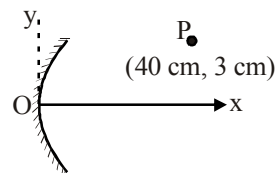
7 Q. [3 M (-1)]

1. The man, whose height is $l = 1.75$ m is standing at a distance $I = 6$ m from a vertical pillar of height $h = 7$ m. At what distance from the man should we put a small flat mirror on the ground horizontally to see in it the image of the top of the pillar?

(A) 4.8 m (B) 1.2 m (C) 5.2 m (D) 4.2 m

2. The co-ordinates of the image of point object P formed by a concave mirror of radius of curvature 20 cm (consider paraxial rays only) as shown in the figure is

(A) (13.33 cm, -1 cm) (B) (13.33 cm, +1 cm)
 (C) (-13.33 cm, +1 cm) (D) (-13.33 cm, -1 cm)

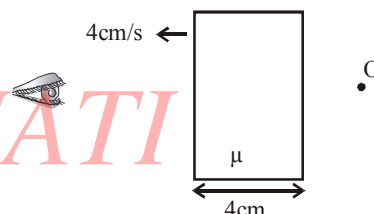


3. A vessel of depth H is filled with a non-homogeneous liquid whose refractive index varies with the depth 'y' as $\mu = (1 + y/H)$. The apparent depth as seen by an observer from above is

(A) $0.693 H$ (B) $0.667 H$ (C) $0.75 H$ (D) $0.500 H$

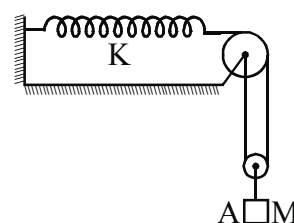
4. A slab of thickness 4 cm is moving towards the observer with velocity 4 cm/s. Find the speed of the object at $t = 1$ sec as observed by the observer. Refractive index of the slab varies as $\mu = (1 + t)$:-

(A) 2 cm/s towards slab
 (B) 2 cm/s away from slab
 (C) 1 cm/s towards slab
 (D) 1 cm/s away from slab

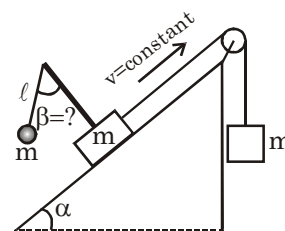


5. Block A in the figure is released from the rest when the extension in the spring is x_0 . The maximum downward displacement of the block.

(A) $Mg/2k - x_0$
 (B) $Mg/2k + x_0$
 (C) $2Mg/k - x_0$
 (D) $2Mg/k + x_0$



6. A box of mass 'm' slides without friction on an inclined plane. On the box, and perpendicular to the ceiling of the box, there is a standing massless rod. A bob of mass 'm' is connected to the upper edge of the rod by a massless thread (see figure). The angle of the inclined plane is α . The mass m is at rest relative to the box (a state of equilibrium). Let ℓ be the length of the thread. A mass m is connected to the box by a pulley. The angle between the thread

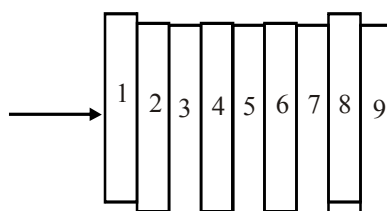


and the rod is β . α and β are in radians. Value of $\frac{\pi}{\beta}$ is :-

(A) 3 (B) 6 (C) 2 (D) 4

7. Take a stack of nine identical books and hold them horizontally by pressing them together with your hands (figure). Then decrease the pressure slowly until the books are just about to start falling down. Which book(s) will start to slide first? Assume that friction coefficient between books is greater than that between books and hands.

(A) 5
 (B) 4, 5 and 6 together
 (C) 2, 3, 4, 5, 6, 7, 8 together
 (D) all 9 books slide together

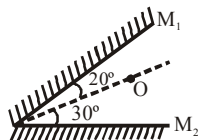


Hold nine books horizontal between your hands. Loosen the grip somewhat. Which book(s) will tend to slide down first?

Multiple Correct Answer Type

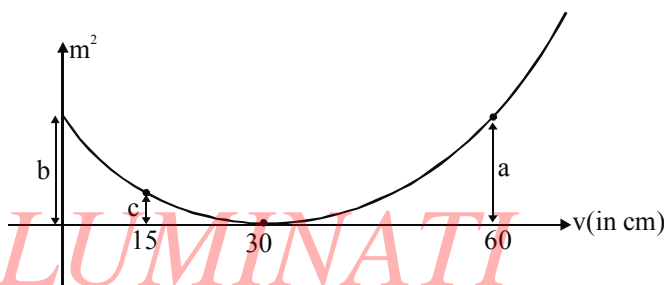
5 Q. [4 M (0)]

8. An object O is placed between two inclined plane mirrors M_1 & M_2 as shown in the figure.

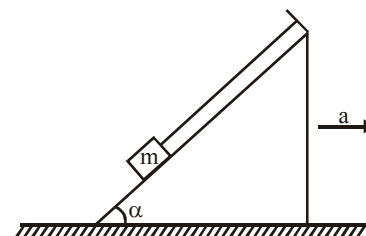


- (A) Eight images are formed
(B) Angular positions of images with respect to mirror M_2 measured anticlockwise are $70^\circ, 130^\circ, 170^\circ, 190^\circ, 330^\circ, 290^\circ, 230^\circ, 280^\circ$
(C) Seven images are formed
(D) Angular position of images with respect to mirror M_2 measured anticlockwise are $70^\circ, 130^\circ, 170^\circ, 190^\circ, 330^\circ, 290^\circ, 230^\circ$

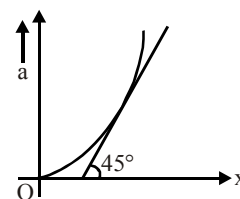
9. For a concave mirror, graph of square of magnification and image distance from pole is given for real object. Choose the correct statement(s).



- (A) This graph is hyperbola.
(B) length a must be equal to length b.
(C) radius of concave mirror is 60 cm
(D) the length c will be half of length a.
10. A body of mass $m = 1.8$ kg is placed on an inclined plane, the angle of inclination is $\alpha = 37^\circ$, and is attached to the top end of the slope with a thread which is parallel to the slope. Then the slope is moved with a horizontal acceleration of a. Friction is negligible.



- (A) Tension in thread for $a = \frac{5}{6} m / \text{sec}^2$ is 12N
(B) Tension in thread for $a = \frac{5}{6} m / \text{sec}^2$ is 24N
(C) The body loses contact with plane at the time when acceleration becomes zero
(D) The body loses contact with plane at the time when acceleration becomes $\frac{40}{3} m / \text{s}^2$
11. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x in parabolic nature as shown in figure. At $x = 1$ m tangent to the graph makes an angle 45° with positive x-axis as shown in diagram. Then at $x = 3$ m :-



- (A) velocity of the particle is $3\sqrt{2}$ m/s
(B) velocity of the particle is $3\sqrt{3}$ m/s
(C) acceleration of the particle is 4.5 m/s^2
(D) acceleration of the particle is 9 m/s^2
12. Two particles are projected from the same point with the same speed, at different angles θ_1 and θ_2 to the horizontal. They have the same horizontal range. Their times of flight are t_1 and t_2 respectively.

- (A) $\theta_1 + \theta_2 = 90^\circ$ (B) $\frac{t_1}{t_2} = \tan \theta_1$ (C) $\frac{t_1}{t_2} = \tan \theta_2$ (D) $\frac{t_1}{\sin \theta_1} = \frac{t_2}{\sin \theta_2}$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) [3 M (0)]

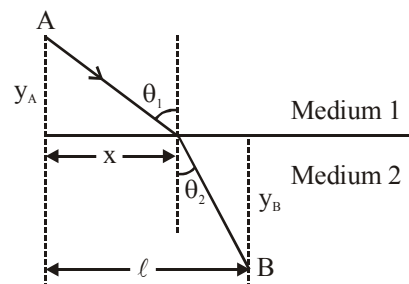
Paragraph for Question no 13 and 14

The refraction of light may be understood by purely kinematic considerations. We need to assume that light takes the shortest (timewise) path between two points (Fermat's principle of least time). Referring to the figure, let the speed of light in medium 1 be v_1 , and in medium 2, v_2
if $n_2 > n_1$ then $0 \leq \theta_1 \leq 90$

Let $v_1 = \frac{c}{n_1}$ and $v_2 = \frac{c}{n_2}$

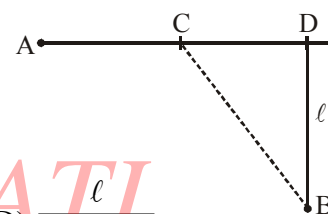
Now if we calculate the time taken by the light to go from point A to point B as a function of the variable x and minimize it with respect to x , we get

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$



13. From point A located on a highway (figure) one has to get by car as soon as possible to point B located in the field at a distance ℓ from the highway. It is known that the car moves in the field n times slower than on the highway. At what distance from point D one must turn off the highway.

(A) $\frac{2\ell}{\sqrt{n^2 - 1}}$ (B) $\frac{\ell}{\sqrt{n^2 + 1}}$ (C) $\frac{\ell}{\sqrt{n^2 - 1}}$ (D) $\frac{\ell}{\sqrt{2n^2 - 1}}$

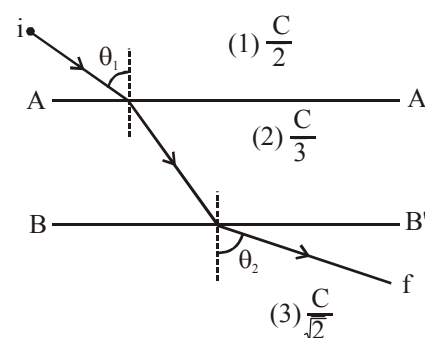


14. Three different regions (1), (2) and (3) are separated by parallel lines AA' & BB'. Speed of a particle in region (1), (2) & (3) are

$\frac{C}{2}$, $\frac{C}{\sqrt{3}}$ and $\frac{C}{\sqrt{2}}$ respectively. The particle starts from point (i)

in region (1) to reach point (f) in region (3) in minimum time. If θ_1 is 30° then value of θ_2 is

- (A) 30° (B) 45°
(C) 60° (D) 90°



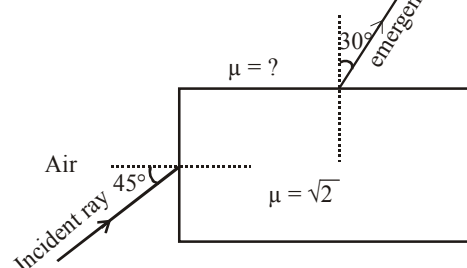
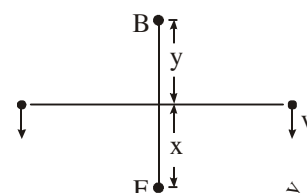
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

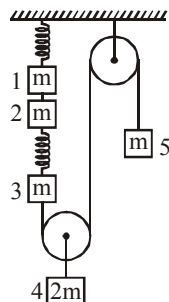
1. Figure shows a bird B in air and a fish F in water ($\mu = 4/3$). If the water level starts coming down with velocity $v = 1\text{cm/sec}$ while fish and bird remain at rest. The ratio of speeds of bird B as seen by fish F and fish F as seen by bird B is $(n\mu)$. Find the value of n .

2. A light ray incidents from air ($\mu = 1$) to a glass slab ($\mu = \sqrt{2}$) as shown in figure & emerges in an unknown medium of refractive index $\mu = \sqrt{k}$ as shown. What is the value of k ?

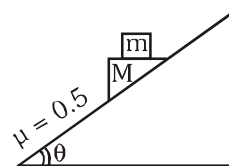
5 Q. [4 M (-1)]



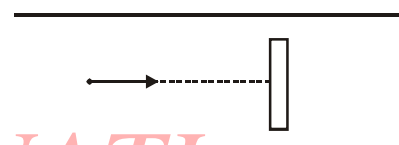
3. All the strings, springs and pulleys shown in figure are ideal. Initially the system is in equilibrium and blocks are at rest. Now, the upper spring is cut. The acceleration of block 1 just after cutting the upper spring is found $(y) \times (g) \text{ m/s}^2$. Find y ?



4. A block of mass 'm' is kept on the horizontal top surface of wedge of mass 'M' which is kept on an incline plane of inclination θ ($\sin \theta = 3/5$) as shown in the figure. Coefficient of friction between the wedge and incline is 0.5. The minimum coefficient of friction between 'm' and 'M' so that 'm' does not slip on 'M' when the system is released, from rest is $\alpha/44$ then value of α is ($g = 10 \text{ m/s}^2$) ($m = M = 1 \text{ kg}$)



5. A small boat is travelling downstream with a speed of 5 m/s relative to river flowing with speed $2\frac{1}{3} \text{ m/s}$. It encounters a log of wood floating on the river at a distance of a from itself. It turns by an angle of 37° and just crosses the edge of the log after 8 sec. If the log was fixed and other data remain same, what would be the time (in sec.) taken by it to just cross the edge of the log.



SECTION-IV

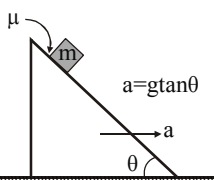
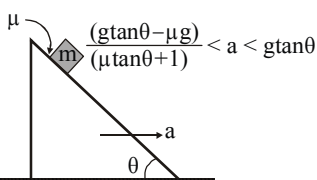
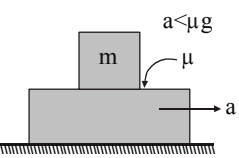
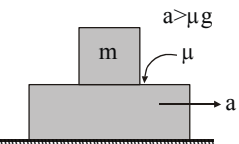
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Column-I shows arrangements having a block and a wedge. Wedge is move with constant acceleration 'a' by applying external force. Match the entries of column-I with the entries of column-II.

Column-I

Column-II

- (A) 
- (B) 
- (C) 
- (D) 

- (P) Net force on block of mass m is ma
- (Q) Net force on block of mass m is more than ma
- (R) Net force on block of mass m is less than ma
- (S) Friction force on block of mass m is ma
- (T) Contact force on block of mass m is more than ma

PHYSICS

QUIZ # 02

SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

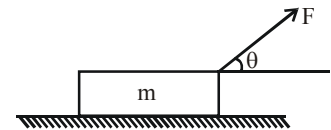
1. A block of mass m is placed on rough horizontal floor. Coefficient of friction between block and floor is μ & force F is applied as shown in diagram & block is accelerating towards right. Then its acceleration will be ($F < mg \operatorname{cosec} \theta$):-

(A) $\frac{F \cos \theta}{m}$

(B) $\frac{\mu F \sin \theta}{m}$

(C) $\frac{F}{m} [\cos \theta + \mu \sin \theta] - \mu g$

(D) $\frac{F \cos \theta - \mu mg}{m}$



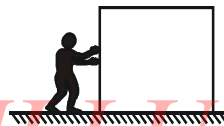
2. A boy of mass 50 kg is pushing a block of mass 20 kg in horizontal direction which produces an acceleration of 2 m/s^2 in block. Boy moves with block in such a way that acceleration of block & boy are equal. Coefficient of friction between block & horizontal surface is 0.2 and coefficient of friction between boy and horizontal surface is 0.5 then frictional force acting on boy will be :-

(A) 40 N

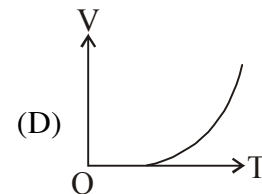
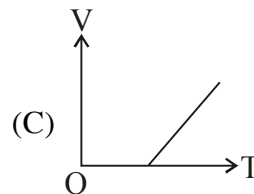
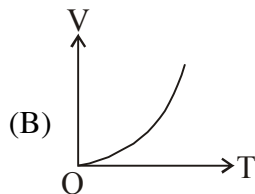
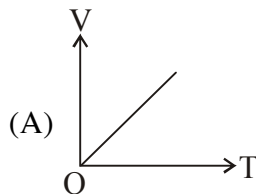
(B) 250 N

(C) 180 N

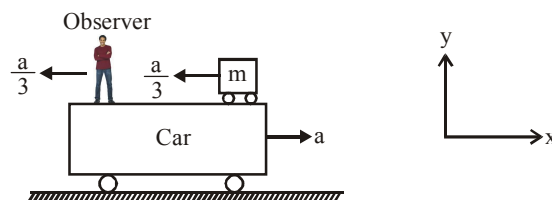
(D) 140 N



3. A block rests on a rough floor. A horizontal force which increases linearly with time T , begins to act on the block at $t = 0$. Its velocity V is plotted against T . Which of the given graph is correct ?



4. A particle of mass m is moving with an acceleration $\frac{a}{3}$ towards left relative to the trolley car. An observer accelerates towards left with an acceleration of magnitude $\frac{a}{3}$. If the car moves with a rightward acceleration a :-



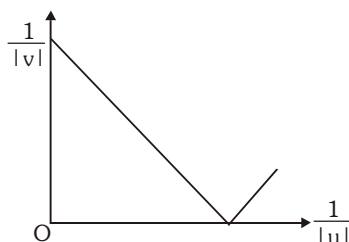
(A) Net force on the particle is $+\frac{2ma}{3}\hat{i}$

(B) Pseudo force on the particle in the frame of trolley is $-\frac{ma}{3}\hat{i}$.

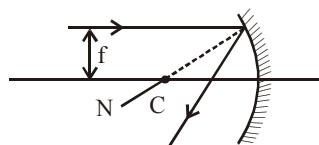
(C) The rate of change of momentum of observer in ground frame will be $ma\hat{i}$

(D) None of the above.

5. Figure shows the variation of $\frac{1}{|v|}$ versus $\frac{1}{|u|}$ for a concave mirror. The line $y=3x$ cuts the graphs at an ordinate 1.5 m^{-1} . The curvature radius of the mirror is



- (A) 25 cm (B) 50 cm (C) 100 cm (D) 200 cm
6. A light ray is coming parallel to principal axis, the distance between ray and axis is equal to focal length (as shown). Find the angle of deviation after reflection :-



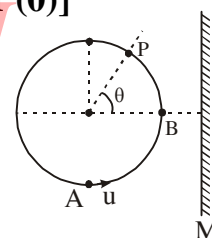
- (A) 60° (B) 90° (C) 120° (D) small deviation

Multiple Correct Answer Type

5 Q. [4 M (0)]

7. A particle is moving along a circle with a constant speed u , as shown in figure. M is a plane mirror, then

- (A) relative velocity of image w.r.t. object becomes zero twice in one revolution
(B) path of image with respect to object is circle
(C) path of image with respect to object is straight line
(D) speed of image with respect to object when particle is at P is $2u \sin \theta$

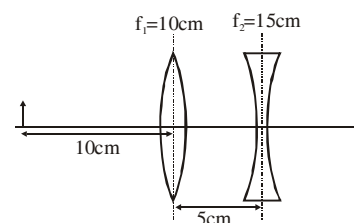


8. Chose the correct option(s) for the given figure. Assume the pulleys and string to be ideal and neglect any friction. (a_A = acceleration of A, a_B = acceleration of B)

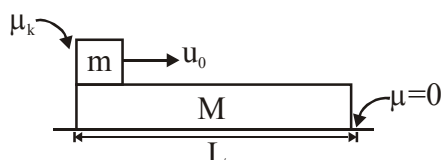


- (A) $a_B = 1.56 \text{ m/s}^2$ (B) $a_A = 3.12 \text{ m/s}^2$ (C) $a_A = 2.34 \text{ m/s}^2$ (D) $a_B = 4.68 \text{ m/s}^2$
9. An object is placed nearly at focus of a converging lens of focal length 10 cm as shown in figure. A diverging lens of focal length of 15 cm is placed 5 cm behind the converging lens as shown in the figure.

- (A) Final image forms at the object
(B) Image forms at a distance 30 cm from concave lens
(C) Magnification of image is 1.5
(D) Magnification can not be determined.



10. A long plank of mass M is initially at rest on a frictionless surface. A small block with mass m [$m = \alpha M$] and initial speed u_0 slides on the top of the larger plank. The coefficient of friction between the block and plank is μ .



(A) At the time when the small block ceases to slide, displacement of smaller block with respect to

ground is $\frac{u_0^2}{2\mu g} \left[1 - \left(\frac{\alpha}{1+\alpha} \right)^2 \right]$

(B) At the time when the small block ceases to slide, displacement of smaller block with respect to

ground is $\frac{u_0^2}{2\mu g} \left(\frac{1+\alpha}{\alpha} \right)^2$

(C) Displacement of bigger plank till same instant is $\frac{u_0^2}{2\mu g} \frac{\alpha}{(1+\alpha)^2}$

(D) Displacement of bigger plank till same instant is $\frac{u_0^2}{2\mu g} \left(\frac{1+\alpha}{\alpha} \right)^2$

11. A certain mirror placed at the origin has $\hat{i}$ as the normal vector to its reflecting surface. The mirror begins to translate with a velocity $\hat{i} + \hat{j} + \hat{k}$. At $t = 0$, at the same time an object which was initially at $\hat{i} + \hat{j}$ starts moving with a velocity $(\hat{i} + \hat{j})$ m/s. Now choose the correct options.

- (A) Initial position of the image will be $-\hat{i} + \hat{j}$.
 (B) The velocity of the image will be $\hat{i} + \hat{j}$.
 (C) The velocity of the image relative to the object will be zero
 (D) The velocity of the image relative to the mirror will be $-\hat{k}$.

Linked Comprehension Type (Single Correct Answer Type)

(2 Para × 2Q.) [3 M (0)]

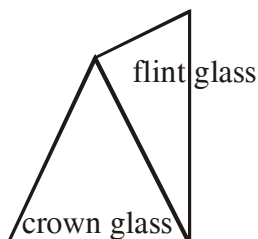
Paragraph for Question no. 12 and 13

A parallel beam of light of intensity I , travelling in a glass medium of refractive index $\mu = 1.5$, is refracted by a spherical air bubble of radius R .

12. The nature of air bubble and final image will be
 (A) converging, real (B) diverging, real (C) converging, virtual (D) diverging, virtual
13. The distance of final image from centre due to refraction by both surfaces of bubble, is
 (A) $3R/2$ (B) $R/2$ (C) R (D) None of these

Paragraph for Question no. 14 and 15

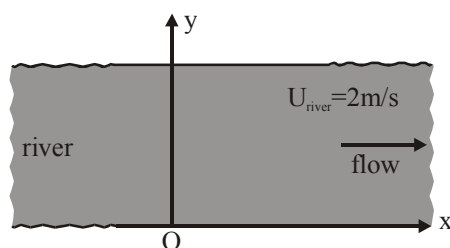
By joining prism of two different glasses (shown in the figure), it is possible to obtain deviation without dispersion. We then say that the prism is achromatic. It is required to fit a 5° crown-glass prism with a flint-glass prism so as to achromatize the wavelength interval between the C and F spectra lines. (The D line is located in the yellow region of the spectrum and is taken as the mean ray between the C and F lines). Indices of refraction of the crown glass for the C, D, F lines are $\mu_C = 1.514$, $\mu_D = 1.517$, $\mu_F = 1.523$; indices of refraction of the flint glass for C, D, F lines are $\mu'_C = 1.644$, $\mu'_D = 1.650$, $\mu'_F = 1.664$.



14. The angle of the flint-glass prism is
 (A) 9° (B) 2.25° (C) 3° (D) none of these
15. The deviation produced by the prism combination, as computed for the D line is nearly
 (A) 4.24° (B) 1.24° (C) 3.33° (D) 1.12°

SECTION-III**Numerical Grid Type (Single digit Ranging from 0 to 9) 3 Q. [4 M(-1)]**

1. A man can swim in still water with a speed of 3 m/s. x and y axis with respect to ground are drawn along and normal to the bank of river flowing to right with a speed of 2 m/s. The man starts swimming from origin O at t = 0 second. Assume size of man to be negligible. Locus of all the possible points where man can reach at t = 2 sec is $(x - a)^2 + y^2 = c^2$. Find value of $\frac{ac}{4}$.



2. A flat glass slab of thickness 6 cm and refractive index 1.5 is placed in front of a plane mirror. An observer is standing behind the glass slab and looking at the mirror. The actual distance of the observer from the mirror is 50 cm. The observer sees the image at a distance of d_1 cm from himself. Now the slab is removed and the observer sees his image in plane mirror at a distance of d_2 cm from himself. What is the value of $d_2 - d_1$?
3. A small piece of wood is floating on the surface of a $\sqrt{7}$ m deep lake. When the sun is vertically above the piece the image is formed at A. When the sun is just setting the image of the piece is formed at B. If the R.I. of water is $\frac{4}{3}$ then the distance between A and B is _____.

SECTION-IV**Matrix Match Type (4 × 5)****1 Q. [8 M (for each entry +2(0))]**

1. A particle is projected with a velocity $\vec{v}_0 = a\hat{i} + b\hat{j}$ from ground with y-axis as vertical and it lands on ground.

Column-I

(A) $\frac{v_x}{v_y}$ at $t = \frac{b}{2g}$

(B) $h_{\max}$

(C) R (Horizontal)

(D) $\frac{v_{av}}{v_0}$ where v_{av} = Average velocity

Column-II

(P) $\frac{a}{\sqrt{a^2 + b^2}}$

(Q) $\frac{2a}{b}$

(R) $\frac{b^2}{2g}$

(S) $\frac{2ab}{g}$

(T) $\frac{a}{b}$

SECTION-I

Single Correct Answer Type

12 Q. [3 M (-1)]

1. Which of these actions will move the real image point P' farther from the boundary ?

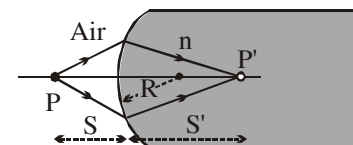
- (1) Decrease the index of refraction n .
(2) Increase the distance S .
(3) Decrease the radius of curvature R .

(A) 1, 2, 3

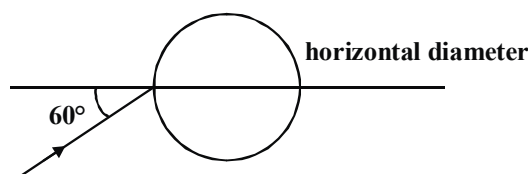
(B) 1 only

(C) 2 & 3 only

(D) 2 only



2. A ray of light falls on a transparent sphere kept in air as shown in figure. If the final ray emerges from the sphere parallel to the horizontal diameter, then the refractive index of the sphere is



(A) $\sqrt{2}$

(B) $\sqrt{3}$

(C) $\frac{3}{\sqrt{2}}$

(D) 2

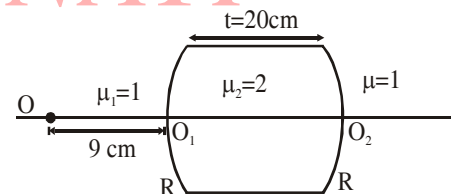
3. In the given figure, if object is placed at a distance 9 cm from O_1 then find the image distance (in cm) from O_2 . ($R = 9$ cm)

(A) 9 cm

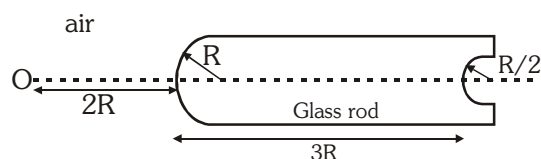
(B) 20 cm

(C) 18 cm

(D) infinity



4. A glass rod has ends as shown in figure. The refractive index of glass is μ . The object O is at a distance $2R$ from the surface of larger radius of curvature. The distance between apexes of ends is $3R$. The range of μ for which the image is real is given by



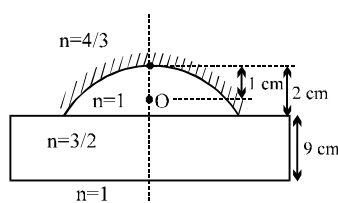
(A) $2-2.5$

(B) $2-2.25$

(C) for any value of $\mu > 1$

(D) for any value of $\mu < 2.25$

5. A concave mirror of focal length 2 cm is placed on a glass slab as shown in the figure. Then the image of object O formed due to reflection at mirror and then refraction by the slab :



(A) will be virtual and will be at 2 cm from the pole of the concave mirror

(B) will be virtual and formed on the pole of the mirror

(C) will be real and on the object itself

(D) none of these

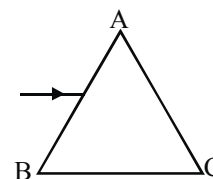
6. A transparent cube of 0.21 m edge contains a small air bubble. Its apparent distance when viewed through one face of the cube is 0.10 m and when viewed from the opposite face is 0.04 m. The actual distance of the bubble from the second face of the cube is :-
 (A) 0.06 m (B) 0.17 m (C) 0.05 m (D) 0.04 m
7. The apparent depth of water in cylindrical water tank of diameter $2R$ cm is reducing at the rate of x cm/minute when water is being drained out at a constant rate. The amount of water drained in c.c. per minute is (n_1 = refractive index of air, n_2 = refractive index of water) :-
 (A) $x \pi R^2 n_1/n_2$ (B) $x \pi R^2 n_2/n_1$ (C) $2 \pi R n_1/n_2$ (D) $\pi R^2 x$
8. A ray of light is incident on an equilateral triangular prism parallel to its base as shown in the figure. The ray just fails to emerge from the face AC. If μ be the refractive index of the prism then the incorrect relation is

(A) $2 \sin^{-1} \left(\frac{1}{\mu} \right) = \frac{\pi}{3}$

(B) $\sin^{-1} \left(\frac{1}{\mu} \right) + \sin^{-1} \left(\frac{1}{2\mu} \right) = \frac{\pi}{6}$

(C) $\sin^{-1} \left(\frac{1}{\mu} \right) + \sin^{-1} \left(\frac{1}{2\mu} \right) = \frac{\pi}{3}$

(D) $\sin^{-1} \left(\frac{\mu}{2} \right) + \sin^{-1} \left(\frac{\mu}{4} \right) = \frac{\pi}{3}$



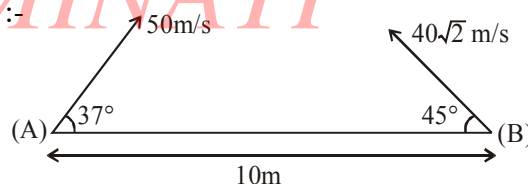
9. A ray of light is incident at a small angle i on a small angle prism ($A = 4^\circ$) and emerges normally from the opposite surface. If the refractive index of the material of the prism is $\mu = 3/2$, the angle of incidence i (in degrees) is nearly equal to
 (A) 0 (B) 4 (C) 6 (D) none of these
10. For the given situation select the **INCORRECT** statement :-

(A) $\frac{d|\vec{r}|}{dt} = 80$ m/s

(B) Change of angular position of (A) w.r.t (B) is 1rad/sec

 (C) Magnitude of rate of change in position vector of (A) w.r.t (B) is $10\sqrt{65}$ m/s

(D) $\frac{d|\vec{r}|}{dt} = 50 + 40\sqrt{2}$ m/s



11. An aeroplane flying at constant speed 115 m/s towards east, makes a gradual turn following a circular path to fly south. The turn takes 15 seconds to complete. The magnitude of the centripetal acceleration during the turn, is

(A) $\frac{23\pi}{8}$ m/s²

(B) $\frac{46\pi}{3}$ m/s²

(C) $\frac{23\pi}{6}$ m/s²

(D) none of these

12. The point particle of mass m interacts with the stationary Center. The potential energy is $U = \frac{a}{r} + br$. At the initial moment the point was at a distance $r_0 = 2\sqrt{\frac{a}{b}}$, from the center and Had zero speed.

 (A) The minimum distance $r_{\min}$ that the mass can approach the center is $\sqrt{\frac{a}{4b}}$.

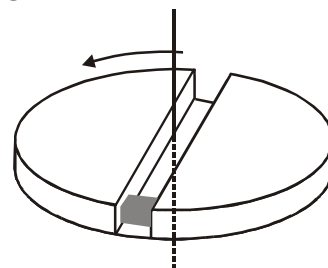
 (B) A stable equilibrium position of the material point is $r = \sqrt{\frac{a}{b}}$.

 (C) The magnitude of the force acting on the material point at the point r_0 is $\frac{3b}{4}$

(D) The particle executes SHM about the stable equilibrium position.

Multiple Correct Answer Type**1 Q. [4 M (0)]**

13. A large, horizontal disk of radius R , shown below, starts to rotate from rest with an angular acceleration of α . The rotation is about a vertical axis through the centre of disk. The disk contains a narrow channel of length $2R$ and rectangular cross-section. Gravity acts in the vertical direction with an acceleration of g . There is a small rectangular puck that just fits easily in the aforementioned channel, as shown. The puck is situated a distance r from the axis of rotation.



(A) If the sides of the channel are frictionless but the bottom of the channel has a static coefficient of

friction μ , then puck begins to slide at $t = \sqrt{\frac{\mu}{\alpha}}$

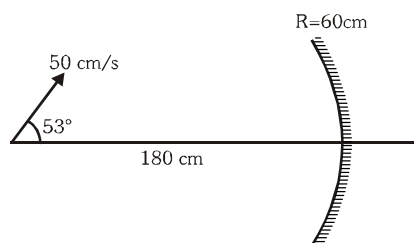
(B) In case mentioned in option (A), $t = \sqrt{\frac{\mu g}{r\alpha^2}}$

(C) Now, instead, the situation is that the bottom of the channel is frictionless but the walls have a static coefficient of friction μ . Now puck begins to slide at $t = \sqrt{\frac{\mu g}{r\alpha^2}}$

(D) In case mentioned in option (C), $t = \sqrt{\frac{\mu}{\alpha}}$

Linked Comprehension Type
(Single Correct Answer Type)
(1 Para $\times$ 3Q. and 1 Para $\times$ 2 Q.) [3 M (0)]**Paragraph for Questions no. 14 to 16**

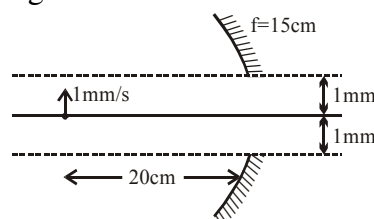
A particle is projected with a speed of 50 cm/sec from a point at a distance of 180 cm from pole. Assuming all rays to be paraxial.



14. Height of the image when object is at the maximum height, is
 (A) 0.25 cm (B) 0.8 cm (C) 0.45 cm (D) none of these
15. Vertical speed of the image when object is at the greatest height, is
 (A) (9/20) cm/s (B) (1/9) cm/s (C) (0.8) cm/s (D) none
16. Direction of velocity the image when particle is at the maximum height, is
 (A)  (B)  (C)  (D) None of these

Paragraph for Question no 17 and 18

A concave mirror of focal length 15 cm is divided into two parts and separated by 2 mm. A point object at a distance of 20 cm starts moving at $t = 0$ with 1mm/s as shown.



17. Distance between images formed by two part at $t = 0$:
 (A) 12 mm (B) 4 mm (C) 6 mm (D) 8 mm

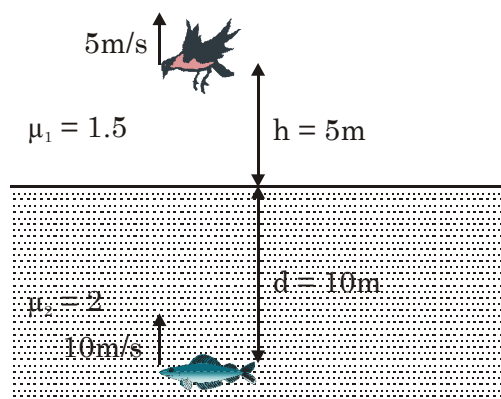
18. Rate of change of distance between images formed by two parts at $t = 0$:

- (A) 0 mm/s (B) 2 mm/s (C) 4 mm/s (D) 8 mm/s

Matching List Type (4×4)

1 Q. [3 M (0)]

19. Diagram shows a bird B and a fish F moving in two different medium of refractive index 1.5 and 2 with speed 10 m/s and 5 m/s respectively. At the given instant bird is located at height 5m above the interface while fish is located at depth of 10m. List-I indicates various parameters while list-II gives its values. Match them correctly :-



List-I

List-II

- | | |
|--|-------------------------|
| (P) Separation of fish as observed by bird | (1) $\frac{50}{3}$ unit |
| (Q) Separation of bird as observed by fish | (2) 12.5 unit |
| (R) Speed of fish as observed by bird | (3) $\frac{10}{3}$ unit |
| (S) Speed of bird as observed by fish | (4) 2.5 unit |

Code :-

	P	Q	R	S
(A)	3	2	1	4
(B)	2	1	4	3
(C)	4	1	2	3
(D)	1	2	3	4

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. In vertical circular motion of a bob, match the entries of column-I with entries of column-II. Here v_0 is the velocity of bob at lowest point & T is tension in string.

Column – I (speed at lowest point)

Column-II (possible situation)

- | | |
|------------------------|--|
| (A) $v_0 = \sqrt{5gl}$ | (P) $T_{\text{lowest}} - T_{\text{highest}} = 6mg$ |
| (B) $v_0 = \sqrt{gl}$ | (Q) string will slack for a finite time |
| (C) $v_0 = 2\sqrt{gl}$ | (R) bob will oscillate |
| (D) $v_0 = 3\sqrt{gl}$ | (S) bob will complete the circle |

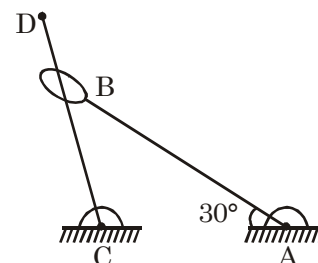
SECTION-I

Single Correct Answer Type

3 Q. [3 M (-1)]

1. A smooth ring is connected to rod AB, while rod CD passes through ring. At the given moment angular velocity of AB about A is 1 rad/s and $AC = CB$. Find angular velocity of rod CD about C at given moment.

- (A) 1 rad/s
(B) $\frac{1}{2}$ rad/s
(C) $\frac{\sqrt{3}}{2}$ rad/s
(D) $\frac{3}{2}$ rad/s

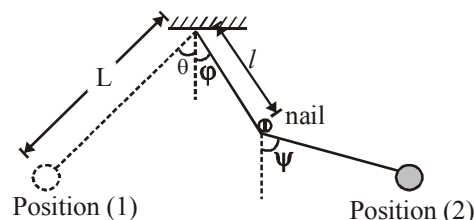


2. Tangential acceleration of a particle moving in a circular path of radius R can be given as : ($\vec{a}$ = total acceleration, $\vec{v}$ = velocity of particle). The most **CORRECT** statement is :

- (A) $\sqrt{a^2 - \frac{v^4}{R^2}}$ (B) $\frac{d|\vec{v}|}{dt}$ (C) $\frac{\vec{v} \cdot \frac{d|\vec{v}|}{dt}}{v}$ (D) All of the above

3. In figure, A sphere suspended from the ceiling starts from rest at position (1) and ends its upward swing at position (2). Ignoring friction and air resistance, the correct relation among the following is :-

- (A) $\cos \psi = \frac{L \sin \theta - \ell \cos \phi}{L - \ell}$
(B) $\sin \psi = \frac{L \cos \theta - \ell \sin \phi}{L - \ell}$
(C) $\cos \psi = \frac{L \cos \theta - \ell \cos \phi}{L - \ell}$
(D) $\sin \psi = \frac{L \sin \theta - \ell \cos \phi}{L - \ell}$

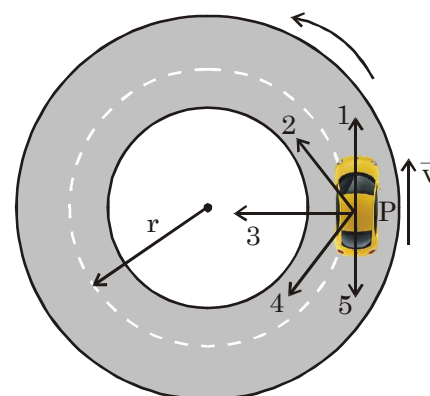


Multiple Correct Answer Type

4 Q. [4 M (0)]

4. A car is moving without slipping anticlockwise sense (top view of track) in a circular path on a rough horizontal surface. $\vec{v}$ and $\vec{a}$ are velocity and acceleration vector at time 't'. When car is at point P, the direction of friction force for different conditions mentioned in the options below are given by arrows as shown in figure. Select the **CORRECT** alternative.

- (A) If $\vec{a} \cdot \vec{v} = 0$, friction force will act along arrow 3.
(B) If $\vec{a} \cdot \vec{v} > 0$, friction force will act along arrow 2.
(C) If $\vec{a} \cdot \vec{v} < 0$, friction force will act along arrow 5.
(D) If $\vec{a} \cdot \vec{v} < 0$ friction force will act along arrow 4.



Top view of the circular track

5. The potential energy of a particle of mass 1kg is given by $U = 4x^3 - 5x^2$, where x is in meters.
- (A) The particle is at stable equilibrium at $x = 0$.
- (B) The particle is at unstable equilibrium at $x = 5/6$ m.
- (C) If kinetic energy of particle at $x = 5/6$ m is 1J, the particle oscillates about its equilibrium position.
- (D) The magnitude of acceleration of the particle at $x = 1$ m is 2 m/s^2 .
6. A particle moves along straight line acted by a force which delivers power $P = Ct^7$. Initially it's velocity is zero. Where C is constant and t is time. Then mark the correct option :-
- (A) Displacement of the particle is proportional to t^5
- (B) Force acting on the particle increases with time.
- (C) Acceleration of the particle is proportional to t^3 .
- (D) Rate of change of kinetic energy is proportional to t^6 .
7. A girl is riding on bicycle with a constant velocity 5m/sec. If mass of girl is 50kg and experience a 500N force from air-resistance. Choose the **CORRECT** statement(s) :-
- (A) Power supply by her on level ground is 2.5 kW
- (B) Power supply by her on inclined plane going up having inclination 37° is 4 kW
- (C) Power supply by her on inclined plane going down having inclination 53° is 0.5 kW
- (D) Power supply by her on inclined plane going down having inclination 53° is 1.0 kW

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (0)]
(Single Correct Answer Type)

Paragraph for question nos. 8 to 10

Focal Length of a Zoom Lens : Figure shows a simple version of a zoom lens. The converging lens has focal length f_1 , and the diverging lens has focal length $f_2 = -|f_2|$. The two lenses are separated by a variable distance d that is always less than f_1 . Also, the magnitude of the focal length of the diverging lens satisfies the inequality $|f_2| > (f_1 - d)$. To determine the effective focal length of the combination lens, consider a bundle of parallel rays of radius r_0 entering the converging lens. It can be shown that the

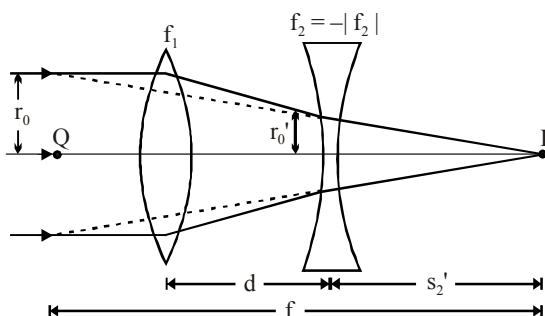
radius of the ray bundle decreases to $r'_0 = \frac{r_0(f_1 - d)}{f_1}$ at the point that it enters the diverging lens. Final

image I' is formed a distance $s'_2 = \frac{|f_2|(f_1 - d)}{(|f_2| - f_1 + d)}$ to the right of the diverging lens.

If the rays that emerge from the diverging lens and reach the final image point are extended backward to the left of the diverging lens, they will eventually expand to the original radius r_0 at some point Q. The distance from the final image I' to the point Q is the effective focal length f of the lens combination ; if the combination were replaced by a single lens of focal length f placed at Q, parallel rays would still be

brought to a focus at I'. The effective focal length is given by $f = \frac{f_1 |f_2|}{(|f_2| - f_1 + d)}$.

Assume that $f_1 = 12.0 \text{ cm}$, $f_2 = -18.0 \text{ cm}$, and the separation d is adjustable between 0 and 3.0 cm.



8. If the incident beam is parallel, the image of the converging lens will act as an object for the diverging lens. This point will be
 (A) Real image for 1st lens and virtual object for 2nd lens.
 (B) Virtual image for 1st lens and virtual object for 2nd lens.
 (C) Real image for 1st lens and real object for 2nd lens.
 (D) Virtual image for 1st lens and real object for 2nd lens.
9. Which of the following can be a value of the effective focal length of the zoom lens?
 (A) 18 cm (B) 12 cm (C) 28 cm (D) 38 cm
10. The final image distance s_2' will be such that
 (A) $0 \leq s_2' \leq |f_2|$ (B) $|f_2| \leq s_2' \leq 2|f_2|$
 (C) $2|f_2| \leq s_2' \leq \infty$ (D) any of the above three options

Paragraph for Question no. 11 and 12

Two identical particles each of mass 4 kg are placed together at the edge of a tall cliff. They are joined by an inextensible string of negligible mass. One of the particles is projected with a speed of $10\sqrt{3}$ m/s at 60° above the horizontal. The string becomes taut after 2 sec. ($g = 10$ m/s²)

11. The length of the string is :
 (A) $5\sqrt{3}$ m (B) 10 m (C) 20 m (D) $10\sqrt{3}$ m
12. Tension in the string when the second particle starts moving is :
 (A) $10\sqrt{3}$ N (B) 7.5 N (C) 15 N (D) 10 N

Matching List Type ($4 \times 4 \times 4$) (1 Table $\times$ 3 Que.) [3 M (0)]

Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table.

In the below question u and v represents distance of object and image from the lens/mirror. Both are measured using usual conventional method i.e. distance in the direction of incident rays are positive and opposite to incident rays are negative. Column 2 gives the nature of optical element and column 3 gives the variation of image distance from lens/mirror as object distance changes. To answer the related questions we have to consider the part of graph from 1 to 2 as shown in column 3.

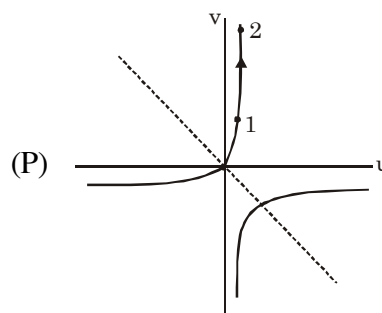
Column-1

Column-2

Column-3

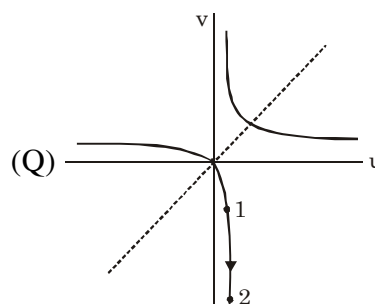
(I) $u < 0$ $v > 0$

(i) Convex mirror



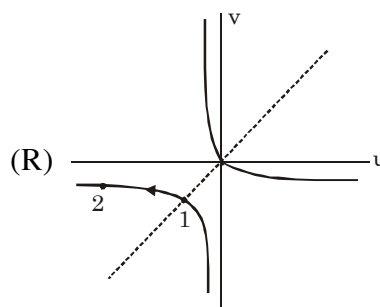
(II) $u > 0$ $v > 0$

(ii) Convex lens



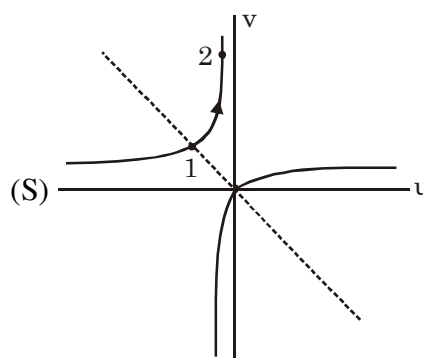
(III) $u < 0$ $v < 0$

(iii) Concave mirror



(IV) $u > 0$ $v < 0$

(iv) Concave lens



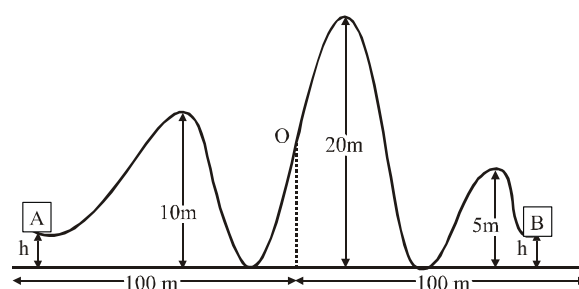
13. In which of the following case. Real image of virtual object is moving away from mirror/lens:-
(A) (I) (iii) (R) (B) (II) (iv) (P) (C) (II) (ii) (P) (D) (III) (i) (S)
14. Which of the following represents real image of real object going towards mirror/lens :-
(A) (IV) (ii) (S) (B) (II) (i) (R) (C) (III) (iii) (R) (D) (I) (iii) (P)
15. Which represents real image of real object going away from mirror/lens :-
(A) (I) (ii) (S) (B) (III) (i) (P) (C) (III) (iv) (R) (D) (II) (iii) (Q)

SECTION-II

Numerical Grid Type

(Upto Second Decimal Place)

1. A point object of mass 2 kg is moved from point A to point B very slowly on a curved path by applying a tangential force on a curved path as shown in figure. Then find the work done (in J) by external force in moving the body. Given that $\mu_s = 0.3$, $\mu_k = 0.1$ for the part OB while AO is smooth.

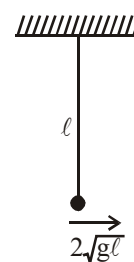


SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)

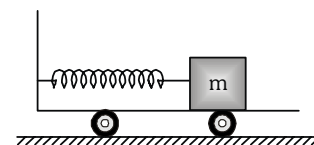
2 Q. [4 M(-1)]

1. One end of a string of length $\ell = \frac{14}{9} \text{ m}$ is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point as shown in figure. The string is cut when the ball becomes horizontal. Find the distance (in m) travelled till it stop for the 1st time (Take $\pi = \frac{22}{7}$).



2. At $t = 0$ spring is relaxed and system is at rest. Now trolley starts moving with constant velocity V_0 towards right. Work done by external agent on trolley

from $t = 0$ till spring attains maximum compression is $\frac{kmv_0^2}{2}$. Then k is:



SECTION-IV

Matrix Match Type (4×5)

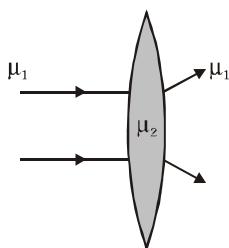
2 Q. [8 M (for each entry +2(0))]

1. Column I shows incident ray, refracted ray associated with different lenses whereas column-II shows refractive index, of the lens surrounding medium and nature of optical instruments. Match the entries of column-I with column-II

Column I

Column II

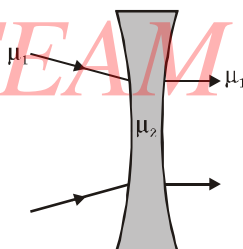
(A)



(P)

$$\mu_2 > \mu_1$$

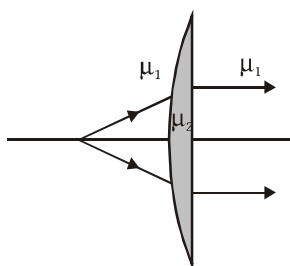
(B)



(Q)

$$\mu_2 < \mu_1$$

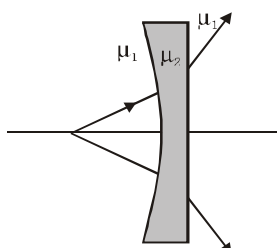
(C)



(R)

diverging

(D)



(S)

converging

(T)

None of these

2. Column-I

Column-II

- (A) Diverging lens
(B) Converging lens
(C) Concave mirror
(D) Convex mirror

- (P) Can form magnified, inverted and virtual image.
(Q) Always forms a virtual, erect and diminished image of a real object
(R) Can form virtual, erect and magnified image of a real object
(S) Can form a real, inverted and diminished images of a real object
(T) Can form virtual image of virtual object.

PHYSICS

QUIZ # 05

SECTION-I

Single Correct Answer Type

1. A wire is bent along an arc with a radius R . A bead is placed on the wire that can move along it without friction. At the initial moment the bead was at point O . The horizontal velocity should be imparted to the bead for it to get on to the wire again at point B after flying a certain distance (AB) through the air, will be

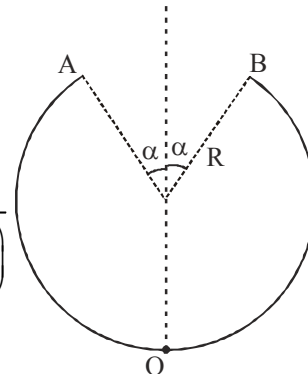
(A) $\sqrt{gR \left(2 + 2 \cos \alpha + \frac{1}{\cos \alpha} \right)}$

(B) $\sqrt{gR \left(2 + \cos \alpha + \frac{1}{\sin \alpha} \right)}$

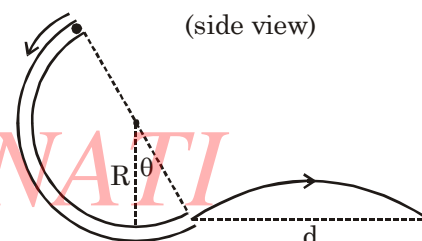
(C) $\sqrt{gR \left(\sin \alpha + \frac{1}{2} \cos \alpha \right)}$

(D) $\sqrt{5gR \cos \alpha}$

2 Q. [3 M (-1)]



2. A frictionless tube is bent into the shape of a semicircle with radius R . The semicircle is tilted so that its diameter makes a fixed angle θ with respect to the vertical, as shown in figure. A small mass is released from rest at the top of the tube and slides down through it. When the mass leaves the tube, it undergoes projectile motion. Let d be the distance traveled in the projectile motion, up to the time when the mass returns to the height it had when it left the tube. What should θ be so that d is as large as possible?



(A) $\sin \theta = \frac{1}{\sqrt{3}}$

(B) $\sin \theta = \frac{1}{\sqrt{2}}$

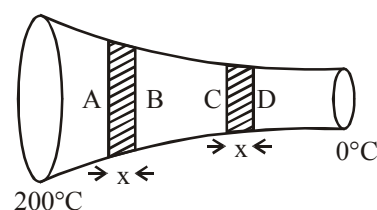
(C) $\cos \theta = \frac{1}{\sqrt{3}}$

(D) $\tan \theta = \frac{1}{\sqrt{3}}$

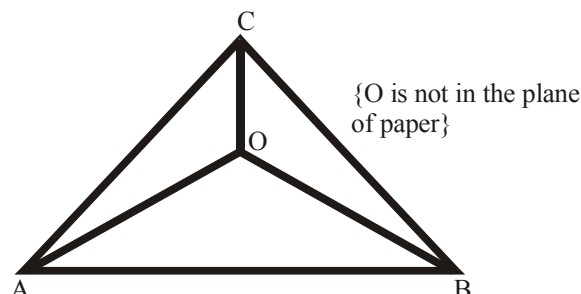
Multiple Correct Answer Type

3. Two ends of conducting rod of varying cross-section are maintained at 200°C and 0°C . In steady state:
- (A) Temperature difference across AB and CD are equal
 (B) Temperature difference across AB is greater than that across CD .
 (C) Temperature difference across AB is less than that across CD .
 (D) Temperature gradient increases as we move from end A to B .

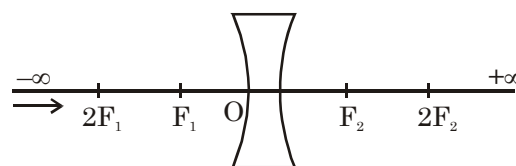
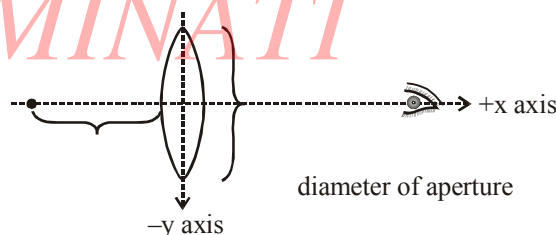
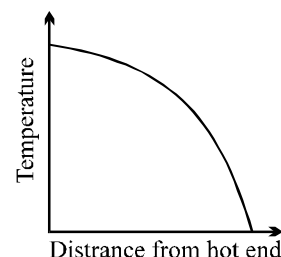
10 Q. [4 M (0)]



4. Six identical rods are joined together to form a network as shown. A temperature difference of ΔT can be maintained between two points. Considering heat flow through conduction only, in steady state :-
- (A) Net rate of heat flow between points where temperature difference is maintained, depends on points chosen.
 (B) Net rate of heat flow between points where temperature difference is maintained, is independent of points chosen.
 (C) In all the cases of points chosen { where ΔT is to be maintained } there is one rod through which no heat flows.
 (D) Heat conduction takes place through all the rods irrespective of choice of points where temperature difference is maintained.

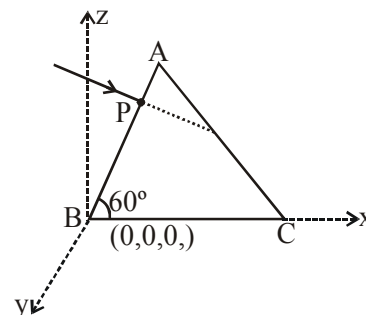


5. The ends of a long homogenous bar are maintained at different temperatures and there is no loss of heat from the sides of the bar due to conduction or radiation. The graph of temperature against distance of the bar when it has attained steady state is shown here. The graph shows
- (A) the temperature gradient is not constant
 (B) the bar has uniform cross-sectional area
 (C) the cross-sectional area of the bar increases as the distance from the hot end increases
 (D) the cross-sectional area of the bar decreases as the distance from the hot end increases.
6. A lens forms 3 times magnified image of an object on screen. When the distance between the screen and the lens is increased by 10 cm and object is also moved to get 5 times magnified image again on the screen. Then
- (A) Focal length of the lens is 5 cm
 (B) Focal length of the lens is -5 cm
 (C) The displacement of the object is $\frac{2}{3}$ cm towards the lens
 (D) The displacement of the object is $\frac{2}{3}$ cm away the lens
7. A convex lens made of glass ($\mu_g = \frac{3}{2}$) has focal length f in air. The image of an object placed in front of it is inverted, real and magnified. Now whole arrangement is immersed in water ($\mu_g = \frac{4}{3}$) without changing distance between object and lens. The :
- (A) the new focal length will become $4f$ (B) the new focal length will become $f/4$
 (C) new image will be virtual and magnified (D) new image will be real, inverted and smaller in size
8. A convex lens of focal length +20 cm and diameter of aperture 1 cm is placed on origin as shown. The lens is free from any defect. A small luminous object is placed 30 cm away on -x axis. An observer is situated on +x side of lens. (Consider only light rays, reaching observers eyes only through the lens.) The least distance of distinct vision for this observer is 30 cm :-
- (A) Observer will see light coming into his eyes for any positive value of x on axis.
 (B) Observer will be able to see a sharp image on x axis for all $x > 60$ cm.
 (C) If x coordinate of observer is +90 cm, he would be able to see light entering his eyes for any value of y.
 (D) If x-coordinate of observer is +90 cm, he would be able to see light entering his eyes for $|y| < 2.5$ mm.
9. A object is moving on the optical axis of a diverging lens from $-\infty$ to O optical centre of the lens. Choose the **CORRECT** statement(s) about the image of particle :
- (A) When object is moving between $(-\infty, 2F_1)$ its image will be virtual & diminished.
 (B) When object is moving between $(2F_1, F_1)$ its image will be real and diminished.
 (C) When object is moving between F_1 to O its image will be virtual & enlarge.
 (D) For complete motion of object, image will lie on same side of the lens.
10. A real object is kept in front of a lens in air. The object is a linear extended object with its length perpendicular to the optical axis of lens. With reference to different cases of image formation by lens, choose the correct options :
- (A) If the image has a magnification - 2.5 then Image is real and power of lens is positive.
 (B) If the magnification of the image is +0.5 then Image is virtual and power of lens is negative
 (C) If length of image is the same as that of object then Image is real and power of lens is positive.
 (D) If length of image is the same as that of object then Image is virtual and power of lens is negative.
11. Which of the following quantities increases when wavelength is increased? Consider only the magnitudes
- (A) The power of a converging lens (B) The focal length of a converging lens
 (C) The power of a diverging lens (D) The focal length of a diverging lens



12. An equilateral prism ABC is placed in air with its base side BC lying horizontally along X-axis as shown in the figure. A ray given by $\sqrt{3}z + x = 10$ is incident at a point P on face AB of prism :-

- (A) For $\mu = \frac{2}{\sqrt{3}}$ the ray grazes the face AC.
(B) For $\mu = \frac{3}{2}$ finally refracted ray is parallel to z-axis.
(C) For $\mu = \frac{2}{\sqrt{3}}$ the ray emerges perpendicular to the face AC.
(D) For $\mu = \frac{3}{2}$ finally refracted ray is parallel to x-axis.



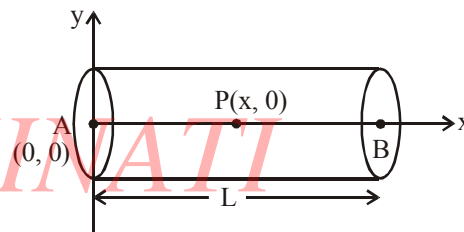
Linked Comprehension Type
(Single Correct Answer Type)

(2 Para x 2Q.) [3 M (0)]

Paragraph for Question no. 13 and 14

The rate of flow of heat through any cross-section is given by,

$$\frac{dQ}{dt} = -KS \left(\frac{dT}{dx} \right)$$



Where K is thermal conductivity, S is cross sectional area and $\frac{dT}{dx}$ is temperature gradient. For given rod, temperature at point P is given by, $T = 100 - 2x^2 - 4x$, where T is in kelvin and x is in metre. $L = 2$ m, $K = 100$ Watt/m-K, $S = 0.1$ m², $M = 5$ kg, $\alpha = 10^{-4}$ /K.

13. Temperature gradient at A and B would be :

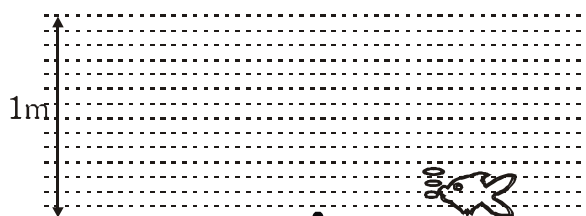
- (A) $-4 \left(\frac{K}{m} \right), -2 \left(\frac{K}{m} \right)$ (B) $-4 \left(\frac{K}{m} \right), -12 \left(\frac{K}{m} \right)$ (C) $100 \left(\frac{K}{m} \right), 84 \left(\frac{K}{m} \right)$ (D) $0, -2 \left(\frac{K}{m} \right)$

14. The rate of heat absorption in the rod would be :

- (A) 160 Watt (B) 60 Watt (C) 0 (D) 80 Watt

Paragraph for Question 15 and 16

An object O is placed at the bottom of a pond of depth 1m. A fish is lying at the bottom of the pond as shown in figure.



15. If refractive index of the liquid is $\sqrt{2}$, what will be minimum distance between fish and the object so that image of the object can be seen by the fish.
(A) 3m (B) 2m (C) 1m (D) none of these
16. Suppose fish is observing the image and object starts moving vertically upwards. Pick the correct option
(A) image will not move (B) image will move vertically upwards
(C) image will move vertically downward (D) none of these

Linked Comprehension Type

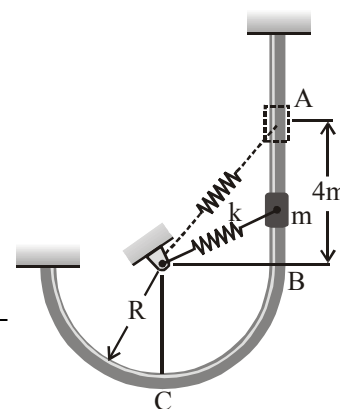
(1 Para × 2 Q.) [4 M (0)]

(Multiple Correct Answer Type)

Paragraph for Questions 17 and 18

The spring of constant $k = 50 \text{ N/m}$ is unstretched when the slider of mass $m = 2 \text{ kg}$ passes through position B. The slider is released from rest in position A. There is friction between slider and guide whose work done from A to B is -11 J and from B to C is -4 J . (Radius of guide $R = 3 \text{ m}$)

17. If speed of slider at position B and C are v_B and v_C respectively then :-
 (A) $v_B = 6.5 \text{ m/s}$ (B) $v_B = 13 \text{ m/s}$
 (C) $v_C = 16 \text{ m/s}$ (D) $v_C = 15 \text{ m/s}$
18. If the normal force at position B and C are N_B and N_C respectively then :-
 (A) $N_B = 112.67 \text{ N}$ (B) $N_B = 216.33 \text{ N}$
 (C) $N_C = 150 \text{ N}$ (D) $N_C = 170 \text{ N}$

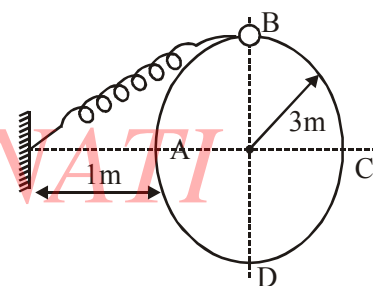


SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)

1 Q. [4 M(-1)]

1. A bead of mass 150 g is constrained to move along a vertical smooth and fixed circular track of radius 3 m as shown in figure is released at B. The spring is in the plane of the track has natural length of 1 m and spring constant of $\frac{3}{16} \text{ N/m}$. It starts from rest at B. What is the normal force (in N) exerted by the track on the bead when it passes through A ?

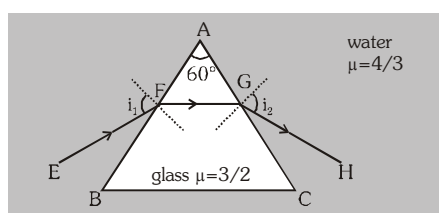


SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. Consider an equilateral prism ABC of glass ($\mu = \frac{3}{2}$) placed in water ($\mu = \frac{4}{3}$)



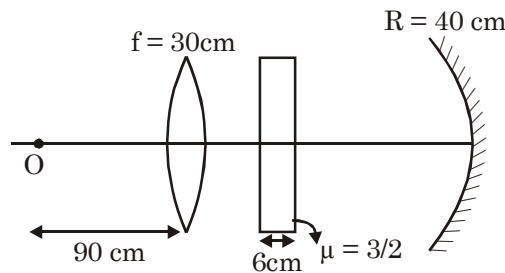
- | Column-I | Column-II |
|---|--|
| (A) FG is parallel to BC | (P) maximum deviation |
| (B) $i_1 = 90^\circ$ | (Q) minimum deviation |
| (C) $i_1 = i_2 = \sin^{-1} \left(\frac{9}{16} \right)$ | (R) TIR will take place at surface AC |
| (D) EF is perpendicular to AB | (S) No TIR will take place at surface BC |
| | (T) Can't say anything |

SECTION-I

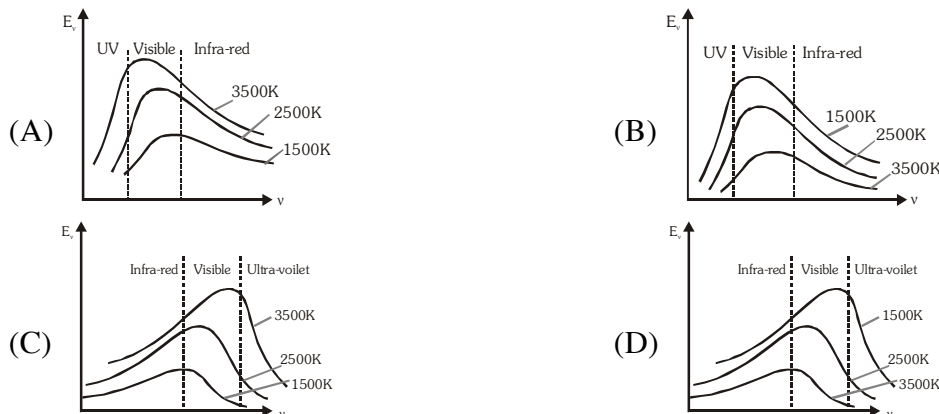
Single Correct Answer Type

8 Q. [3 M (-1)]

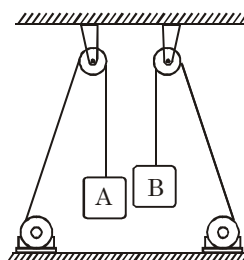
1. In the situation shown, slab is close to the lens. The distance (in cm) between the lens & the mirror so that final image of object is formed on object itself may be :-



- (A) 83 (B) 87 (C) 85 (D) 86
2. A body is in thermal equilibrium with surrounding. Absorptive power of the surface of body is 0.5. E is the radiant energy incident in unit time on the surface of the body. How much energy is propagate from it's surface in unit time.
(A) $2E$ (B) E (C) $3E$ (D) $4E$
3. An Electric kettle is used to heat $m = 1.8$ kg of water. During the time $\tau_1 = 2$ min, water temperature increased from $t_1 = 85^\circ\text{C}$ to $t_2 = 90^\circ\text{C}$. After turning off the kettle water temperature decreased by one degree during the time $\tau_2 = 1$ min. What is the approximate power P of the boiler? Specific heat of water $C = 4200$ J/(kg-K).
(A) 341 W (B) 441 W (C) 541 W (D) 641 W
4. Following graph shows the correct variation in intensity of heat radiations by black body and frequency at a fixed temperature—



5. Two equal masses are raised at constant velocity by ropes that run over pulleys as shown. Mass B is raised twice as fast as mass A. The magnitudes of the forces are F_A and F_B , while the power Supplied is respectively P_A and P_B . Which of the following statements is correct?

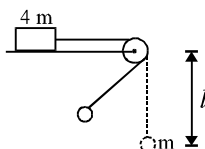


- (A) $F_B = F_A$; $P_B = P_A$ (B) $F_B = F_A$; $P_B = 2P_A$ (C) $F_B = 2F_A$; $P_B = P_A$ (D) $F_B = 2F_A$; $P_B = 2P_A$

6. A body of mass 2 kg is moving along a straight line. Power of net force acting on body is given by as $P = 3V^2$, where V is speed of the particle. Initial speed of particle is V_0 . Then determine speed of the particle as a function of time (Initial velocity is along the direction of force) :

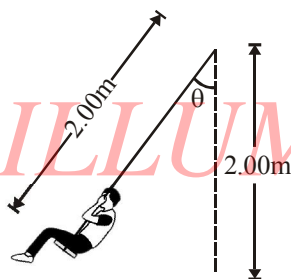
(A) $V = V_0 e^{3t/2}$ (B) $V = V_0 e^{3t}$ (C) $V = V_0 \ln t$ (D) $V = \frac{V_0}{2}$

7. Two bodies of mass m and $4m$ are attached to a light string as shown in the figure. The body of mass m hanging from string is executing oscillation with angular amplitude 60° while other body is at rest on a horizontal surface. The minimum coefficient of friction between the mass $4m$ and the horizontal surface is (here pulley is light & smooth)



(A) $1/4$ (B) $3/4$ (C) $1/2$ (D) $1/8$

8. The length of each of the ropes on a playground swing is 2.00 m. What is the maximum speed attainable on the swing if the maximum value of θ is 45.0° ?



(A) 1.41 m/s (B) 2.00 m/s (C) 3.39 m/s (D) 8.85 m/s

Multiple Correct Answer Type

3 Q. [4 M (0)]

9. A bi-convex thin lens of glass ($\mu = \frac{3}{2}$) has focal length in air is 20 cm. Now one of its convex face is silvered, to make equivalent mirror. In case (i) this equivalent mirror is kept in air and in case (ii) it is kept in water ($\mu = \frac{4}{3}$). Then :

- (A) Radius of curvature of convex side of lens is 10 cm
(B) Focal length of equivalent mirror in both case is same.
(C) Focal length in case (i) is 5 cm.
(D) Focal length in case (ii) is 8 cm.

10. A thin copper tube of outer radius 0.5 cm carries a liquid flowing at $T = 100^\circ\text{C}$. The copper tube loses heat according to Newton's law with constant of proportionality $3 \times 10^{-3} \text{ cal/cm}^2 \text{ sec}^\circ\text{C}$. The temperature of surrounding is 20°C . Now we coat a layer with thermal conductivity $2.8 \times 10^{-3} \text{ cal/cm}^\circ\text{C sec}$. The layer is 0.5 cm thick. Assume that outer surface of layer loses heat with same constant of proportionality : (Take : $\ln 2 = 0.7$)

- (A) The rate of heat loss becomes $\frac{8}{7}$ times
(B) The rate of heat loss becomes $\frac{7}{8}$ times
(C) The temperature T of outer surface of layer is approximately 65.7°C
(D) The temperature T of outer surface of layer is approximately 42.2°C

11. The potential energy of a diatomic molecule having the two atoms at a separation (r) is given by

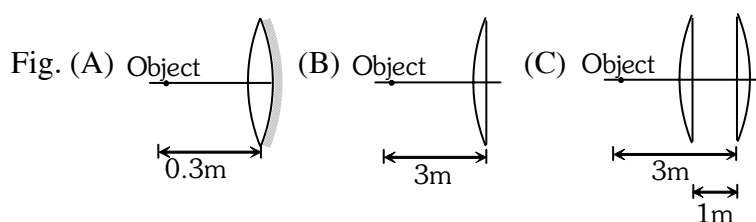
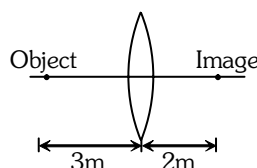
$$U(r) = -\frac{a}{r^6} + \frac{b}{r^{12}} \text{ where } r \text{ is the separation between the two atoms and } a \text{ \& } b \text{ are positive constants:-}$$

- (A) For $r < \left(\frac{2b}{a}\right)^{1/6}$, $F > 0$ (B) For $r > \left(\frac{2b}{a}\right)^{1/6}$, $F < 0$
- (C) At $r = \left(\frac{2b}{a}\right)^{1/6}$, potential energy is minimum
- (D) At $r = \left(\frac{b}{a}\right)^{1/6}$, potential energy is maximum.

Linked Comprehension Type
(2 Para × 3Q.) [3 M (0)]
(Single Correct Answer Type)
Paragraph for Question 12 to 14

An object is placed at a distance 3 meter from an equi-convex lens of refractive index $\mu = \frac{3}{2}$. Its image

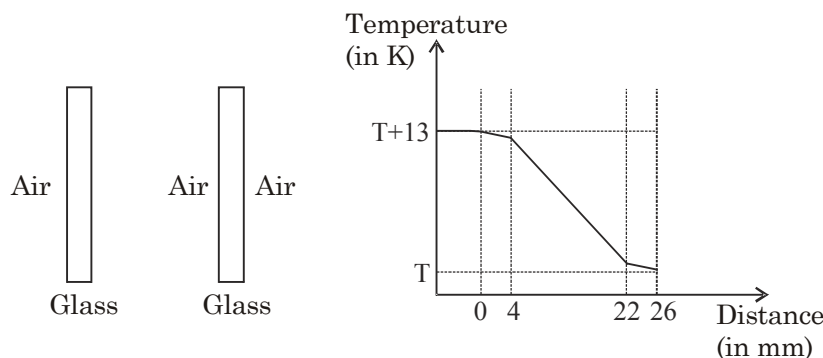
is formed at 2 m from the lens as shown in the figure.



12. Find the position of image if one face of the lens is silvered [Fig.(A)]
- (A) 12 m from the lens towards right (B) 12 m from the lens towards left
- (C) 6 m from the lens towards right (D) at infinity
13. Find the position of image if lens is cut into two symmetrical plano-convex lenses and one of the plano-convex lens is removed [Fig.(B)].
- (A) 12 m from the lens towards right (B) 12 m from the lens towards left
- (C) 6 m from the lens towards right (D) 6 m from the lens towards left
14. Find the position of image if plano-convex surfaces are displaced by 100 cm. [Fig.(C)]
- (A) 2.94 m towards right of second lens (B) 2.94 m towards right of first lens
- (C) 3.31 m towards right of second lens (D) 4.31 m towards right of first lens

Paragraph for question nos. 15 to 17

The figure shows a cross section of a double glass unit and a graph of the temperatures at different points within the unit. The temperature difference across the unit is 13K. It has a cross sectional area of 1.3 m^2 and the rate of heat flow through it is 65 W. The temperature distance graph does not change with time. Glass has a thermal conductivity of 1 W/mk .



15. Select the correct statement

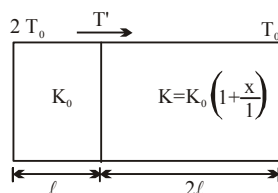
- (A) The unit is in steady state and in thermal equilibrium
(B) The unit is in steady state but not in thermal equilibrium
(C) The unit is not in steady state but is in thermal equilibrium
(D) The unit is neither in steady state nor in thermal equilibrium

16. The thermal conductivity of air is

- (A) $\frac{1}{10} \text{ W/mk}$ (B) $\frac{1}{12} \text{ W/mk}$ (C) $\frac{1}{14} \text{ W/mk}$ (D) $\frac{1}{18} \text{ W/mk}$

17. What is the temperature T' at the interface if heat conductivity of first slab of length ' ℓ ' is K_0 and is uniform every where but the conductivity of second slab of length ' 2ℓ ' varies with the distance ' x '

measured from the interface according to the law $K = K_0 \left(1 + \frac{x}{\ell}\right)$. The temperature at the boundaries of the composite slab are $2T_0$ and T_0



- (A) $T' = \left(\frac{2\ell n 2 + 1}{\ell n 2 + 1}\right) T_0$ (B) $T' = \left(\frac{2\ell n 3 + 1}{\ell n 3 + 1}\right) T_0$
(C) $T' = \left(\frac{\ell n 3 + 1}{2\ell n 3 + 1}\right) T_0$ (D) $T' = \left(\frac{4\ell n 3 + 1}{\ell n 3 + 1}\right) T_0$

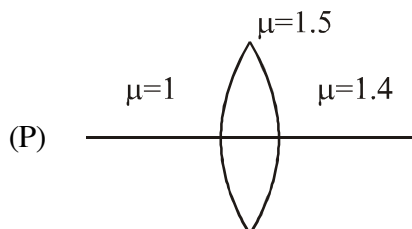
Matching List Type (4 × 4)

1Q.[3 M (0)]

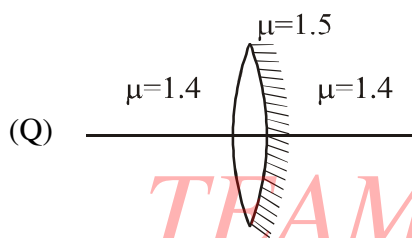
- 18.** List-I gives different lens configurations. The radius of curvature of each surface is R . Rays of light parallel to the axis of lens from left of lens traversing through the lens get focused at distance f from the lens. List-II gives corresponding values of magnitudes of f (μ represent refractive index) :-

List-I

List-II

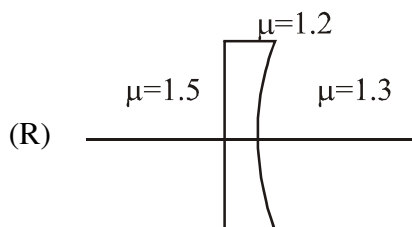


(1) $13R$



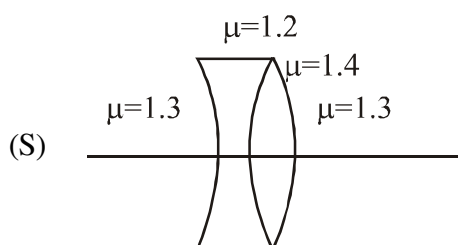
(2) $\frac{13R}{4}$

Lens is silvered



(3) $\frac{7R}{3}$

Plano concave lens



(4) $\frac{7R}{16}$

Codes :

	P	Q	R	S
(A)	1	3	2	4
(B)	1	2	3	4
(C)	2	3	4	1
(D)	3	4	1	2

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 1 Q. [4 M(-1)]

1. A convex lens forms a real image 4 cm long on a screen. When the lens is shifted to a new position without disturbing the object or the screen, again real image is formed on the screen which is 16 cm long. The length of the object (in cm) is ' ℓ '. Fill the value of $\frac{\ell}{2}$ in OMR sheet.

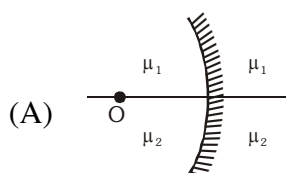
SECTION-IV

Matrix Match Type (4 × 5)

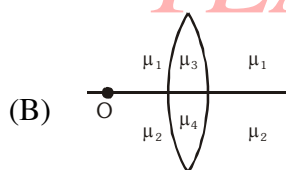
1 Q. [8 M (for each entry +2(0))]

1. **Column-I**
(Optical system)

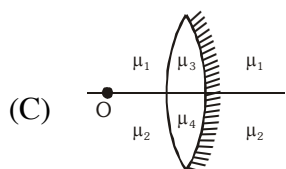
Column-II
(Number of images formed of object O)



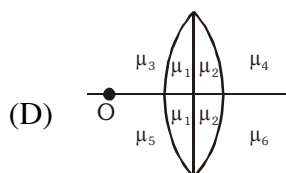
(P) 1



(Q) 2



(R) 3



(S) 4

TEAM ILLUMINATI

PHYSICS

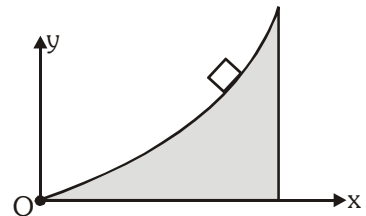
QUIZ # 07

SECTION-I

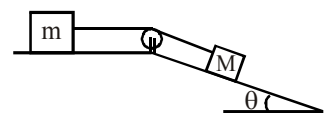
Single Correct Answer Type

14 Q. [3 M (-1)]

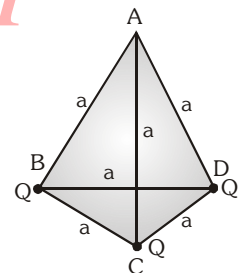
1. A wedge as shown in figure has a rough curved surface of coefficient of kinetic friction $\frac{1}{\sqrt{3}}$ whose shape can be given by equation $y=x^2$, where the horizontal direction is x-axis, vertical direction as y axis and the end of surface O as origin and x & y are measured in meter. A small block is released, from a certain point on the surface. The maximum height of the point from where block should be released so that it does not slip.



- (A) $\frac{1}{4}$ m (B) $\frac{1}{6}$ m (C) $\frac{1}{12}$ m (D) $\frac{1}{16}$ m
2. Find the maximum value of (M/m) in the situation shown in figure so that the system remains at rest. Friction coefficient of both the contacts is μ , string is massless and pulley is frictionless.

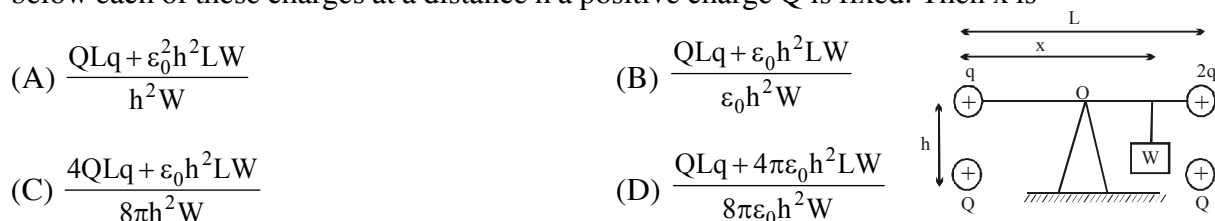


- (A) $\frac{\cos \theta}{\sin \theta - \mu \cos \theta}$ (B) $\frac{\sin \theta}{\sin \theta - \mu \cos \theta}$ (C) $\frac{\mu \cos \theta}{\sin \theta - \mu \cos \theta}$ (D) $\frac{\mu}{\sin \theta - \mu \cos \theta}$
3. A, B, C, D are the corners of a regular tetrahedron of side a. Three identical charges each Q are fixed at three corners B, C and D. The magnitude of electric field intensity at the corner A is

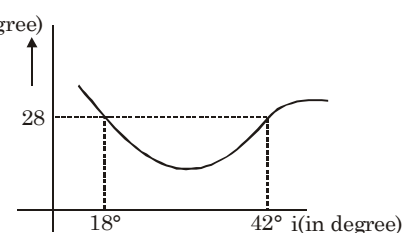


- (A) $\frac{\sqrt{6}Q}{4\pi\epsilon_0 a^2}$ (B) $\frac{2\sqrt{3}Q}{2\pi\epsilon_0 a^2}$ (C) $\frac{\sqrt{6}Q}{2\pi\epsilon_0 a^2}$ (D) $\frac{2\sqrt{3}Q}{4\pi\epsilon_0 a^2}$
4. A point charge $25 \mu\text{C}$ is located in the XY plane at the point of position vector $\vec{r}_0 = (\hat{i} + \hat{j})\text{m}$. What is the magnitude of electric field at the point of position vector $\vec{r}_1 = (4\hat{i} + 5\hat{j})\text{m}$?

- (A) $900 \frac{\text{V}}{\text{m}}$ (B) $9 \frac{\text{kV}}{\text{m}}$ (C) $90 \frac{\text{V}}{\text{m}}$ (D) can't be determined
5. An insulating long light rod of length L pivoted at its centre O and balanced with a weight W at a distance x from the left end as shown in figure. Charges q and 2q are fixed to the ends of the rod. Exactly below each of these charges at a distance h a positive charge Q is fixed. Then x is

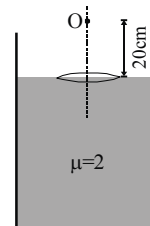


- (A) $\frac{QLq + \epsilon_0 h^2 LW}{h^2 W}$ (B) $\frac{QLq + \epsilon_0 h^2 LW}{\epsilon_0 h^2 W}$ (C) $\frac{4QLq + \epsilon_0 h^2 LW}{8\pi h^2 W}$ (D) $\frac{QLq + 4\pi\epsilon_0 h^2 LW}{8\pi\epsilon_0 h^2 W}$
6. The plot of deviation (δ) v/s angle of incidence (i) for a prism is shown in figure. Find the angle of the prism (A)



7. A point object O is placed at a distance of 20 cm in front of a equi-convex lens ($\mu_g = 1.5$) of focal length 10 cm. The lens is placed on a liquid of refractive index 2 as shown. Image will be formed at a distance h from lens. The value of h is

(A) 5 cm (B) 10 cm
(C) 20 cm (D) 40 cm



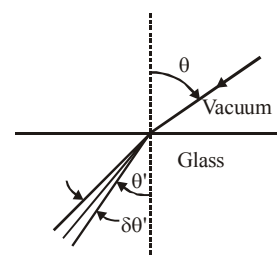
8. Chromatic aberration of a lens can be corrected by :
(A) providing different suitable curvature of its two surfaces
(B) proper polishing of its two surface
(C) suitably combining it with another lens
(D) reducing its aperture
9. In telescope, if the powers of an objective and eye lens are + 1.25 D and + 20 D respectively, then for relaxed vision, the length and magnification will be :-
(A) 85 cm and 25 (B) 85 cm and 16 (C) 21.25 cm and 16 (D) 21.25 cm and 25
10. Consider the formation of primary rainbow and secondary as seen by an observer on earth. The order of colours is
(A) VIBGYOR from inside to outside for primary rainbow and ROYGBIV from inside to outside for secondary rainbow
(B) VIBGYOR from inside to outside for primary rainbow and VIBGYOR from inside to outside for secondary rainbow
(C) ROYGBIV from inside to outside for primary rainbow and VIBGYOR from inside to outside for secondary rainbow
(D) ROYGBIV from inside to outside for primary rainbow and ROYGBIV from inside to outside for secondary rainbow
11. The least distance of distinct vision for a farsighted person is 1m. The optical power of the lens of his specs which effectively reduces his LDDV to 25 cm, is :-
(A) +3D (B) +2D (C) -3D (D) None
12. A beam of light has a small wavelength spread $\delta\lambda$ about a central wavelength λ . The beam travels in vacuum until it enters a glass plate at an angle θ relative to the normal to the plate, as shown in the figure. The index of refraction of the glass is given by $n(\lambda)$. The angular spread $\delta\theta'$ of the refracted beam is given by :-

(A) $\delta\theta' = \left| \frac{1}{n} \delta\lambda \right|$

(B) $\delta\theta' = \left| \frac{dn(\lambda)}{d\lambda} \delta\lambda \right|$

(C) $\delta\theta' = \left| \frac{1}{\lambda} \frac{d\lambda}{dn} \delta\lambda \right|$

(D) $\delta\theta' = \left| \frac{\tan \theta'}{n} \frac{dn(\lambda)}{d\lambda} \delta\lambda \right|$



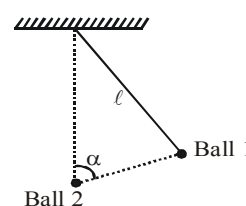
13. A small charged ball 'q' of mass 'm' is suspended on an insulating string of length ℓ . Another identical charged ball is slowly moved by a student towards the first one from a large distance. Eventually, the second ball is placed at the original location of the first one as shown in figure. At that moment, the first ball is elevated a small distance h above its original position and remains at rest. Then

(A) $\cos\alpha = \frac{kq^2}{mg\ell h}$

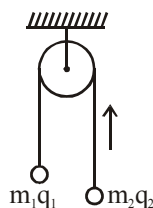
(B) $\cos\alpha = \frac{kq^2}{2mgh\ell}$

(C) $\cos\alpha = \frac{kq^2}{4mgh\ell}$

(D) $\cos\alpha = \frac{kq^2}{8mgh\ell}$



14. Two small balls carrying masses and charges m_1, q_1 & m_2, q_2 respectively. The entire system is placed in uniform electric field $\vec{E}$ directed upward. If $(m_1 + m_2)g = (q_1 + q_2)E$, then the acceleration of centre of mass. (Neglect the interaction between balls).

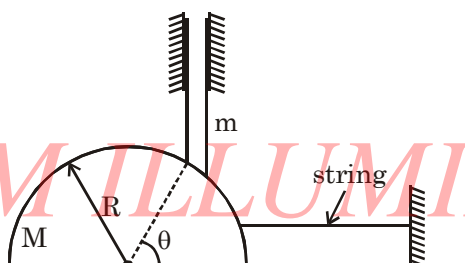


- (A) is necessarily zero
(B) is not necessarily zero.
(C) is directed downward
(D) none

Multiple Correct Answer Type

1 Q. [4 M (0)]

15. A hemisphere of mass M and radius R rests on smooth horizontal ground. A vertical rod of mass m is held between two smooth guide walls supported on sphere as shown. A horizontal string is tied to sphere keeps the system at rest then :



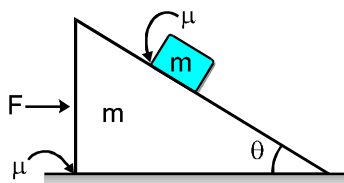
- (A) Tension in string is $mg \cot \theta$
(B) Tension in string is $Mg \cot \theta$
(C) Acceleration of hemisphere is just after string is cut $\frac{mg}{M \tan \theta + m \cot \theta}$
(D) Acceleration of hemisphere is just after string is cut $\frac{mg}{M \tan \theta - m \cot \theta}$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para x 2Q.) [3 M (0)]

Paragraph for Question No. 16 and 17

In the situation shown in figure a wedge of mass m is placed on a rough surface, on which a block of equal mass is placed on the inclined plane of wedge. Friction coefficient between plane and the block and the ground and the wedge is (μ) . An external force F is applied horizontally on the wedge. Given that m does not slide on incline due to its weight.



16. The value of F at which wedge will start slipping is :
(A) $F \geq \mu mg$ (B) $F \geq (3/2) \mu mg$ (C) $F \geq 2 \mu mg$ (D) $F \leq \mu mg$
17. The value of F at which no friction will act on block on inclined plane, is :
(A) $2 \mu mg$ (B) $2 \mu mg + 2 mg \tan \theta$
(C) $2 \mu mg + mg \tan \theta$ (D) $2 \mu mg + mg \sin \theta$

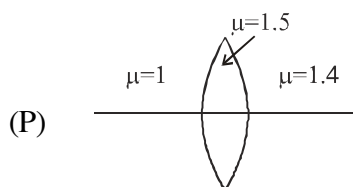
Matching List Type (4 × 4)

1Q.[3 M (0)]

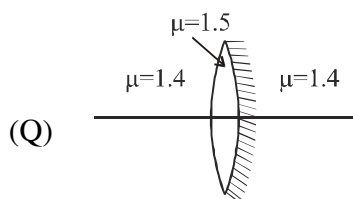
18. List-I gives different lens configurations. The radius of curvature of each curved surface is R. Parallel paraxial rays of light parallel to the axis of lens traversing through the lens get focused at distance f from the lens. List-II gives corresponding values of f. (μ represent refractive index of medium)

List-I

List-II

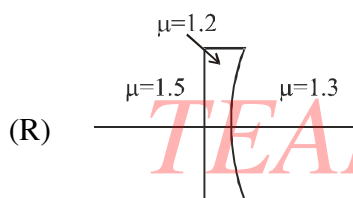


(1) 13 R



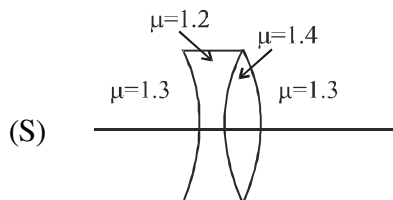
(2) $\frac{13R}{4}$

Lens is silvered



(3) $\frac{7R}{3}$

Plano concave lens



(4) $\frac{7R}{16}$

Codes :

	P	Q	R	S		P	Q	R	S
(A)	4	1	3	2	(B)	2	3	4	1
(C)	1	2	4	3	(D)	3	4	1	2

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q.[3(0)]

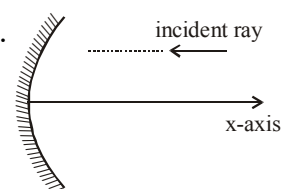
1. A ray of light enters a rectangular glass slab of refractive index $\sqrt{3}$ at an angle of incidence 60° . It travels a distance of 5cm inside the slab and emerges out of the slab. Find out the perpendicular distance (in cm) between the incident and the emergent ray.

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)

1 Q. [4 M(-1)]

1. A parabolic reflecting surface given by $x = \frac{y^2}{2}$, is placed at origin, as shown. An incident ray moving along $y = 1$ is incident on it. The ray gets reflected by the surface twice. The deviation suffered by the ray is $\frac{\pi}{n}$ radians. Find n.



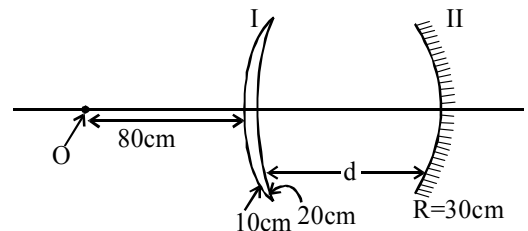
SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

1. If final image formed after two refractions through the lens of refractive index 1.5 and one reflection from the mirror forming at same point 'O' then d is equal to :-

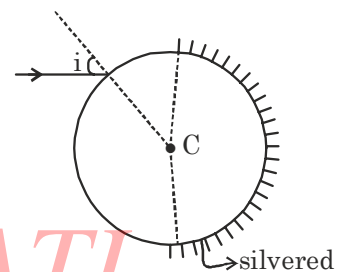
(A) 100 cm
(B) 120 cm
(C) 90 cm
(D) 80 cm



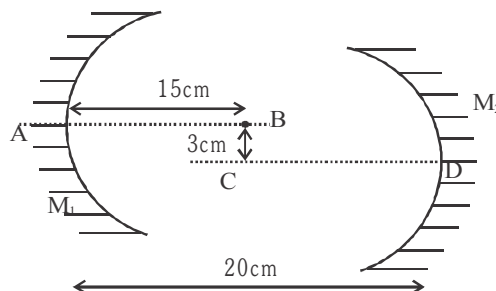
2. A ray of light is incident on the surface of a sphere of refractive index $\frac{\sqrt{7}}{2}$ other half of sphere is silvered. After refraction it is reflected and then refracted out of sphere again such that the total deviation is minimum then total deviation

of the ray will be $\left(\sin 41^\circ = \sqrt{\frac{3}{7}} \right)$

(A) 102° (B) 98°
(C) 136° (D) 163°



3. M_1 and M_2 are two concave mirrors of the same focal length 10 cm. AB & CD are their principal axes respectively. An object is kept on the line AB at distance 15 cm from M_1 . The distance between the mirrors is 20 cm. Considering two successive reflections, first on M_1 and then on M_2 . The distance of final image from the line AB is :-



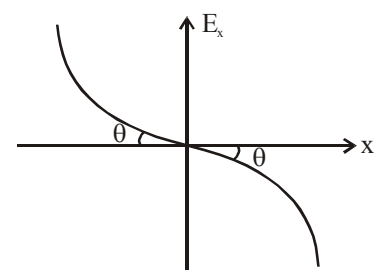
(A) 3 cm (B) 1.5 cm (C) 4.5 cm (D) 1 cm

4. A sphere of radius R carries charge density proportional to the square of the distance from the center: $\rho = Ar^2$, where A is a positive constant. At a distance of $R/2$ from the center, the magnitude of the electric field is

(A) $A/(4\pi\epsilon_0)$ (B) $AR^3/(40\epsilon_0)$ (C) $AR^3/(24\epsilon_0)$ (D) $AR^3/(5\epsilon_0)$
(E) $AR^3/(3\epsilon_0)$

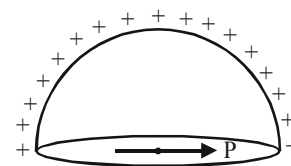
5. Variation of x-component of electric field with x-coordinate in a region is given in the graph. A small electric dipole having dipole moment $P_0 \hat{i}$ is placed at origin, then net force on the dipole is :

(A) $P_0 \tan \theta$ along positive x-axis
(B) Zero
(C) $P_0 \tan \theta$ along negative x-axis
(D) $P_0 \tan \theta$ along positive y-axis



6. A dipole is kept in circular plane of a uniformly charged hemispherical shell. Center of dipole coincide with center of hemispherical shell and dipole is restricted to move in given circular plane. Identify it's type of equilibrium if dipole is slightly rotate in this plane :-

(A) Neutral (B) Unstable
(C) Stable (D) Depends upon the direction of rotation

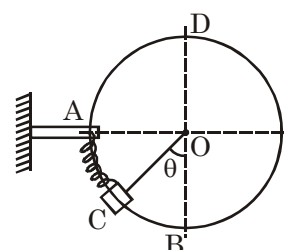


Multiple Correct Answer Type

6 Q. [4 M (0)]

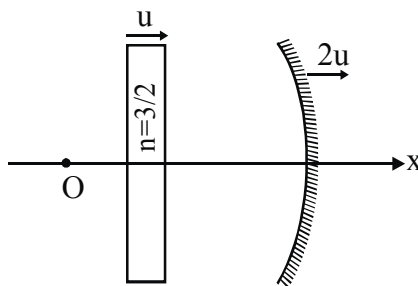
7. The potential energy U of a particle is given by $U = \{20 + (x - 4)^2\}J$. Total mechanical energy of the particle is 36 J. Select the correct alternative(s)
- (A) The particle oscillates about point $x = 4$ m
(B) The amplitude of the particle is 4 m
(C) The kinetic energy of the particle at $x = 2$ m is 12 J
(D) The motion of the particle is periodic but not simple harmonic

8. A thin circular rod is supported by a hinge at A. Attached to point A is a spring of spring constant k , and natural length equal to arc length AB. A bead of mass m , not attached to the spring, can slide without friction along the rod. The bead is released at rest making an angle θ with the vertical.

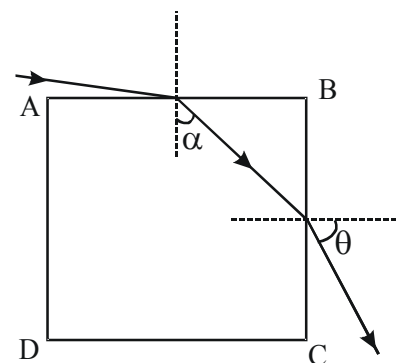


- (A) For the bead to complete the vertical circle, $k > \frac{2mg}{R\theta^2}(1 + \cos\theta)$
(B) For all the values of k , normal will remain radially inwards for the complete motion.
(C) Tangential acceleration of block is maximum at point B.
(D) Normal force on the bead at point B just after losing contact with spring is $N = \frac{19}{9}mg$ for $k = \frac{mg}{\pi^2 R}$ and $\theta = 60^\circ$.

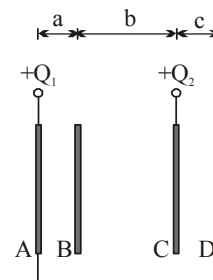
9. For the system shown in the figure, the image formed by the concave mirror



- (A) may have speed greater than the speed of the object
(B) will have positive velocity.
(C) may have speed greater than the speed of the mirror
(D) sign of velocity will depend upon thickness of slab.
10. ABCD is plane glass cube. A nearly horizontal beam of light enters the face AB at grazing incidence. Then,
- (A) $\sin\theta = \cot\alpha$, where α is critical angle
(B) Critical angle $\alpha \leq 45^\circ$, for TIR at second surface
(C) Refracting index $\mu \leq \sqrt{2}$ for TIR at second surface
(D) None of these



11. Figure shows an arrangement of four identical rectangular plates A, B, C and D each of area S. Find the charges appearing on each face (from left to right) of the plates. Ignore the separation between the plates in comparison to the plate dimensions.



- (A) Potential difference between plates A & B is independent of Q_1 .
(B) Potential difference between plates C & D is independent of Q_1 .
(C) Potential difference between plates A & B is independent of Q_2 .
(D) Potential difference between plates C & D is independent of Q_2 .

12. Two concentric conducting shells have radii R and 2R; charges q_A and q_B and potentials 2V and $\frac{3V}{2}$ respectively if reference for electric potential is taken at infinity. Now shell B is earthed and let charges on them become q'_A and q'_B . Then:

(A) $\frac{q_A}{q_B} = \frac{1}{2}$

(B) $\frac{q'_A}{q'_B} = 1$

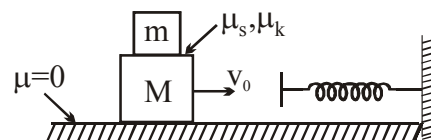
(C) Potential of A after earthing becomes $\frac{3V}{2}$

(D) Potential difference ($V_A - V_B$) between A and B after earthing becomes V/2

Linked Comprehension Type (2 Para × 3Q. & 1 Para × 2Q.) [3 M (0)]
(Single Correct Answer Type)

Paragraph for question nos. 13 to 15

A block of mass M slides on a frictionless surface with an initial speed of v_0 . On top of block is a small box of mass m. The coefficients of friction between box and block are μ_s and μ_k . The sliding block encounters an ideal spring with force constant k. Answer following questions.



13. Assuming no relative motion between box and block what is the maximum possible acceleration of block and box at the instant of maximum compression ?

(A) $\mu_s g$ (B) $\frac{\mu_s Mg}{m}$ (C) $\frac{\mu_s (M+m)g}{m}$ (D) $\frac{\mu_s mg}{M}$

14. What is maximum value of k for which it remains true that box does not slide ?

(A) $\left(\frac{\mu_s g}{v_0}\right)^2 \frac{M}{(M+m)}$ (B) $\left(\frac{\mu_s g}{v_0}\right)^2 (M+m)$ (C) $\left(\frac{\mu_s g}{2v_0}\right)^2 \frac{(M+m)^2}{M}$ (D) None of these

15. Suppose the value of k is just slightly greater than the value found in previous question, so that the box begins to slide just as the spring reaches maximum compression. The accelerations of box and block are respectively.

(A) $a_{\text{Box}} = \mu_k g$ $a_{\text{Block}} = \frac{m(\mu_s - \mu_k) + \mu_s M}{M} g$

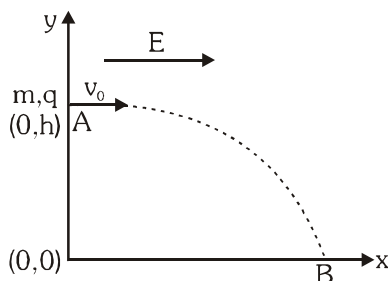
(B) $a_{\text{Box}} = (\mu_s - \mu_k)g$ $a_{\text{Block}} = (\mu_s - \mu_k) \frac{mg}{M}$

(C) $a_{\text{Box}} = \mu_k g$ $a_{\text{Block}} = (\mu_s - \mu_k) \frac{mg}{M}$

(D) None of these

Paragraph for Questions no. 16 to 18

A particle of mass m and positive charge ' q ' is projected horizontally from height ' h ' with velocity v_0 as shown in figure. A uniform electric field is acting in +ve x -direction. When the particle reaches at points B then find out



16. Time taken by the particle to reaches at point B

- (A) $v_0 \sqrt{\frac{qE}{m}}$ (B) $\sqrt{\frac{2h}{g}}$ (C) $v_0 \sqrt{\frac{m}{qE}}$ (D) $\sqrt{\frac{h}{2g}}$

17. The x -coordinate of point B

- (A) $v_0 \sqrt{\frac{2h}{g}}$ (B) $\frac{qE}{m} \left(\frac{2h}{g} \right)$ (C) $v_0 \sqrt{\frac{2h}{g}} + \frac{qEh}{2mg}$ (D) $v_0 \sqrt{\frac{2h}{g}} + \left(\frac{qEh}{mg} \right)$

18. Work done by electric field

- (A) $qE \left(v_0 \sqrt{\frac{2h}{g}} + \frac{qEh}{mg} \right)$ (B) $qE \left(v_0 \sqrt{\frac{h}{g}} + \frac{2qEh}{mg} \right)$ (C) $qE \left(v_0 \sqrt{\frac{h}{2g}} + \frac{2qEh}{mg} \right)$ (D) $qE \left(v_0 \sqrt{\frac{h}{g}} + \frac{qEh}{2mg} \right)$

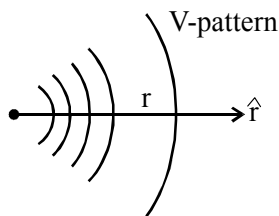
Paragraph for Question Nos. 19 & 20

Equipotential surfaces are the planes where potential is constant for any point in the plane. The potential at any point on the equipotential surface is the net potential of all charges of the system given as $V = \int dV$;

since $V = \text{constant}$, no work is done on moving a charge on equipotential surface because $\vec{F}_{el} \cdot d\vec{s} = 0$.

Based on the above facts answer the following questions :-

19. The circular equipotential lines are given such that $Vr = -60$. Then, the field at a distance r is :-



- (A) $\frac{60}{r^2} \hat{r}$ (B) $\frac{60}{r^2} (-\hat{r})$ (C) $\frac{60}{r} \hat{r}$ (D) $\frac{60}{r} (-\hat{r})$

20. Since $V = \text{constant}$ on an equipotential surface, $\vec{E}$ at any point of the equipotential surface is :-

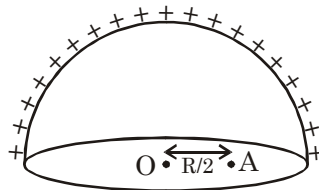
- (A) Zero (B) Perpendicular to the equipotential surface
(C) Non-zero along equipotential surface (D) Not defined

SECTION-I

Single Correct Answer Type

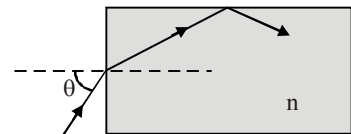
3 Q. [3 M (-1)]

1. A hemispherical bowl of mass m is uniformly charged with charge density ' σ '. Electric potential due to charge distribution at a point 'A' is (which lies at centre of radius as shown).



- (A) $\frac{\sigma R}{4\epsilon_0}$ (B) $\frac{\sigma R}{3\epsilon_0}$ (C) $\frac{\sigma R}{2\epsilon_0}$ (D) $\frac{\sigma R}{\epsilon_0}$

2. A model of an optical fiber is shown in the figure below. The optical fiber has an index of refraction, n , and is surrounded by free space. What angles of incidence, θ , will result in the light being totally internally reflected at the second surface?



- (A) $\theta > \sin^{-1}(\sqrt{n^2 - 1})$ (B) $\theta < \sin^{-1}(\sqrt{n^2 - 1})$
 (C) $\theta < \sin^{-1}(\sqrt{n^2 + 1})$ (D) $\sin^{-1}(\sqrt{n^2 - 1}) < \theta < \sin^{-1}(\sqrt{n^2 + 1})$

3. What must be the lengths of steel and copper rods at 0°C for the difference in their lengths to be 10 cm at any common temperature? ($\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{K}^{-1}$ and $\alpha_{\text{copper}} = 1.8 \times 10^{-5} \text{K}^{-1}$)

- (A) 30 cm for steel and 20 cm for copper (B) 20 cm for steel and 30 cm for copper
 (C) 40 cm for steel and 30 cm for copper (D) 30 cm for steel and 40 cm for copper

Multiple Correct Answer Type

15 Q. [4 M (0)]

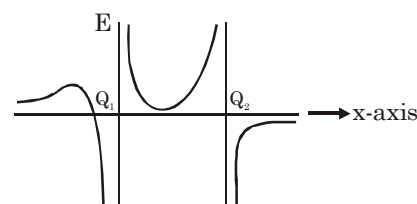
4. Three infinite line charges of charge per unit length λ , 2λ and $-\lambda$ are kept in xy -plane parallel to the y -axis. They pass through $x = -a$, $x = a$ and $x = 2a$ respectively. Find the electric field at different points on the x -axis :-

- (A) At $x = -2a$, $E = -\frac{25k\lambda}{6a} \hat{i}$ (B) At $x = 0$, $E = -\frac{k\lambda}{a} \hat{i}$
 (C) At $x = \frac{3a}{2}$, $E = \frac{64k\lambda}{5a} \hat{i}$ (D) At $x = 0$, $E = -\frac{2k\lambda}{3a} \hat{i}$

5. A point charge $+Q$ is fixed at a point O. Another point charge $+q$ & mass m is projected from infinity towards $+Q$ with speed v_0 . Minimum distance of approach is r . If it is now given speed $2v_0$ at infinity then

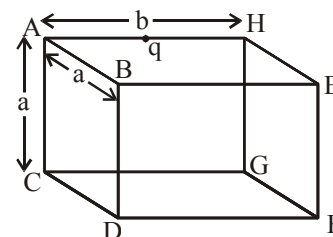
- (A) Minimum distance of approach becomes $\frac{r}{2}$
 (B) Minimum distance of approach becomes $\frac{r}{4}$
 (C) Angular momentum conservation will hold true about O.
 (D) Mechanical energy will remain conserved throughout the motion.

6. The graph shown below is relationship between electric field and distance. The electric field is due to 2 point charges Q_1 & Q_2 on x-axis. Select the correct statements from the following statement(s) :



- (A) Q_1 and Q_2 are unlike charges
(B) Q_1 and Q_2 are like charges
(C) $|Q_1| < |Q_2|$
(D) $|Q_2| < |Q_1|$

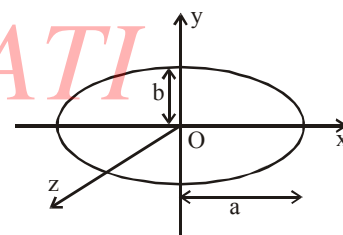
7. A cuboid of dimension $[a \times a \times b]$. Charge q is placed at the centre of edge having length 'b'. If flux through face 'ABCD' is $\frac{q}{32\epsilon_0}$ then select the correct statement(s) :-



- (A) Flux through the entire cuboid is $\frac{q}{4\epsilon_0}$
(B) Flux through the face 'ABEH' is zero.
(C) Flux through the face 'BEFD' is $\frac{3q}{32\epsilon_0}$
(D) Flux through the face 'BEFD' is $\frac{q}{16\epsilon_0}$

8. An ellipsoidal gaussian surface with semi major axis a and minor axis b is

shown in figure. An electric field $\vec{E} = \frac{C(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$ exist in space.



- (A) Net flux through constant gaussian surface is $4\pi C$
(B) Net charge enclosed by gaussian surface is $4\pi C\epsilon_0$
(C) If given surface is a conductor electric potential at all its inside points may be constant for given field function
(D) If given surface is a conductor electric field at all its inside points may be constant for given field function

9. A spherical volume centered at the origin of coordinates with radius a , has a volume charge density

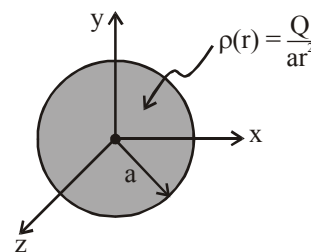
$\rho(r) = +\frac{Q}{a}\left(\frac{1}{r^2}\right)$. The electric field :-

(A) at $r = \frac{a}{2}$; $E = \frac{2Q}{a^2\epsilon_0}$

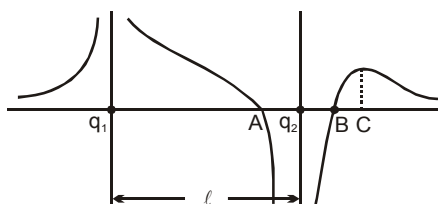
(B) at $r = a$; $E = \frac{Q}{a^2\epsilon_0}$

(C) at $r = 2a$; $E = \frac{Q}{2a^2\epsilon_0}$

(D) at $r = 4a$; $E = \frac{Q}{16a^2\epsilon_0}$



10. The curve represents distribution of potential along a line joining two charges q_1 and q_2 (separated by distance ℓ) then which of the following statements are incorrect ?



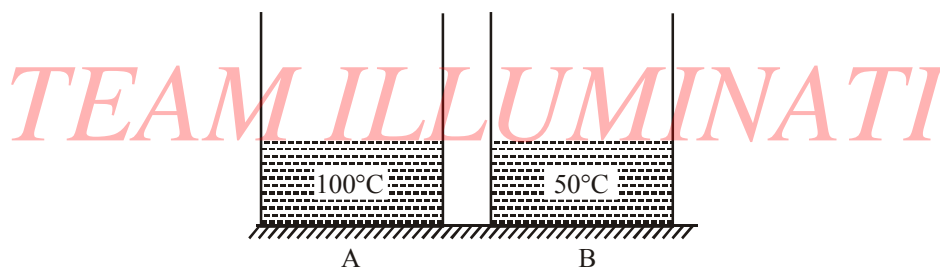
(A) $|q_1| < |q_2|$

(B) q_1 is negative in nature

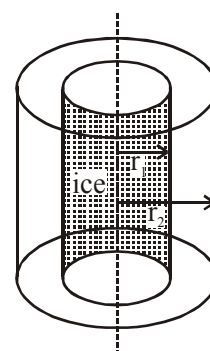
(C) A and B are equilibrium points

(D) C is point of unstable equilibrium

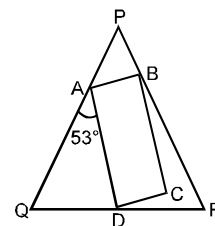
11. The electric field in x-y plane is given by $\vec{E} = x^2\hat{i} + y^2\hat{j}$. Potential at point (3, 3) is taken as zero. Then :-
 (A) Work done in moving a test charge q slowly from a point (a, b) to a point (b, a) is zero.
 (B) Potential at any point on x-y plane can't exceed 18 unit for x & y in 1st quadrant.
 (C) Potential can't be defined as field is non- conservative in nature.
 (D) Equipotential curve are concentric circles with centre (3, 3)
12. A rod 'PQ' of length ' ℓ ' is pivoted at an end P and freely rotated in a horizontal plane at an angular speed ' ω ' about a vertical axis passing through P. If coefficient of linear expansion of material of rod is α and temperature of rod is increased by ΔT . Select the **CORRECT** alternatives.
 (A) Angular momentum of rod about axis of rotation will increase.
 (B) Angular momentum of rod about axis of rotation will remains constant.
 (C) Final angular velocity of rod after heating is $\omega[1 + 2\alpha\Delta T]$.
 (D) Final angular velocity of rod after heating is $\omega[1 - 2\alpha\Delta T]$.
13. 2 kg water at 100°C and 2.5 kg water at 50°C is kept in two identical containers A and B respectively of water equivalent 0.5 kg. If water of container A is poured into container B the final temperature of mixture is T_1 and if water of container B is poured into container A the final temperature is T_2 . (heat loss is negligible)



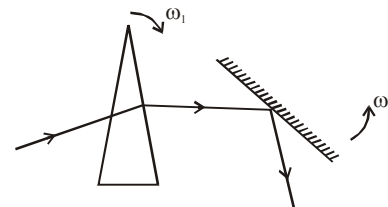
- (A) $15T_1 = 14T_2$ (B) $T_1 = T_2$ (C) $T_1 = 70^\circ\text{C}$ (D) $T_2 = 70^\circ\text{C}$
14. A 100 cm long cylindrical flask with inner and outer diameter 2 cm and 4 cm respectively is completely filled with ice as shown in the figure. The constant temperature outside the flask is 40°C. (Thermal conductivity of the flask is 0.693 W/m°C, $L_{\text{ice}} = 80 \text{ cal/gm}$ & $\ln 2 = 0.693$).
 (A) Rate of heat flow from outside to the flask is $80\pi \text{ J/s}$
 (B) The rate at which ice melts is $\frac{\pi}{4200} \text{ kg/s}$
 (C) The rate at which ice melts is $100\pi \text{ kg/s}$
 (D) Rate of heat flow from outside to flask is $40\pi \text{ J/s}$
15. The two ends of a well lagged uniform rod (thermal conductivity K) are maintained at different but constant temperatures. The temperature gradient at any point on the rod is $\frac{dT}{d\ell}$ in steady state. The heat flow per unit time per unit cross-section of the rod is I. Which statements is/are correct :-
 (A) $\frac{dT}{d\ell}$ is the same for all points on the rod.
 (B) I will increase as we move from higher to lower temperature.
 (C) $I = K \frac{dT}{d\ell}$
 (D) In steady state the temperature gradient is zero at the ends of the rods.



16. A rectangular cavity ABCD is carved inside an equilateral prism PQR of refractive index $\sqrt{2}$ shown in the figure. For a ray entering at face PQ and emerging at face PR, without passing through AB and CD:

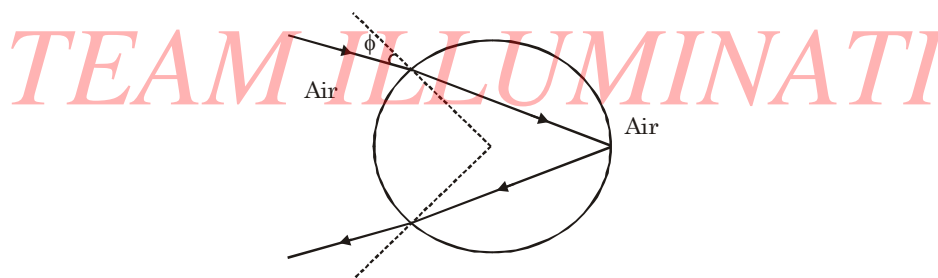


- (A) there will not be any effect of cavity on δ
(B) there may be effect of cavity on δ
(C) for cavity filled with water $\delta_{\min} = 30^\circ$
(D) for cavity filled with air $\delta_{\min} = 30^\circ$
17. For the fixed incident beam shown in figure. Choose the correct option regarding the emergent beam at the given instant, given that prism is thin, and light is incident on the prism at a very small angle. Prism and mirror are rotated at angular velocity ω_1 and ω_2 in directions as shown :-



- (A) Emergent beam will rotate at $\omega_1 + \omega_2$
(B) Emergent beam will rotate at $\omega_1 + 2\omega_2$
(C) Emergent beam will rotate at $2\omega_2$
(D) Emergent beam will rotate at $2\omega_1$

18. A ray of light enters a spherical drop of a liquid of refractive index $\mu = \frac{\sqrt{7}}{2}$. This ray after getting reflected from back surface comes out as shown. Given $\sin 41^\circ = \sqrt{\frac{3}{7}}$:-



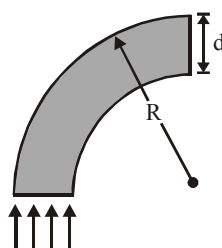
- (A) Ray suffers total internal reflection at back surface.
(B) Total internal reflection at back surface is not possible for any value of ϕ .
(C) Total deviation suffered by ray (Refraction + Reflection + Refraction) is minimum, when $\phi = 60^\circ$
(D) When angle of incidence is chosen such that total deviation is minimum, then value of total deviation is 136° .

SECTION-III

Numerical Grid Type (Ranging from 0 to 999)

2 Q. [4 M(-1)]

1. A solid hemisphere has radius R and uniform charge density ρ . The electric field at the centre is E. The value of $\frac{\rho R}{E \epsilon_0}$ is
2. An optical fiber has index of refraction n and diameter d. It is surrounded by air. Light is sent into the fiber along its axis as shown in figure. Find the smallest outside radius R permitted for a bend in the fiber if no light is to escape. Evaluate R (in μm) assuming the fiber diameter is $100 \mu\text{m}$ and its index of refraction is 1.40.



PHYSICS

QUIZ # 10

SECTION-I

Single Correct Answer Type

9 Q. [3 M (-1)]

1. Three charges with magnitudes q , $2q$, $3q$ are placed at the vertices of an equilateral triangle whose each side is of length a . Only by changing the signs of charges, we change the electrostatic energy of the configuration. Then the energy gap between the most stable and most unstable configuration is :

(A) $24 \left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \right)$ (B) $12 \left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \right)$ (C) $18 \left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \right)$ (D) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{a}$

2. The potential (in volts) of a charge distribution is given by

$$V(z) = 30 - 5z^2 \text{ for } |z| \leq 1 \text{ m}$$

$$V(z) = 35 - 10|z| \text{ for } |z| \geq 1 \text{ m.}$$

$V(z)$ does not depend on x and y . If this potential is generated by a uniform charge per unit volume ρ_0 (in units of ϵ_0) which is spread over a certain region, then choose the correct statement.

(A) $\rho_0 = 20 \epsilon_0$ in the entire region

(B) $\rho_0 = 10 \epsilon_0$ for $|z| \leq 1 \text{ m}$ and $\rho_0 = 0$ elsewhere

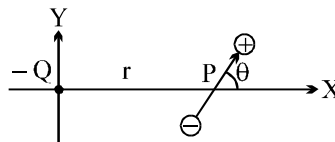
(C) $\rho_0 = 40 \epsilon_0$ in the entire region

(D) $\rho_0 = 20 \epsilon_0$ for $|z| \leq 1 \text{ m}$ and $\rho_0 = 0$ elsewhere

3. Two small dipoles of dipole moment $\vec{p}_1$ and $\vec{p}_2$ are at a distance x from each other and $\vec{p}_1 \parallel \vec{p}_2$. The force between the dipoles is :

(A) $\frac{1}{4\pi\epsilon_0} \frac{4p_1p_2}{x^4}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{3p_1p_2}{x^3}$ (C) $\frac{1}{4\pi\epsilon_0} \frac{6p_1p_2}{x^4}$ (D) $\frac{1}{4\pi\epsilon_0} \frac{8p_1p_2}{x^4}$

4. A point negative charge $-Q$ is placed at a distance r from a dipole with dipole moment P as shown in figure. The x component of force acting on the charge $-Q$ is -



(A) $-\frac{PKQ}{r} \cos \theta \hat{i}$ (B) $\frac{PKQ}{r} \cos \theta \hat{i}$ (C) $-\frac{2PKQ}{r^3} \cos \theta \hat{i}$ (D) $\frac{2PKQ}{r^3} \cos \theta \hat{i}$

5. The equation of equipotential lines in x - y plane for a dipole kept at the origin & oriented along x -axis is

(A) $\frac{x}{(x^2 + y^2)^{3/2}} = \text{const}$

(B) $\frac{y}{(y^2 + x^2)^{3/2}} = \text{const}$

(C) $\frac{y^2}{(x^2 + y^2)^{3/2}} = \text{const}$

(D) $\frac{x^2}{(x^2 + y^2)^{3/2}} = \text{const}$

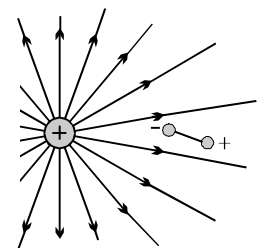
6. An electric dipole, consisting of two equal and opposite point charges at the ends of a insulating rod, is placed in the electric field of a stationary positive point charge, as shown. The dipole is free to move. The dipole will

(A) rotate clockwise and move to the left

(B) rotate counterclockwise and move to the right

(C) rotate clockwise and move to the right

(D) rotate counterclockwise and move to the left



7. Satellites A and B are orbiting around the earth in orbits of ratio R and $4R$ respectively. The ratio of their areal velocities is:

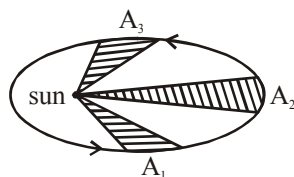
(A) $1 : 2$

(B) $1 : 4$

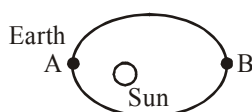
(C) $1 : 8$

(D) $1 : 16$

8. A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days. Then the relation between A_1 , A_2 and A_3 is :-



- (A) $3A_1 = 2A_2 = A_3$ (B) $2A_1 = 3A_2 = 6A_3$ (C) $3A_1 = 2A_2 = 6A_3$ (D) $6A_1 = 3A_2 = 2A_3$
9. The earth is moving around the sun in an elliptical orbit. Point A is the closest and point B is the farthest point in the orbit, as shown. In comparison to the situation when the earth passes through point B :



- (A) total energy of the earth-sun system is greater when the earth passes through point A.
(B) gravitational potential energy of the earth-sun system is greater when the earth passes through point A.
(C) kinetic energy of the earth due to the motion around the sun is greater when it passes through the point A.
(D) magnitude of angular momentum of the earth about the sun is greater when the earth passes through point A.

Multiple Correct Answer Type

11 Q. [4 M (0)]

10. If instead of coulomb's law, the electric force between two point charges q and Q placed at r is given by

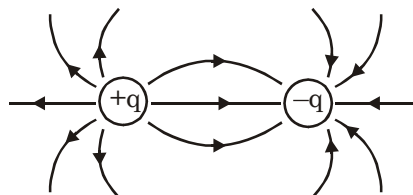
$$\vec{F} = \frac{kQq(\alpha - r)}{r^3} \vec{r}, \quad \vec{r} \text{ is position vector of one charge relative to other, } \alpha \text{ is constant:-}$$

- (A) $\vec{E}$ due to point charge Q is $\vec{E} = \frac{kQ(\alpha - r)}{r^3} \vec{r}$.
(B) Gauss law $\oint \vec{E} \cdot d\vec{s} = \frac{q_{\text{enclosed}}}{\epsilon_0}$ holds true.
(C) $\oint \vec{E} \cdot d\vec{\ell}$, line integral over a closed path is zero.
(D) For isolated charges the field would vanish at finite distance from charge.
11. A satellite is moving in circular orbit of r with speed v_0 and time period T_1 . A particle of small mass is projected from satellite in direction of its motion with relative speed $\left(\frac{2}{\sqrt{3}} - 1\right)v_0$:-
- (A) Maximum distance of particle from center of earth is $2r$
(B) Maximum distance of particle from center of earth is $(\sqrt{5} - 1)r$
(C) Time period of particle around earth $> T_1$
(D) Time period of particle around earth $< T_1$

12. Two equal and opposite charges are separated by a distance as shown.

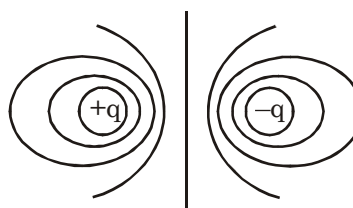


- (A) Electric field lines for this system can be represented by

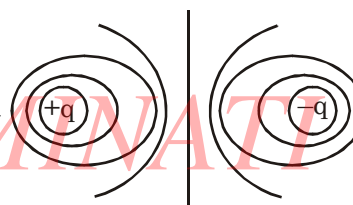


- (B) There can not be any null point for this system {exclude points at ∞ }

- (C) Equipotential lines for the system can be represented by



- (D) Equipotential lines for the system can be represented by



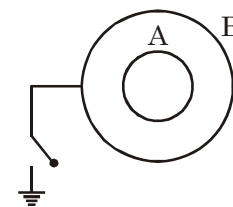
13. Two concentric shells having radius R and $2R$ have charges q_A and q_B and potentials $2V$ and $(3/2)V$ respectively. Now shell B is earthed and let new charges be q'_A and q'_B . Then

(A) $q'_A/q'_B = 1/2$

(B) $q'_A/q'_B = 1$

(C) potential of A after earthing becomes $(3/2)V$

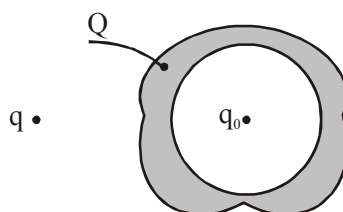
(D) potential difference between A and B after earthing becomes $V/2$



14. In an electrostatic field $\vec{E}(x, y, z) = [2y\hat{i} + (2x + 3z)\hat{j} + 3y\hat{k}]$ V/m where x, y, z are in m, the points A = $(2, 3, -2)$ m, B = $(1, 2, -1)$ m, C = $(4, 3, 2)$ m, O = $(0, 0, 0)$ m

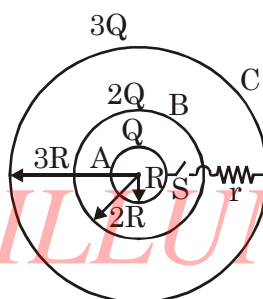
(A) $V_A - V_B = -4V$ (B) $V_A - V_O = 6V$ (C) $V_O - V_C = -46V$ (D) $V_B - V_C = 44V$

15. Consider a conductor of arbitrary shape with a spherical cavity inside it. A charge q_0 is placed inside cavity, a charge Q is given to conductor and a charge q is placed outside conductor. Mark the **CORRECT** statement (s)

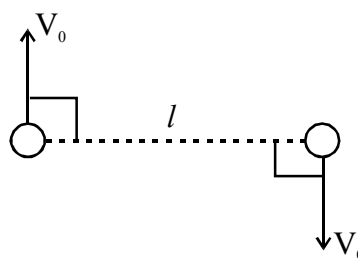


- (A) If cavity is spherical and charge q_0 is placed asymmetrically inside it, it experiences a resultant force
(B) Force due to charges on outer surface of conductor on q_0 is non zero.
(C) Field of external charge q is cancelled off for all inside points by charges appearing on outer surface of conductor
(D) Charges on inner surface of cavity cancel off field of charge q_0 for all points outside cavity

16. Two conducting spheres carry equal charges. The distance between spheres is not large as compared to the size of spheres :-
- If they are like charges then the magnitude of force of interaction is less in comparison to force between point charges of same charge magnitude
 - If they are unlike charges then the magnitude of force of interaction is more in comparison to force between point charges of same charge magnitude
 - If they are like charges then the magnitude of force of interaction is same in comparison to force between point charges of same charge magnitude
 - If they are unlike charges then the magnitude of force of interaction is same in comparison to force between point charges of same charge magnitude
17. Three thin concentric conducting spherical shells A, B, C of radius R , $2R$ and $3R$ are given charges Q , $2Q$, $3Q$ respectively as shown. The inner most and outermost can be joined through switch 'S' which is initially opened. Choose the **INCORRECT** options :-



- After closing switch S, potential of shell C does not change.
 - After closing switch S, charge on shell A becomes zero.
 - After closing switch S, potential of shell B does not change.
 - After closing switch S, charge on shell C does not change.
18. Two masses m separated by a distance l are given initial velocities V_0 , as shown in the diagram. The masses interact only through mutual universal gravitation. Choose **CORRECT** statement(s)



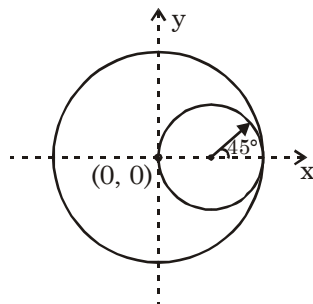
- For non zero value of V_0 , particles will never collide
- Minimum distance between particles can be l for a specific value of V_0

(C) If $\frac{Gm}{V_0^2 l} > 2$, then minimum distance between particles is $d = \frac{l}{\frac{Gm}{V_0^2 l} - 1}$

(D) If $\frac{Gm}{V_0^2 l} < 2$, then minimum distance between particles is $d = \frac{l}{\frac{Gm}{V_0^2 l} - 2}$

19. A spherical cavity of radius $\frac{R}{2}$ is made in a sphere of mass M and radius R as shown in figure. An

object is projected at an angle of 45° with speed $\sqrt{\frac{GM}{4R}}$ in the X-Y plane :-



- (A) The co-ordinate where the particle will hit the walls of cavity is $\left(\frac{R}{2}, \frac{R}{2}\right)$
 (B) The path of the particle in the cavity will be parabola
 (C) The time after which its velocity is parallel to y-axis is $\sqrt{\frac{R^3}{2GM}}$
 (D) The field inside the cavity is uniform
20. A particle of mass m is thrown in a parabolic path towards earth having mass M and radius R . It comes to a greatest approach distance $R/2$ from surface of earth. Magnitude of total energy change required to make the particle moving in an elliptical path having eccentricity $1/2$, with the greatest approach point either aphelion or perihelion are :

- (A) $\frac{1}{3} \frac{GMm}{R}$ (B) $\frac{1}{6} \frac{GMm}{R}$ (C) $\frac{1}{2} \frac{GMm}{R}$ (D) $\frac{GMm}{R}$

SECTION-I

Single Correct Answer Type

10 Q. [3 M (-1)]

1. A conducting open pipe has shape of a half cylinder of length ℓ . Its semicircular cross section has radius r and thickness of the conducting wall is $t (< r)$. The resistance of the conductor when the current enters and leaves as shown in Figure(a) is R_1 and its resistance is R_2 when the current is as shown in

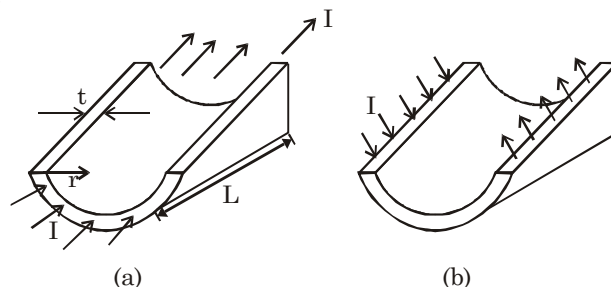
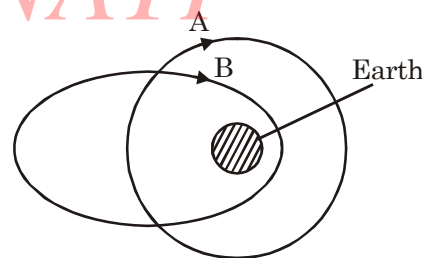


Figure (b). Find $\frac{R_1}{R_2}$

- (A) $\frac{L^2}{\pi^2 r^2}$ (B) $\frac{L}{\pi r}$ (C) $\frac{tL}{\pi r^2}$ (D) $\frac{rL}{\pi t^2}$

2. Satellite A is following a circular path of radius a around the earth. Another satellite B of same mass follows an elliptical path around the earth. Both of them have same mechanical energy. Their orbits intersect as shown. The speed of A and B are v_A & v_B at the intersecting point.



- (A) $v_A = v_B$ (B) $v_A = \frac{1}{\sqrt{2}} v_B$
 (C) $v_A = \frac{v_B}{2}$ (D) $v_A = \frac{v_B}{\sqrt{3}}$

3. A copper wire carries a current of 10 A. It has cross-sectional area of 0.05 cm^2 . Density of copper is 8.9 g/cm^3 . Molar mass of copper is $63.5 \times 10^{-3} \text{ kg/mol}$. The drift speed of the electrons is closest to

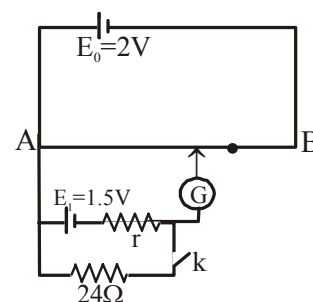
- (A) $1.5 \times 10^{-4} \text{ m/s}$ (B) $1.5 \times 10^4 \text{ m/s}$ (C) $3.5 \times 10^{-4} \text{ m/s}$ (D) 1.5 m/s

4. Two wires of same material and identical lengths are joined in parallel to a battery of constant voltage. If the radius of cross section of both the wires are in the ratio of 3:5 then the ratio of drift speed of electrons in the wire is given by :

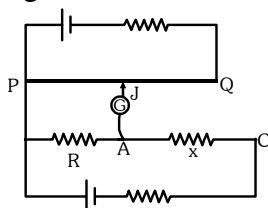
- (A) 9:25 (B) 5:3 (C) $\sqrt{3} : \sqrt{5}$ (D) None of these

5. For the arrangement of the potentiometer shown in the figure, the balance point is obtained at a distance 75 cm from A when the key k is open. The second balance point is obtained at 60 cm from A when the key k is closed. Find the internal resistance (in Ω) of the battery E_1 .

- (A) 3 (B) 6
 (C) 9 (D) 12



6. Circuit for the measurement of resistance by potentiometer is shown. The galvanometer is first connected at point A and zero deflection is observed at length $PJ = 10$ cm. In second case it is connect at point C and zere deflection is observed at a length 30cm from P. Then the unknown reitance X is

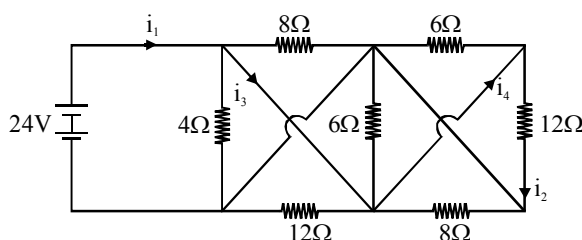


- (A) $2R$ (B) $\frac{R}{2}$ (C) $\frac{R}{3}$ (D) $3R$
7. Calculate energy needed for moving a mass of 4kg from the centre of the earth to its surface. If radius of the earth is 6400 km and acceleration due to gravity at the surface of the earth is $g = 10 \text{ m/sec}^2$
(A) $1.28 \times 10^8 \text{ J}$ (B) $1.28 \times 10^6 \text{ J}$ (C) $2.56 \times 10^8 \text{ J}$ (D) $2.56 \times 10^{10} \text{ J}$
8. Two satellites are launched at a distance R from a planet of negligible radius. Both satellites are launched in the tangential direction. The first satellite launches correctly at a speed v_0 and enters a circular orbit. The second satellite, however, is launched at a speed $\frac{1}{2}v_0$. What is the minimum distance between the second satellite and the planet over the course of its orbit?
(A) $\frac{1}{2}R$ (B) $\frac{1}{3}R$ (C) $\frac{1}{4}R$ (D) $\frac{1}{7}R$
9. Imagine a smooth tunnel along a chord of non rotating fixed earth at a distance $R/2$ from the centre. R is the radius of the earth. A projectile is fired along the tunnel from the centre of the tunnel at a speed $\sqrt{gR}$. g is acceleration due to gravity at the surface of the earth.
(A) The maximum speed of the projectile is at the center of the tunnel.
(B) The angular momentum of the projectile remains constant as it moves along the tunnel.
(C) The path of the projectile is a straight line.
(D) The speed of the projectile at its maximum distance from earth's centre is zero.
10. Imagine a planet whose diameter and mass are one half of those of the earth. The day's temperature of this planet reaches up to 800K. (Escape velocity on the surface of the earth is 11.2 km/sec, $k = 1.38 \times 10^{-23} \text{ J/K}$. Consider that the molecules possess average kinetic energy. Among the following, choose the wrong statement
(A) Oxygen molecules escape from the planet
(B) Oxygen molecules cannot escape from the planet
(C) Nitrogen molecules cannot escape from the planet
(D) Argon molecules cannot escape from planet.

Multiple Correct Answer Type

4 Q. [4 M (0)]

11. For the circuit shown in figure



- (A) $i_1 = 24 \text{ A}$ (B) $i_2 = 2 \text{ A}$ (C) $i_3 = 15 \text{ A}$ (D) $i_4 = 6 \text{ A}$

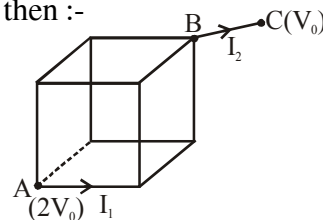
12. All 13 roads are identical & potential at A is maintained $2V_0$ and at C is V_0 then :-

(A) Potential at B is $\frac{6V_0}{5}$

(B) Potential at B is $\frac{17V_0}{11}$

(C) Ratio $\frac{I_1}{I_2}$ is $\frac{1}{3}$

(D) Ratio $\frac{I_1}{I_2}$ is $\frac{5}{18}$



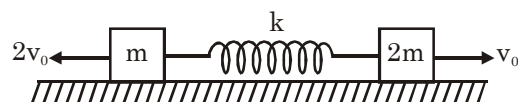
13. Two blocks of mass m and $2m$ are connected by a spring of force constant k on a smooth surface. At $t = 0$ the velocities are imparted with spring in natural length as shown in figure. Identify the **INCORRECT** statements. [v_{cm} = velocity of centre of mass of two blocks]

(A) The maximum elongation in the spring is $\sqrt{\frac{m}{k}} 6v_0$

(B) At maximum elongation both the blocks comes to rest.

(C) Kinetic energy of system remains constant = $\frac{1}{2} 3mv_{cm}^2$

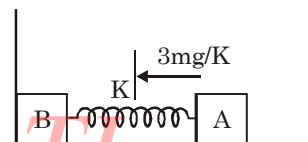
(D) Centre of mass moves with constant velocity towards left



14. A system consists of block A and B each of mass m connected by a light spring as shown with block B in contact with a wall. The block A compresses

the spring by $\frac{3mg}{K}$ from natural length of spring and then released from rest.

Neglect friction everywhere.



(A) Acceleration of centre of mass of system comprising A and B just after A is released, is $\frac{3g}{2}$

(B) Velocity of centre of mass of system comprising A and B when block B just loses contact with the

wall is $\frac{3g}{2} \sqrt{\frac{m}{k}}$

(C) Maximum extension in the spring after system loses contact with wall is $\frac{3mg}{\sqrt{2k}}$

(D) Maximum extension in the spring after system loses contact with wall is $\frac{3mg}{2k}$

Linked Comprehension Type

(1 Para × 2Q.) [3 M (0)]

(Single Correct Answer Type)

Paragraph for Question no. 15 and 16

When a battery of emf E and internal resistance r is connected to a load resistance R , current $I = \frac{E}{r + R}$

start in the circuit and battery start working. Chemical energy of battery start converting into heat, some part through load resistance and other part through internal resistance. Power supplied by battery to load resistance is $= I \Delta V$; ΔV is potential difference across the load resistance.

For two different value of resistance R_1 and R_2 , power supplied by battery is same. Internal resistance of battery is 8Ω . We can choose only integer value of load resistance.

15. Maximum value of $R_1 + R_2$ is :

(A) 16Ω

(B) 65Ω

(C) 8Ω

(D) 13Ω

16. $R_1 + R_2$ can't be equal to :
(A) 34Ω (B) 65Ω (C) 20Ω (D) 32Ω

SECTION-III

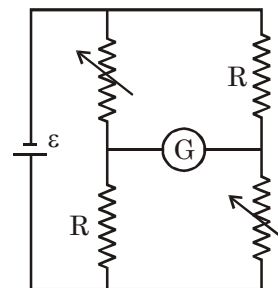
Numerical Grid Type (Single digit Ranging from 0 to 9)

4 Q. [4 M(-1)]

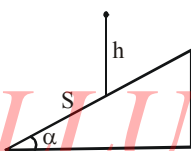
1. The circuit consists of two identical resistors with resistance R and two identical nonlinear elements, whose voltage - current

relationship looks like $U = \alpha I^2$, where $\alpha = 2 \frac{\text{volt}}{\text{Ampere}^2}$. At which

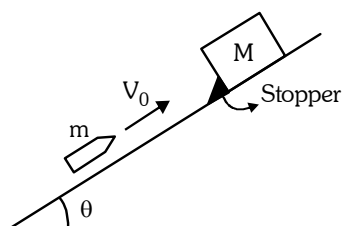
emf $\mathcal{E}$ of the source the current through the galvanometer is equal to zero? Resistance of the source can be neglected. $R = 3 \Omega$.



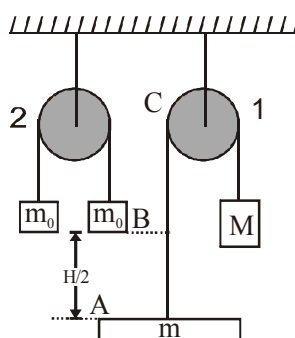
2. As the figure shows, a small object is dropped on a fixed, frictionless slope of angle of elevation $\alpha = 30^\circ$ from a height h . The time of the fall is t_1 . After a perfectly inelastic collision the object slides down the slope. It covers the same distance h on the slope during the time t_2 . If the ratio $\frac{t_1}{t_2}$ equals $\left(\frac{1 + \sqrt{x}}{2} \right)$ find x .



3. A block of mass M is kept at rest on a long smooth inclined plane by a stopper. A bullet of mass m travelling at velocity v_0 parallel to inclined plane collides with block at $t = 0$ and gets embedded. The time elapsed since $t = 0$ when block hits the stopper is $\frac{2}{n}$ sec. Find n .
(Given $m = 5 \text{ kg}$; $M = 10 \text{ kg}$; $v_0 = 5 \text{ ms}^{-1}$; $g = 10 \text{ ms}^{-2}$; $\theta = 30^\circ$)



4. Consider the system shown in figure. We know the masses of bodies : $m = 1 \text{ kg}$, $M = 3 \text{ kg}$. The distance $AB = BC = H/2$ are covered in equal time intervals by the body of mass m and at B it undergoes an elastic collision with a stationary body of mass m_0 (unknown). Time after this collision when the string in pulley 2 becomes tight again is t_0 . t_0 is in second. t_0^2 is equal to $1/x$. Value of $x/10$ is.
(Given : $H = 1 \text{ m}$)



SECTION-I

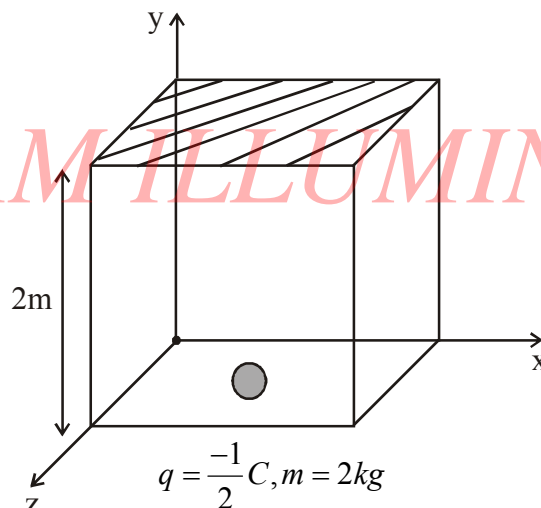
Single Correct Answer Type

4 Q. [3 M (-1)]

1. A charge q is uniformly distributed on the surface of a disc of radius R . Then a circular hole of radius $R/2$ is punched taking one of the radii as the diameter of the hole. Calculate the potential V due to

remaining portion of the disc at the center of the disc. $\left[V_0 = \frac{q}{2\pi\epsilon_0 R} \left(1 - \frac{1}{\pi} \right) \right]$

- (A) $V = 2V_0$ (B) $V = V_0$ (C) $V = 3V_0$ (D) $V = \frac{V_0}{2}$
2. Consider a gravity free container as shown. System is initially at rest and electric potential in the region is $V = (y^3 + 2)$ J/C. A ball of charge q and mass m is released from rest from base starts to move up due to electric field and collides with the shaded face as shown.

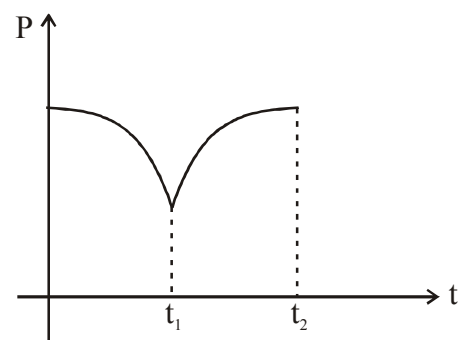


If its speed just after collision is 1.5 m/s and time for which ball is in contact with shaded face is 0.1 sec, find external force required to hold the container fixed in its position during collision assuming ball exerts constant force on wall during entire span of collision.

- (A) 70 N (B) 72 N (C) 74 N (D) 76 N
3. Work done on charge of mass 2kg, due to external force against electrostatic force is -10 J if charge is displaced from A to B. Velocity of charge at point A is 4 m/s and at B is 2m/s then find difference in electric potential energy ($U_B - U_A$) :
- (A) + 10J (B) - 10 J (C) - 2J (D) + 2 J

4. Graph between power & time is given below. Then which of the following option is correct for work done by force [from $t = 0$ to time t]

- (A) First increases till t_1 & then decreases till t_2 .
 (B) First decreases till t_1 & then increase till t_2 .
 (C) Always increases
 (D) Always decreases.



Multiple Correct Answer Type

4 Q. [4 M (0)]

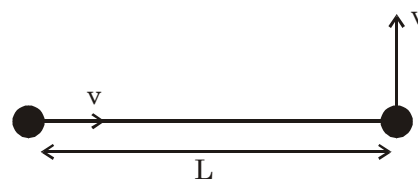
5. Two identical small balls are interconnected with a light and inextensible thread having length L . The system is on a smooth horizontal table with the thread just taut. Each ball is given the velocity as shown at $t = 0$.

(A) The thread will become taut again after time elapsed $\frac{L}{v}$.

(B) The thread will become taut again after time elapsed $\frac{L}{\sqrt{2}v}$.

(C) The kinetic energy of the system after the string gets taut will be $\frac{3}{4}mv^2$.

(D) The kinetic energy of the system after the string gets taut will be $\frac{1}{4}mv^2$.



6. A large insulating thick sheet of thickness $2d$ carries a uniform charge per unit volume ρ . A particle of mass m , carrying a charge q having a sign opposite to that of the sheet, is released from the surface of the sheet. The sheet does not offer any mechanical resistance to the motion of the particle. Find the oscillation frequency ν of the particle inside the sheet.

(A) $\nu = \frac{1}{2\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$ (B) $\nu = \frac{1}{2\pi} \sqrt{\frac{2q\rho}{m\epsilon_0}}$ (C) $\nu = \frac{1}{4\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$ (D) $\nu = \frac{1}{2\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$

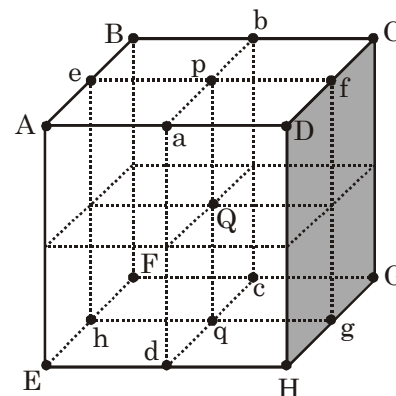
7. The following figure shows a charge Q kept at the centre of a cube. Let ϕ represent the flux of field due to the charge Q , the correct options are :

(A) $\phi_{ABCD} = \frac{Q}{6\epsilon_0}$

(B) $\phi_{cbpf} = \frac{Q}{24\epsilon_0}$

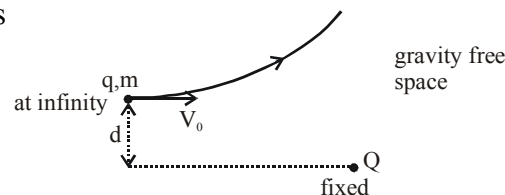
(C) $\phi_{abcd} = 0$

(D) $\phi_{abcd} = \frac{Q}{2\epsilon_0}$



8. A charged body A (having charge q and mass m) moves towards a fixed charge Q with speed v_0 (when far from Q) and aiming parameter ℓ as shown in the figure.

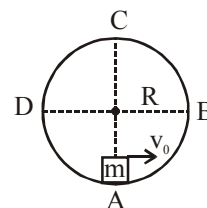
- (A) The path traveled by q is parabola.
(B) The path traveled by q is hyperbola.



(C) The minimum distance of approach is $\frac{mv_0^2 d^2}{kQq \left[\sqrt{1 + \left(\frac{mv_0^2 d}{kQq} \right)^2} - 1 \right]}$

(D) The minimum distance of approach is $\frac{mv_0^2 d^2}{kQq \left[\sqrt{1 + \left(\frac{mv_0^2 d}{kQq} \right)^2} + 1 \right]}$

9. A small block of mass m can move inside a fixed spherical rough surface of radius R . Block is given a horizontal velocity V_0 at lower most point sufficient for block to complete one round of vertical circular motion. Choose the **CORRECT** statement/s for first round of vertical circular motion :-



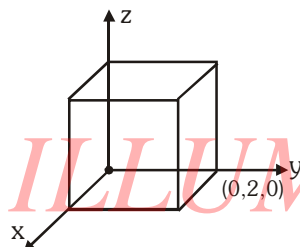
- (A) Work done by friction in part AB is equal to that in part DA
(B) Work done by friction in part ABC is greater than that in part CDA.
(C) Velocity at point B must be greater than $\sqrt{3gR}$.
(D) Velocity at D is equal to velocity at B

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para x 3Q.) [3 M (0)]

Paragraph for Question 10 to 12

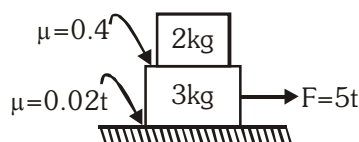
Electric field in a region is given by following equation $\vec{E} = \frac{K(\vec{r} - \vec{r}_0)}{|\vec{r} - \vec{r}_0|^2}$ where $\vec{r} = x\hat{i} + y\hat{j}$ and $\vec{r}_0 = \hat{i} + \hat{j}$ and an imaginary cube is made as shown in the diagram.



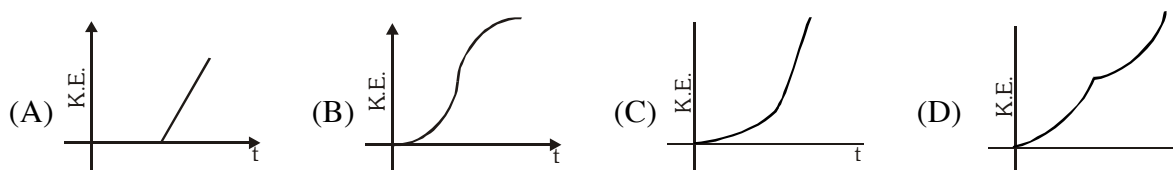
10. Find total flux through this cube due to the given electric field.
(A) $4\pi K$ (B) $6\pi K$ (C) $16K$ (D) $8K$
11. Find total charge encloses in this imaginary shown cube.
(A) $4K\pi\epsilon_0$ (B) $8K\pi\epsilon_0$ (C) $6K\pi\epsilon_0$ (D) $2K\pi\epsilon_0$
12. If $\vec{r}_0 = (2\hat{i} + 2\hat{j})$ then find the value of total flux through the walls of the shown cube
(A) $2K\pi$ (B) $K\pi$ (C) $4K\pi$ (D) $4K$

Paragraph for Question 13 to 15

Two blocks of mass 2kg and 3kg are arranged as shown in the figure. The value of friction coefficient between 2kg and 3kg surface is 0.4 and 0.02t between the surface of 3kg block and ground. A time varying horizontal external force $F = 5t$ is acting on 3kg block, where t is time in second



13. Work done by the friction force on 2kg block upto 5 sec with respect to 3 kg block is :
(A) 100 J (B) 400 J (C) zero (D) 160 J
14. Choose the nature of the graph of KE of block 2kg with time :

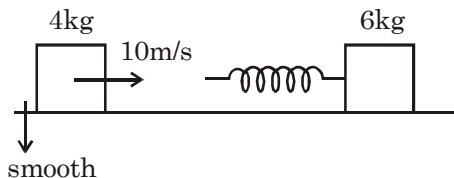


15. Kinetic energy of the system at $t = 5$ sec is :
(A) 62.5 J (B) 125 J (C) 112.5 J (D) 250 J

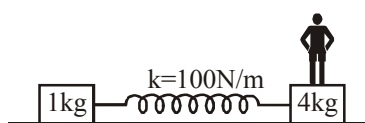
SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 2 Q. [4 M(-1)]

1. Initially 6kg block is at rest and 4kg block is moving with velocity 10 m/s as shown. What is final velocity (in m/s) of 6kg block when spring will come again to its natural length.



2. A boy of mass 10 kg is standing on a 4 kg block as shown. The whole system is on smooth level ground. The boy jumps to the right with a velocity of 7 m/s relative to the 4 kg block. Find the maximum velocity 'v' of 1 kg block (in m/sec) in the subsequent motion.

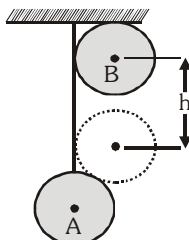


SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A smooth ball A of mass m is attached to one end of a light inextensible string, and is suspended from fixed point O. Another identical ball B, is dropped from a height h, so that the string just touches the surface of the sphere.



Column-I

- (A) If collision between balls is completely elastic then speed of ball A just after collision is
- (B) If collision between balls is completely elastic then impulsive tension provided by string is
- (C) If collision between balls is completely inelastic then speed of ball A just after collision is
- (D) If collision between balls is completely inelastic then impulsive tension provided by string is

Column-II

- (P) $\frac{3m}{5}\sqrt{2gh}$
- (Q) $\frac{\sqrt{6gh}}{5}$
- (R) $\frac{6m}{5}\sqrt{2gh}$
- (S) $\frac{2\sqrt{6gh}}{5}$

SECTION-I

Single Correct Answer Type

31 Q. [3 M (-1)]

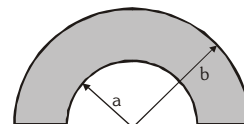
1. A non conducting semicircular disc (as shown in figure) has a uniform surface charge density σ . The electric potential at the centre of the disc :-

(A) $\frac{\sigma}{2\pi\epsilon_0} \frac{\ln(b/a)}{(b-a)}$

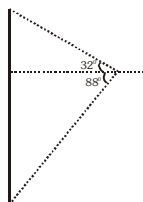
(B) $\frac{\sigma(b-a)}{2\epsilon_0}$

(C) $\frac{\sigma(b-a)}{4\epsilon_0}$

(D) $\frac{\sigma(b-a)}{4\pi\epsilon_0}$



2. Consider a finite charged rod. Electric field at Point P (shown) makes an angle θ with horizontal dotted line then angle θ is :-



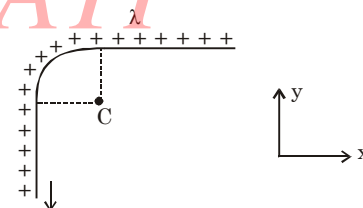
3. Two semi-infinite wires are joined with a quarter circular arc as shown in diagram. The charge density is same on both the wires and the arc. The electric field intensity at point C, the centre of arc is :

(A) $-\frac{k\lambda}{R}\hat{i} - \frac{k\lambda}{R}\hat{j}$

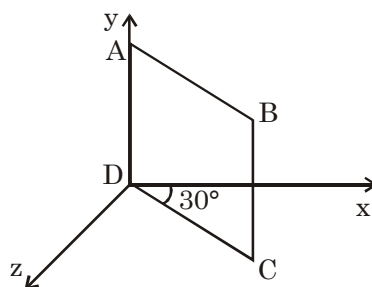
(B) $\frac{k\lambda}{R}\hat{i} + \frac{k\lambda}{R}\hat{j}$

(C) $\frac{k\lambda}{R}\hat{i} - \frac{k\lambda}{R}\hat{j}$

(D) $-\frac{k\lambda}{R}\hat{i} + \frac{k\lambda}{R}\hat{j}$



4. A square cross-section ABCD of side 'a' is kept in the direction shown in the diagram. An electric field $E_0\hat{i} + E_0\hat{j}$ exists in the region. The flux of this field through the area cross-section is :



(A) $E_0 a^2$

(B) $\frac{E_0 a^2}{3}$

(C) $\frac{E_0 a^2}{2}$

(D) $\frac{\sqrt{3}}{2} E_0 a^2$

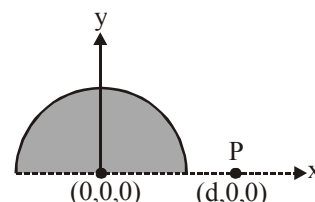
5. A solid hemispherical uniform charged body having charge Q is kept symmetrically along the y-axis as shown in figure. The electric potential at a distance d from the origin along the x-axis at point P will be

(A) $\frac{1}{4\pi\epsilon_0} \frac{Q}{d}$

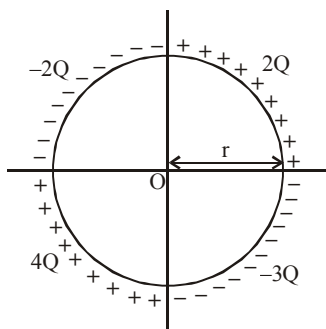
(B) less than $\frac{1}{4\pi\epsilon_0} \frac{Q}{d}$

(C) more than $\frac{1}{4\pi\epsilon_0} \frac{Q}{d}$ and less than $\frac{2}{4\pi\epsilon_0} \frac{Q}{d}$

(D) more than $\frac{2}{4\pi\epsilon_0} \frac{Q}{d}$



6. Charge distribution on a ring in x-y plane is shown in the figure. Then electric potential at origin is :



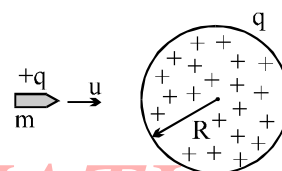
- (A) $\frac{Q}{4\pi\epsilon_0 r}$ (B) Zero (C) $\frac{Q}{2\pi\epsilon_0 r}$ (D) $\frac{3Q}{4\pi\epsilon_0 r}$
7. A bullet of mass m and charge q is fired towards a solid uniformly charged sphere of radius R and total charge $+q$. If it strikes the surface of sphere with speed u , find the minimum speed u so that it can penetrate through the sphere. (Neglect all resistance forces or friction acting on bullet except electrostatic forces)

(A) $\frac{q}{\sqrt{2\pi\epsilon_0 mR}}$

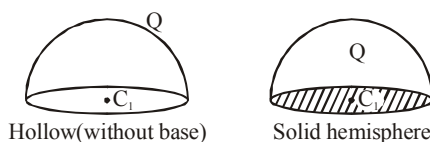
(B) $\frac{q}{\sqrt{4\pi\epsilon_0 mR}}$

(C) $\frac{q}{\sqrt{8\pi\epsilon_0 mR}}$

(D) $\frac{\sqrt{3}q}{\sqrt{4\pi\epsilon_0 mR}}$



8. Charges Q each are distributed uniformly on a nonconducting hemispherical shell (on the surface) and in the volume of a nonconducting solid hemisphere. A point charge is brought at the centre C_1 of shell and at the centre C_2 of solid hemisphere from infinity. If work done in these process are W_1 and W_2 the W_1/W_2 is equal :



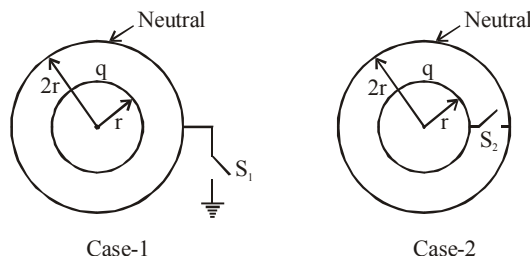
(A) $\frac{3}{4}$

(B) $\frac{3}{2}$

(C) $\frac{4}{3}$

(D) $\frac{2}{3}$

9. In given two cases two concentric conducting spherical shells are arranged as shown. In case-1 when switch S_1 is closed energy loss is Q_1 . In case-2 when switch S_2 is closed energy loss is Q_2 then the value of $\frac{Q_1}{Q_2}$ is :



(A) 5

(B) 0.5

(C) 1

(D) 0

10. A solid conducting sphere of radius a has charge $+q$. The sphere is surrounded concentrically by a conducting shell of inner and outer radii r and R respectively. The electric potential at a point which is at a distance x , ($r < x < R$), from the centre of the sphere is :-

(A) $\frac{q}{4\pi\epsilon_0 R}$

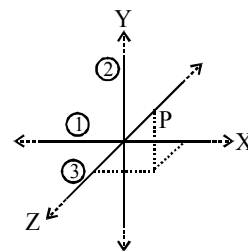
(B) $\frac{3q}{4\pi\epsilon_0 R}$

(C) $\frac{8q}{4\pi\epsilon_0 R}$

(D) $\frac{5q}{4\pi\epsilon_0 R}$

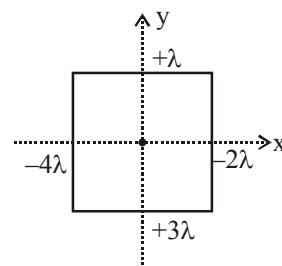
11. The electric field vector at point P (a, a, a) due to three uniformly charged infinite wires 1, 2 and 3 with uniform charge density λ each kept along the x, y and z-axes, respectively as shown in figure is :-

(A) $\frac{\lambda}{3\pi\epsilon_0 a}(\hat{i} + \hat{j} + \hat{k})$ (B) $\frac{\lambda}{2\pi\epsilon_0 a}(\hat{i} + \hat{j} + \hat{k})$
(C) $\frac{\lambda}{2\sqrt{2}\pi\epsilon_0 a}(\hat{i} + \hat{j} + \hat{k})$ (D) $\frac{\sqrt{2}\lambda}{\pi\epsilon_0 a}(\hat{i} + \hat{j} + \hat{k})$



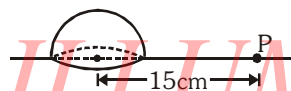
12. Electric field at the centre of the square of side a :

(A) $\frac{4\sqrt{2}}{a}k\lambda(-\hat{i} + \hat{j})$ (B) $\frac{4\sqrt{2}}{a}k\lambda(\hat{i} + \hat{j})$
(C) $\frac{6\sqrt{2}}{a}k\lambda(-\hat{i} + \hat{j})$ (D) None of these



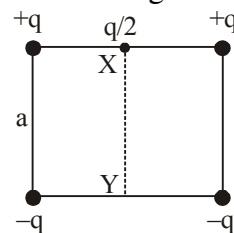
13. Figure shows a solid hemisphere with a charge of 5nC distributed uniformly through its volume. The hemisphere lies on a plane and point P is located on this plane, along a radial line from the centre of curvature at distance 15cm. The electric potential at point P due to the hemisphere, is :

(A) 150V (B) 300V (C) 450V (D) 600V



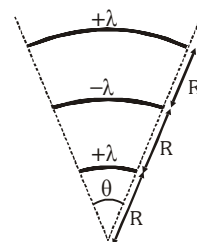
14. Work done in shifting a charge $q/2$ from a point X to a point Y in the diagram shown in figure is:

(A) $\frac{8Kq^2}{a} \left(\frac{\sqrt{5}-1}{\sqrt{5}} \right)$ (B) $\frac{8Kq^2}{a} \left(\frac{1-\sqrt{5}}{\sqrt{5}} \right)$
(C) $\frac{4Kq^2}{a} \left(\frac{\sqrt{5}-1}{\sqrt{5}} \right)$ (D) $\frac{4Kq^2}{a} \left(\frac{1-\sqrt{5}}{\sqrt{5}} \right)$



15. Even number of infinite concentric circular arcs of same angular span θ carry uniform linear charge densities $+\lambda$ and $-\lambda$ alternatively as shown in figure. Their radii are R, 2R, 3R..... respectively. The potential at their common centre is-

(A) $\frac{\lambda\theta \ln 2}{4\pi\epsilon_0}$ (B) $\frac{\lambda\theta \ln 2}{2\pi\epsilon_0}$
(C) $\frac{\lambda\theta \ln 1}{4\pi\epsilon_0}$ (D) None of these



16. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are +q and -q. The potential difference between the centres of the two rings is :

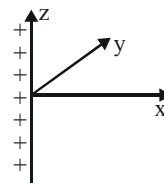
(A) $\frac{qR}{4\pi\epsilon_0 d^2}$ (B) $\frac{q}{2\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$ (C) zero (D) $\frac{q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$

17. The maximum electric field intensity on the axis of a uniformly charged ring of charge q and radius R will be :-

(A) $\frac{1}{4\pi\epsilon_0} \frac{q}{3\sqrt{3}R^2}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{2q}{3R^2}$ (C) $\frac{1}{4\pi\epsilon_0} \frac{2q}{3\sqrt{3}R^2}$ (D) $\frac{1}{4\pi\epsilon_0} \frac{3q}{2\sqrt{2}R^2}$

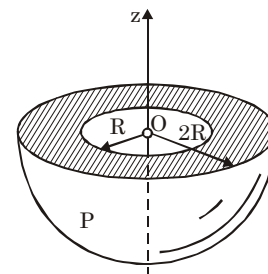
18. An infinite thread of charge density λ lies along z-axis. The potential difference between points A (4, 3, 4) and B(3, 4, 0) is :-

(A) $\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{\sqrt{41}}{5}\right)$ (B) $\frac{\lambda}{2\pi\epsilon_0} \ln 5$ (C) zero (D) $\frac{\lambda}{2\pi\epsilon_0}$



19. A hemispherical bowl of inner radii R and outer radii $2R$ is charged with density $\rho = \rho_0 \frac{r}{R}$. Find the electric field at point O.

(A) $\frac{2\rho_0 R}{3\epsilon_0}$ (B) $\frac{3\rho_0 R}{4\epsilon_0}$
(C) $\frac{\rho_0 R}{2\epsilon_0}$ (D) $\frac{3\rho_0 R}{2\epsilon_0}$

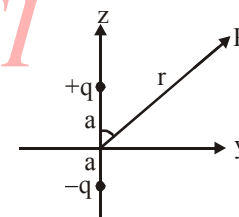


20. An electric dipole is placed at the origin O such that its equator is y-axis. At a point P far away from dipole, the electric field direction is along y-direction. OP makes an angle α with the x-axis such that

(A) $\tan \alpha = \sqrt{3}$ (B) $\tan \alpha = \sqrt{2}$ (C) $\tan \alpha = 1$ (D) $\tan \alpha = \frac{1}{\sqrt{2}}$

21. Figure shows an electric dipole. For $r \gg a$, what is potential of dipole at a point P with coordinates (x, y, z)

(A) $v = \frac{p}{4\pi\epsilon_0} \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$ (B) $v = \frac{p}{4\pi\epsilon_0} \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$
(C) $v = \frac{p}{4\pi\epsilon_0} \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$ (D) $v = \frac{p}{4\pi\epsilon_0} \frac{xyz}{(x^2 + y^2 + z^2)^{3/2}}$



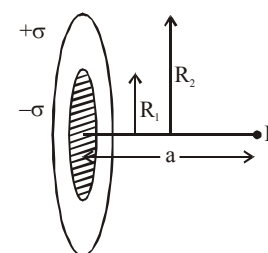
22. A non-conducting disc is charged by two different uniform charge densities $-\sigma$ and $+\sigma$. $-\sigma$ is upto radius R_1 and $+\sigma$ is from radius R_1 to radius R_2 . $\hat{E}$ is unit vector along direction of net electric field at point P then ($R_2 > R_1$)

(A) if $R_1 > \frac{a}{\sqrt{3}}$ then for all values of R_2 , $\hat{E}$ is $-\hat{i}$

(B) $\hat{E}$ may equal to $\hat{i}$ or $-\hat{i}$

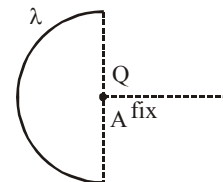
(C) $\hat{E}$ must equal to $-\hat{i}$ for all values of R_1 and R_2

(D) $\hat{E}$ must equal to $\hat{i}$ for all values of R_1 and R_2



23. A charge is kept fixed at point A as shown in the figure in a gravity free space. A semicircular ring of linear charge density λ , mass m & radius R is released from rest. Initially centre of semicircular ring coincides with the fixed charge. The initial acceleration of the semicircular ring is :-

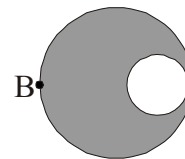
(A) $\frac{2K\lambda Q}{mR}$ (B) $\frac{K\lambda Q}{2mR}$ (C) $\frac{K\lambda Q}{mR}$ (D) 0



24. Two identical non-conducting thin hemispherical shells each of radius R are brought in contact to make a complete sphere. If a total charge Q is uniformly distributed on them, how much minimum force F will be required to hold them together.

(A) $F = \frac{Q^2}{16\pi\epsilon_0 R^2}$ (B) $F = \frac{Q^2}{32\pi\epsilon_0 R^2}$ (C) $F = \frac{Q^2}{64\pi\epsilon_0 R^2}$ (D) $\frac{Q^2}{32\pi\epsilon_0 R^2} > F > \frac{Q^2}{64\pi\epsilon_0 R^2}$

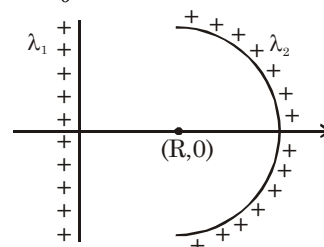
25. A positively charged sphere of radius r_0 carries a volume charge density ρ_E (Figure). A spherical cavity of radius $r_0/2$ is then scooped out and left empty, as shown. What is the direction and magnitude of the electric field at point B ?



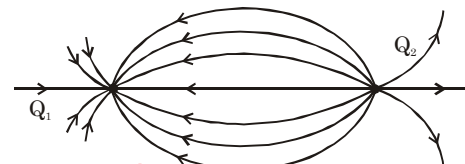
- (A) $\frac{17\rho r_0}{54\epsilon_0}$ left (B) $\frac{\rho r_0}{6\epsilon_0}$ left (C) $\frac{17\rho r_0}{54\epsilon_0}$ right (D) $\frac{\rho r_0}{6\epsilon_0}$ right

26. A uniformly charged infinite wire is placed along 'y' axis having linear charge density ' λ_1 '. A semicircle wire having linear charge density ' λ_2 '

centred at $(R, 0)$ is placed as shown. Find the ratio of $\frac{\lambda_1}{\lambda_2}$. If electric field at $(R, 0)$ is zero.



- (A) $\frac{\lambda_1}{\lambda_2} = 1$ (B) $\frac{\lambda_2}{\lambda_1} = 2$ (C) $\frac{\lambda_2}{\lambda_1} = \frac{1}{2}$ (D) $\frac{\lambda_1}{\lambda_2} = \sqrt{2}$
27. A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on the x axis are shown in the figure. These lines suggest that.



- (A) $\frac{Q_1}{Q_2} = \frac{6}{5}$
(B) $Q_1 > 0$
(C) At a finite distance to the left of Q_1 , the electric field will be zero
(D) At a finite distance to the right of Q_2 , the electric field will be zero.

28. A charge particle is placed at a point on the axis of a circular disc of radius R . If its distance from the disc increases from $\frac{3R}{4}$ to $\frac{4R}{3}$, then flux through the disc becomes :

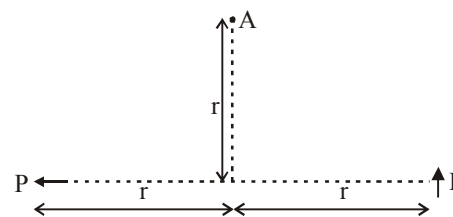
- (A) half (B) double (C) one-fourth (D) four times

29. Dipole having dipole moment $4C\cdot m$ is placed on origin along x-axis. Point charge of $8\mu C$ is fixed at $(4, 0, 0)$. Now dipole is rotated by an angle of $\frac{\pi}{2}$. Find the work done (in mJ) in rotating the dipole.

- (A) 4 mJ (B) 16 mJ (C) 18 mJ (D) 32 mJ

30. P is the dipole moment of two small dipoles shown in the figure. What is the potential at point A?

- (A) 0 (B) $\frac{Kp}{\sqrt{2}r^2}$
(C) $\sqrt{2} \frac{KP}{r^2}$ (D) $2 \frac{KP}{r^2}$



31. Two planes of charge with no thickness, A and B, are parallel and vertical. The electric field in region-1 to the left of plane A has magnitude $\frac{3\sigma}{2\epsilon_0}$ and points to the left. The electric field in region-2 to the right of B has magnitude $\frac{3\sigma}{2\epsilon_0}$ and points to the right. The electric field in region-3 between the two planes has magnitude $\frac{\sigma}{2\epsilon_0}$ and points to the right. The surface charge density on planes A and B respectively is

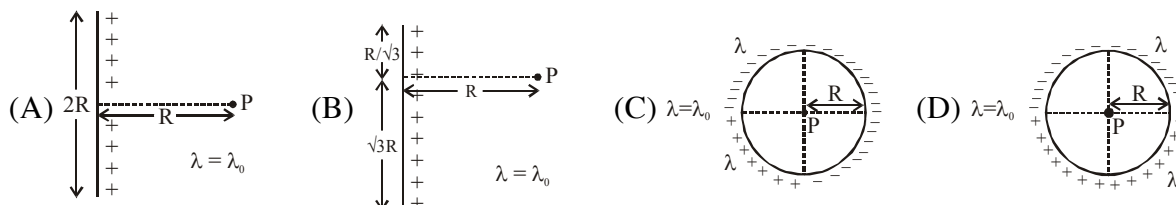
- (A) $-\frac{\sigma}{2}, \sigma$ (B) $+\frac{\sigma}{2}, \sigma$ (C) $\sigma, -\frac{\sigma}{2}$ (D) $2\sigma, \sigma$

Multiple Correct Answer Type

6 Q. [4 M (0)]

32. For which of the following continuous charge system magnitude of electric field at point P is $\frac{\sqrt{2}k\lambda_0}{R}$

where λ_0 is linear charge density.



33. The given two fixed rings of radius R lie in XY plane as shown in the figure. Linear charge density of the rings A and B varies as per the relation $\lambda = \lambda_0 \sin \theta$ and $\lambda = \lambda_0 \cos \theta$ respectively where θ is measured from +x axis. Distance between the centre of the two rings is $r \gg R$.

- (A) Force of interaction between the two ring is $\frac{3}{4} \times \frac{\lambda_0^2 \pi}{\epsilon_0} \left(\frac{R}{r} \right)^4$.

- (B) Force of interaction between the two ring is $\frac{3}{2} \times \frac{\lambda_0^2 \pi}{\epsilon_0} \left(\frac{R}{r} \right)^4$.

- (C) Potential energy of interaction between the two ring is $\frac{\pi \lambda_0^2 R^4}{4\epsilon_0 r^3}$.

- (D) Potential energy of interaction between the two ring is zero.

34. A dipole of dipole moment $8 \times 10^{-10} \hat{i}$ C-m is kept at the origin of coordinates :-

- (A) the flux through the y-z plane passing through origin is zero
(B) the potential at (1, 1, 1) m is 0.8 volt
(C) the potential at (2, 3, -6) m is positive
(D) the electric field at any point on the z-axis is in the negative x-direction

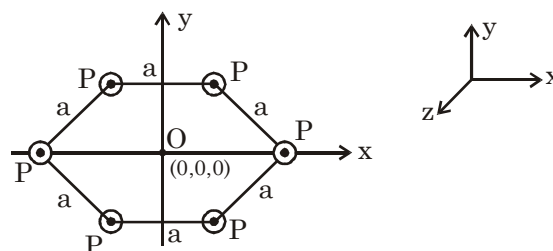
35. Six dipoles of dipole moment $P \hat{k}$ are arranged at corners of a hexagon of side a (value of a is much more greater than size of dipole). Coordinates of point A and B are (0, 0, a) and (0, 0, -a):

- (A) Electric field at origin O is $\frac{6kP}{\sqrt{2}a^3} (-\hat{k})$

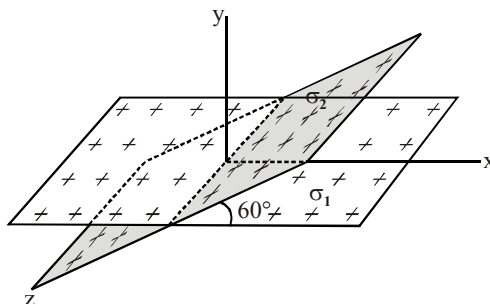
- (B) Potential at origin O is zero.

- (C) Potential at point A is $+\frac{3kP}{\sqrt{2}a^2}$

- (D) Potential at point B is $+\frac{3kP}{\sqrt{2}a^2}$



36. An infinite plane in the xz plane carries a uniform surface charge density $\sigma_1 = 8.85 \text{ nC/m}^2$. A second infinite plane carrying a uniform charge density $\sigma_2 = 17.7 \text{ nC/m}^2$ intersects the xz plane at the z axis and makes an angle of 60° with the xz plane as shown in figure. The electric field in the xy plane.



- (A) at $x = 6\text{m}$, $y = 2\text{m}$ is $500\sqrt{3} \text{ N/C}$ (B) at $x = -5\text{m}$, $y = 0$ is $500\sqrt{3} \text{ N/C}$
(C) at $x = 2\text{m}$, $y = 6\text{m}$ is $500\sqrt{7} \text{ N/C}$ (D) at $x = -1\text{m}$, $y = -1\text{m}$ is $500\sqrt{7} \text{ N/C}$
37. An electric field in free space is $\vec{E} = \left(\frac{5z^3}{\epsilon_0} \right) \hat{k} \text{ V/m}$. consider a sphere of 3-m radius, centered at the origin.

- (A) The charge stored in this sphere is $972 \pi \text{ C}$ (B) The energy stored in this sphere is $\frac{425\pi}{28\epsilon_0} \times 3^8 \text{ J}$
(C) The charge stored in this sphere is $432 \pi \text{ C}$ (D) The energy stored in this sphere is $\frac{425\pi}{16\epsilon_0} \times 3^8 \text{ J}$

Matching list Type ($4 \times 4 \times 4$) single option correct

(Three Columns and Four Rows)

1 Table $\times$ 3 Q. [4(-1)]

Answer Q.38, Q.39 and Q.40 by appropriately matching the information given in the three columns of the following table.

Column-I & II represents the magnitude of electric field at points A & B respectively. Column-III represents the magnitude of potential difference between the two points. (Take potential at

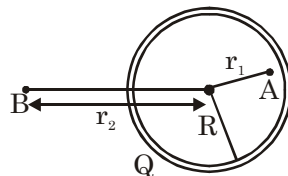
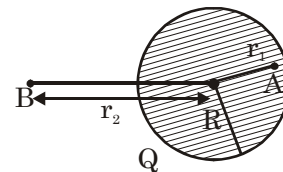
infinity to be zero) $\left(k = \frac{1}{4\pi\epsilon_0} \right)$

Column-I Electric field at point A	Column-II Electric field at point B	Column-III Potential difference between two points
(I) $\frac{\rho r_1}{2\epsilon_0}$	(i) $\frac{\rho R^2}{4\epsilon_0 r_2}$	(P) $ V_A - V_B = \frac{kQ}{2R} \left(3 - \frac{r_1^2}{R^2} \right) - \frac{kQ}{r_2}$
(II) $\frac{kQr_1}{R^3}$	(ii) $\frac{\rho R^2}{2\epsilon_0 r_2}$	(Q) $ V_A - V_B = \frac{3kQ}{2R} - \frac{kQ}{r_2}$
(III) 0	(iii) $\frac{kQ}{r_2^2}$	(R) $ V_C - V_B = \frac{\rho R^2}{2\epsilon_0} \ln \left(\frac{r_3}{r_2} \right)$
(IV) $\frac{\rho r_1}{4\epsilon_0}$	(iv) $\frac{kQr_2}{R^3}$	(S) $ V_A - V_B = \frac{kQ}{R} - \frac{kQ}{r_2}$

38. Which of the following combination represents the situation due to a uniformly charged non-conducting solid sphere having total charge Q & radius R :-

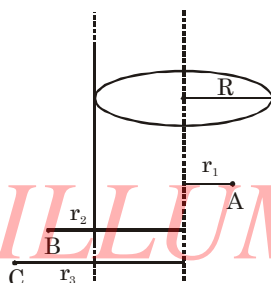
(A) (II) (iii) (Q) (B) (III) (iii) (S)
 (C) (II) (iii) (P) (D) (III) (iv) (P)

39. Which of the following combination represents the situation due to a uniformly charged non-conducting hollow thin sphere having total charge Q & radius R .



(A) (III) (iii) (S) (B) (II) (iii) (P) (C) (III) (iv) (Q) (D) (II) (iv) (S)

40. Which of the following combination represents the situation due to a uniformly charged infinite solid cylinder of radius R and charge density ρ .



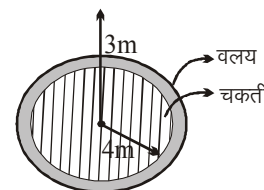
(A) (III) (i) (P) (B) (I) (i) (R) (C) (IV) (ii) (P) (D) (I) (ii) (R)

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q.[3(0)]

1. A disc of radius 4 m is uniformly charged with the charge of $1\mu\text{C}$. On the periphery of the disc, we stick a thin ring which has a uniform charge of $-1\mu\text{C}$. What is the Electric field (in N/C) due to the combination at a point 3 m away from the centre of combination on the common axis ?

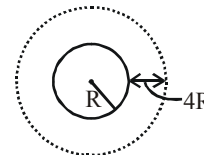


SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 3 Q. [4 M(-1)]

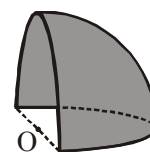
1. An uniformly charged non conducting solid sphere has a radius R . Total energy of system (inside + outside) is U_1 . If this sphere is now covered by a conducting spherical

layer of thickness $4R$, the total energy of system reduces to U_2 . Find value of $\frac{U_1}{U_2}$.



2. Half hemispherical shell (one fourth of total spherical shell) of radius r is charged uniformly such that surface charge density is σ . Charge is on the curved surface

only. Then the electric field intensity at the O as shown in figure is given by $\frac{\sigma}{2\sqrt{\alpha}\epsilon_0}$ value of α is



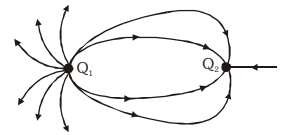
3. Two small dipoles $2p\hat{k}$ and $p\hat{k}$ are located at $(0, 5, 0)$ and $(1, 5, 2)$ respectively. The magnitude of resultant electric field due to the two dipoles at the point $(1, 5, 0)$ is given as $n \left[\frac{kp}{4} \right]$ find value of n .

SECTION-I

Single Correct Answer Type

9 Q. [3 M (-1)]

1. From the field pattern shown in figure, identify the correct statement related to the charges Q_1 and Q_2 :-



(A) $\frac{Q_1}{Q_2} = +2.5$

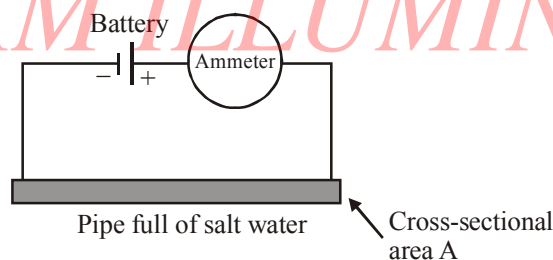
(B) $\frac{Q_1}{Q_2} = +2$

(C) $\frac{Q_1}{Q_2} = -2$

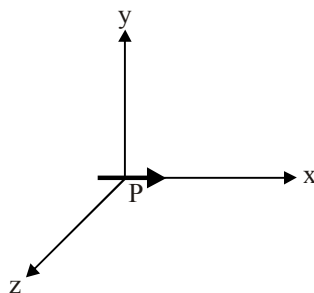
(D) There is no relation exist between Q_1 & Q_2

2. Salt water contains n sodium ions (Na^+) per cubic meter and n chloride ions (Cl^-) per cubic meter. A battery is connected to metal rods that dip into a narrow pipe full of salt water. The cross sectional area of the pipe is A . The magnitude of the drift velocity of the sodium ions is V_{Na} and the magnitude of the drift velocity of the chloride ions is V_{Cl} . Assume that $V_{\text{Na}} > V_{\text{Cl}}$ (+e is the charge of a proton). What is the magnitude of the ammeter reading?

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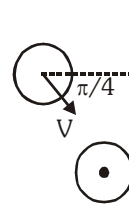


- (A) $enAV_{\text{Na}} - enAV_{\text{Cl}}$ (B) $enAV_{\text{Na}} + enAV_{\text{Cl}}$
 (C) $enAV_{\text{Na}}$ (D) Zero
3. There are four concentric shells A, B, C and D of radii a , $2a$, $3a$ and $4a$ respectively. Shells B and D are given charges $+q$ and $-q$ respectively. Shell C is now earthed. The potential difference $V_A - V_C$ is :
- (A) $\frac{Kq}{2a}$ (B) $\frac{Kq}{3a}$ (C) $\frac{Kq}{4a}$ (D) $\frac{Kq}{6a}$
4. A short electric dipole is oriented along x-direction at origin. At which of following point the electric field have no x component.



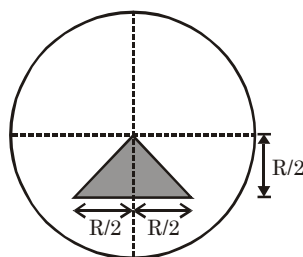
- (A) $(1, 1, 0)$ (B) $(\sqrt{2}, 1, 0)$ (C) $(1, \sqrt{2}, 0)$ (D) $(1, 0, 0)$

5. In a carom-board game the striker and the coins are identical and of mass m . In a particular hit the coin is hit when it is placed close to the edge of the board as shown in figure such that the coin travels parallel to the edge. If the striker is moving with speed v before the strike, then the net impulse on the striker during collision if it moves perpendicular to the edge is (assume all collisions to be perfectly elastic)

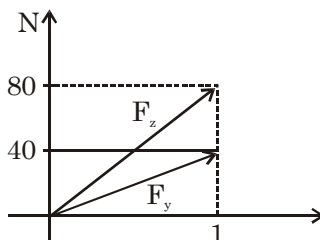


- (A) $mv\sqrt{\frac{5}{2}}$ (B) $2mv$ (C) $\frac{mv\sqrt{3}}{2}$ (D) mv

6. In the figure as shown, a triangular portion is cut from a circular disc of radius R . The distance of centre of mass of the remaining portion from the centre of disc is $\frac{R}{\alpha(\beta\pi - 1)}$. Calculate $\left(\frac{\alpha\beta}{2}\right)$.



- (A) 4 (B) 6 (C) 2 (D) 8
7. A particle is acted upon by a force for 1 sec. Whose x component remains constant at $F_x = 30\text{N}$ but y and z components vary with time as shown. The magnitude of change in momentum (in kg m/s) of particle in 1 sec.



- (A) 90 (B) 100 (C) $29\sqrt{10}$ (D) $10\sqrt{29}$
8. The wire shown in figure has a uniform cross-section A .

$$\overline{\quad\quad\quad} \quad x=0 \quad \quad \quad x=L$$

Resistivity of the material of wire is given by $\rho = \rho_0 \left(\frac{L}{L+x} \right)$

A potential difference V is applied across the wire :-

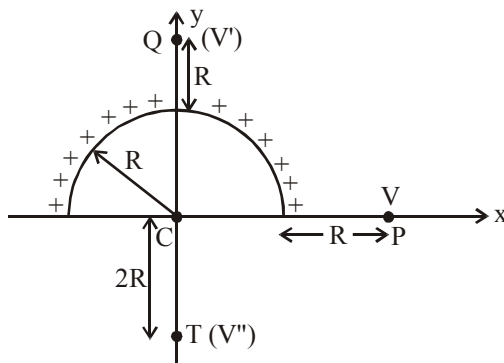
- (A) Resistance of wire is $\frac{\rho_0 L}{A} \cdot \ln(2)$ (B) Current density is variable inside the wire
- (C) Electric field at $x=0$ is $\frac{2V}{(\ln 2) \cdot L}$ (D) Electric field at $x=L$ is $\frac{V}{(\ln 2) \cdot L}$

9. A hollow conducting sphere of inner radius R and outer radius $2R$ has resistivity ' ρ ' as a function of the distance ' r ' from the centre of the sphere: $\rho = kr^2/R$. The inner and outer surfaces are painted with a perfectly conducting 'paint' and a potential difference ΔV is applied between the two surfaces. Then, as ' r ' increases from R to $2R$, the electric field inside the sphere—
- (A) increases (B) decreases (C) remains constant (D) passes through a maxima

Multiple Correct Answer Type

7 Q. [4 M (0)]

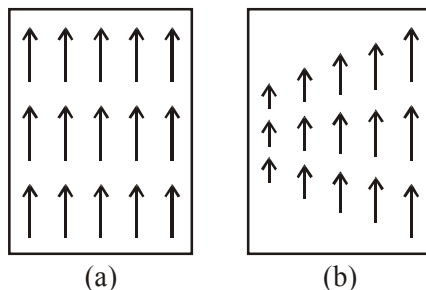
10. Charge is uniformly distributed on the curved surface of hemisphere. Potential at P, Q and T is V, V' and V'' respectively :-



- (A) $V < V' < 2V$ (B) $0 < V'' < V$ (C) $V' > V > V''$ (D) $V' = V'' = V$
11. Figure shows force vectors at different points in space for two forces.

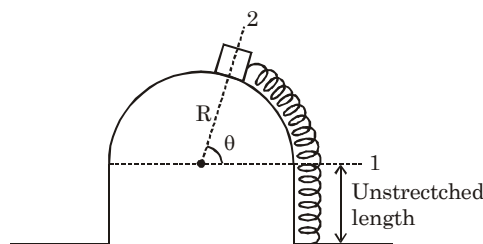
In figure (a) the force function is $\vec{F}_1 = F_0 \hat{j}$, where F_0 is a constant. The force in figure (b) is given by

$\vec{F}_2 = F_0 \left(\frac{x}{\alpha} \right) \hat{j}$, where origin is taken at center of left edge of the box, α is width of the square box.

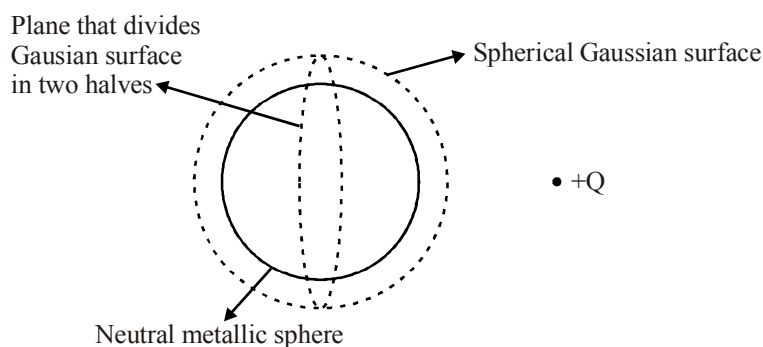


- (A) $\vec{F}_1$ is conservative force (B) $\vec{F}_1$ is non conservative force
(C) $\vec{F}_2$ is conservative force (D) $\vec{F}_2$ is non-conservative force
12. A block of mass m is moved slowly from position 1 to position 2 by applying a tangential force F of variable magnitude. The spring has negligible mass and force constant k . End of the spring moves in an arc of radius R . When block is at position 1, the spring is at natural length. Mark the **CORRECT** statement(s) :

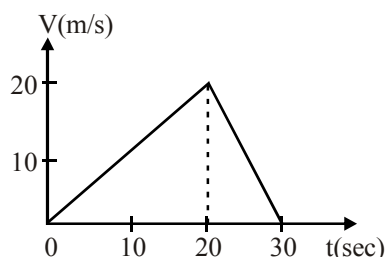
- (A) Work done by force $F = mgR \sin \theta + \frac{1}{2} k(R\theta)^2$
(B) Work done by gravity $= -mgR \sin \theta$
(C) Work done by spring force $= \frac{1}{2} k(R\theta)^2$
(D) Normal reaction from surface at position (2) is $mg \sin \theta$ in magnitude.



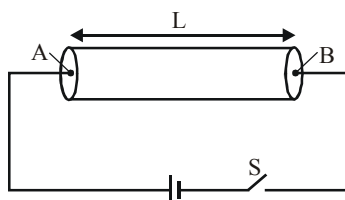
13. Figure shows a neutral metallic sphere with a point charge $+Q$ placed near its surface. Electrostatic equilibrium conditions exist on metallic sphere. Mark the **CORRECT** statement(s) :



- (A) Net flux through Gaussian surface due to charge Q is zero
 (B) Net flux through Gaussian surface due to charges appearing on the outer surface of metallic sphere must be zero
 (C) If point charge Q is displaced towards metallic sphere, magnitude of flux through right hemispherical part of Gaussian surface due to this charge increases.
 (D) If point charge Q is displaced towards metallic sphere, charge distribution on outer surface of sphere will change
14. V-t graph of an object of mass 1 kg is shown :



- (A) net work done on the object in 30 sec. is zero
 (B) the average acceleration from 0 to 30 sec. of the object is zero
 (C) the average velocity from 0 to 30 sec. of the object is zero
 (D) the average force from 0 to 30 sec. on the object is zero
15. Figure shows a battery, a conductor and a switch. After the switch is closed



- (A) The electric field is set up inside the conductor.
 (B) The time taken by the current to travel from A to B is $\frac{L}{v_d}$. [v_d is drift speed of electrons]
 (C) The random motion of the electron stops and they start drifting to the left.
 (D) The random motion of the electron continues along with the drifting.

16. Which of the following statements are true for a metallic conductor?
- (A) The electrical conductivity depends on the density of atoms
 - (B) The electrical conductivity decreases with rise in temperature
 - (C) The current density depends upon the drift velocity of electrons
 - (D) The electrical conductivity increases with increase in voltage across it

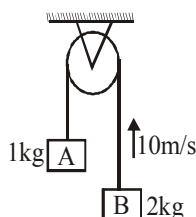
Linked Comprehension Type

(1 Para × 3Q.) [3 M (0)]

(Single Correct Answer Type)

Paragraph for question nos. 17 to 19

Two block of A and B of mass 1 kg and 2 kg are hung from light pulley. Initially the block B is held stationary. At $t = 0$ block B is given velocity 10 m/s in upward direction. String and pulley are light and there is no friction any where.



17. Time till which acceleration (magnitude and direction both) of the two block remains same
- (A) $\frac{1}{2}$ sec.
 - (B) 1 sec.
 - (C) 2 sec.
 - (D) Acceleration of the blocks always remains same.
18. Velocity of block 'B' as block 'A' has ascended by distance 5m from its original position
- (A) $10\sqrt{7}$ m/s
 - (B) $\frac{10}{3}\sqrt{2}$ m/s
 - (C) $10\sqrt{\frac{5}{3}}$ m/s
 - (D) $10\frac{\sqrt{7}}{3}$ m/s
19. Total mechanical energy lost when the string becomes taut is
- (A) $\frac{100}{3}$ J
 - (B) 100 J
 - (C) 300 J
 - (D) $\frac{200}{3}$ J

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)

1 Q. [4 M(-1)]

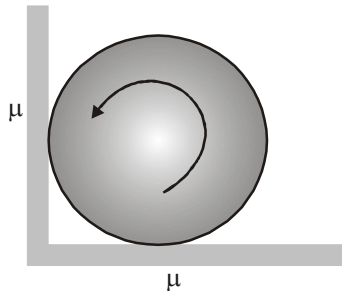
1. A beam of fast moving electrons having cross-sectional area $A = 1 \text{ cm}^2$ falls normally on a flat surface. The electrons are absorbed by the surface and the average pressure exerted by the electrons on this surface is found to be $P = 9.1 \text{ Pa}$. If the electrons are moving with a speed $v = 8 \times 10^7 \text{ m/s}$, then find the effective current (in A) through any cross-section of the electron beam.
(mass of electron = $9.1 \times 10^{-31} \text{ kg}$)

SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

1. A uniform sphere of mass m and radius r is placed in a corner with an initial counter clockwise angular velocity ω_0 as shown in fig. Coefficient of kinetic friction between sphere and both surfaces is μ . The time required for sphere to come to rest.

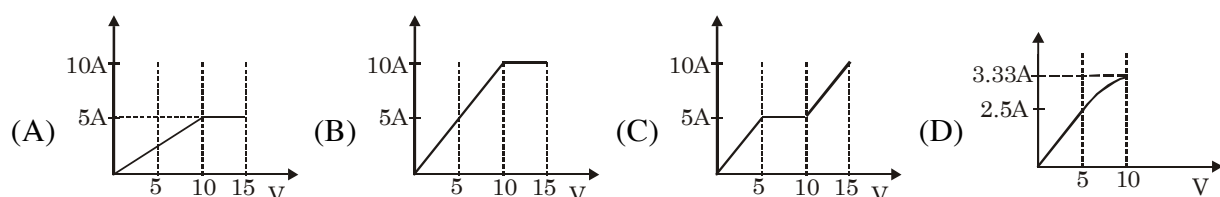
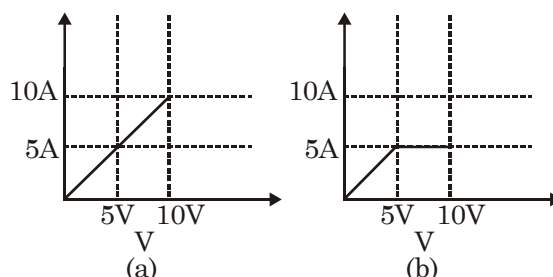


- (A) $T = \frac{2\omega_0 r(1+\mu^2)}{3\mu(1+\mu)g}$ (B) $T = \frac{5\omega_0 r(1+\mu^2)}{7\mu(1+\mu)g}$ (C) $T = \frac{2\omega_0 r(1+\mu^2)}{5\mu(1+\mu)g}$ (D) None of these

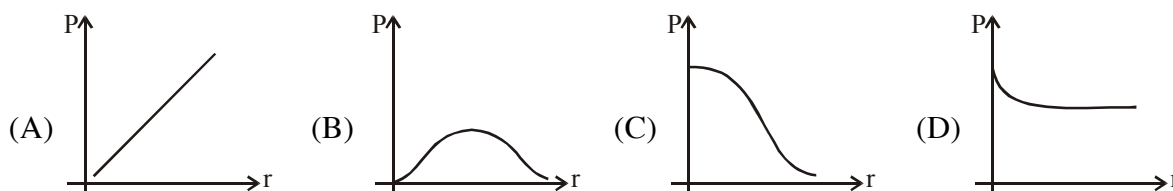
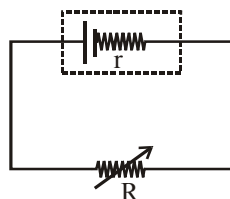
2. A projectile is fired with a speed of $\sqrt{\frac{GM}{r}}$ at certain angle ($\theta \neq 0^\circ, 90^\circ, 180^\circ$) with the radius vector.

Here r is distance of projectile from center of Earth and M is mass of Earth. Choose the **INCORRECT** statement.

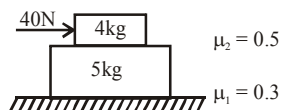
- (A) The projectile will follow elliptical path.
(B) The projectile is at semi-minor axis of its elliptical path.
(C) Earth is at nearer focus if the projectile is initially approaching the Earth.
(D) Earth could be at any focus of the ellipse.
3. Figure (a) and (b) show the I versus V graph for an ohmic and a nonohmic conductor respectively. If the two are joined in series, then which of the following graphs would correspond to the I versus V graph of their combination ?



4. Which of the following graphs represent the variation of power loss in the external load with external resistance R ?



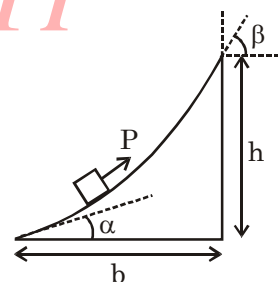
5. Find the acceleration of the two blocks of 4 kg & 5 kg mass if a force of 40 N is applied on 4 kg block. Friction coefficient between respective surfaces are shown in fig.



- (A) 5 m/sec², 5 m/sec² (B) 5 m/sec², 2 m/sec² (C) 2 m/sec², 5 m/sec² (D) 5 m/sec², 0 m/sec

6. A block of mass M has to be dragged up a hill of height h that does not have uniform slope from bottom to top. The externally applied force (P) is always applied to parallel to the surface, block moves with negligible velocity and the coefficient of kinetic friction (μ_k) is same everywhere. Slope of hill varies from α to β as shown in figure. Work done by force P is given by :

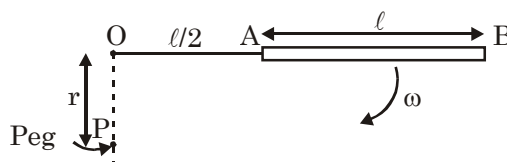
- (A) $Mg(\mu_k h \cot \alpha + \mu_k b \cot \beta)$
 (B) $Mg(\mu_k h \tan \alpha + \mu_k b \tan \beta)$
 (C) $Mgh + \mu_k Mgb$
 (D) $Mgh \left(\frac{\sin \alpha + \sin \beta}{2} \right) + \mu_k Mgb \left(\frac{\cos \alpha + \cos \beta}{2} \right)$



Multiple Correct Answer Type

8 Q. [4 M (0)]

7. A uniform thin rod AB mass M of length ℓ attached to a string OA of length $\frac{\ell}{2}$ is supported by a smooth horizontal plane and rotates with angular velocity ω around a vertical axis through O. A peg P is inserted in the plane in order that on striking it the bar will come exactly to rest :-



- (A) Magnitude of angular momentum of rod about O is $\frac{4}{3} M \ell^2 \omega$
 (B) Magnitude of tension in string is $M \ell \omega^2$
 (C) Location of peg for rod coming to rest is $r = \frac{13}{12} \ell$
 (D) Magnitude of angular impulse by peg on the rod is $\frac{4}{3} M \ell^2 \omega$

8. A binary star system consists of two stars having masses M and $2M$. The distance between their centers is equal to R . They revolve about their common center of mass under their mutual gravitational force of interaction. Which of the following is/are **CORRECT** ?

(A) Heavier star revolves in orbit of radius $\frac{R}{3}$

(B) Both the stars will revolve with the same time period which is equal to $\frac{2\pi}{\sqrt{3Gm}} R^{3/2}$

(C) Kinetic energy of lighter star is twice that of the other

(D) Kinetic energy of lighter star is 4 times that of the other

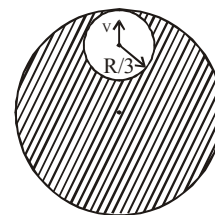
9. Inside a uniform sphere of mass M (M is mass of complete sphere) and radius R , a cavity of radius $R/3$ is made in the sphere as shown :-

(A) Gravitational field inside the cavity is uniform

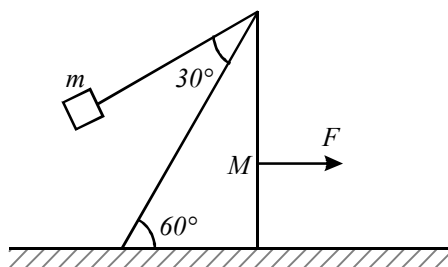
(B) Gravitational field inside the cavity is non-uniform

(C) The escape velocity of a particle projected from centre of cavity is $\sqrt{\frac{88GM}{45R}}$

(D) The escape velocity of a particle projected from centre of cavity is $\sqrt{\frac{20GM}{9R}}$



10. Figure shows a block of mass $m = 1$ kg attached to the top of a wedge of mass $M = 2$ kg through a string. M is pulled on a horizontal smooth surface by an external force such that string attached to it makes an angle 30° with inclined plane as shown. Choose the correct option from below. (Take $g = 10$ m/s²)



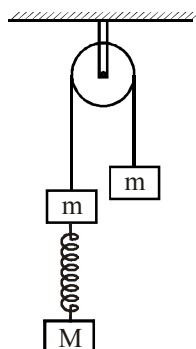
(A) External force $F = 30\sqrt{3}$ N

(B) String tension $T = 20$ N

(C) Normal reaction on ground is 30 N

(D) Normal reaction on ground is $(20 + 10\sqrt{3})$ N

11. The system shown in figure is released from rest with spring at natural length. The spring starts extending as soon as it is released **if** (Neglect friction and masses of pulley, string and spring)



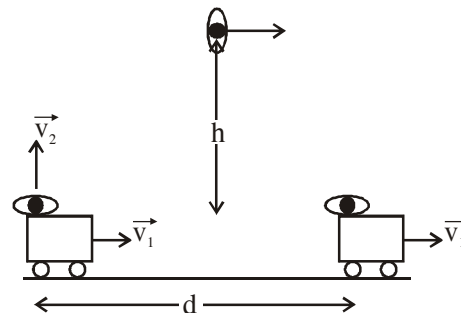
(A) $M > m$

(B) $M > 2m$

(C) $M > m/2$

(D) $M > m/3$

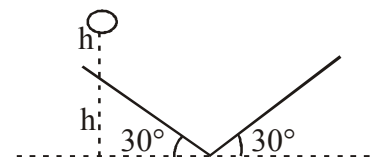
12. A cart moving with horizontal velocity $\vec{v}_1$ ejects a ball with a velocity $\vec{v}_2$ in the vertical direction relative to the moving cart. The ball must pass horizontally through a small stationary hoop appropriately oriented and located at a height 'h' above the launch point. Later the ball passes through a second hoop that is fixed horizontally on the cart and centered at the launch point. You may ignore air resistance. Which of the following is/are correct?



- (A) The height h is $\left(\frac{v_2^2}{2g}\right)$ (B) The cart has traveled a distance $2v_1 v_2 / g$
(C) The cart has travelled a distance $v_1 v_2 / g$ (D) The height h is $2v_2^2 / g$
13. A person, standing on a plank which is kept over a smooth surface, as shown. Person jumps towards right, during the jump, plank also acquires some velocity :-



- (A) Work done by friction, on person is negative.
(B) Work done by friction, on plank is negative.
(C) Position of centre of mass of system (man + plank) will remain same, with respect to a fixed point, before and after the jump.
(D) Increase in mechanical energy of system (man + plank) will be equal to kinetic energy of system in c-frame.
14. A ball is dropped as shown in figure. Both the inclined planes are smooth and ball makes perfect inelastic collision with both planes
- (A) maximum height of ball on second plane is $\frac{5}{16}h$
(B) maximum height of ball on second plane is $\frac{5}{4}h$
(C) Total impulse on ball from start to maximum height on second plane is zero.
(D) Total distance covered by ball from start to maximum height on second plane is $\frac{29}{8}h$



Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 2Q.) [3 M (0)]

Paragraph for Questions 15 and 16

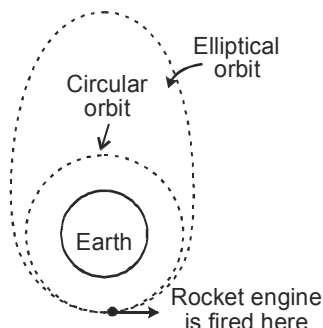
Changing from a circular to An elliptical orbit

Let us identify the system as the spacecraft and the Earth but not the portion of the fuel in the spacecraft that we use to change the orbit. In a given orbit, the mechanical energy of the spacecraft – Earth system is given by

$$E = - \frac{GMm}{2r}$$

This energy includes the kinetic energy of the spacecraft and the potential energy associated with the gravitational force between the spacecraft and the Earth. If the rocket engines are fired, the thrust force moves the spacecraft through a displacement. As a result, the mechanical energy of the spacecraft – Earth system increases.

The spacecraft has a new higher energy but is constrained to be in an orbit that includes the original starting point. It can not be in a higher energy circular orbit having a larger radius because this orbit would not contain the starting point. The only possibility is that the orbit is elliptical as shown in the figure.



$$E = - \frac{GMm}{2a}$$

Above equation gives the energy of the spacecraft – Earth system for an elliptical orbit where a is semimajor axis. Thus if we know the new energy of the orbit, we can find the semi-major axis of the elliptical orbit. Conversely, if we know the semi-major axis of an elliptical orbit we would like to achieve, we can calculate how much additional energy is required from the rocket engines.

A spacecraft is moving in a circular orbit around the Earth (Radius 6400 km), at a height of 300 km from the surface. To place the spacecraft in an elliptical orbit, the magnitude of the mechanical energy of the spacecraft- Earth system is decreased by 10.0%.

15. If the spacecraft-earth system had initial energy $(-E_0)$, then the total mechanical energy of the system after firing the rocket will be :
 (A) $-1.1 E_0$ (B) $-0.9 E_0$ (C) $-E_0$ (D) None of these
16. Minimum distance of the spacecraft from the surface of the Earth is :
 (A) 900 km (B) 1474 km (C) 300 km (D) 1774 km

Linked Comprehension Type

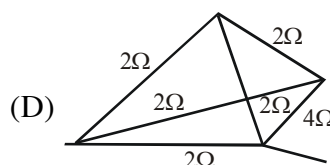
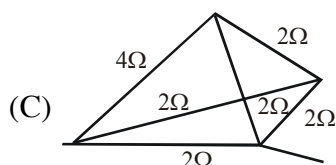
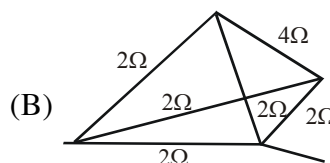
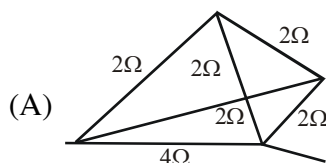
(1 Para × 2 Q.) [4 M (0)]

(Multiple Correct Answer Type)

Paragraph for Question no. 17 and 18

We have five resistances of resistance 2Ω each and one resistance of resistance 4Ω . We arrange these resistances to make the six sides of a tetrahedron. We have a multimeter with the help of which we can measure the equivalent resistance between any two vertices of the tetrahedron.

17. Which of the following values of resistances can we get by our measurement?
 (A) $\frac{4}{3}\Omega$ (B) 1Ω (C) $\frac{4}{11}\Omega$ (D) $\frac{13}{12}\Omega$
18. If our measurement gives the resistance as $4/3\Omega$, which of the following arrangement could be possible?



SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 1 Q. [4 M(-1)]

- To study a solar battery, a multi-range voltmeter is used, which consists of a sensitive ideal micro-ammeter and a set of additional resistances. The voltmeter is connected across the battery. When 10 V range is selected on the voltmeter, it shows 1.4 V and when 100 V range is selected, the voltmeter shows 2.8 V. If 1000 V range is selected, the voltmeter shows 3.x, where x is a digit after the decimal point. Calculate value of x.

SECTION-IV

Matrix Match Type (4 × 5)

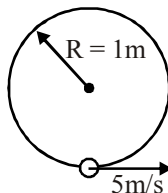
1 Q. [8 M (for each entry +2(0))]

- In column-I, a situation is depicted each of which is in vertical plane. The surfaces are frictionless. Match with appropriate entries in column-II.

Column-I

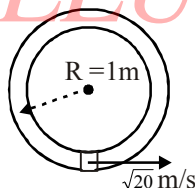
Column-II

- (A) Bead is threaded on a circular fixed wire and is projected from the lowest point.



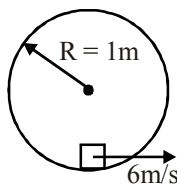
- (P) Normal force is zero at top most point of its trajectory.

- (B) Block loosely fits inside the fixed small tube and is projected from lowest point



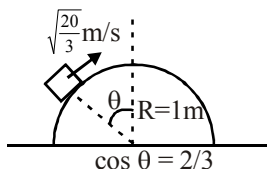
- (Q) Velocity of the body is zero at top most point of its trajectory

- (C) Block is projected horizontally from lowest point of a smooth fixed cylinder.



- (R) Acceleration of the body is zero at the top most point of its trajectory

- (D) Block is projected on a fixed hemisphere from angular position θ .



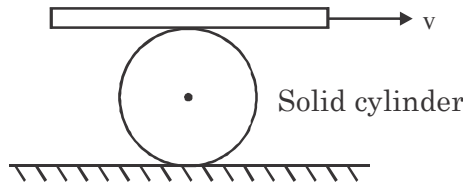
- (S) Normal force is radially outward at top most point of trajectory.

SECTION-I

Single Correct Answer Type

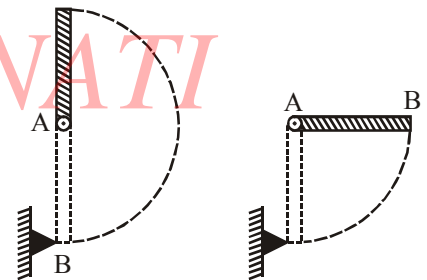
9 Q. [3 M (-1)]

1. Plank is pulled with constant speed v , find KE. of solid cylinder (assume no sliding at any surface, m : mass of cylinder)

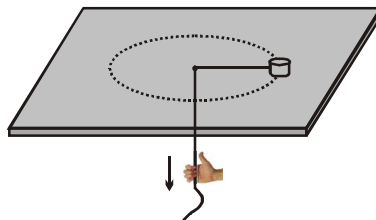


- (A) $\frac{3}{4}mv^2$ (B) $\frac{3}{16}mv^2$ (C) $\frac{1}{16}mv^2$ (D) $\frac{1}{8}mv^2$
2. A rod AB is free to rotate in a vertical plane about a horizontal axis through A as shown in figure. It is slightly disturbed from rest in its position of unstable equilibrium and when it is next vertical the end B collides with a fixed peg and rebounds. If the rod comes to instantaneous rest when AB is horizontal (as shown in figure) then :-

- (A) the coefficient of restitution between the rod and the peg is $\frac{1}{\sqrt{3}}$
(B) the coefficient of restitution between the rod and the peg is $\frac{1}{\sqrt{2}}$
(C) the angular momentum of the rod is constant except for a sudden change at the instant of impact with the peg.
(D) the coefficient of restitution between the rod and the peg is $\frac{1}{\sqrt{6}}$

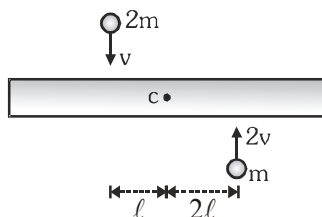


3. A small disk is attached to the end of a light inextensible string, which passes through a hole in a frictionless horizontal tabletop. Initially the disk moves on a circle of radius R with kinetic energy K_0 . The string is then slowly pulled so that the disk finally rotates on a circle of radius R/η . What is the work W done in pulling the string ?



- (A) $W = \eta^2 K_0$ (B) $W = (\eta^2 - 1)K_0$ (C) $W = (\eta - 1)K_0$ (D) $W = 0$
4. A ring of radius R is rolling purely on the outer surface of a pipe of radius $4R$. At some instant, the center of the ring has a constant speed $= v$. Then, the acceleration of the point on the ring which is in contact with the surface of the pipe is
(A) $4v^2/5R$ (B) $3v^2/5R$ (C) $v^2/4R$ (D) zero
5. In the previous question, the acceleration of the point on the ring which is farthest from the center of the pipe, at the given moment, is
(A) $4v^2/5R$ (B) $3v^2/5R$ (C) $3v^2/4R$ (D) $6v^2/5R$

6. A ring of mass m is placed on a rough horizontal surface with its plane vertical. A horizontal impulse J is applied on the ring along a line passing through its centre in its plane. The linear velocity of the centre of the ring once it starts rolling is
- (A) $\frac{J}{m}$ (B) $\frac{J}{2m}$ (C) $\frac{J}{4m}$ (D) $\frac{J}{3m}$
7. A uniform bar of length ' 6ℓ ' and mass ' $8m$ ' lies on a smooth horizontal table. Two point masses m and $2m$ moving in the same horizontal plane with speed $2v$ and v respectively, strike the bar (as shown in the fig.) and stick to the bar after collision. Total energy (about the center of mass, c) will be :-

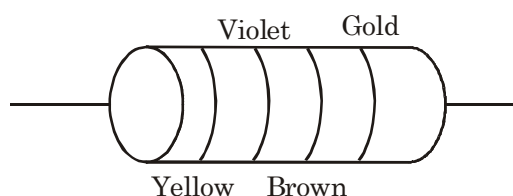


- (A) $\frac{2mv^2}{5}$ (B) $\frac{mv^2}{5}$ (C) $\frac{3mv^2}{5}$ (D) mv^2

8. Current density in a cylindrical wire of radius R is given as $J = \begin{cases} J_0 \left(\frac{x}{R} - 1 \right) & \text{for } 0 \leq x < \frac{R}{2} \\ J_0 \frac{x}{R} & \text{for } \frac{R}{2} \leq x \leq R \end{cases}$. The current flowing in the wire is:

- (A) $\frac{7}{24} \pi J_0 R^2$ (B) $\frac{1}{6} \pi J_0 R^2$ (C) $\frac{7}{12} \pi J_0 R^2$ (D) $\frac{5}{12} \pi J_0 R^2$

9. Figure shows a colour-coded resistor. What is the resistance of this resistor?



- (A) $47 \times 10^1 \Omega \pm 5\%$ (B) $23 \times 10^1 \Omega \pm 5\%$ (C) $47 \times 10^2 \Omega \pm 5\%$ (D) $23 \times 10^2 \Omega \pm 10\%$

Multiple Correct Answer Type

5 Q. [4 M (0)]

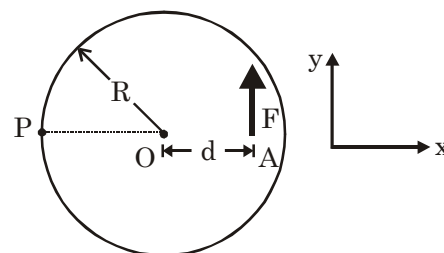
10. A uniform thin flat isolated disc is floating in gravity free space. It has radius R and mass m . A force of constant magnitude ' F ' is applied to it at a distance $d = \frac{R}{2}$ from the center at point A which is always perpendicular to line OA as shown in figure. Treat this problem as two-dimensional. Choose the **CORRECT** statements :-

(A) Magnitude of acceleration of the center of the disc is $\frac{F}{m}$.

(B) Angular acceleration of the disk is $\frac{F}{mR}$.

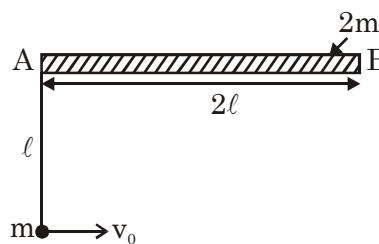
(C) Angular velocity when disc rotate by angle $\pi/2$ is $\sqrt{\frac{F\pi}{mR}}$

(D) Acceleration of point P at shown instant is equals to zero.



11. A uniform rod AB has mass $2m$, length 2ℓ kept on a smooth horizontal plane. Rod AB is orthogonally connected to a taut string whose other end has a particle of mass m . If particle is given initial velocity v_0 , perpendicular to string then just after giving velocity. Mark the **INCORRECT** options :-

- (A) Tension in string is $\frac{2mv_0^2}{9\ell}$
(B) Acceleration of rod is $\frac{v_0^2}{6\ell}$
(C) Angular acceleration of rod is $\frac{2v_0^2}{3\ell^2}$
(D) From the frame of centre of mass of rod, particle moves in a circular path.



12. A student used potentiometer and galvanometer to make variable range voltmeter (V) as shown. He used this voltmeter in figure-2.

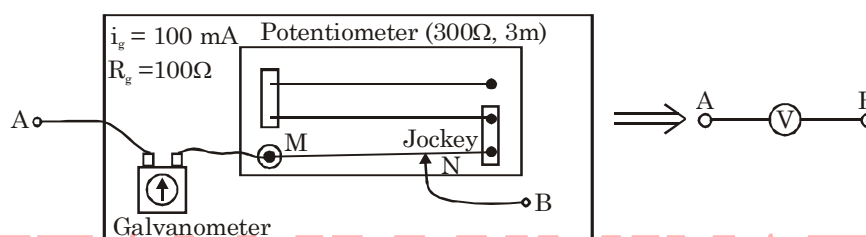


Figure-1 : MN – length of potentiometer wire between point M and N

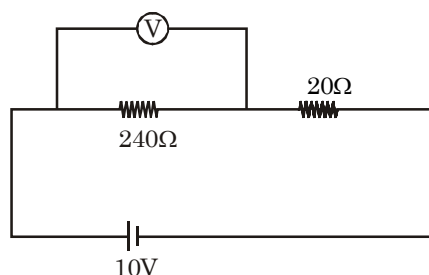
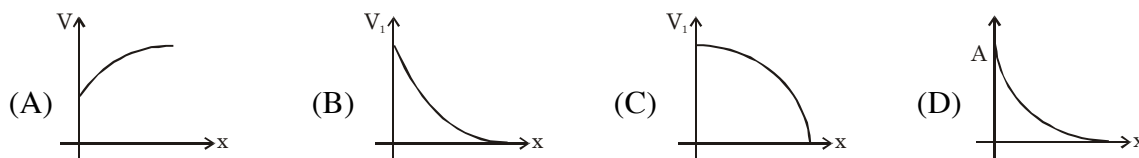
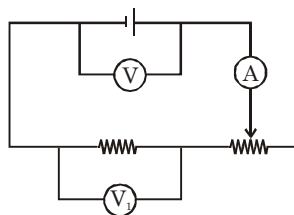
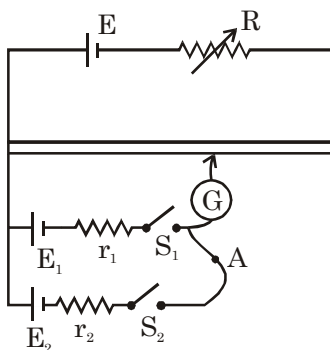


Figure-2

- (A) Reading of voltmeter is 8V when $MN = 20$ cm
(B) Reading of voltmeter is 7V when $MN = 140$ cm
(C) Maximum range of voltmeter is 40 volt
(D) Sensitivity of voltmeter decreases with increase in length MN
13. Two ideal voltmeters & an ideal ammeter are connected as shown. The battery is non ideal. Find the correct plot of A vs x , V vs x & V_1 vs x , where x is distance of slider on rheostat from left end.



14. Two batteries of EMF E_1 and E_2 and internal resistance r_1 and r_2 respectively are connected across an potentiometer as shown. When only switch S_1 is closed, reading of potentiometer is ℓ_1 . When only switch S_2 is closed, reading of potentiometer is ℓ_2 . When both switches are closed, reading is ℓ_3 . Mark the **CORRECT** statement(s) :



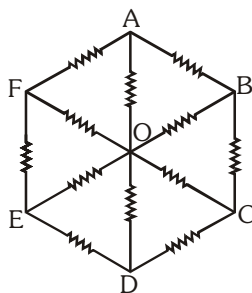
- (A) $\ell_3 = \frac{\ell_1 + \ell_2}{2}$
 (B) $\ell_1 < \ell_3 < \ell_2$ or $\ell_2 < \ell_3 < \ell_1$
 (C) If another galvanometer is attached at point A in series and both switches are closed, it will show zero deflection.
 (D) A change in primary circuit rheostat R changes all the reading of potentiometer.

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q.) [3 M (0)]

Paragraph for Questions no. 15 to 17

Twelve wires of equal resistance R are placed in a given arrangement as shown in figure. Then find



15. Equivalent resistance between A & B is
 (A) $\frac{11R}{20}$ (B) $\frac{10R}{20}$ (C) $\frac{R}{20}$ (D) $\frac{3R}{2}$
16. Equivalent resistance between A & C is
 (A) $\frac{3R}{4}$ (B) $\frac{R}{4}$ (C) $\frac{5R}{6}$ (D) $\frac{3R}{5}$
17. Equivalent resistance between A & O is
 (A) $\frac{19R}{41}$ (B) $\frac{20R}{41}$ (C) $\frac{18R}{23}$ (D) $\frac{9R}{20}$

Linked Comprehension Type

(1 Para × 2 Q.) [4 M (0)]

(Multiple Correct Answer Type)

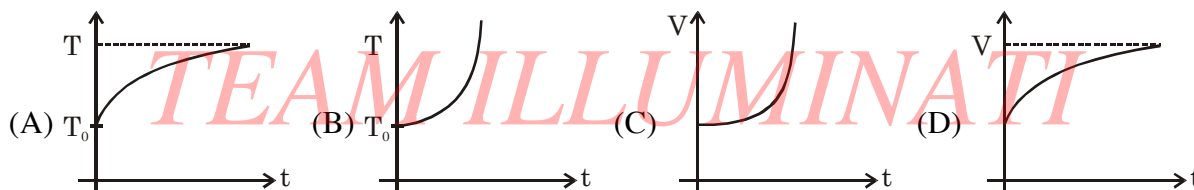
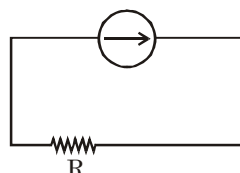
Paragraph for Question no. 18 and 19

A current source is a device which maintains a constant current in the device connected to it. If we connect a variable resistor to it, the voltage across the current source will automatically change to maintain the current. If it is connected to a capacitor, the capacitor will go on charging at a constant rate. The current source cannot be open circuited, or the current will drop to zero. Assume $R = R_0 (1 + \alpha(T - T_0))$ for all temperatures. Its symbol is

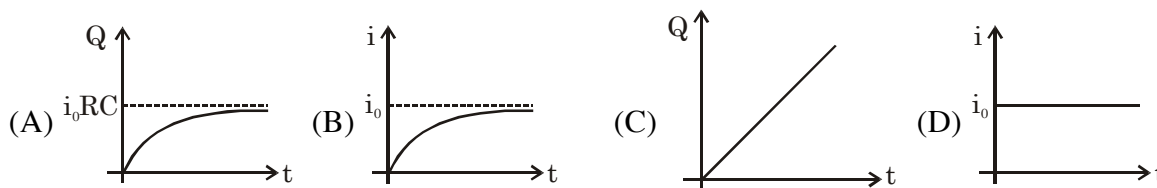
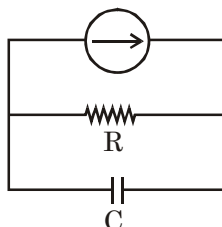


Here the arrow indicates the direction of current flow.

18. As the resistance gets heated up, its resistance changes. The temperature vs time graph, voltage vs time graph can be (Assume no heat is lost to surroundings)



19. The graph of charge on the capacitor as a function of time and current in resistor vs time graph is :

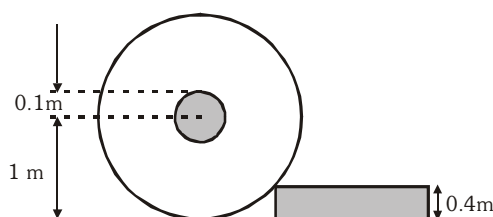


SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)

1 Q. [4 M(-1)]

1. As shown, a wheel of weight W and radius 1.0 m is placed against a 0.4 m height rectangular block fixed on the ground. The wheel has an axle of radius 0.1 m . A force F is applied tangentially to the axle to lift the wheel. The minimum value of F is given by expression $F = \alpha W/11$. Then find the value of α .





QUIZ

JEE (Main + Advanced) 2020
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PHASE : TOAS, TEAS & T-AS

PHYSICS

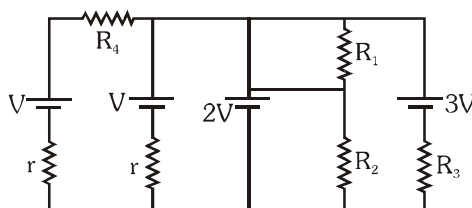
QUIZ # 17

SECTION-I

Single Correct Answer Type

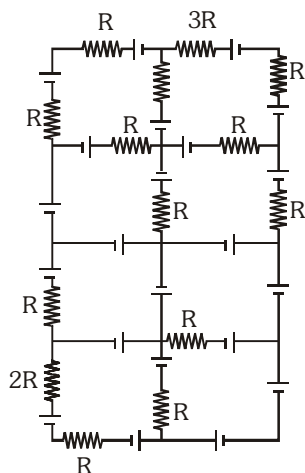
11 Q. [3 M (-1)]

- A spherical cavity of radius $\frac{R}{2}$ is made in a sphere of radius R and mass M . The centre of the cavity is at a distance $\frac{R}{2}$ from the centre of the sphere. The intensity of gravitational field at the centre of cavity is :-
 (A) 0 (B) $\frac{GM}{R^2}$ (C) $\frac{GM}{2R^2}$ (D) $\frac{3GM}{2R^2}$
- Consider a spherical homogeneous cloud of mass M made by an explosion. Due to energy received from the explosion, it is expanding and the expansion is spherically symmetric. At an instant its radius is R_0 and all the particle on the surface are moving away from its center with velocity v_0 . What would be its radius when expansion ceases :-
 (A) $\frac{GMR_0}{2GM - R_0 v_0^2}$ (B) $\frac{2GMR_0}{2GM - R_0 v_0^2}$ (C) $\frac{GMR_0}{2GM + R_0 v_0^2}$ (D) None of these
- The masses and radii of the earth and the moon are M_1, R_1 and M_2, R_2 respectively. Their centres are distance d apart. The minimum speed with which particle of mass m should be projected from a point midway between the two centres so as to escape to infinity is :
 (A) $v = \sqrt{\frac{4g(M_1 + M_2)}{d}}$ (B) $v = \sqrt{\frac{4G(M_1 + M_2)}{d}}$
 (C) $v = \sqrt{4G(M_1 M_2)}$ (D) $v = \sqrt{4Gd(M_1 + M_2)}$
- A binary star system consists of two stars one of which has *double* the mass of the other. The stars rotate about their common centre of mass
 (A) both the stars have same angular momentum about centre of mass
 (B) star having the smaller mass has larger angular momentum about the centre of mass
 (C) the lighter star has smaller linear speed
 (D) the heavier star has higher kinetic energy
- A steady current in a straight conducting wire produces a surface charge on it. Let E_{out} and E_{in} be the magnitudes of the electric fields just outside and just inside the wire, respectively. Which of the following statements is true for these fields?
 (A) E_{out} is always greater than E_{in} (B) E_{out} is always smaller than E_{in}
 (C) E_{out} could be greater or smaller than E_{in} (D) E_{out} is equal to E_{in}
- In the circuit shown, if $R_1 : R_2 : R_3 : R_4 = 1 : 2 : 3 : 4$ then



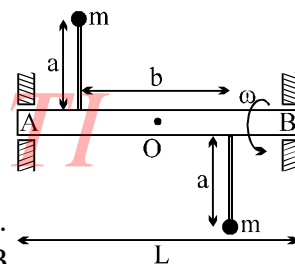
- (A) Ratio of current in R_2 and R_3 is 3 : 1 (B) Ratio of current in R_1 and R_3 is 1 : 2
 (C) Ratio of current in R_1 and R_2 is 4 : 5 (D) current in R_1 is more than one in R_2

7. In the shown circuit all batteries have equal emf E . Current through the $2R$ resistance is



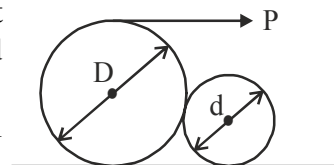
- (A) $\frac{3E}{2R}$ (B) $\frac{2E}{3R}$ (C) $\frac{E}{3R}$ (D) zero

8. A horizontal rod AB which is held by frictionless bearings at its ends, can rotate freely around its horizontal axis. Two equal masses held in position, as shown, by rigid rods of negligible mass are symmetrically located relative to the centre of rod. If the system is rotating with angular velocity ω in absence of gravity, then choose the correct alternative(s).



- (A) The angular momentum of system about point O is along axis of rotation.
(B) The reaction force at bearings **on rod** is upwards at A & downwards at B.
(C) The reaction force at bearings **on rod** is downwards at A & upwards at B.
(D) There would be no reaction forces at bearings.

9. A heavy and a light cylindrical rollers of diameters D and d respectively rest on a horizontal plane as shown. The larger roller has a string wound around it to which a horizontal force 'P' can be applied as shown. Assuming ' μ ' (coefficient of friction) is same at all surfaces of contact, find the critical value of ' μ ' such that larger roller can be just pulled over smaller one.



- (A) $\mu = \sqrt{\frac{D}{d}}$ (B) $\mu = 2\sqrt{\frac{d}{D}}$ (C) $\mu = \sqrt{\frac{d}{D}}$ (D) $\mu = 2\sqrt{\frac{D}{d}}$
10. Two parallel plate capacitors of capacitance C_0 and $2C_0$ are connected in parallel and charged to a potential difference V_0 . The battery is disconnected and the region between the capacitor plates of capacitance C_0 is completely filled with a dielectric of dielectric constant K . The potential difference across the capacitors, now becomes :

- (A) $\frac{V_0}{(3K+1)}$ (B) $\frac{3V_0}{(K+2)}$ (C) $\frac{2V_0}{K}$ (D) $\frac{V_0}{(2K+1)}$
11. Capacity of an isolated sphere of radius R_1 is increased n times when it is enclosed by an earthed concentric sphere of radius R_2 ($R_2 > R_1$). The ratio $\frac{R_2}{R_1}$ is

- (A) $\frac{n^2}{n-1}$ (B) $\frac{n}{n-1}$ (C) $\frac{2n}{n+1}$ (D) $\frac{2n+1}{n+1}$

Multiple Correct Answer Type

7 Q. [4 M (0)]

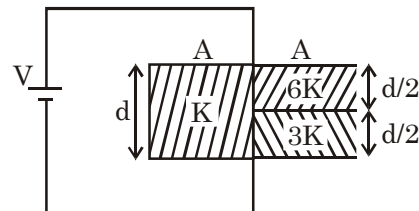
12. 3 dielectric slabs (each of area A) are filled between capacitor of plate area 2A. Capacitor is connected with battery of emf V. Electric field and electric field energy stored in dielectric k, 6k, 3k are E_1, E_2, E_3 and U_1, U_2, U_3 respectively :

(A) Total energy stored in capacitors is $\frac{5Ak\epsilon_0 V^2}{2d}$

(B) $E_1 : E_2 : E_3 :: 3 : 2 : 4$

(C) $U_1 : U_2 : U_3 :: 3 : 4 : 8$

(D) Capacitance of capacitor is $\frac{5Ak\epsilon_0}{d}$.



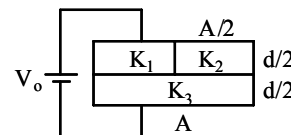
13. An ideal parallel plate capacitor of area A is filled with three dielectric slabs having dielectric constants $K_1 = 3.0$, $K_2 = 5.0$ and $K_3 = 2.0$ as shown in the figure. Choose the **CORRECT** option(s) :-

(A) the equivalent capacitance of the system is $\frac{8\epsilon_0 A}{3d}$

(B) the electric field in K_1 is more than the electric field in K_2 .

(C) the electric field on the left half of K_3 is less than the electric field on the right half of K_3 .

(D) the electric field on the left half of K_3 is more than the electric field on the right half of K_3 .



14. Two square plates of sides ℓ are placed parallel to each other with separation d as suggested in figure. You may assume d is much less than ℓ . The plates carry uniformly distributed static charges $+Q_0$ and $-Q_0$. A block of metal has width ℓ , length ℓ , and thickness slightly less than d. It is inserted at distance x into the space between the plates. The charges on the plates remain uniformly distributed as the block slides in. In a static situation, a metal prevents an electric field from penetrating inside it. The metal can be thought of as a perfect dielectric, with $\kappa \rightarrow \infty$.

(A) The stored energy as a function of x is given by $\frac{Q_0^2 d (\ell - x)}{(2\ell^3 \epsilon_0)}$

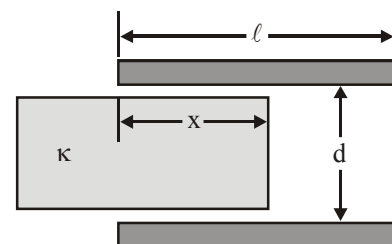
(B) The magnitude of the force that acts on the metallic block is

given by $\frac{Q_0^2 d}{(\ell^3 \epsilon_0)}$ to the right .

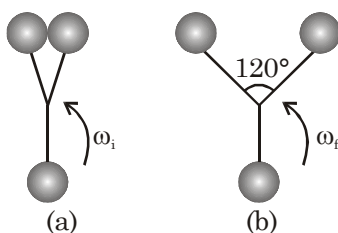
(C) The area of the advancing front face of the block is essentially equal to ℓd , the stress (force per unit area) on it is $\frac{Q_0^2}{(2\ell^4 \epsilon_0)}$.

(D) The energy density in the electric field between the charged plates in terms of Q_0, ℓ, d , and ϵ_0 is

$\frac{Q_0^2}{(2\ell^4 \epsilon_0)}$



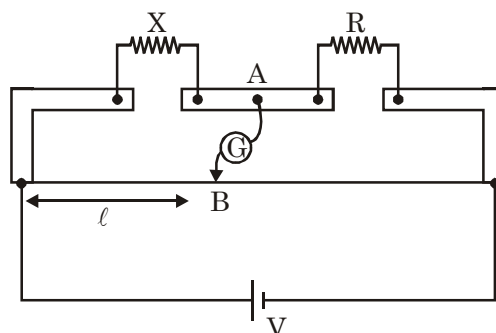
15. Native people used a bola to hunt for birds and animals. A bola consists of three stones, each with mass m , at the ends of three light cords, each with length l . The other ends of the cords are tied together to form a Y. The hunter holds one stone and swings the other two stones. Both stones move together in a horizontal circle of radius $2l$ with speed v_0 . At moment when the horizontal component of their velocity is directed toward the target, the hunter releases the stone from her hand. As the bola flies through the air, the cords quickly take a stable arrangement with constant 120-degree angles between them as shown in figure (b). In the vertical direction, the bola is in free fall. You may ignore the vertical motion as you proceed to describe the horizontal motion of the bola. In terms of m , l , and v_0 , mark the correct statement(s) :



- (A) Angular momentum about the centre of mass is $\frac{4m\ell v_0}{3}$
- (B) Angular speed of the bola about its centre of mass in final configuration is $\frac{4v_0}{9\ell}$
- (C) Kinetic energy of the bola at the instant of release is mv_0^2
- (D) Kinetic energy of the bola in its stable Y shape is $\frac{26mv_0^2}{27}$
16. The radius of the tire of a car is R . The valve cap is at a distance r from the axis of the wheel. The car starts from rest without skidding, at constant acceleration. It's found that the valve cap has no acceleration in the $\frac{1}{8}$ turn preceding the bottom most position. Select the correct alternatives.
- (A) The ratio $\frac{R}{r} = \sqrt{2}$
- (B) The car has travelled a distance $\frac{R}{2}$ upto this time
- (C) The wheel of the car has turned by 0.5 rad upto this time
- (D) The car has travelled a distance $\frac{R}{3}$ upto this time.
17. A small area is removed from a uniform spherical shell of mass M and radius R . Then the gravitational field intensity near the hollow portion is :-

- (A) $\frac{GM}{R^2}$ (B) $\frac{GM}{2R^2}$ (C) $\frac{3GM}{2R^2}$ (D) 0

18. A meter bridge is used to predict the value of unknown resistance X , it is observed that current is flowing from A to B initially (when jockey is fixed at a point B). If the temperature of X is increased, then select possible option/s :



- (A) Current reduces in magnitude
(B) Current remain constant (for any raise in temperature)
(C) Current become zero
(D) Current flows in opposite direction.

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 2 Q. [4 M(-1)]

1. A cell of emf ϵ and internal resistance r is connected to an external resistance R . Efficiency of the power supplied by cell increases to double when R resistance changes from 1Ω to 3Ω , find r (in Ω)
2. A solid sphere of mass m , radius r undergo pure rolling inside a fixed vertical stationary circular track of radius R . The sphere just completes vertical circle. The normal at the lowest point of circular track is

$\frac{34}{\alpha} mg$. The value of α is



QUIZ

**JEE (Main + Advanced) 2020
ENTHUSIAST COURSE
PHASE : TOAS, TEAS & T-AS**

PHYSICS

QUIZ # 18

TIME ALLOTTED : 90 MIN.

SECTION-I

Single Correct Answer Type

3 Q. [3 M (-1)]

1. A current carrying wire heats a metal rod. The wire provides a constant power (P) to the rod. The metal rod is enclosed in an insulated container. It is observed that the temperature (T) in the metal rod changes with time (t) as :

$$T(t) = T_0 (1 + \beta t^{1/4})$$

where β is a constant with appropriate dimension while T_0 is a constant with dimension of temperature. The heat capacity of the metal is :

(1) $\frac{4P(T(t) - T_0)^3}{\beta^4 T_0^4}$ (2) $\frac{4P(T(t) - T_0)}{\beta^4 T_0^2}$ (3) $\frac{4P(T(t) - T_0)^4}{\beta^4 T_0^5}$ (4) $\frac{4P(T(t) - T_0)^2}{\beta^4 T_0^3}$

2. A thin spherical insulating shell of radius R carries a uniformly distributed charge such that the potential at its surface is V_0 . A hole with a small area $\alpha 4\pi R^2$ ($\alpha \ll 1$) is made on the shell without affecting the rest of the shell. Which one of the following statements is correct ?

- (1) The ratio of the potential at the center of the shell to that of the point at $\frac{1}{2}R$ from center towards

the hole will be $\frac{1-\alpha}{1-2\alpha}$

- (2) The magnitude of electric field at the center of the shell is reduced by $\frac{\alpha V_0}{2R}$

- (3) The magnitude of electric field at a point, located on a line passing through the hole and shell's center on a distance $2R$ from the center of the spherical shell will be reduced by $\frac{\alpha V_0}{2R}$

- (4) The potential at the center of the shell is reduced by $2\alpha V_0$

3. Consider a spherical gaseous cloud of mass density $\rho(r)$ in free space where r is the radial distance from its center. The gaseous cloud is made of particles of equal mass m moving in circular orbits about the common center with the same kinetic energy K . The force acting on the particles is their mutual gravitational force. If $\rho(r)$ is constant in time, the particle number density $n(r) = \rho(r)/m$ is :
[G is universal gravitational constant]

(1) $\frac{K}{\pi r^2 m^2 G}$ (2) $\frac{K}{6\pi r^2 m^2 G}$ (3) $\frac{3K}{\pi r^2 m^2 G}$ (4) $\frac{K}{2\pi r^2 m^2 G}$

Multiple Correct Answer Type

4. One mole of a monoatomic ideal gas goes through a thermodynamic cycle, as shown in the volume versus temperature (V - T) diagram. The correct statement(s) is/are :

[R is the gas constant]

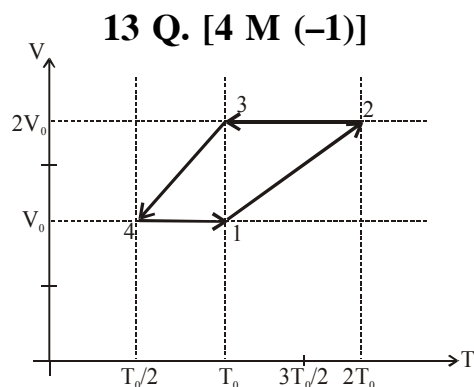
- (1) Work done in this thermodynamic cycle

$$(1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1) \text{ is } |W| = \frac{1}{2} RT_0$$

- (2) The ratio of heat transfer during processes $1 \rightarrow 2$ and $2 \rightarrow 3$ is $\left| \frac{Q_{1 \rightarrow 2}}{Q_{2 \rightarrow 3}} \right| = \frac{5}{3}$

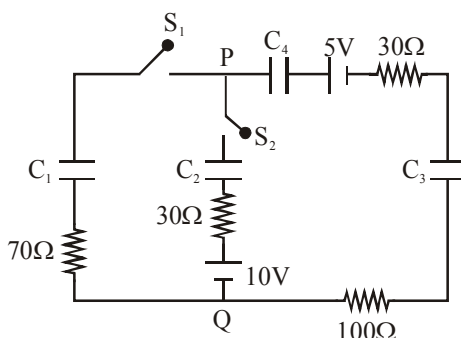
- (3) The above thermodynamic cycle exhibits only isochoric and adiabatic processes.

- (4) The ratio of heat transfer during processes $1 \rightarrow 2$ and $3 \rightarrow 4$ is $\left| \frac{Q_{1 \rightarrow 2}}{Q_{3 \rightarrow 4}} \right| = \frac{1}{2}$



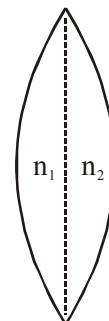
13 Q. [4 M (-1)]

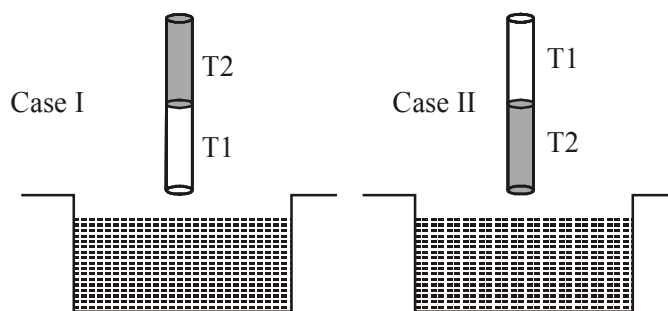
5. In the circuit shown, initially there is no charge on capacitors and keys S_1 and S_2 are open. The values of the capacitors are $C_1 = 10 \mu\text{F}$, $C_2 = 30 \mu\text{F}$ and $C_3 = C_4 = 80 \mu\text{F}$.



Which of the statement(s) is/are correct ?

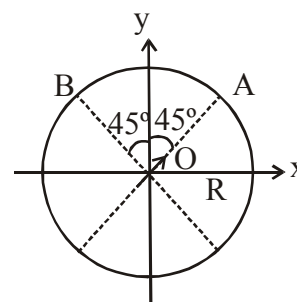
- (1) The keys S_1 is kept closed for long time such that capacitors are fully charged. Now key S_2 is closed, at this time, the instantaneous current across 30Ω resistor (between points P and Q) will be 0.2 A (round off to 1st decimal place).
 - (2) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage difference between points P and Q will be 10 V .
 - (3) At time $t = 0$, the key S_1 is closed, the instantaneous current in the closed circuit will be 25 mA .
 - (4) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage across the capacitors C_1 will be 4 V .
6. A thin convex lens is made of two materials with refractive indices n_1 and n_2 , as shown in figure. The radius of curvature of the left and right spherical surfaces are equal. f is the focal length of the lens when $n_1 = n_2 = n$. The focal length is $f + \Delta f$ when $n_1 = n$ and $n_2 = n + \Delta n$. Assuming $\Delta n \ll (n-1)$ and $1 < n < 2$, the correct statement(s) is/are :
- (1) The relation between $\frac{\Delta f}{f}$ and $\frac{\Delta n}{n}$ remains unchanged if both the convex surfaces are replaced by concave surfaces of the same radius of curvature.
 - (2) $\left| \frac{\Delta f}{f} \right| < \left| \frac{\Delta n}{n} \right|$
 - (3) For $n = 1.5$, $\Delta n = 10^{-3}$ and $f = 20 \text{ cm}$, the value of $|\Delta f|$ will be 0.02 cm (round off to 2nd decimal place)
 - (4) If $\frac{\Delta n}{n} < 0$ then $\frac{\Delta f}{f} > 0$
7. Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L , which of the following statement (s) is/are correct ?
- (1) The dimension of force is L^{-3}
 - (2) The dimension of energy is L^{-2}
 - (3) The dimension of power is L^{-5}
 - (4) The dimension of linear momentum is L^{-1}
8. A cylindrical capillary tube of 0.2 mm radius is made by joining two capillaries T1 and T2 of different materials having water contact angles of 0° and 60° , respectively. The capillary tube is dipped vertically in water in two different configurations, case I and II as shown in figure. Which of the following option(s) is/are correct ?
- (Surface tension of water = 0.075 N/m , density of water = 1000 kg/m^3 , take $g = 10 \text{ m/s}^2$)





- (1) The correction in the height of water column raised in the tube, due to weight of water contained in the meniscus, will be different for both cases.
- (2) For case I, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be more than 8.75 cm. (Neglect the weight of the water in the meniscus)
- (3) For case I, if the joint is kept at 8 cm above the water surface, the height of water column in the tube will be 7.5 cm. (Neglect the weight of the water in the meniscus)
- (4) For case II, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be 3.75 cm. (Neglect the weight of the water in the meniscus)
9. Two identical moving coil galvanometer have $10\ \Omega$ resistance and full scale deflection at $2\ \mu\text{A}$ current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1 mA full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with $R = 1000\ \Omega$ resistor by using an ideal cell. Which of the following statement(s) is/are correct ?
- (1) The measured value of R will be $978\ \Omega < R < 982\ \Omega$.
- (2) The resistance of the Voltmeter will be $100\ \text{k}\Omega$.
- (3) The resistance of the Ammeter will be $0.02\ \Omega$ (round off to 2nd decimal place)
- (4) If the ideal cell is replaced by a cell having internal resistance of $5\ \Omega$ then the measured value of R will be more than $1000\ \Omega$.
10. A charged shell of radius R carries a total charge Q . Given Φ as the flux of electric field through a closed cylindrical surface of height h , radius r and with its center same as that of the shell. Here, center of the cylinder is a point on the axis of the cylinder which is equidistant from its top and bottom surfaces. Which of the following option(s) is/are correct ? [ϵ_0 is the permittivity of free space]
- (1) If $h > 2R$ and $r > R$ then $\Phi = \frac{Q}{\epsilon_0}$
- (2) If $h < \frac{8R}{5}$ and $r = \frac{3R}{5}$ then $\Phi = 0$
- (3) If $h > 2R$ and $r = \frac{4R}{5}$ then $\Phi = \frac{Q}{5\epsilon_0}$
- (4) If $h > 2R$ and $r = \frac{3R}{5}$ then $\Phi = \frac{Q}{5\epsilon_0}$
11. A mixture of ideal gas containing 5 moles of monatomic gas and 1 mole of rigid diatomic gas is initially at pressure P_0 , volume V_0 and temperature T_0 . If the gas mixture is adiabatically compressed to a volume $V_0/4$, then the correct statement(s) is/are, (Give $2^{1.2} = 2.3$; $2^{3.2} = 9.2$; R is gas constant)
- (1) The final pressure of the gas mixture after compression is in between $9P_0$ and $10P_0$
- (2) The average kinetic energy of the gas mixture after compression is in between $18RT_0$ and $19RT_0$
- (3) The work $|W|$ done during the process is $13RT_0$
- (4) Adiabatic constant of the gas mixture is 1.6

12. An electric dipole with dipole moment $\frac{p_0}{\sqrt{2}}(\hat{i} + \hat{j})$ is held fixed at the origin O in the presence of an uniform electric field of magnitude E_0 . If the potential is constant on a circle of radius R centered at the origin as shown in figure, then the correct statement(s) is/are: (ϵ_0 is permittivity of free space, $R \gg$ dipole size)



- (1) $R = \left(\frac{p_0}{4\pi\epsilon_0 E_0} \right)^{1/3}$
 - (2) The magnitude of total electric field on any two points of the circle will be same
 - (3) Total electric field at point A is $\vec{E}_A = \sqrt{2}E_0(\hat{i} + \hat{j})$
 - (4) Total electric field at point B is $\vec{E}_B = 0$
13. A thin and uniform rod of mass M and length L is held vertical on a floor with large friction. The rod is released from rest so that it falls by rotating about its contact-point with the floor without slipping. Which of the following statement(s) is/are correct, when the rod makes an angle 60° with vertical ? [g is the acceleration due to gravity]

- (1) The radial acceleration of the rod's center of mass will be $\frac{3g}{4}$

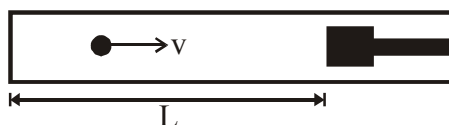
- (2) The angular acceleration of the rod will be $\frac{2g}{L}$

- (3) The angular speed of the rod will be $\sqrt{\frac{3g}{2L}}$

- (4) The normal reaction force from the floor on the rod will be $\frac{Mg}{16}$

14. A small particle of mass m moving inside a heavy, hollow and straight tube along the tube axis undergoes elastic collision at two ends. The tube has no friction and it is closed at one end by a flat surface while the other end is fitted with a heavy movable flat piston as shown in figure. When the distance of the piston from closed end is $L = L_0$ the particle speed is $v = v_0$. The piston is moved inward at a very

low speed V such that $V \ll \frac{dL}{L} v_0$, where dL is the infinitesimal displacement of the piston. Which of the following statement(s) is/are correct ?

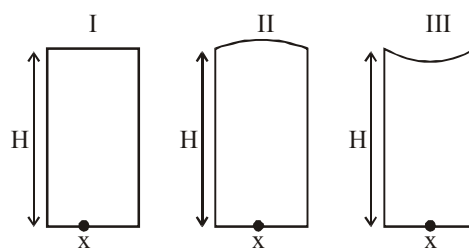


- (1) The rate at which the particle strikes the piston is v/L
- (2) After each collision with the piston, the particle speed increases by $2V$
- (3) The particle's kinetic energy increases by a factor of 4 when the piston is moved inward from

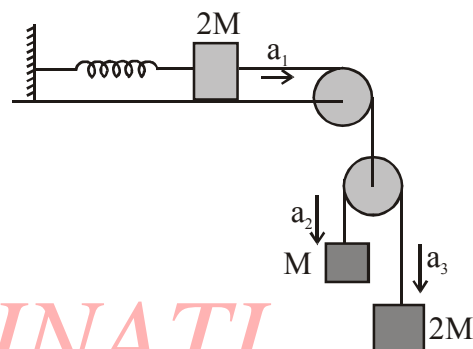
$$L_0 \text{ to } \frac{1}{2}L_0$$

- (4) If the piston moves inward by dL, the particle speed increases by $2v \frac{dL}{L}$

15. Three glass cylinders of equal height $H = 30$ cm and same refractive index $n = 1.5$ are placed on a horizontal surfaces shown in figure. Cylinder I has a flat top, cylinder II has a convex top and cylinder III has a concave top. The radii of curvature of the two curved tops are same ($R = 3$ m). If H_1 , H_2 and H_3 are the apparent depths of a point X on the bottom of the three cylinders, respectively, the correct statement(s) is/are



- (1) $H_3 > H_1$ (2) $0.8 \text{ cm} < (H_2 - H_1) < 0.9 \text{ cm}$
(3) $H_2 > H_3$ (4) $H_2 > H_1$
16. A block of mass $2M$ is attached to a massless spring with spring-constant k . This block is connected to two other blocks of masses M and $2M$ using two massless pulleys and strings. The accelerations of the blocks are a_1 , a_2 and a_3 as shown in figure. The system is released from rest with the spring in its unstretched state. The maximum extension of the spring is x_0 . Which of the following option(s) is/are correct ? [g is the acceleration due to gravity. Neglect friction]



(1) $x_0 = \frac{4Mg}{k}$

- (2) When spring achieves an extension of $\frac{x_0}{2}$ for the first time, the speed of the block connected to

the spring is $3g\sqrt{\frac{M}{5k}}$

(3) $a_2 - a_1 = a_1 - a_3$

- (4) At an extension of $\frac{x_0}{4}$ of the spring, the magnitude of acceleration of the block connected to the

spring is $\frac{3g}{10}$

Matching List Type (4×6)

2Q.[3 M (-1)]

17. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In a thermodynamics process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas

$$X = \frac{3}{2} R \ln \left(\frac{T}{T_A} \right) + R \ln \left(\frac{V}{V_A} \right). \text{ Here, } R \text{ is gas constant, } V \text{ is volume of gas, } T_A \text{ and } V_A \text{ are constants.}$$

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List-I

- (I) Work done by the system in process $1 \rightarrow 2 \rightarrow 3$
(II) Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$
(III) Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$
(IV) Heat absorbed by the system in process $1 \rightarrow 2$

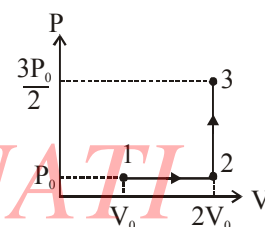
List-II

- (P) $\frac{1}{3}RT_0 \ln 2$
(Q) $\frac{1}{3}RT_0$
(R) RT_0
(S) $\frac{4}{3}RT_0$
(T) $\frac{1}{3}RT_0(3 + \ln 2)$
(U) $\frac{5}{6}RT_0$

If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram

with $P_0 V_0 = \frac{1}{3}RT_0$, the correct match is,

- (1) $I \rightarrow Q, II \rightarrow R, III \rightarrow P, IV \rightarrow U$
(2) $I \rightarrow S, II \rightarrow R, III \rightarrow Q, IV \rightarrow T$
(3) $I \rightarrow Q, II \rightarrow R, III \rightarrow S, IV \rightarrow U$
(4) $I \rightarrow Q, II \rightarrow S, III \rightarrow R, IV \rightarrow U$



18. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas

$X = \frac{3}{2}R \ln \left(\frac{T}{T_A} \right) + R \ln \left(\frac{V}{V_A} \right)$. Here, R is gas constant, V is volume of gas, T_A and V_A are constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List-I

- (I) Work done by the system in process $1 \rightarrow 2 \rightarrow 3$
(II) Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$
(III) Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$
(IV) Heat absorbed by the system in process $1 \rightarrow 2$

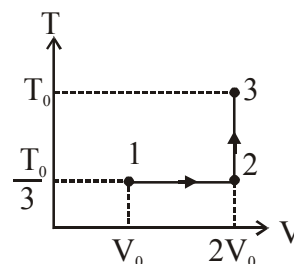
List-II

- (P) $\frac{1}{3}RT_0 \ln 2$
(Q) $\frac{1}{3}RT_0$
(R) RT_0
(S) $\frac{4}{3}RT_0$
(T) $\frac{1}{3}RT_0(3 + \ln 2)$
(U) $\frac{5}{6}RT_0$

If the process on one mole of monatomic ideal gas is as shown in the TV-diagram with

$$P_0 V_0 = \frac{1}{3} RT_0, \text{ the correct match is}$$

- (1) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ U
- (2) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ S
- (3) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ T
- (4) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ P



SECTION-II

Numerical Answer Type Question

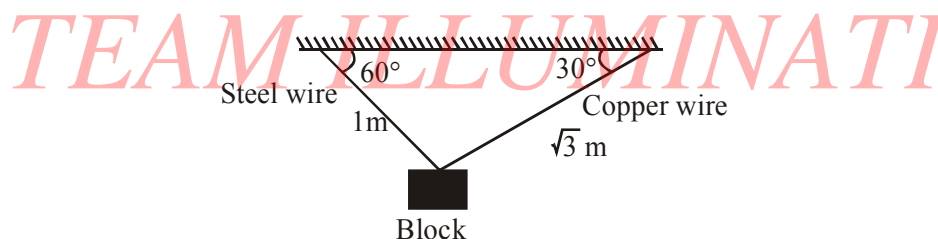
7Q.[3(0)]

(upto second decimal place)

1. A block of weight 100 N is suspended by copper and steel wires of same cross sectional area 0.5 cm^2 and, length $\sqrt{3} \text{ m}$ and 1 m , respectively. Their other ends are fixed on a ceiling as shown in figure. The angles subtended by copper and steel wires with ceiling are 30° and 60° , respectively. If elongation

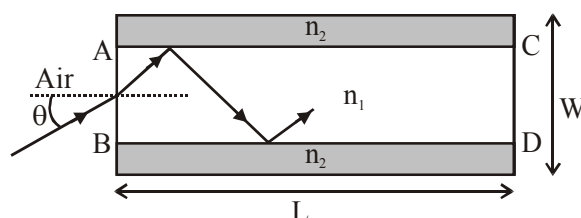
in copper wire is $(\Delta \ell_C)$ and elongation in steel wire is $(\Delta \ell_S)$, then the ratio $\frac{\Delta \ell_C}{\Delta \ell_S}$ is ____.

[Young's modulus for copper and steel are $1 \times 10^{11} \text{ N/m}^2$ and $2 \times 10^{11} \text{ N/m}^2$ respectively]



2. A planar structure of length L and width W is made of two different optical media of refractive indices $n_1 = 1.5$ and $n_2 = 1.44$ as shown in figure. If $L \gg W$, a ray entering from end AB will emerge from end CD only if the total internal reflection condition is met inside the structure. For $L = 9.6 \text{ m}$, if the incident angle θ is varied, the maximum time taken by a ray to exit the plane CD is $t \times 10^{-9} \text{ s}$, where t is ____.

[Speed of light $c = 3 \times 10^8 \text{ m/s}$]

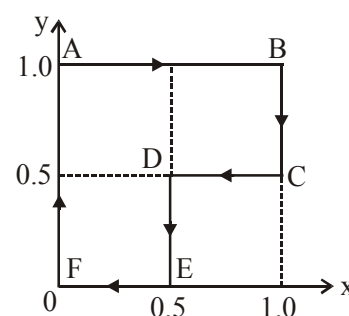


3. A particle is moved along a path AB-BC-CD-DE-EF-FA, as shown

in figure, in presence of a force $\vec{F} = (\alpha y \hat{i} + 2\alpha x \hat{j}) \text{ N}$, where x and

y are in meter and $\alpha = -1 \text{ N/m}^{-1}$. The work done on the particle

by this force $\vec{F}$ will be ____ Joule.



4. A parallel plate capacitor of capacitance C has spacing d between two plates having area A . The region between the plates is filled with N dielectric layers, parallel to its plates, each with thickness $\delta = \frac{d}{N}$.

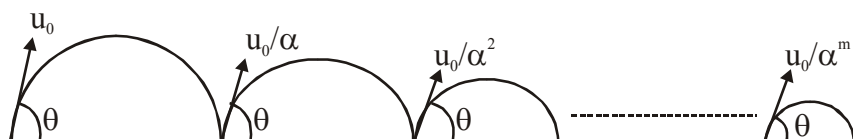
The dielectric constant of the m^{th} layer is $K_m = K \left(1 + \frac{m}{N} \right)$. For a very large $N (> 10^3)$, the capacitance C is $\propto \left(\frac{K \epsilon_0 A}{d \ln 2} \right)$. The value of α will be ____.

$[\epsilon_0$ is the permittivity of free space]

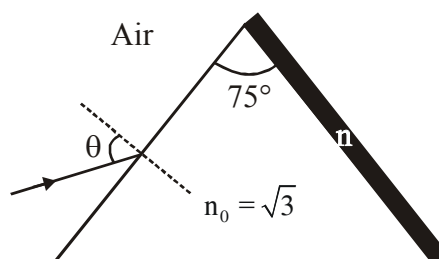
5. A liquid at 30°C is poured very slowly into a Calorimeter that is at temperature of 110°C . The boiling temperature of the liquid is 80°C . It is found that the first 5 gm of the liquid completely evaporates. After pouring another 80 gm of the liquid the equilibrium temperature is found to be 50°C . The ratio of the Latent heat of the liquid to its specific heat will be ____ $^\circ\text{C}$.

[Neglect the heat exchange with surrounding]

6. A ball is thrown from ground at an angle θ with horizontal and with an initial speed u_0 . For the resulting projectile motion, the magnitude of average velocity of the ball up to the point when it hits the ground for the first time is V_1 . After hitting the ground, ball rebounds at the same angle θ but with a reduced speed of u_0/α . Its motion continues for a long time as shown in figure. If the magnitude of average velocity of the ball for entire duration of motion is $0.8 V_1$, the value of α is ____



7. A monochromatic light is incident from air on a refracting surface of a prism of angle 75° and refractive index $n_0 = \sqrt{3}$. The other refracting surface of a prism is coated by a thin film of material of refractive index n as shown in figure. The light suffers total internal reflection at the coated prism surface for an incidence angle of $\theta \leq 60^\circ$. The value of n^2 is ____.



PHYSICS

QUIZ # 19

SECTION-I

Single Correct Answer Type

1 Q. [3 M (-1)]

1. A parallel plate capacitor of value $1.77 \mu\text{F}$ is to be designed using a dielectric material of dielectric constant = 200, break through strength = $3 \times 10^6 \text{ Vm}^{-1}$. In order to make such a capacitor, which can withstand a potential difference of 20V across the plates, the separation between the plates d and the area of the plates A (each) can be
- (A) $d = 10^{-6} \text{ m}$, $A = 10^{-3} \text{ m}^2$ (B) $d = 10^{-5} \text{ m}$, $A = 10^{-2} \text{ m}^2$
 (C) $d = 10^{-4} \text{ m}$, $A = 10^{-4} \text{ m}^2$ (D) $d = 10^{-4} \text{ m}$, $A = 10^{-5} \text{ m}^2$

One or More than one correct answer type

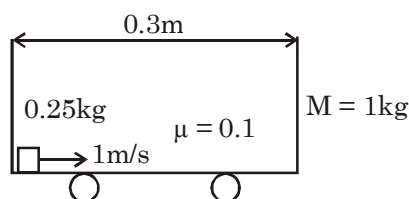
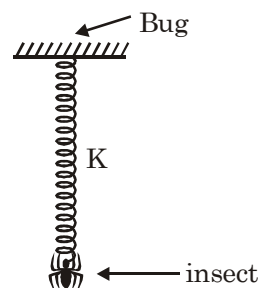
7 Q. [4; +1(-2)]

2. A large horizontal turntable is rotating with constant angular speed ω_0 in counterclockwise sense. A person standing at the centre, begins to walk eastward with a constant velocity V relative to the table. Taking origin at the centre and X direction to be eastward calculate the possible X co-ordinate of the person.
- (A) $\frac{2V}{\omega_0}$ (B) $\frac{V}{\omega_0}$ (C) $\frac{V}{2\omega_0}$ (D) $\frac{V}{3\omega_0}$
3. The ends of a metallic bar are maintained at different temperature. In the steady state the temperature variation along the length of the bar is as shown in the figure. This could possibly be due to :



- (A) No heat loss to the surroundings along the length of the bar, the area of cross-section is decreasing as we move from hot end to cold end.
 (B) No heat loss to the surroundings along the length of the bar, the coefficient of thermal conductivity is decreasing as we moved from hot end to cold end.
 (C) No heat loss to the surroundings along the length of the bar, the area of cross-section is increasing as we move from hot end to cold end.
 (D) The bar is uniform but heat is being lost from the sides along the length of the bar to the surroundings.
4. Two particles of mass m_1 and m_2 are connected by a spring of natural length L , and force constant k . The masses are brought close enough so as to compress the spring completely and a string is used to tie the system. Assume that length of spring in this position is close to zero. The system is projected with a velocity V_0 along the positive X direction. At the instant it reaches origin at time $t = 0$, the string snaps and the spring starts opening.
- (A) Both the particles execute SHM from centre of mass frame of reference with the same time period
 (B) mass m_1 has amplitude of $A_1 = \frac{m_2 L}{m_1 + m_2}$
 (C) mass m_2 has amplitude of $A_2 = \frac{m_1 L}{m_1 + m_2}$
 (D) spring will be in the natural length at time $t = \frac{T}{2}$.

5. Massless spring of force constant K and natural L is hanging from a ceiling. An insect of mass m is hanging at the lower end of the spring and the system is in equilibrium. The insect starts slowly climbing up the spring so as to eat a bug sitting on the ceiling. Assume that insect climbs without slipping on the spring and $K = \frac{mg}{L}$.
- (A) The initial length of the spring is $2L$
- (B) When the insect is at a distance L from the bug, the extension in the spring is $\frac{L}{2}$
- (C) When the insect is at $\frac{L}{4}$ from the bug, the extension in the spring is $\frac{L}{8}$
- (D) When the insect is at $\frac{L}{2}$ from the bug, the length of the spring is $\frac{5L}{4}$.
6. A stationary person observes that rain is falling vertically down at 30 kmph. A cyclist is moving up an inclined plane making an angle 30° with horizontal at 10 kmph. In which direction should the cyclist hold his umbrella to prevent himself from the rain?
- (A) At an angle $\tan^{-1}\left(\frac{3\sqrt{3}}{5}\right)$ with inclined plane
- (B) At an angle $\tan^{-1}\left(\frac{3\sqrt{3}}{5}\right)$ with horizontal
- (C) At an angle $\tan^{-1}\left(\frac{\sqrt{3}}{7}\right)$ with inclined plane
- (D) At an angle $\tan^{-1}\left(\frac{\sqrt{3}}{7}\right)$ with vertical
7. The potential energy 'U' in Joule, of a particle of mass 1kg, moving in x-y plane, obeys the law $U = 3x + 4y$, where (x, y) are the co-ordinate of the particle in metre. If the particle is at rest at (6, 4) at $t = 0$, then
- (A) The particle has constant acceleration
- (B) The work done by the force, from the position of rest of the particle to the instant of crossing the x-axis is 25 J
- (C) The speed of the particle when it crosses the y-axis is 10 m/s
- (D) The co-ordinates of the particle at time $t = 4$ s are $(-18, -28)$
8. A small block of mass 0.25 kg is projected on the rough inner surface of a trolley of length 30cm. The trolley has a mass of 1 kg and can move frictionlessly on ground. Initially trolley is at rest & velocity of projection is 1 m/s. Collision of block with walls of trolley are elastic :



- (A) Finally, both of them are moving at 0.2 m/s.
- (B) Block undergoes two collision with walls.
- (C) The block is finally at a distance of 2.5 cm from rear wall.
- (D) The velocity of trolley continuously increases till the block finally comes to rest with respect to it.

Matching list based comprehension Type ($4 \times 4 \times 4$)**Single option correct****(Three Columns and Four Rows)****2 Table $\times$ 3 Q. [3(-1)]**

Answer Q.9, Q.10 and Q.11 by appropriately matching the information given in the three columns of the following table.

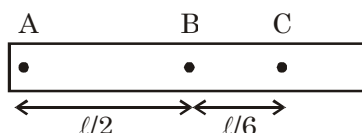
A particle of mass 1 kg is confined in a one dimensional force field. Column-I gives the dependence of potential energy on the x-coordinate. Column-II gives initial position of the particle and column-III gives initial velocity of the particle.

Column-1 Potential energy	Column-2 Initial position	Column-3 Initial velocity
(I) $U = x + \frac{1}{x}$	(i) $x = 0$	(P) 0
(II) $U = 1 - e^{-x^2}$	(ii) $x = +1$	(Q) +0.01 m/s
(III) $U = x^2 + x$	(iii) $x = +\frac{2}{\sqrt{3}}$	(R) -1 m/s
(IV) $U = x \left(1 - \frac{x^2}{4} \right)$	(iv) $x = -1$	(S) +2 m/s

9. In which of the combination, particle will remain at rest ?
 (A) (I) (iv) (P) (B) (II) (iii) (P) (C) (III) (i) (P) (D) (IV) (ii) (P)
10. In which of the combination, particle will not turn back?
 (A) (I) (iv) (R) (B) (II) (i) (R) (C) (III) (i) (R) (D) (IV) (ii) (Q)
11. In which of the combination, particle will execute SHM ?
 (A) (II) (i) (S) (B) (I) (ii) (Q) (C) (III) (iv) (P) (D) (II) (ii) (P)

Answer Q.12, Q.13 and Q.14 by appropriately matching the information given in the three columns of the following table.

Consider a uniform rod of mass m and length ℓ . A is the end of rod. B is centre of rod.



Column-1	Column-2 Force F is applied perpendicular to rod in upward direction at	Column-3
(I) Rod is hinged at A	(i) A	(P) Initial angular velocity = 0 and velocity of COM = 0
(II) Rod is hinged at B	(ii) B	(Q) Initial angular velocity is clockwise
(III) Rod is hinged at C	(iii) C	(R) Initial angular velocity is anticlockwise
(IV) Rod is not hinged at all	(iv) A & C simultaneously (C in downward & A in upward direction)	(S) Initial angular velocity is zero but velocity of COM $\neq 0$

12. In which of the cases, force exerted by hinge is zero.
 (A) (I) (i) (P) (B) (II) (iv) (Q) (C) (III) (iii) (P) (D) (I) (iii) (P)

13. In which of the following cases, angular velocity certainly increases.
(A) (I) (i) (Q) (B) (II) (ii) (R) (C) (IV) (iv) (R) (D) (II) (iv) (P)
14. In which of the following cases, velocity of centre of mass will not change ?
(A) (I) (iii) (P) (B) (III) (iii) (P) (C) (IV) (i) (S) (D) (I) (i) (Q)

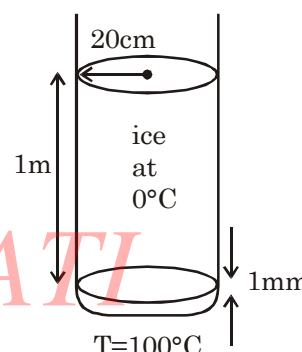
Numerical Grid Type

6 Q. [3(-0)]

(Single digit Ranging from 0 to 9)

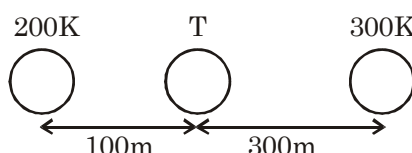
1. A column of mercury of length 10 cm is contained in the middle of a horizontal tube of length 1m which is closed at both the ends. The two equal parts contain air at standard atmospheric pressure of 76 cm of mercury. The tube is now turned to vertical position. Then the column of mercury will be displaced by x cm. The value of x after rounding off to nearest integer.
2. 1 litre water at 1°C, 2 litre water at 2°C, 3 litre water at 3°C, 4 litre water at 4°C, 5 litre water at 5°C, 6 litre water at 6°C, 7 litre water at 7°C, 8 litre water at 8°C, & 9 litre water at 9°C are mixed. If there is no heat loss, what is the final temperature (in °C) of the mixture? Round off to nearest integer.

3. A cylinder of ice is made inside an insulated cup as shown. The bottom of cup is a plate of thermal conductivity 3.35 W/mK. It's other side is at 100°C. Density of ice = 0.9 gm/cc, density of water = 1 gm/cc. What is the downward speed v of the top surface

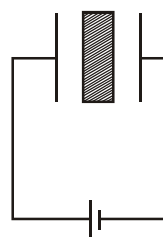
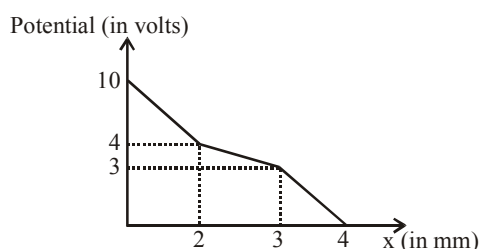


(in mm/s). $L = 3.35 \times 10^5 \text{ J/kg}$. Fill $\frac{1}{v}$ in OMR sheet.

4. Three spheres of radius 0.1 m each are painted black. They are kept on a straight line such that distance between two successive spheres are 100 m and 300 m. They are in vacuum. If the temperature of left most sphere is maintained at 200 K & rightmost at 300 K. What is the steady state temperature of sphere (in K) in between.



5. An elevator is hung by a steel cable ($\sigma_b = 2 \times 10^9 \text{ N/m}^2$) of area 1 cm^2 . For safety purpose, you don't want tension in cable to be more than 10% of it's maximum permissible value. The mass of elevator is 1000 kg. What is the minimum time (in sec) in which the elevator can be moved 10 m vertically up starting from rest & stopping at the destination.
6. A capacitor filled partially with dielectric material of dielectric constant 'k'. Its electric potential v/s position graph is as shown. Distance between the two plates is 4mm. Find the dielectric constant of medium.



PHYSICS

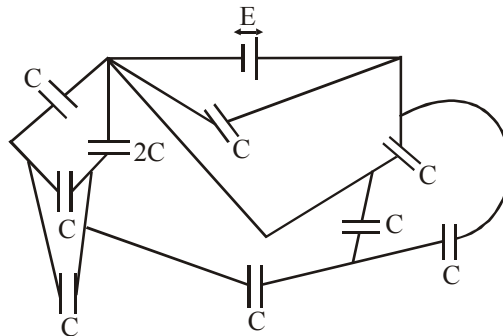
QUIZ # 20

SECTION-I

Single Correct Answer Type

13 Q. [3 M (-1)]

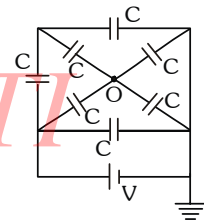
1. What is the equivalent capacitance across the battery ?



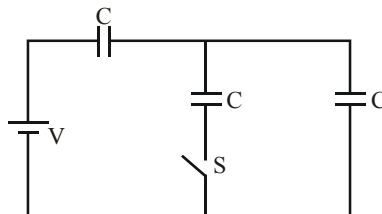
- (A) $\frac{79}{30}C$ (B) $\frac{59}{30}C$ (C) $\frac{41}{30}C$ (D) $\frac{21}{30}C$

2. In the given circuit find potential at point 'O' :-

- (A) $\frac{4V}{11}$ (B) $\frac{3V}{11}$
(C) $\frac{9V}{11}$ (D) None of these

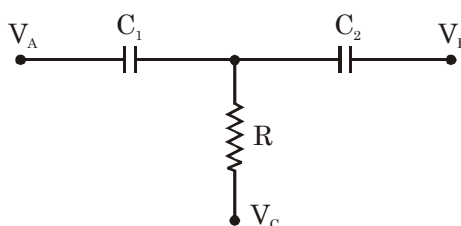


3. In the circuit shown, find heat dissipated in the circuit after switch is opened (Initially switch is closed) :-



- (A) $\frac{CV^2}{3}$ (B) $\frac{CV^2}{6}$ (C) $\frac{CV^2}{12}$ (D) 0

4. For given circuit charge on capacitor C_1 and C_2 in steady state will be equal to :-



- (A) $C_1(V_A - V_C)$, $C_2(V_C - V_B)$ respectively (B) $C_1(V_A - V_B)$, $C_2(V_A - V_B)$ respectively
(C) $(C_1 + C_2)(V_A - V_B)$ on each capacitor (D) $\left(\frac{C_1 C_2}{C_1 + C_2}\right)(V_A - V_B)$ on each capacitor

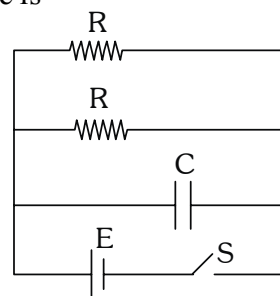
5. Time constant of charging of the capacitor shown in the adjoining figure is

(A) $2RC$

(B) $\frac{CR}{2}$

(C) zero

(D) infinite



6. In the following RC circuit, the capacitor was charged in two different ways.

- (i) The capacitor was first charged to 5V by moving the toggle switch to position P and then it was charged to 10V by moving the toggle switch to position Q.
(ii) The capacitor was directly charged to 10V, by keeping the toggle switch at position Q.

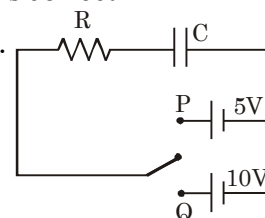
Assuming the capacitor to be ideal, which one of the following statement is correct?

(A) The energy dissipation in cases (i) and (ii) will be equal and non-zero.

(B) The energy dissipation for case (i) will be more than that for case (ii).

(C) The energy dissipation for case (i) will be less than that for case (ii).

(D) The energy will not be dissipated in either case.



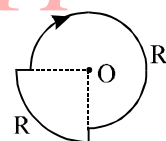
7. A current of i ampere is flowing through each of the bent wires as shown the magnitude and direction of magnetic field at O is

(A) $\frac{\mu_0 i}{4} \left(\frac{1}{R} + \frac{2}{R'} \right)$

(B) $\frac{\mu_0 i}{4} \left(\frac{1}{R} + \frac{3}{R'} \right)$

(C) $\frac{\mu_0 i}{8} \left(\frac{1}{R} + \frac{3}{2R'} \right)$

(D) $\frac{\mu_0 i}{8} \left(\frac{1}{R} + \frac{3}{R'} \right)$



8. Equal current i is flowing in three infinitely long wires along positive x, y and z directions. The magnetic field at a point $(0, 0, -a)$ would be:

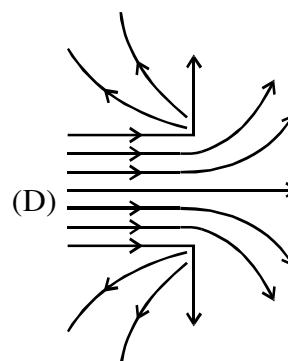
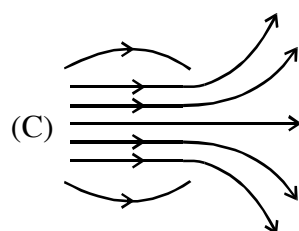
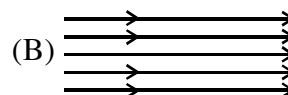
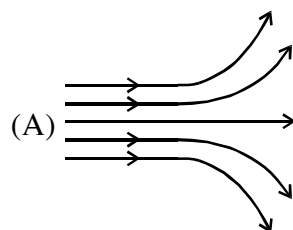
(A) $\frac{\mu_0 i}{2\pi a} (\hat{j} - \hat{i})$

(B) $\frac{\mu_0 i}{2\pi a} (\hat{i} + \hat{j})$

(C) $\frac{\mu_0 i}{2\pi a} (\hat{i} - \hat{j})$

(D) $\frac{\mu_0 i}{2\pi a} (\hat{i} + \hat{j} + \hat{k})$

9. Which of the following most appropriately represent magnetic field lines near the end of a long solenoid



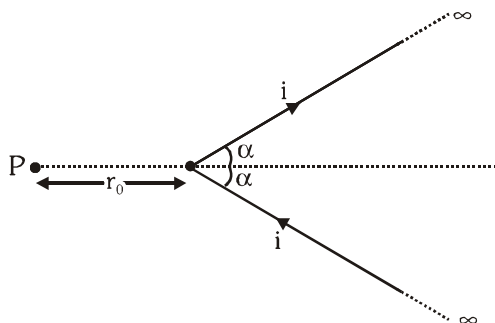
10. A current carrying wire in the form of 'V' alphabet is kept as shown in the figure. Magnetic field intensity at point P which lies on the angular bisector of V is

(A) $\frac{\mu_0 i}{4\pi r_0} [1 - \cos \alpha]$

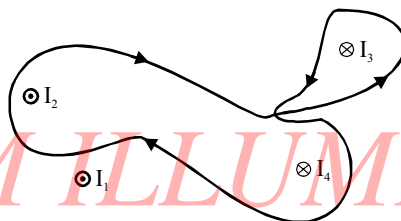
(B) $\frac{\mu_0 i}{2\pi r_0} [1 - \cos \alpha]$

(C) $\frac{\mu_0 i}{4\pi r_0} \frac{[1 - \cos \alpha]}{\sin \alpha}$

(D) $\frac{\mu_0 i}{2\pi r_0} \frac{[1 - \cos \alpha]}{\sin \alpha}$

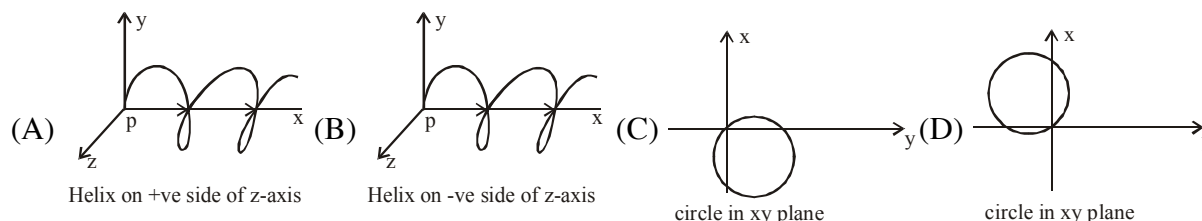


11. Four wires carrying current $I_1 = 2A$, $I_2 = 4A$, $I_3 = 6A$ and $I_4 = 8A$ respectively cut the page perpendicularly as shown in figure. The value of $\int \vec{B} \cdot d\vec{\ell}$ for the loop shown would be :-



- (A) $+2 \mu_0$ -wb/m (B) $-2 \mu_0$ -wb/m (C) $+10 \mu_0$ -wb/m (D) $-10 \mu_0$ -wb/m

12. An electron is projected from origin with velocity $3.2\hat{i} + \hat{j}$ (m/s). A uniform magnetic field of $2\hat{k}$ tesla is present. The path followed by electron is correctly shown in ($m_e = 9 \times 10^{-31}$ kg)



13. A charged particle enters a uniform magnetic field with velocity vector making an angle of 30° with the magnetic field. The particle describes a helical trajectory of pitch x . The radius of the helix is :-

- (A) $\frac{x}{2\pi}$ (B) $\frac{x}{2\sqrt{2}\pi}$ (C) $\frac{x}{2\sqrt{3}\pi}$ (D) $\frac{\sqrt{3}x}{2\pi}$

Multiple Correct Answer Type

5 Q. [4 M (-1)]

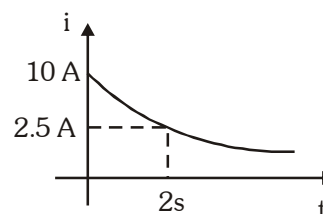
14. The figure, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance of resistance 10Ω .

(A) The initial potential difference across the capacitor is 100 volt.

(B) The capacitance of the capacitor is $\frac{1}{10\ln 2}$ F.

(C) The total heat produced in the circuit will be $\frac{500}{10\ln 2}$ joules.

(D) The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ln 2}$ second.



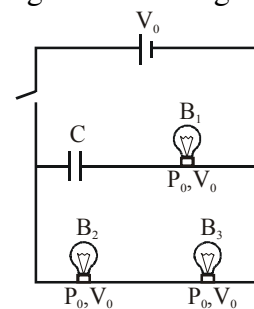
15. An uncharged capacitor is connected in circuit as shown in figure. Power ratings of bulbs are given. At $t = 0$ switch is closed :-

(A) At $t = 0$, power consumption in circuit is $\frac{3P_0}{2}$

(B) After a long time power consumption in circuit is $\frac{P_0}{2}$

(C) Brightness of B_1 decreases with time

(D) Initially brightness of B_2 is less than B_1 but later B_2 will be brighter



16. An infinite current carrying wire carries current i in negative x direction. Force on a charged particle at a point having coordinate $(6, 2, 3)$ is zero if velocity of the particle is

(A) $-4\hat{i} + 8\hat{j}$

(B) $4\hat{j} + 3\hat{k}$

(C) $-6\hat{j} + 4\hat{k}$

(D) $6\hat{i} + 3\hat{j} - 2\hat{k}$

17. A particle ($q/m=1$) is released from rest at origin. Choose correct statement [Given $\vec{B} = 4e^{-6x}\hat{j} + \sin 2y\hat{i}$ and $\vec{E} = 4x\hat{i}$]

(A) Speed of the particle at $x=1$ is $v=2$ m/s

(B) Particle will move in x - y plane only

(C) Speed of the particle at $x=2$ is $v=2$ m/s

(D) None of these

18. A particle of specific charge $\frac{q}{m} = \pi \times 10^{10} \text{ Ckg}^{-1}$ is projected from the origin along the positive x -axis with a velocity of 10^5 ms^{-1} in a uniform magnetic field $\vec{B} = -2 \times 10^{-3} \hat{k}$ tesla. Choose correct alternative(s)

(A) The centre of the circle lies on the y -axis

(B) The time period of revolution is 10^{-7} s.

(C) The radius of the circular path is $\frac{5}{\pi}$ mm

(D) The velocity of the particle at $t = \frac{1}{4} \times 10^{-7}$ s is $10^5 \hat{j} \text{ ms}^{-1}$

Linked Comprehension Type (Single Correct Answer Type) (1 Para $\times$ 2Q.) [3 M (0)]

Paragraph for Question no. 19 and 20

A parallel plate capacitor placed in a cylindrical tank is filled with a liquid of dielectric constant k . The area of cross section of tank is A and height of liquid is equal to the length of the square plate of plate area ℓ^2 . The separation of plates is d . A very small hole of area ' a ' is opened at the bottom of the tank at $t = 0$. If the capacitor in the process remains connected with a battery of emf E . Assume that the level of liquid in the capacitor remains same as outside and Bernoulli's equation is valid for the situation.

19. Choose the **CORRECT** statement(s) :

(A) The current in circuit as a function of time is $i = \frac{\epsilon_0 \ell E}{d} (k-1) \frac{a}{A} \left(\sqrt{2g\ell} - \frac{a}{A} gt \right)$

(B) The current in circuit as a function of time is $i = \frac{\epsilon_0 \ell E}{2d} (k-1) \frac{a}{A} \sqrt{2g\ell}$.

(C) The capacitance of capacitor decreases with time.

(D) The potential difference between the plates of capacitor remains constant.

20. Choose the **CORRECT** statement(s) :

(A) The rate at which the level of liquid decreases is $\frac{dh}{dt} = \frac{a}{2A} \sqrt{2g\ell}$

(B) The rate at which the level of liquid decreases is $\frac{dh}{dt} = \frac{a}{A} \left(\sqrt{2g\ell} - \frac{a}{A} gt \right)$

(C) The power supplied by the battery increases with time.

(D) When the container gets empty, the power delivered by the battery becomes zero.

PHYSICS

QUIZ # 21

SECTION-I

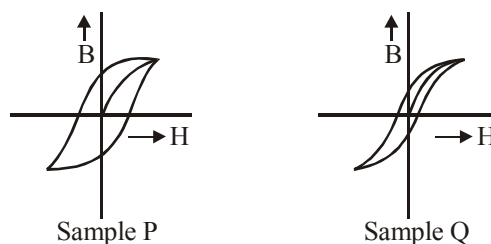
Single Correct Answer Type

20 Q. [3 M (-1)]

- Every iron atom in a ferromagnetic domain in iron has a magnetic dipole moment equal to $9.27 \times 10^{-24} \text{ A-m}^2$. A ferromagnetic domain in iron has the shape of a cube of side $1 \mu\text{m}$. The maximum dipole moment occurs when all the dipoles are aligned. The molar mass of iron is 56g, and its specific gravity is 8. The approximate magnetization of the domain is :-
 (A) $8.0 \times 10^5 \text{ A/m}$ (B) $8.0 \times 10^8 \text{ A/m}$ (C) $8.0 \times 10^{11} \text{ A/m}$ (D) $8.0 \times 10^{14} \text{ A/m}$
- Mark the **INCORRECT** statement :
 (A) The magnetisation of a paramagnetic material is inversely proportional to the absolute temperature T for temperatures above curie temperature.
 (B) For a paramagnetic material as the external field is increased or the temperature is lowered, the magnetisation increased until it reached the saturation value
 (C) The susceptibility above the Curie temperature in the paramagnetic phase is described by

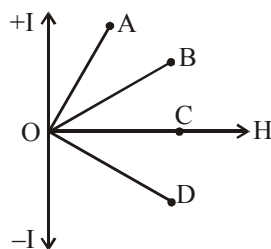
$$\chi = \frac{C}{T - T_c} \quad (T > T_c)$$

 (D) When a ferromagnetic material is placed in non-uniform magnetic field the sample tends to move towards the region of high field.
- Select the **CORRECT** statement :
 (A) Paramagnetic materials are attracted towards weak magnetic field region in variable magnetic field
 (B) Diamagnetic material have atomic dipoles grouped together.
 (C) Permeability of ferromagnetic material decreases with increase in temperature
 (D) Paramagnetic material get oppositely magnetized when placed in external magnetic field.
- If the B-H curves of two samples of P and Q of iron are as shown below, then which one of the following statements is **CORRECT** ?

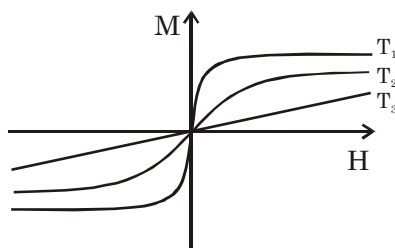


- (A) Both P and Q are suitable for making permanent magnet
 (B) P is suitable for making permanent magnet and Q for making electromagnet
 (C) P is suitable for making electromagnet and Q is suitable for permanent magnet
 (D) Both P and Q are suitable for making electromagnets
- A thin rectangular magnet suspended freely has a period of oscillation equal to T . Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of oscillation is T' , the ratio T'/T is :-
 (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$

6. The variable of intensity of magnetisation (I) with respect to the magnetising field (H), in a diamagnetic substance, is described by which line in given graph :

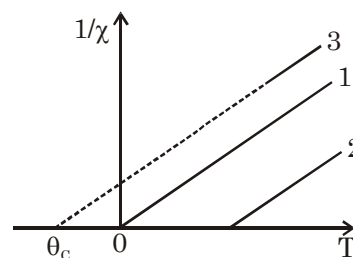


- (A) OA (B) OB (C) OC (D) OD
7. The magnetic susceptibility of a paramagnetic substance at -73°C is 0.0060, then its value at -173°C will be :
(A) 0.0030 (B) 0.0120 (C) 0.0180 (D) 0.0045
8. If a diamagnetic solution is poured into a U-tube and one arm of this U-tube is placed between the poles of a strong magnet, with the meniscus in line with the field, then the level of solution will -
(A) rise (B) fall (C) oscillate slowly (D) remain as such
9. Nickel is ferromagnetic at room temperature. At a value of H field of $5 \times 10^5 \text{ A/m}$, nickel saturates. The magnetic field B is 1.25 T. What is the magnetisation vector $\vec{M}$ of nickel sample?
(A) $5 \times 10^5 \text{ A/m}$ (B) $+3.744 \times 10^5 \text{ A/m}$ (C) $9.31 \times 10^4 \text{ A/m}$ (D) $15 \times 10^5 \text{ A/m}$
10. The following graph shows the response of magnetization M of paramagnetic material to the applied magnetic field strength H for three different temperatures T_1, T_2, T_3



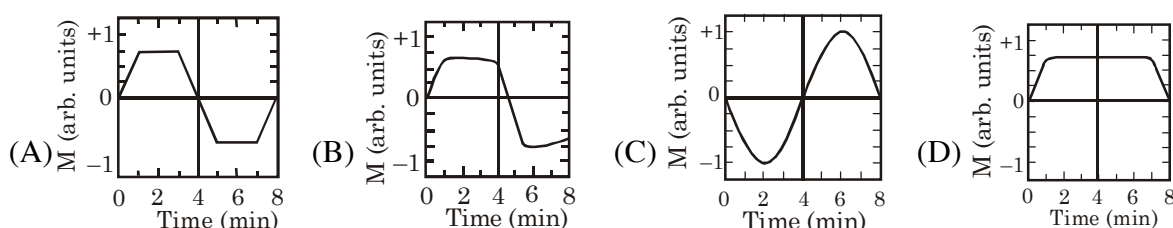
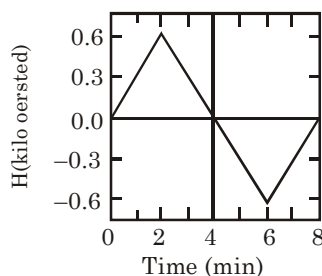
Which of the following is correct?

- (A) $T_1 < T_2 < T_3$ (B) $T_3 > T_1 > T_2$ (C) $T_2 > T_3 > T_1$ (D) $T_1 > T_2 > T_3$
11. The temperature dependence of the reciprocal magnetic susceptibility $1/\chi$ is represented by:
- (A) 1 for paramagnet and 2 for ferromagnet
(B) 1 for paramagnet and 3 for ferromagnet
(C) 2 for paramagnet and 3 for ferromagnet
(D) 3 for paramagnet and 2 for ferromagnet

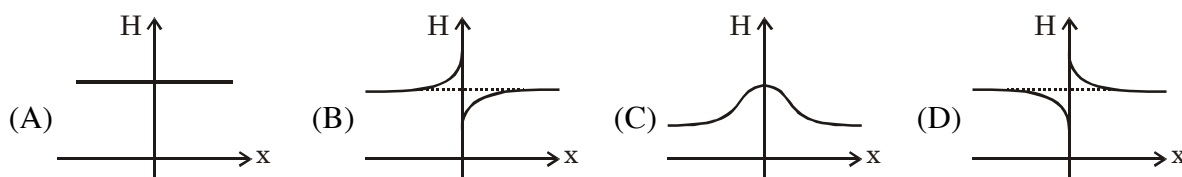
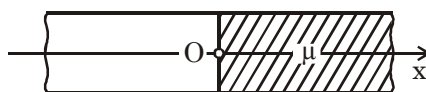


12. The value of reciprocal of magnetic susceptibility for gadolinium (a ferromagnetic substance) above its Curie point are found to be 10 and 20 corresponding to temperature 720 K and 1160 K respectively. Find the Curie constant :-
(A) 44 K (B) 22 K (C) 280 K (D) 56 K

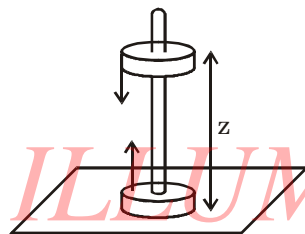
13. A demagnetized sample of iron is placed in a magnetic field H which varies with time as shown in the plot below. The magnetization M of this iron sample is continuously measured and also plotted as a function of time. The appearance of this plot will be closest to



14. A small magnetic dipole is kept at the origin in the x - y plane. One long wire L_1 is located at $z = -a$ in the x - z plane with a current I flowing in the positive x direction. Another long wire L_2 is at $z = +a$ in y - z plane with the same current I as in L_1 , flowing in the positive y direction. The angle ϕ made by the magnetic dipole with respect to the positive x -axis is :-
(A) 225° (B) 120° (C) 45° (D) 270°
15. Suppose a small cylinder of a paramagnetic substance and a small cylinder of a diamagnetic substance are placed between the (parallel) pole pieces of a strong magnet, so that they are aligned along the direction of the field lines. Will they attract or repel each other?
(A) Attract when they are placed such that their axis is common and repel when they are placed side by side.
(b) Repel when they are placed such that their axis is common and attract when they are placed side by side.
(C) Attract when they are placed such that their axis is common as well as when they are placed side by side.
(D) Repel when they are placed such that their axis is common as well as when they are placed side by side.
16. Half of an infinitely long straight current-carrying solenoid is filled with magnetic substance as shown in Fig. Draw the approximate plot of magnetizing field intensity H on the axis as functions of x .



17. A paramagnetic sample shows a net magnetisation of 8 Am^{-1} when placed in an external magnetic field of 0.6 T at a temperature of 4 K . When the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K , the magnetisation will be :
- (A) $\frac{32}{3} \text{ Am}^{-1}$ (B) $\frac{2}{3} \text{ Am}^{-1}$ (C) 6 Am^{-1} (D) 2.4 Am^{-1}
18. A circular coil of radius 0.157 m has 50 turns. It is placed such that its axis is in magnetic meridian. A dip needle is supported at the centre of the coil with its axis of rotation horizontal & in the plane of coil. The angle of dip is 30° when a current flows through the coil. The angle of dip becomes 60° on reversing the current. Find the current in the coil assuming that the magnetic field due to the coil is smaller than the horizontal component of earth's magnetic field, $H = 3 \times 10^{-5} \text{ T}$:-
- (A) 0.75 amp (B) 7.5 amp (C) 0.075 amp (D) 1.75 amp
19. A donut-shaped permanent magnet (magnetization parallel to the axis) can slide frictionlessly on a vertical rod. Treat the magnets as dipoles, with mass m_d and dipole moment M . When we put two back to back magnets on the rod the upper one will float. At what height z does it float?



- (A) $\left[\frac{2\mu_0 M^2}{3\pi m_d g} \right]^{1/4}$ (B) $\left[\frac{6\mu_0 M^2}{\pi m_d g} \right]^{1/4}$ (C) $\left[\frac{3\mu_0 M^2}{2\pi m_d g} \right]^{1/4}$ (D) $\left[\frac{\mu_0 M^2}{6\pi m_d g} \right]^{1/4}$
20. A long solenoid has 1000 turns per metre and carries a current of 1 A . It has a soft iron core of $\mu_r = 1000$. The core is heated beyond the Curie temperature, T_C .
- (A) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically.
 (B) The H and B fields in the solenoid are nearly unchanged.
 (C) The magnetisation in the core reverses direction.
 (D) The magnetisation in the core increases by a factor of about 10^8 .

PHYSICS

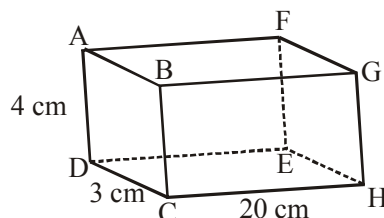
QUIZ # 22

SECTION-I

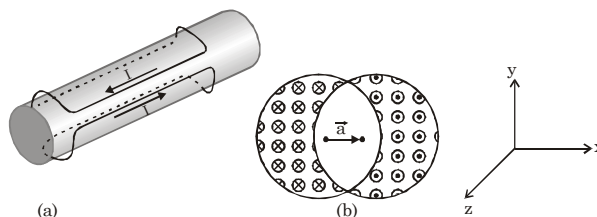
Single Correct Answer Type

7 Q. [3 M (-1)]

1. A rectangular block of lead has dimensions $4 \text{ cm} \times 3 \text{ cm} \times 20 \text{ cm}$. A temperature difference of 100°C can be applied to any pair of opposite faces that we choose.



- (A) The largest amount of heat flows if it flows parallel to line BG.
 (B) The largest amount of heat flows if it flows parallel to line AD.
 (C) The smallest amount of heat flows if it flows parallel to AF.
 (D) The smallest amount of heat flows if it flows parallel to AB.
2. A current I flows along the length of an infinitely long, straight, solid pipe. The current is uniformly distributed on the cross section. Then incorrect statement -
 (A) The magnetic field is zero only on the axis of the pipe
 (B) The magnetic field is different at different points on the same cross section inside the pipe
 (C) The magnetic field is maximum on surface
 (D) The magnetic field at all points inside the pipe is the same, but not zero
3. To produce a uniform magnetic field directed parallel to a diameter of a cylindrical region, one can use the saddle coils illustrated in figure. The loops are wrapped over a somewhat flattened tube. Assume the straight sections of wire are very long. The end view of the tube shows how the windings are applied. The overall current distribution is the superposition of two overlapping, circular cylinders of uniformly distributed current, one toward you and one away from you. The current density J is the same for each cylinder. The position of the axis of one cylinder is described by a position vector $\vec{a}$ relative to the other cylinder. The magnetic field inside the hollow tube is.

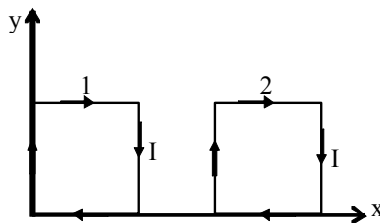


- (A) $\frac{\mu_0 J a}{2}$ along x-axis
 (B) $\frac{\mu_0 J a}{2}$ along y-axis
 (C) $\mu_0 J a$ along z-axis
 (D) $\mu_0 J a$ along y-axis
4. Two current carrying coil having radius ' r ' are separated by distance ' d ' as shown in diagram. If $r \ll d$. Then find the force between two ring. Current in both the ring is clockwise and equal to i .

- (A) $\frac{6\mu_0 i^2 \pi^2 r^4}{4\pi d^4}$
 (B) $\frac{\mu_0 i^2 \pi^2 r^4}{4\pi d^4}$
 (C) $\frac{3\mu_0 i^2 \pi^2 r^4}{4\pi d^4}$
 (D) $\frac{\mu_0 i^2 \pi^2 r^4}{2\pi d^4}$



5. Magnetic field in a region is given by $\vec{B} = B_0 \hat{k}$. Two loops each of side a is placed in this magnetic region in the x - y plane with one of its sides on x -axis. If F_1 is the force on loop 1 and F_2 be the force on loop 2, then :-



- (A) $F_1 = F_2 = 0$ (B) $F_1 > F_2$ (C) $F_2 > F_1$ (D) $F_1 = F_2 \neq 0$
6. Mark the **INCORRECT** statement :
- (A) Every magnetic configuration must have a North pole and a South pole.
(B) Cutting a bar magnet changes the magnetic moment of the each part of the magnet.
(C) A system can have a magnetic moment even if the net charge is zero.
(D) The magnetic flux through any closed surface must be zero.
7. A compass needle made of pure iron (with density 7900 kg/m^3) has a length L of 3.0 cm , a width of 1.00 mm , and a thickness of 0.50 mm . The magnitude of the magnetic dipole moment of an iron atom is $M_{\text{Fe}} = 2.1 \times 10^{-23} \text{ J/T}$. If the magnetisation of the needle is equivalent to the alignment of 10% of the atoms in the needle, what is the magnitude of the needle's magnetic dipole moment $\vec{M}$?
(A = 56)
- (A) $2.7 \times 10^{-5} \text{ J/T}$ (B) $2.7 \times 10^{-7} \text{ J/T}$ (C) $2.7 \times 10^{-3} \text{ J/T}$ (D) $2.7 \times 10^{-10} \text{ J/T}$

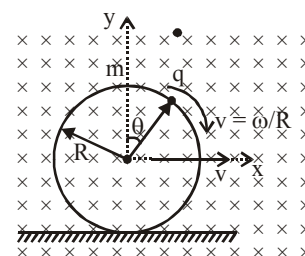
Multiple Correct Answer Type

5 Q. [4 M (-1)]

8. A sphere is placed in room at temperature 30°C where room temperature is also 30°C . Now sphere is heated by an electric heater of 20 watt temperature of sphere rises to 50°C and becomes constant. Now heater is switch off. Let during heating temperature of sphere rises linearly from 30°C to 40°C in 5 minutes and assuming newton law of cooling is valid. Choose the **CORRECT** statements.
- (A) Net heat absorbed by sphere from 30°C to 40°C is 4500 J .
(B) Heat capacity of sphere is $600 \text{ J/}^\circ\text{C}$.
(C) Rate of cooling decreases as temperature of sphere decreases after heater is switched off.
(D) Heat loss by the sphere from 30°C to 40°C is 1500 J .
9. A ring of mass m and radius R is set into pure rolling on horizontal rough surface, in a uniform magnetic field of strength B as shown in the figure. A point charge q of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y -axis) :-

- (A) Ring will continue to move with constant velocity
(B) The value of friction acting on ring is $Bqv \cos \theta$
(C) The value of friction acting on ring is $Bqv \sin \theta$

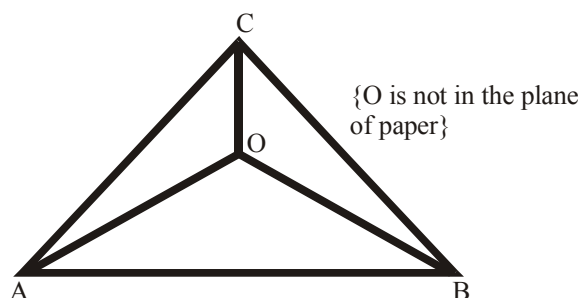
- (D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB} \right)$



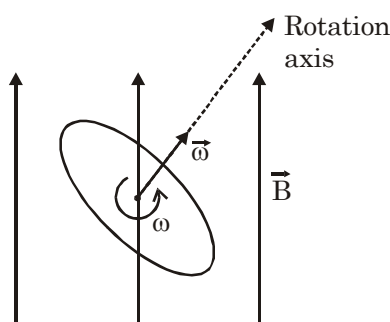
10. Three geometrically identical rods A, B and C are placed end to end. A temperature difference is maintained between the free ends of A and C. The thermal conductivity of B is THRICE that of C and HALF of that of A. The effective thermal conductivity of the system will be (K_A is the thermal conductivity of rod A).

(A) $\frac{1}{3}K_A$ (B) $3K_A$ (C) $\frac{5}{9}K_A$ (D) $\frac{2}{3}K_A$

11. Six identical rods are joined together to form a network as shown. A temperature difference of ΔT can be maintained between two points. Considering heat flow through conduction only, in steady state :-



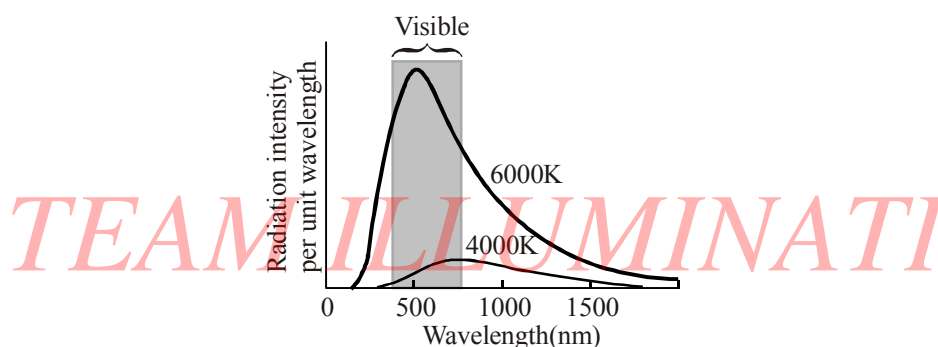
- (A) Net rate of heat flow between points where temperature difference is maintained, depends on points chosen.
(B) Net rate of heat flow between points where temperature difference is maintained, is independent of points chosen.
(C) In all the cases of points chosen {where ΔT is to be maintained} there is one rod through which no heat flows.
(D) Heat conduction takes place through all the rods irrespective of choice of points where temperature difference is maintained.
12. A uniform, thin, uniformly charged disk of mass m radius R and uniform surface charge density σ rotates with angular speed ω about an axis through its centre and perpendicular to disc. The disk is in region with a uniform magnetic field B that makes angle θ with rotation axis. Mark the **CORRECT** statement :-



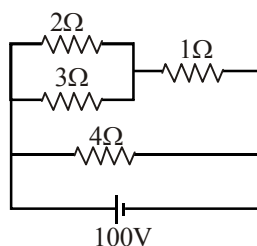
- (A) Torque exerted on the disk by magnetic field is $\frac{1}{4}\pi\sigma R^4\omega B \sin \theta$
(B) Frequency with which angular velocity vector rotates is given by $\frac{\pi\sigma R^2 B}{4m} \sin \theta$.
(C) For an observer looking from above the angular velocity vector begins to rotate anticlockwise sense.
(D) For an observer looking from above the angular velocity vector begins to rotate clockwise sense.

Linked Comprehension Type
(Single Correct Answer Type)
(1 Para × 4Q.) [3 M (0)]
Paragraph for question nos. 13 to 16

All bodies, no matter how hot or cold, continuously radiate photons. At a given temperature, the intensities of the electromagnetic waves emitted by an object vary from wavelength to wavelength throughout the visible, infrared, and other regions of the spectrum. Figure illustrates how the intensity per unit wavelength depends on wavelength for a perfect blackbody emitter. Although this figure can strictly be applied only to a black body, yet this will approximately describe the behavior of many of the self radiating systems. For example, sun has an approximate temperature of 6000K. It is not a black body; it has an emissivity of nearly 0.6. But its peak almost occurs at that predicted by the Wein's law. Suppose we have a bulb of power 100W. It emits only about 5W as visible light. Rest is emitted as infrared radiation. Assume that the bulb filament has a surface area of 10 mm^2 . ($hc = 1250 \text{ eV-nm}$).



13. What is the approximate temperature of the filament ?
 (A) 500 K (B) 350 K (C) 2500 K (D) 10000 K
14. Assume that the light emitted by the bulb in the visible region is entirely of wavelength 500 nm. What is the number of photons emitted per second in the visible region ?
 (A) 1.25×10^{19} (B) 5×10^{19} (C) 2.5×10^{19} (D) 4×10^{19}
15. If we want of increase the number of photons emitted by the bulb in the visible region without changing the wattage, which method would be most appropriate ?
 (A) increasing emissivity by a factor of 2
 (B) increasing the radius of the filament by a factor of 2 and the length by a factor of 4.
 (C) decreasing the radius of the filament by a factor of 2 and the length by a factor of 4.
 (D) doubling the voltage and decrease the length of the filament by a factor of 2.
16. Which of the following resistances would have maximum surface temperature. All of them have a surface area of 10 mm^2 and same emissivity.



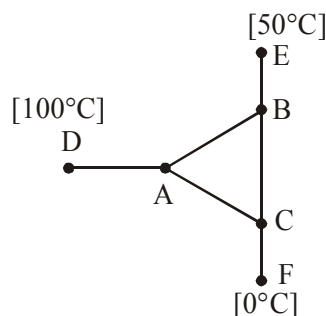
- (A) 1Ω (B) 2Ω (C) 3Ω (D) 4Ω

SECTION-III

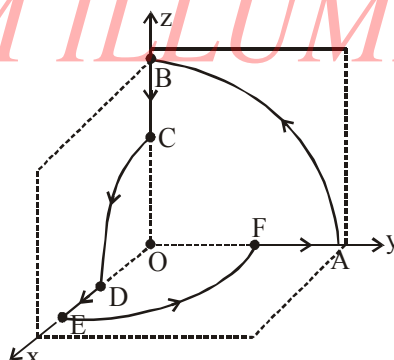
Numerical Grid Type (Single digit Ranging from 0 to 9) 4 Q. [4 M(-1)]

1. Identical six rods are used to form given figure. AB, BC and AC form equilateral triangle.

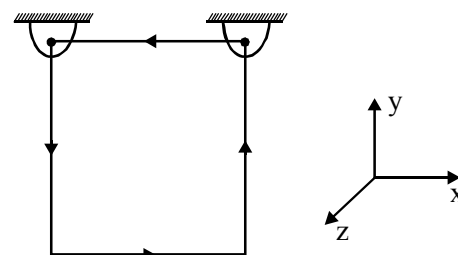
Find $\frac{T_B}{10}$ (in $^{\circ}\text{C}$). [T_B = Temperature at junction B]



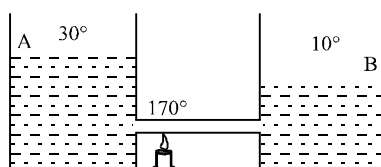
2. In the rigid wire frame ABCDEF, shown in figure, a current $I_0 = 4\text{A}$ flows through the wire from F to A and radii of curves AB, CD and EF are a , b and c respectively with centres at O. If this loop is immersed in a uniform magnetic field of magnitude 1.3 T and directed along the line joining $(2, 3, 4)$ to $(-7, 2, 0)$ and allowed to rotate about the x-axis only, then find the value of torque τ (N-m) exerted on the loop by the axis. (Given : $a = 3\text{m}$, $b = 1\text{m}$, $c = 2\text{m}$).



3. A current carrying uniform square frame is suspended from hinged supports as shown in the figure such that it can freely rotate about its upper side. The length and mass of each side of the frame is 2m and 4kg respectively. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j})$ is applied. When the wire frame is rotated to 45° from vertical and released it remains in equilibrium. If the magnitude of current (in A) in the wire frame is I then find $\left(\frac{3}{5}\right)I$.



4. Two thermally insulated vessel are filled with water of mass m each is kept at temperature 30° and 10°C . They are connected by means of a lagged rod of thermal conductivity K , Area A and length L . At $L/4$ from vessel A, temperature of 170°C is maintained. The temperature of B when water in A starts boiling is T . Assume no heat loss to the surrounding. Find $T/7$. Take $2^{1/3} = 5/4$



PHYSICS

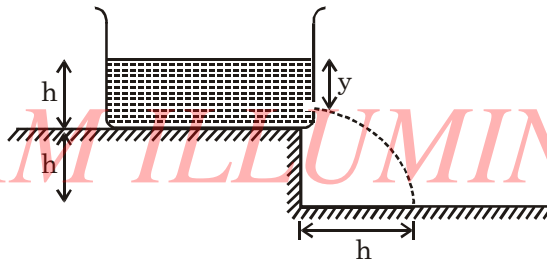
QUIZ # 23

SECTION-I

Straight Objective Type

4 Q. [3(-1)]

- A solid sphere of mass 2 kg and density 10^5 kg/m^3 hanging from a string is lowered into a vessel of uniform cross-section area 10 m^2 containing a liquid of density $0.5 \times 10^5\text{ kg/m}^3$, until it is fully immersed. The increase in pressure of liquid at the base of the vessel is (Assume liquid does not spill out of the vessel and take $g = 10\text{ m/s}^2$)
 (A) 2 N/m^2 (B) 1 N/m^2 (C) 4 N/m^2 (D) $\frac{1}{2}\text{ N/m}^2$
- A carnot refrigeration cycle absorbs heat at 270 K and rejects heat at 300 K . If the cycle is absorbing 1260 KJ/min at 270 K , then work required per second is :
 (A) 2.33 KJ/sec. (B) 4.66 KJ/sec. (C) 1 KJ/sec. (D) 4 KJ/sec.
- A tank is filled upto a height h with a liquid is placed on a platform of height h from the ground. To get range of h , a small hole is punched at a distance of y from the free surface of the liquid. Then y is :



- (A) 0.5 h (B) 0.13 h (C) 1.86 h (D) 0.75 h
- A simple pendulum clock is designed to keep correct time at 30°C . Its is taken to a desert where during day (for 12 hours) temperature is 50°C and during night (for 12 hours) temperature is 10°C . After 1 full day. (Take $\alpha = 10^{-5}/^\circ\text{C}$)
 (A) The clock will show correct time
 (B) The clock will run slow by approx $1.3 \times 10^{-4}\text{ sec.}$
 (C) The clock will run fast by approx $3.2 \times 10^{-4}\text{ sec.}$
 (D) The clock will run slow by approx $4.3 \times 10^{-4}\text{ sec.}$

One or More than one correct answer type

8 Q. [4; +1(-1)]

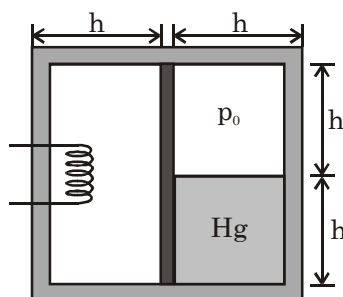
- The cooling of water is investigated in alike containers of negligible heat capacity. In each case the initial temperature of the sample of water is 80°C , and the aimed final temperature is 40°C . The temperature of the environment is 30°C , which does not change during the measurements.
 (i) Firstly it was measured that 2 litres of water cooled from 80°C to 40°C in a time of t_0 .
 (ii) Secondly, we waited only until the 2 litres of water initially having a temperature of 80°C cooled to a temperature of 50°C (which took a time of t_1). Then 1 litre of water was quickly poured out and was replaced with 1 litre of water at a temperature of 30°C .
 (iii) Then the measurement was repeated such that 1 litre of the initial 2 litres of water at a temperature of 80°C was poured out and immediately replaced by 1 litre water at a temperature of 30°C . The mixture reached the temperature of 40°C in a time of t_2 .
 (iv) Finally the 2 litres water of initial temperature 80°C , was left to cool to 60°C , and then quickly 1 litre of water was poured out and replaced by 1 litre of water of temperature 30°C , and then the mixture was left to cool to a temperature of 40°C . The total time of cooling in this case was t_3 .

Take $\ln 5 = 1.6$, $\ln 2 = 0.7$, $\ln 3 = 1.1$

- (A) $t_1 = 0.5625\text{ } t_0$ (B) $t_2 = 0.225\text{ } t_0$ (C) $t_3 = 0.5625\text{ } t_0$ (D) $t_1 + t_2 < 2t_3$

6. There is a piston having good thermal conductivity in the middle of a cuboid-shaped container having thermally insulating walls. On the left side of the piston there is a sample of air of volume V_0 , whilst on the right side there is a sample of air of volume $V_0/2$ at a pressure of $p_0 = 76 \text{ Hgcm} \approx 10^5 \text{ Pa}$, and a mercury column of height $h = 38 \text{ cm}$. The total width of the container is $2h$ (not considering the width of the piston), and its height is also $2h$.

By means of a built-in electric heater the left side of the system is slowly heated. The temperatures of the two samples of gases are the same at any time. Expansion of the container, the piston and the mercury is neglected.



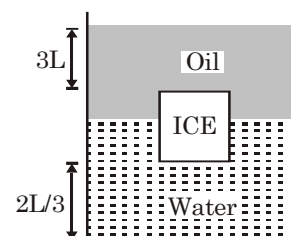
- (A) The initial pressure on the left hand side is 85.5 cm of mercury column
(B) If the piston displaces by a distance x to right hand side, the sample of air on right hand side becomes shorter by distance of $2x$ in height
(C) If the piston displaces by a small distance of dx cm to the right, the pressure on the left hand side increases by $28 dx \text{ Hgcm}$
(D) if the piston displaces by a small distance of dx cm to the right, the pressure of the sample of air on right hand side increases by $1142.5 dx \text{ Hgcm}$

7. In the diagram shown, an ice cube of dimension $L = 10 \text{ cm}$ is submerged half in water and half in oil. Using the details given, find out the density of oil. If the ice cube were to melt completely then find out the final heights of the oil and water columns.

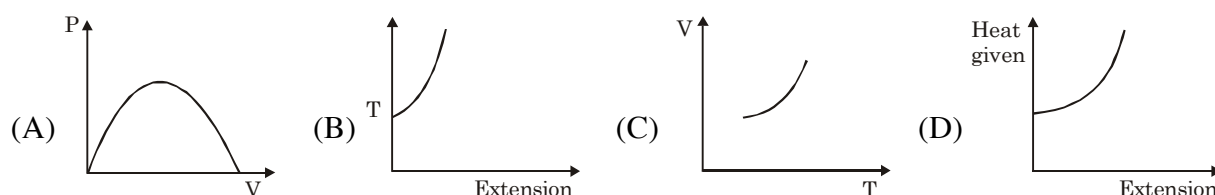
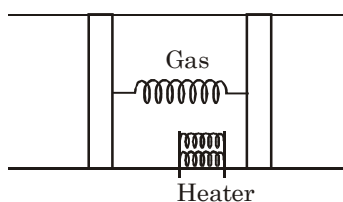
Density of water = 1 gm/cc

Density of ice = 0.9 gm/cc

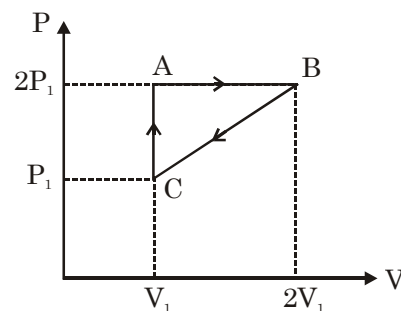
Area of cylinder is 1000 cm^2 .



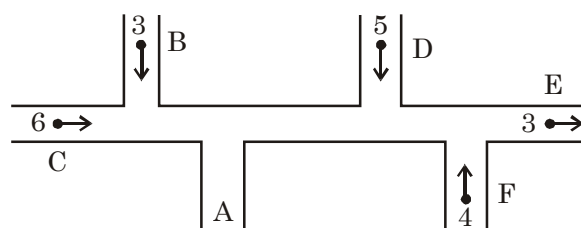
- (A) density of oil is 0.5 gm/cc (B) final height of water is 12.06 cm
(C) Final height of oil column is 34.5 cm (D) The top surface descends by 0.2 cm
8. The two pistons have been connected by a spring. The piston and the cylinder are both insulated. The gas is monoatomic and has n moles. The heater is slowly giving heat. Which of the following graphs most appropriately represent the process performed by the gas? Neglect the heat absorption by the spring and the spring is initially undeformed.



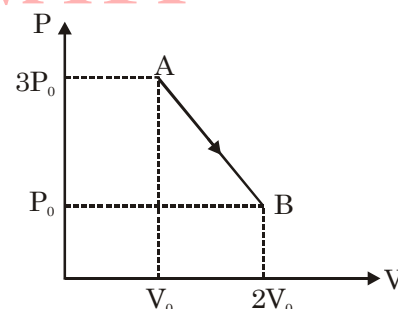
9. Consider the PV cycle shown. If the heat given to one mole of the monatomic gas in the step AB is Q , then answer the following questions:



- (A) Work done in the process BC on the gas is $3Q/10$
(B) The heat released in the process BC is $2Q/5$
(C) The efficiency of the cycle is 10%
(D) The heat given to the gas during CA is $3Q/10$
10. The diagram shows a horizontal pipe of uniform cross section in which water is flowing. The directions of flow and the volume flow rates (in cm^3/s) are shown for various portions of the pipe. Area of cross-section = 1 cm^2 , $\rho_w = 10^3 \text{ kg/m}^3$:



- (A) $V_A = 15 \text{ cm}^3/\text{sec}$ ↓
(B) $p_C - p_A = 9.45 \text{ Pa}$
(C) the pressure at A is minimum
(D) $p_E - p_F = -0.35 \text{ Pa}$
11. An ideal diatomic gas undergoes a process AB, for which PV indicator diagram is shown. Temperature at state A is T_0 .
- (A) Temperature at B is $2T_0/3$
(B) The temperature is maximum at volume of $5V_0/4$
(C) The heat flow changes its direction at temperature $525 T_0/864$
(D) Net heat is given to the gas during the process.
12. Which among the following forces is conservative?
- (A) $\vec{F} = 2xy\hat{i} + x^2\hat{j}$ (B) $\vec{F} = y^3\hat{i} + xy^2\hat{j}$ (C) $\vec{F} = y\hat{i} + x\hat{j}$ (D) $\vec{F} = 2xy^2\hat{i} - x^2y^2\hat{j}$



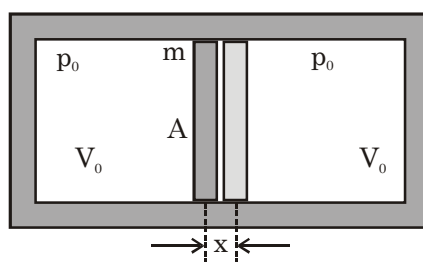
SECTION-II

Numerical-Answer Type

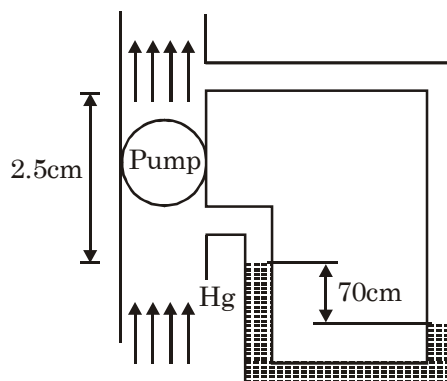
6 Q. [3(-0)]

(Upto Second Decimal Place)

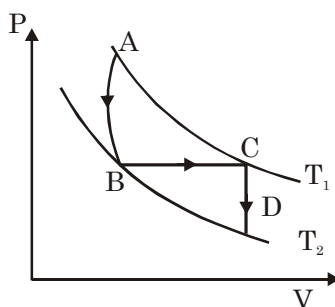
1. An easily moveable piston of mass m and cross sectional area A is made of a material which is also thermally insulating. The piston divides a horizontal, fixed, and thermally insulated cylinder into two parts, having the same volume of V_0 . There is the same amount of helium gas in the two parts, at the pressure of p_0 . The piston is displaced a bit, and then released. What is the angular frequency (in rad/s) of the resulting oscillation? Take : $\gamma = 1.5$, $p_0 = 10^5 \text{ Pa}$, $A = 10 \text{ cm}^2$, $V = 300 \text{ l}$, $m = 1 \text{ kg}$



2. Water flows through a pipeline of uniform diameter at a rate of $2.5\text{ m}^3/\text{min}$. What is the power (in W) delivered by the pump?
Density of water = 10^3 kg/m^3
Density of Hg = 13555 kg/m^3



3. A person of surface area 1.75 m^2 is lying out in the sunlight to get sun bath. If the intensity of the incident sunlight is $7 \times 10^2\text{ W/m}^2$, at what rate must heat be lost (in watt) by the person in order to maintain a constant body temperature? (Assume the effective area of skin exposed to the Sun is 40% of the total surface area and that internal metabolic processes contribute another 90W for an inactive person.) emissivity of the body is 0.6.
4. Between two isotherms T_1 & T_2 , a process ABCD has been carried out with 2 moles of a monoatomic ideal gas. AB is adiabatic, BC is isobaric and CD is isochoric process as shown. The average heat capacity (in J/K) of process is _____. Take : $R = 25/3\text{ J/mol K}$.
(The upper isotherm is at T_1 and the lower is at T_2)



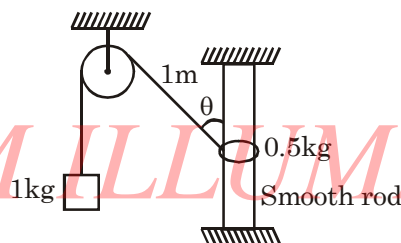
5. A small black sphere (radius = r_1) is located at the centre of a spherical asbestos black shell (thickness = 5.0 cm, outer radius = r_2). The thickness of the shell is small compared to the inner and outer radii of the shell. The temperature of the small sphere is 800 K, while the temperature of the inner surface of the shell is 600 K. The temperature of small sphere is maintained constant. Assuming that $\frac{r_2}{r_1} = 10.0$ and ignoring any air inside the shell, find the temperature (in K) of the outer surface of the shell.
Take : $K_{\text{asbestor}} = 0.085\text{ W/m}^\circ\text{C}$, $\sigma = \frac{17}{3} \times 10^{-8}\text{ W/m}^2\text{K}^4$.
6. We have 61 gm chlorine gas and 2gm Hydrogen gas in a closed container at atmospheric pressure (1 atm). By explosion, 10,000J energy is released and hydrochloric acid is formed. Find the pressure of the gas after the explosion (in atm). $R = \frac{25}{3}\text{ J/mol K}$. volume of container is 50 litres.

SECTION-I

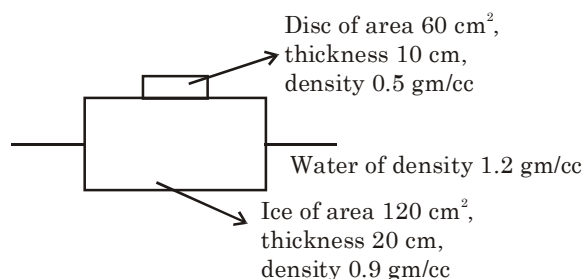
One or More than one correct answer type

8 Q. [4; +1(-1)]

- A small ball is dropped from a height of 80 cm above the ground. Simultaneously, another small ball is projected from the ground such that they collide at some point in between. It is seen that the first ball just reaches the same height repeatedly (80 cm) after collision with second ball :
(A) Both balls have same mass.
(B) The second ball was projected with a velocity of 4 m/s.
(C) The collision must be elastic.
(D) The period of 1st ball's motion is 0.4 sec.
- A ring is connected to 1kg mass and threaded onto a vertical smooth rod as shown. Choose the correct statement(s) for the equilibrium position.

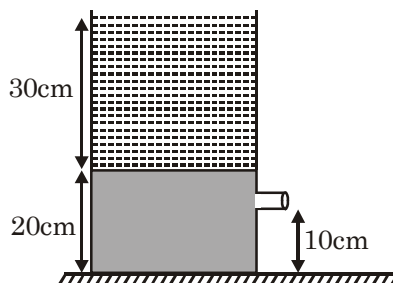


- $\theta = 60^\circ$
 - For small displacement from equilibrium, ring executes SHM
 - The time period of oscillation of ring from equilibrium is $\frac{2\pi}{\sqrt{10}}$ sec.
 - If the ring is given a velocity v in the downward direction, the velocity of 1 kg mass is $2v$ in upward direction.
- A wooden disc is kept on ice floating in water as shown. Considerably mean position.

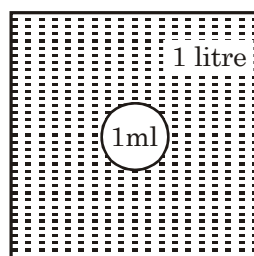


- The depth to which ice block is submerged in water is 17.08 cm
- If the disc is gently lifted, the ice block will be submerged by 2.08 cm less.
- If 30% of ice melts (keeping same base area and disc on it), the depth to which ice is submerged will be 5.75 cm less.
- If 30% of ice melts (keeping same thickness and disc on it), the depth to which ice is submerged will be 5.75 cm less.

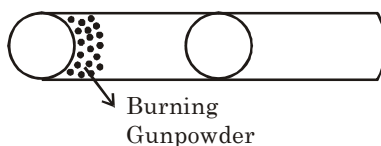
4. A cylindrical vessel of a very large cross-sectional area 1000 cm^2 is containing two immiscible liquids of density $\rho_1 = 600 \text{ kg/m}^3$ & $\rho_2 = 1200 \text{ kg/m}^3$ as shown in figure. A small hole having cross-sectional area 5 cm^2 is made in right side vertical wall. (Neglect the mass of container) For the given situation, mark the correct statement (s). (Take $g = 10 \text{ m/s}^2$).



- (A) If the surface is smooth then a right ward force of magnitude 4 N is to be applied on vessel to maintain its equilibrium
(B) If the surface is smooth then a right ward force of magnitude 3 N is to be applied on vessel to maintain its equilibrium
(C) If the surface is rough ($\mu = 0.04$), then minimum force (horizontal) needed to be applied on the vessel to maintain equilibrium is zero
(D) If the surface is rough ($\mu = 0.04$), then the maximum force (horizontal) needed to be applied on the vessel to maintain equilibrium is 19.8 N.
5. A tightly closed vessel is filled with glycerin. (Neglect surface tension). The inside volume of vessel is 1 litre. The walls of vessel has negligible thermal expansion coefficient, but the volume expansion coefficient of glycerine is $5 \times 10^{-4} / ^\circ\text{C}$. Inside glycerin, there is a bubble of air of volume 1 ml. At temperature of 27°C , the pressure in vessel is 10^5 Pa . The temperature increases by ΔT ($\Delta T \ll 300 \text{ K}$)



- (A) The pressure in vessel increases in proportion to ΔT .
(B) The volume of bubble decreases & $\Delta V \propto \Delta T^2$.
(C) If temperature decreases to 150 K, pressure in vessel is approx 660 Pa.
(D) If temperature decreases to 150K, volume of bubble is 76 ml.
6. A cannon is fired using an explosive. The muzzle of cannon is a cylinder of area $1 \times 10^{-3} \text{ m}^2$. The muzzle is horizontal. The burning gunpowder produces gas, which pushes the ball at constant pressure of $51 \times 10^5 \text{ Pa}$. Atmospheric pressure is 10^5 Pa . The gas being formed is at constant temperature. The mass of ball is 2.5 kg and length of muzzle is 2.5m.



- (A) The muzzle velocity of the ball is 100 m/s.
(B) The rate of burning of gunpowder (Rate at which gas is produced) is constant.
(C) The rate of burning of gunpowder increases as the ball moves along the muzzle.
(D) The work done by gases is completely converted into the kinetic energy of the ball.

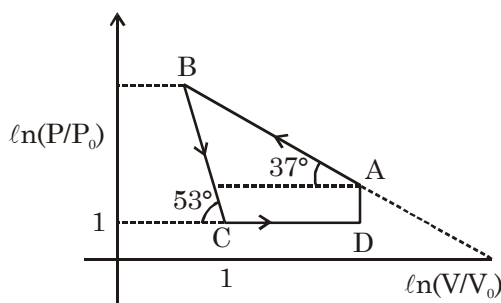
7. A monatomic gas enclosed under a piston is heated such that $PT^{-1/2} = \text{constant}$. During the process,
- (A) the molar heat capacity is $3R$.
 - (B) the average intermolecular separation increases.
 - (C) the rate of collision of molecules with the walls increases
 - (D) the density of gas increases.
8. A helium balloon has rubber envelope of weight 2.6 kg and is used to lift a weight of m kg. The volume of fully inflated balloon is $V \text{ m}^3$. The pressure of helium inside is always equal to atmospheric pressure. The average molar mass of air = 30 gm/mole. If the ambient temperature is 27°C , (Neglect thickness of balloon). Atmospheric pressure is 100kPa.
- (A) Minimum volume of balloon V needed is 2.5 m^3
 - (B) To lift $m = 12 \text{ kg}$ volume of balloon needed is 10 m^3
 - (C) If balloon is in equilibrium at height above ground, it will be in equilibrium near ground also.
 - (D) If $V = 15 \text{ m}^3$ and $m = 1\text{kg}$, balloon will accelerate up with acceleration of 33.33 m/s^2 .

**"Matching list based comprehension type (4×6) single option correct
“(Two Columns with List I having 4 options & List II contains 6 options)”**

2 List match Sets $\times$ 2 Q 3(-1,0)

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For a thermodynamic cycle ABCD on a non-linear polyatomic gas, we draw $\ln\left(\frac{P}{P_0}\right)$ versus $\ln\left(\frac{V}{V_0}\right)$ diagram.



Match the process.

List-I

- (I) AB
- (II) BC
- (III) CD
- (IV) DA

List-II

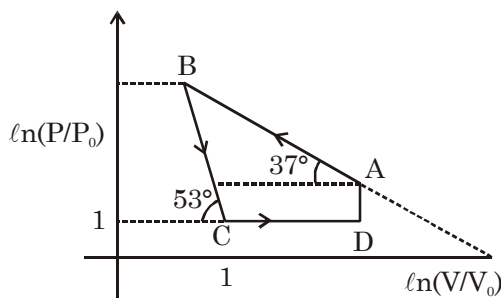
- (P) Polytropic $PV^m = \text{constant}$. ($m \neq \gamma$ or 1 or 0)
- (Q) Adiabatic
- (R) Isochoric
- (S) Isobaric
- (T) Isothermal
- (U) $P = a + bV$

- (A) $I \rightarrow P, II \rightarrow U, III \rightarrow R, IV \rightarrow S$
- (C) $I \rightarrow P, II \rightarrow Q, III \rightarrow S, IV \rightarrow R$

- (B) $I \rightarrow P, II \rightarrow P, III \rightarrow T, IV \rightarrow R$
- (D) $I \rightarrow Q, II \rightarrow P, III \rightarrow S, IV \rightarrow R$

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For a thermodynamic cycle ABCD on a non-linear polyatomic gas, we draw $\ln\left(\frac{P}{P_0}\right)$ versus $\ln\left(\frac{V}{V_0}\right)$ diagram.



Match the process.

List-I

- (I) W_{A-B}
(II) Q_{B-C}
(III) ΔU_{CD}
(IV) Q_{D-A}

List-II

- (P) Positive
(Q) Negative
(R) Zero
(S) Can't be said
(T) Initially positive and negative after that
(U) Initially negative and positive after that

(A) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ P, IV $\rightarrow$ P

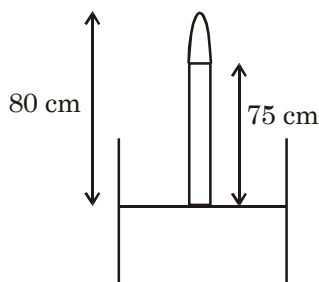
(C) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ R

(B) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ P

(D) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ P

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An ideal barometer has tube length of 80 cm and mercury column of length 75 cm. Take density of mercury = 13.6 gm/cc, $\gamma_{Hg} = 10^{-3} / ^\circ C$



List-I

- (I) Water layer of thickness 1.36 cm added over mercury in cup
(II) Whole apparatus is accelerated downwards with acceleration of 0.1 m/s^2 .
(III) The temperature increases by $10^\circ C$
(IV) Atmospheric pressure decreases by 1 %.

List-II

Length of mercury column

- (P) Decreases by 0.75 cm
(Q) Decreases by 0.1 cm
(R) Increases by 0.1 cm
(S) Increases by 0.75 cm
(T) Remains same
(U) Increases by 1 cm

(A) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ U, IV $\rightarrow$ P

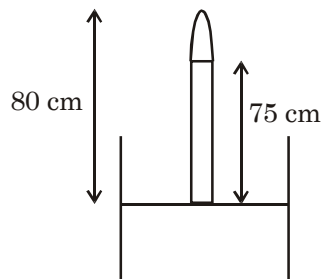
(C) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ P

(B) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ S, IV $\rightarrow$ P

(D) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ U

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A barometer has tube length of 80 cm and mercury column of length 75 cm. Take density of mercury = 13.6 gm/cc , $\gamma_{\text{Hg}} = 10^{-3} / ^\circ\text{C}$



In another situation, there is a little air above the mercury column. If the atmospheric pressure is 76 cm of mercury column & other data remain same. If the air is heated. (In this part, assume that mercury does not expand).

List-I

- (I) Pressure of air
- (II) Volume of air
- (III) Temperature of air
- (IV) Length of mercury column

List-II

- (P) Decreases
- (Q) Increases
- (R) First increases then decreases
- (S) First decreases then increases
- (T) Remains constant
- (U) Can't be predicted

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ P
- (C) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ P, IV $\rightarrow$ Q

- (B) I $\rightarrow$ Q, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ P
- (D) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ U

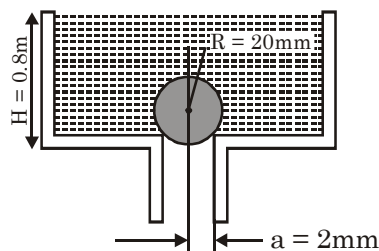
SECTION-II

Numerical-Answer Type

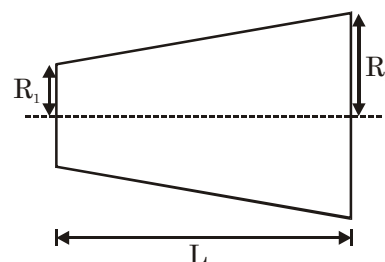
6 Q. [3(-0)]

(Upto Second Decimal Place)

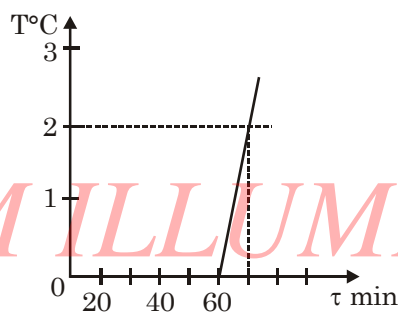
1. A sphere of radius R , made from material of specific gravity S_G , is submerged in a tank of water. The sphere is placed over a hole, of radius a , in the tank bottom. For the dimensions given, determine the minimum S_G required for the sphere to remain in the position shown.



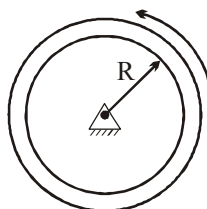
2. A conical cylinder of length $L = 100 \text{ cm}$ and radii $R_1 = 5 \text{ cm}$ and $R_2 = 10 \text{ cm}$ (see figure) is fully insulated along the outer surface. The surface of R_1 is maintained at temperature $T_1 = 227^\circ\text{C}$ and surface of R_2 is maintained at temperature $T_2 = 27^\circ$. The thermal conductivity of the material is $40 \text{ Wm}^{-1}\text{K}^{-1}$. Considering the heat flow along the axis of the cylinder, find the heat (in W) conducted through the cylinder per second.



3. A power cable of copper is just stretched (initial tension zero) straight between two fixed towers. If the temperature decreases, the cable tends to contract. The amount of contraction for a free copper cable or rod is 0.0002% per degree Celsius. Estimate what temperature decrease (in $^{\circ}\text{C}$) will cause the cable to snap. Assume that the cable obeys Hooke's law until it reaches its breaking point, which for copper occurs at a tensile stress of $2.2 \times 10^8 \text{ N/m}^2$. Ignore the weight of the cable and the sag and stress produced by the weight. Young's modulus for copper is $1.1 \times 10^{11} \text{ N/m}^2$.
4. A bucket contains a mixture of water and ice of mass $m = 10\text{ kg}$. The bucket is brought into a room, after which the temperature of the mixture is immediately measured. The obtained $T(\tau)$ dependence is plotted in figure. The specific heat of water is $c_w = 4.2 \text{ J/g K}$ and the latent heat of fusion of ice is $L = 340 \text{ kJ/kg}$. Determine the mass m_{ice} of ice in the bucket at the moment it is brought in the room, neglecting the heat capacity of the bucket. Express your answer in gm



5. A body is floating in a liquid such that at 80°C , 95% of it's volume is under water. Find the minimum temperature (in $^{\circ}\text{C}$) such that the body is completely, submerged in liquid $\gamma_{\text{liquid}} = 6 \times 10^{-4} / ^{\circ}\text{C}$, $\gamma_{\text{body}} = 8 \times 10^{-5} / ^{\circ}\text{C}$.
6. An elastic cord having an unstretched length ℓ , force constant k and mass per unit length m_0 is stretched around the drum of radius r ($2\pi r > \ell$). Determine the speed of the cord, due to rotation of the drum, which will allow the cord to loosen its contact with the drum. Express value of v . (in m/s) Cord lies in horizontal plane. (for the given data $m_0 = 40 \text{ g/cm}$, $k = \frac{25}{11} \text{ N/m}$, $\pi = \frac{22}{7}$, $\ell = 40 \text{ cm}$, $r = 70 \text{ cm}$).



PHYSICS

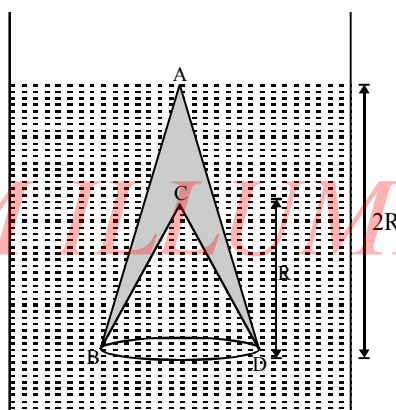
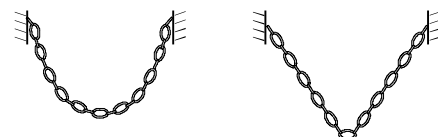
QUIZ # 25

SECTION-I

Single Correct Answer Type

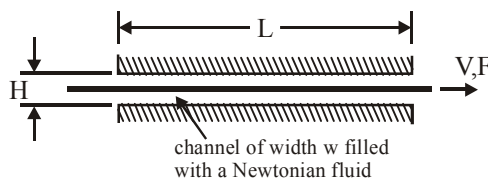
9 Q. [3 M (-1)]

- Pull down in the middle a heavy chain, fixed at both ends, in such a way that it forms a triangle. Does the centre of mass takes up a higher or lower position?
 (A) Centre of mass comes down
 (B) Centre of mass goes up
 (C) Centre of mass remains stationary
 (D) Cannot be predicted
- A solid cone of uniform density and height $2R$ and base radius R has a conical portion scooped out from its base with the same base radius but height R as shown in the figure. The solid cone is floating in a liquid of density ρ with vertex A touching the fluid surface. If atmospheric pressure is P_0 , the force acting on the surface BCD due to liquid is :



- (A) $\frac{\pi R^3 \rho g}{3\sqrt{2}} \left[2 + \frac{2P_0}{R\rho g} \right]$ (B) $\frac{\pi R^3 \rho g}{3\sqrt{2}} \left[5 + \frac{3P_0}{R\rho g} \right]$ (C) $\frac{\pi R^3 \rho g}{3} \left[1 + \frac{3P_0}{R\rho g} \right]$ (D) $\frac{\pi R^3 \rho g}{3} \left[5 + \frac{3P_0}{R\rho g} \right]$

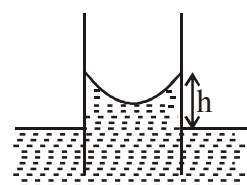
- During a coating process, a thin, flat tape of width w is pulled through a channel of length L containing a Newtonian fluid of density ρ and dynamic viscosity μ . The fluid is in contact with both sides of the tape. Estimate the force required to pull the tape through the channel if the tape has velocity V and the channel has height H . You may assume that the tape is much thinner than H .



- (A) $\frac{4\mu VLw}{H}$ (B) $\frac{2\mu VLw}{H}$ (C) $\frac{\mu VLw}{H}$ (D) $\frac{\mu VLw}{2H}$

- Two vertical parallel plates are partially submerged in water. The distance between plate is equal to d . Water rises due to surface tension T , the width of plate is l , and contact angle of water with glass is 0° . Find the force of attraction between the plates.

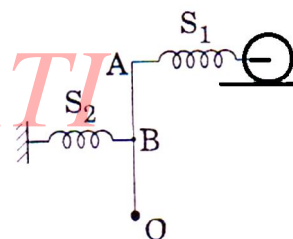
- (A) $\frac{T^2 \ell}{\rho g d^2}$ (B) $\frac{2T^2 \ell}{\rho g d^2}$
 (C) $\frac{T^2 \ell}{2\rho g d^2}$ (D) $\frac{T^2 \ell}{4\rho g d^2}$



5. Horizontal magnetic field of earth at Mumbai is 1.5×10^{-4} T in North direction. A small bar magnet of magnetic moment 10 Am^2 is kept on a horizontal table such that its North pole points due North. What is the magnetic field at 10 cm from centre of magnet at a point on its axis due North of it :-
 (A) 3.5×10^{-4} T (B) 5×10^{-5} T (C) 2.15×10^{-3} T (D) 1.15×10^{-3} T
6. A compass needle is free to rotate in a horizontal plane. It's magnetic moment is 60 Am^2 . When it is pointing geographical north, it experiences a torque of 1.2×10^{-3} Nm due to earth's magnetic field. If earth's magnetic field in horizontal direction is 40×10^{-6} T, what is declination at that place ?
 (A) 60° (B) 45° (C) 30° (D) 0°
7. Two pendulums of time periods 3 s and 7 s respectively start oscillating simultaneously from two opposite extreme positions. After how much time they will be in phase
 (A) $\frac{21}{8}$ s (B) $\frac{21}{4}$ s (C) $\frac{21}{2}$ s (D) $\frac{21}{10}$ s
8. A system is shown in the figure. The time period for small oscillations of the two blocks will be :-



- (A) $2\pi\sqrt{\frac{3m}{k}}$ (B) $2\pi\sqrt{\frac{3m}{2k}}$ (C) $2\pi\sqrt{\frac{3m}{8k}}$ (D) $2\pi\sqrt{\frac{3m}{4k}}$
9. A solid sphere (m kg, R meter) placed on a horizontal rough plane, is attached to a spring S_1 , fixed to a light rod OA of length L meter, as shown in the figure. The other end O of the rod is hinged to rotate in the plane about vertical axis passing through it. A spring S_2 is fixed to the midpoint B of the rod. If the spring constants of both springs are $k \text{ Nm}^{-1}$ each, then the angular frequency of oscillations of the mass would be (in rad s^{-1})

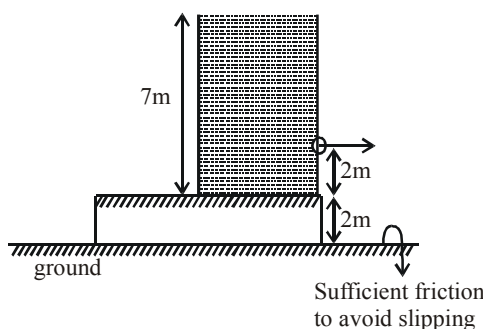


- (A) $\omega = \sqrt{\frac{k}{7m}}$ (B) $\omega = \sqrt{\frac{11k}{28m}}$ (C) $\omega = \sqrt{\frac{2k}{7m}}$ (D) $\omega = \sqrt{\frac{2k}{5m}}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

10. A cylindrical vessel is full of water (density $\rho = 1 \text{ gm/cc}$) is placed on horizontal ground as shown. A small hole of area ($= 1 \text{ cm}^2$) is made on a curved wall at a height of 2m from the bottom of container (container does not move) choose **CORRECT** statement(s) ($g = 10 \text{ m/s}^2$) :-



- (A) Horizontal range of water jet is maximum when water level is at height of 8 m from ground
 (B) Maximum horizontal force on container by table is 1 Newton
 (C) Maximum horizontal range of water jet on ground is $4\sqrt{5}$ meter
 (D) Horizontal range of water jet will decrease then increase with decrease of water level.

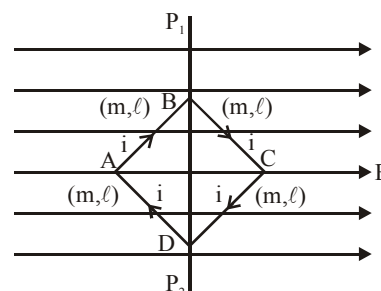
11. Rod AB, BC, CD & DA form a square loop having current i , mass and length of each rod is m and ℓ respectively, is situated in a uniform magnetic field B as shown in the figure and it can rotate about axis P_1P_2 , then

(A) Angular acceleration of the square loop at $t = 0$, is $\frac{3}{2} \frac{Bi}{m}$

(B) Angular velocity when square loop rotated by 30° , is $\sqrt{\frac{3iB}{2m}}$

(C) Torque on the loop when it rotated by 60° , is $\frac{iB\ell^2}{2}$

(D) Angular acceleration of the square loop when it is rotated from starting to 90° , decreases



12. The energy of a particle executing simple harmonic motion is given by $E = Ax^2 + Bv^2$ where x is the displacement from mean position $x = 0$ and v is the velocity of the particle at x then choose the **CORRECT** statement(s).

(A) Amplitude of SHM is $\sqrt{\frac{2E}{A}}$

(B) Maximum velocity of the particle during SHM is $\sqrt{\frac{E}{B}}$

(C) Time period of motion is $2\pi\sqrt{\frac{B}{A}}$

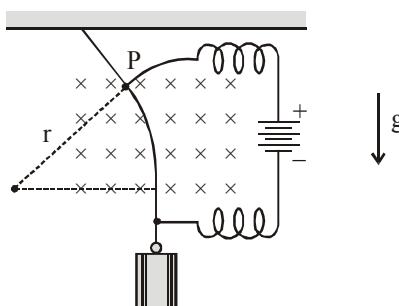
(D) Displacement of the particle is directly proportional to the velocity of the particle.

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para $\times$ 2Q.) [3 M (0)]

Paragraph for Questions 13 and 14

A thin flexible wire carrying a current I hangs in a uniform magnetic field B (Figure). A block of mass m attached to one end of the wire provides a tension T . Within the magnetic field, the wire will finally adopt the shape of an arc of circle. P is the fix support point.



13. Then the radius of arc of the circle is.

(A) $\frac{mg}{4BI}$

(C) $\frac{mg}{2BI}$

(C) $\frac{mg}{BI}$

(D) $\frac{2mg}{BI}$

14. If we remove the wire and launch a particle of charge $-q$ from the point P with speed v along the tangent of wire at point P then we observe that, it will move along the same arc of circle. Then the mass of particle is (Neglect the effect of gravity on particle)

(A) $\frac{2qmg}{Iv}$

(B) $\frac{qmg}{Iv}$

(C) $\frac{qmg}{2Iv}$

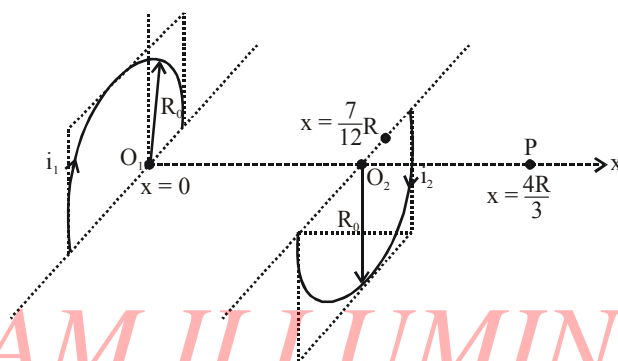
(D) $\frac{3qmg}{Iv}$

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 6 Q. [4 M(-1)]

1. Two semicircular ring of same radius (R) having current i_1 & i_2 are kept parallel to each other in y - z plane as shown in the figure. Centre of first ring is kept at $x = 0$ and that of second ring is kept at $x = \frac{7}{12}R$.

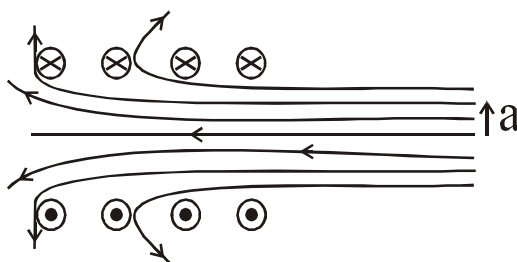
If magnetic field at point P ($x = \frac{4R}{3}$) is along the x -axis then its value is $\frac{n\mu_0 i_1}{20R}$. Find the value of n .



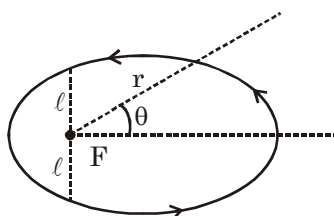
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2. A direct current flows in a solenoid of length L and radius R , ($L \gg R$), producing a magnetic field of magnitude B_0 inside the solenoid. Magnetic field line which leaves end perpendicularly is at distance

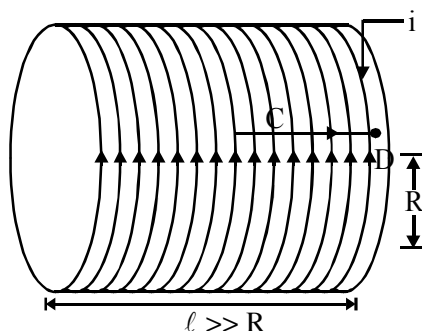
' a ' from axis of solenoid. If $a = \frac{R}{\sqrt{n}}$, find n



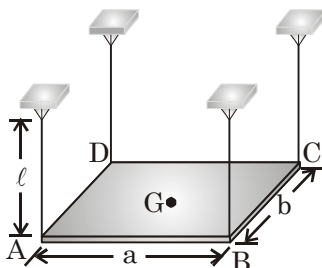
3. Find the magnetic field due to current i flowing in an elliptical loop at its focus. The equation of ellipse (in polar coordinates as shown) is $\frac{\ell}{r} = (1 + e \cos \theta)$. Here e is eccentricity which is a constant. Take $\ell = 50$ cm, $e = 0.8$, $i = 2$ A, if your answer is $n\pi \times 10^{-7}$ T, fill n in OMR sheet.



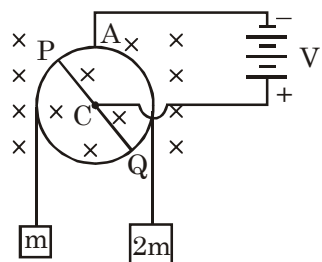
4. A small circular loop of radius $a = 2\text{cm}$ carrying a current $i_0 = 1\text{A}$ is placed co-axially with an ideal solenoid of radius $R = 5\text{cm}$, and current $i = 1\text{A}$ flowing through its windings. Number of turns per unit length is $n = 100$. If the loop is displaced from the centre C of the solenoid to the end of the solenoid slowly by an external agent, find the magnitude of work done by the external agent. If answer is $x \left(\frac{\pi\mu_0 \times 10^{-2}}{3} \right)$ (in J). Then the value of x is.



5. A thin rectangular plate of mass M , sides a and b is suspended from four vertical wires of the same length ℓ . Determine the period of small oscillations of the plate when it is given a small horizontal displacement in a direction perpendicular to AB for $a = b = \ell$. If your answer is $T = 2\pi\sqrt{\frac{x\ell}{yg}}$ fill value of $x + y$.



6. A conducting ring of mass m and radius r has a weightless conducting rod PQ of length $2r$ and resistance $2R$ attached to it along its diameter. It is pivoted at its centre C with its plane vertical, and two blocks of mass m and $2m$ are suspended by means of a light in-extensible string passing over it as shown in the figure. The ring is free to rotate about C and the system is placed in a magnetic field B (into the plane of the ring). A circuit is now completed by connecting the ring at A and C to a battery of emf V . Find the value of V (in volt) so that the system remains static. Take $B = 0.1\text{ T}$, $m = 10\text{ gm}$, $r = 0.5\text{ m}$, $R = 1\Omega$.



PHYSICS

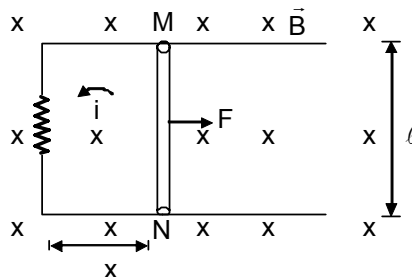
QUIZ # 26

SECTION-I

Single Correct Answer Type

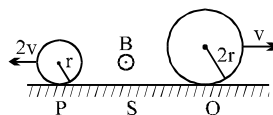
7 Q. [3 M (-1)]

1. Two long parallel horizontal rails, a distance ℓ apart and each has a resistance λ per unit length are joined at one end by a resistance R . A perfectly conducting rod MN of mass m is free to slide along the rails without friction. There is a uniform magnetic field of induction B normal to the plane of paper and directed into the paper. A variable force F is applied to the rod MN such that, as the rod moves, a constant current i flows through P . The applied force F as function of distance x of the rod from R is

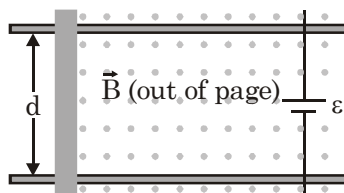


- (A) $i\ell B - \frac{2m\lambda i^2}{B^2 \ell^2} (R + 2\lambda x)$ (B) $i\ell B + \frac{4m\lambda i^2}{B^2 \ell^2} (R - 2\lambda x)$
(C) $i\ell B + \frac{2m\lambda i^2}{B^2 \ell^2} (R + 2\lambda x)$ (D) none of these

2. Two conducting rings P and Q of radii r and $2r$ rotate uniformly in opposite directions with centre of mass velocities $2v$ and v respectively on a conducting surface S . There is a uniform magnetic field of magnitude B perpendicular to the plane of the rings. The potential difference between the highest points of the two rings is

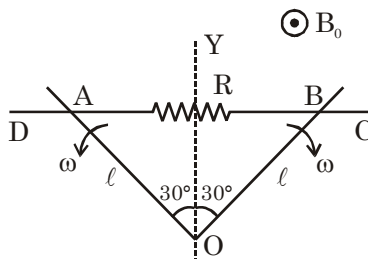


- (A) 0 (B) $4 Bvr$ (C) $8 Bvr$ (D) $16 Bvr$
3. A bar of mass m , length d and resistance R slides without friction in a horizontal plane, moving on parallel rails as shown in figure. A battery that maintains a constant emf ϵ is connected between the rails, and a constant magnetic field $\vec{B}$ is directed perpendicularly to the plane of the page. Assuming the bar starts from rest, Find speed at time t :-



- (A) $v = \frac{\epsilon}{Bd} \left(1 - e^{-\frac{B^2 d^2 t}{mR}} \right)$ (B) $v = \frac{\epsilon}{Bd} \left(1 - e^{-\frac{B^2 d^2 t}{2mR}} \right)$
(C) $v = \frac{\epsilon}{Bd} \left(1 - e^{-\frac{B^2 d^2 t}{mR}} \right)$ (D) $v = \frac{\epsilon}{Bd} \left(1 - e^{-\frac{B d t}{mR}} \right)$

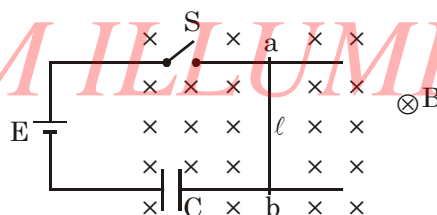
4. In the shown figure, uniform magnetic field B_0 is pointing out of the plane in the region. Wire CD is fixed and has resistance R , while OA and OB are conducting wires rotating with angular velocity ω about O as shown. If at some instant $OA = OB = \ell$ and each wire makes angle $\theta = 30^\circ$ with y-axis, then current through resistance R is



- (A) $\frac{B_0 \omega \ell^2}{R}$ (B) $\frac{\sqrt{3} B_0 \omega \ell^2}{2R}$ (C) $\frac{\sqrt{3} B_0 \omega \ell^2}{4R}$ (D) Zero

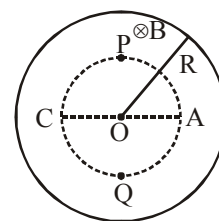
5. A conducting rod ab of mass m and length ℓ is placed on parallel smooth conducting rails connected to an uncharged capacitor of capacitance C and a battery of emf E and as shown. A uniform magnetic field B is existing perpendicular to the plane of the rails. The steady state velocity acquired by the conducting rod ab after closing switch S is (neglect resistance) :

TEAM ILLUMINATI



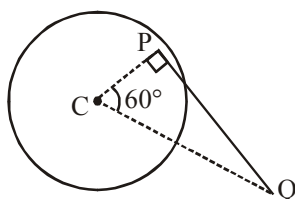
- (A) $\frac{2CB\ell E}{m + CB^2\ell^2}$ (B) $\frac{CB\ell E}{m + CB^2\ell^2}$ (C) $\frac{CB\ell E}{2(m + CB^2\ell^2)}$ (D) $\frac{CB\ell E}{4(m + CB^2\ell^2)}$

6. A uniform magnetic field B increasing with time exists in a cylindrical region of centre O and radius R . The direction of magnetic field is inwards the paper as shown. The work done by external agent in taking a unit positive charge slowly from A to C via paths APC , AOC and AQC be W_{APC} , W_{AOC} and W_{AQC} respectively. Then :



- (A) $W_{APC} = W_{AOC} = W_{AQC}$ (B) $W_{APC} > W_{AOC} > W_{AQC}$
(C) $W_{APC} < W_{AOC} < W_{AQC}$ (D) $W_{APC} = W_{AQC} < W_{AOC}$

7. A magnetic field, confined in a cylindrical region of radius R , is changing at the rate of 4 T/s . A conducting rod PQ of length $\sqrt{\frac{3}{2}}R$ is placed in the region as shown. The induced emf across the rod will be

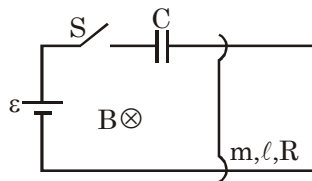


- (A) $\frac{R^2}{24}(\pi + 6)V$ (B) $\frac{R^2\pi}{24}V$ (C) $\frac{R^2\pi}{6}V$ (D) $\frac{R^2}{6}(\pi + 6)V$

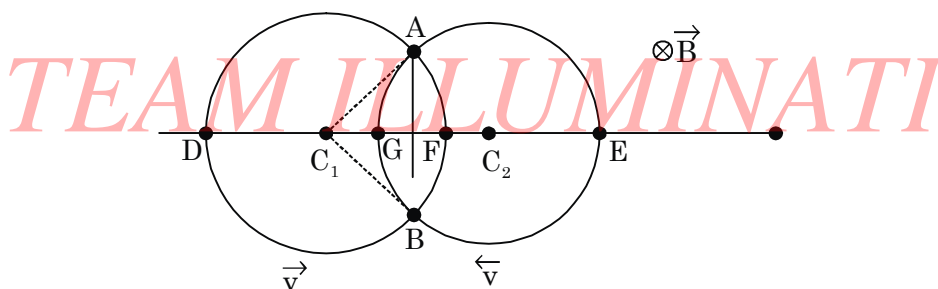
Multiple Correct Answer Type

4 Q. [4 M (-1)]

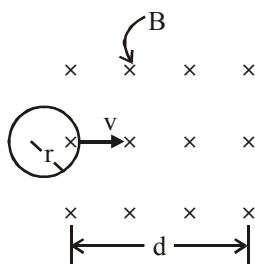
8. In the figure shown a conducting rod of mass m , length ' ℓ ' and resistance ' R ' can smoothly move along parallel rails in horizontal plane. Initially the rod is at rest. A uniform magnetic field $\vec{B}$ perpendicular to the plane of motion exists in the region. Now switch S is closed at $t = 0$, then :



- (A) The charge on the capacitor in steady state is $m c \epsilon / (m + c B^2 \ell^2)$
 (B) The charge on the capacitor in steady state is $2 m c \epsilon / (m + c B^2 \ell^2)$
 (C) Velocity of conducting rod in steady state is $2 B \ell c \epsilon / (m + c B^2 \ell^2)$
 (D) Velocity of conducting rod in steady state is $B \ell c \epsilon / (m + c B^2 \ell^2)$
9. Two circular rings of radius a , having resistance per unit length λ are moving with uniform speed v along x-axis one over other as shown in the figure in a uniform magnetic field $\vec{B} = B_0(-\hat{k})$. At $t = 0$, separation between centre is just greater than $2a$.



- (A) the magnetic force on any circular ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{72B^2av}{5\pi\lambda}$.
 (B) the magnetic force on any circular ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{18B^2av}{5\pi\lambda}$.
 (C) the current in the branch BDA of the ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{3Bv}{5\pi\lambda}$.
 (D) the current in the branch BDA of the ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{2Bv}{5\pi\lambda}$.
10. A conducting loop is pulled with a constant velocity towards a region of uniform magnetic field of induction B as shown in the figure. Then the current involved in the loop is ($d > r$)



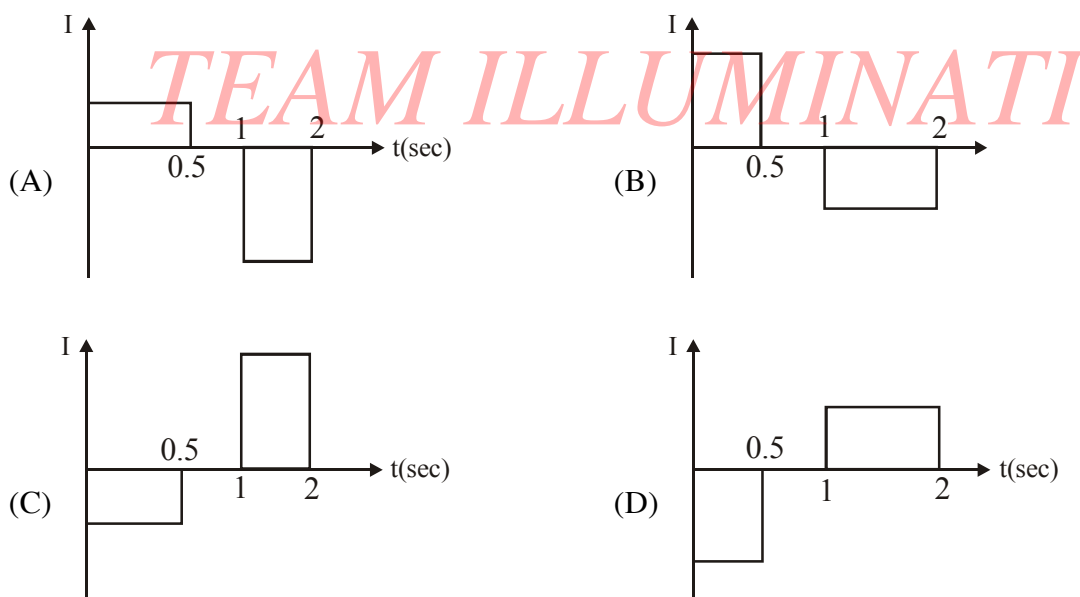
- (A) clockwise while entering
 (B) anti-clockwise while entering
 (C) zero when completely inside
 (D) clockwise while leaving

11. A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction. The beam's axis is coincident with the z axis and the electron charge density is given by $\rho = cr^2$ (C/m^3) where c is a constant and r is the radial distance from the axis of the beam :-
- (A) The current crossing the xy -plane is $\pi c u a^4$
 (B) The electric field at distance r ($r < a$) from z axis is proportional to r^3
 (C) The magnetic field at distance r ($r < a$) from z axis is proportional to r^3
 (D) The electric field at distance r ($r < a$) from z axis is inversely proportional to r .

Linked Comprehension Type
(Single Correct Answer Type)
(2 Para × 2Q. & 1 Para × 3Q.) [3 M (0)]
Paragraph for Question no. 12 and 13

A circular coil of wire has a radius of 0.5m and a total resistance of 1.57Ω . The coil lies in the x - y plane. The coil is in uniform magnetic field $\vec{B}$ that is in $-z$ direction, which is directed away from you as you view the coil. The magnitude B of the field depends on time as follows : It increases at a constant rate from 0 at $t = 0$ to 0.8 T at $t = 0.5$ sec, is constant at 0.8T from $t = 0.5$ sec to $t = 1$ sec and decreases at a constant rate from 0.8T at $t = 1$ sec to 0 at $t = 2$ sec.

12. Draw a graph of the current I induced in the coil versus t for 0 to 2 sec. Let counter clockwise currents be positive and clockwise currents be negative.



13. What is the maximum induced electric field magnitude in the coil during 0 to 2 sec interval :-
- (A) 0.4 V/m (B) 0.2 V/m (C) 0.6 V/m (D) 8 V/m

Paragraph for Question No. 14 and 15

A smooth insulating ring of radius R , with a bead of mass m and charge q is placed horizontally and in a uniform magnetic field of strength B_0 and perpendicular to the ring plane. Starting from $t = 0$, the magnetic field is changed to $B(t) = B_0 + \alpha t$, where α is a positive constant & t is time. Neglect gravity

14. The contact force between the ring and bead as a function of time t is :

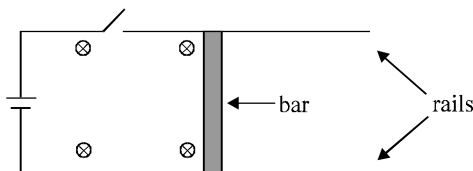
(A) $\frac{\alpha q^2 R t}{4m} (2B_0 + \alpha t)$ (B) $\frac{\alpha q^2 R t}{2m} (B_0 + \alpha t)$ (C) $\frac{\alpha q^2 R t}{m} (B_0 + \alpha t)$ (D) $\frac{qR}{2m} (2B_0 + \alpha t)$

15. The velocity of bead as function of time t

(A) $\frac{qR}{m} \alpha t$ (B) $\frac{qR}{2m} \alpha t$ (C) $\frac{2qR}{m} \alpha t$ (D) $\frac{qR}{2m} (B_0 + \alpha t)$

Paragraph for Question nos. 16 to 18

A type of projectile launcher is shown in the figure below. A large current moves in a closed loop composed of fixed rails, a battery and a very light bar touching the rails. The loop lies in a horizontal plane. A magnetic field of 1.70 T is perpendicular to the plane of the circuit. The bar has a length of 0.200m, a mass of 1.52×10^{-3} kg. and a resistance of $4.00 \times 10^{-3} \Omega$. The bar is made of copper which has a resistivity of $1.70 \times 10^{-8} \Omega \cdot \text{m}$. When the switch is closed, the bar experiences an acceleration of $4.50 \times 10^2 \text{ m/s}^2$.

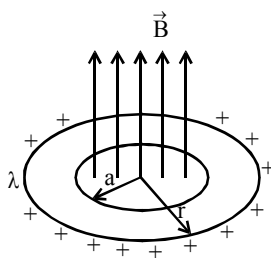


16. What is the cross-sectional area of the bar?
(A) $6.72 \times 10^{-7} \text{ m}^2$ (B) $7.13 \times 10^{-7} \text{ m}^2$ (C) $7.96 \times 10^{-7} \text{ m}^2$ (D) $8.50 \times 10^{-7} \text{ m}^2$
17. In which direction does the bar move when the switch is closed?
(A) to the right
(B) to the left
(C) out of the page, i.e. the bar jumps off the fixed rails
(D) Can't predict
18. When the switch is closed, what is the current through the bar?
(A) 1.00 A (B) 2.01 A (C) 3.02 A (D) 4.03 A

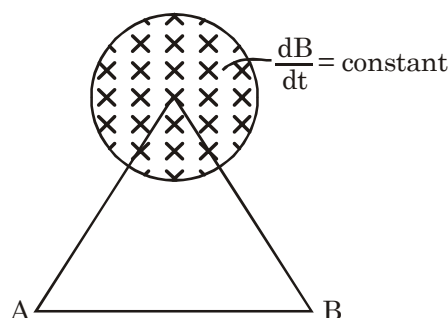
TEAM ILLUMINATI
SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 2 Q. [4 M(-1)]

1. A non conducting ring of mass 4 kg is uniformly charged with $\lambda = 4 \text{ C/m}$ and kept on rough horizontal surface with friction coefficient $\mu = \pi/4$. A time varying magnetic field $B = B_0 t^2$ is applied in a circular region of radius a ($a < r$) perpendicular to the plane of ring as shown in figure. Find out the time (in sec) when ring just starts to rotate on surface. (take $a = 5 \text{ cm}$ and $g = 10 \text{ m/s}^2$ $B_0 = 125 \text{ SI Unit}$).



2. An uniform magnetic field is present in a cylindrical region as shown. This field is increasing uniformly with time. An equilateral loop is placed in such a way that its vertex coincide with centre of cylindrical region. Resistances of sides BC and CA are negligible whereas that of AB is 2Ω . If a current of magnitude 2A ampere flows in the triangular loop, due to induced emf, find potential difference between points A and B (in volt).





QUIZ

**JEE (Main + Advanced) 2020
ENTHUSIAST COURSE
PHASE : TOAS, TEAS & T-AS**

PHYSICS

QUIZ # 27

SECTION-I

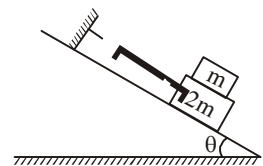
Single Correct Answer Type

8 Q. [3 M (-1)]

1. A particle performs harmonic oscillations along a straight line with a period T and amplitude a . The mean velocity of the particle averaged over the time interval during which it travels a distance $\frac{a}{2}$ starting from the extreme position is :

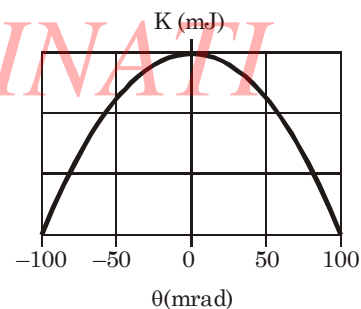
(A) $\frac{a}{T}$ (B) $\frac{2a}{T}$ (C) $\frac{3a}{T}$ (D) $\frac{a}{2T}$

2. The coefficient of friction between block of mass m and $2m$ is $\mu = 2 \tan \theta$. There is no friction between block of mass $2m$ and inclined plane. The maximum amplitude of two block system for which there is no relative motion between both the blocks.



(A) $\frac{2mg \sin \theta}{k}$ (B) $\frac{3mg \sin \theta}{k}$ (C) $\frac{6mg \sin \theta}{k}$ (D) $\frac{8mg \sin \theta}{k}$

3. Figure shows the kinetic energy of a simple pendulum versus its angle θ from the vertical. The maximum kinetic energy is 15mJ. The pendulum bob has mass 0.200 kg.



- (A) The angular amplitude is 0.1 rad/s
(B) The angular frequency is 8 rad/s approx
(C) The length of the pendulum is 1.5 m
(D) The angular frequency is 12 rad/s approx
4. A simple pendulum is inside a container which is filled with non-viscous liquid of density σ . The density of material of the pendulum bob is ρ . If the container is placed in a lift which is moving down with retardation a_0 , then time period of simple pendulum is

(A) $2\pi \sqrt{\frac{L}{g\left(1 - \frac{\sigma}{\rho}\right) - a_0}}$

(B) $2\pi \sqrt{\frac{L}{g\left(1 - \frac{\sigma}{\rho}\right) + a_0}}$

(C) $2\pi \sqrt{\frac{L}{(g + a_0)\left(1 - \frac{\sigma}{\rho}\right)}}$

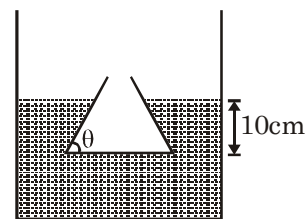
(D) $2\pi \sqrt{\frac{L}{(g - a_0)\left(1 - \frac{\sigma}{\rho}\right)}}$

5. A uniform straight rod of mass ' m ' kg & length L is hinged at one end. It is free to oscillate in vertical plane. A point mass (m kg) is attached to it at a distance " x " from the hinge. The value of x

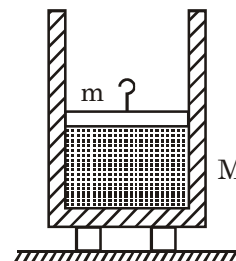
for which time period of oscillations will be minimum is (use: $\sqrt{\frac{7}{3}} = 1.5$)

(A) $2L/3$ (B) $L/2$ (C) $L/4$ (D) $2L/7$

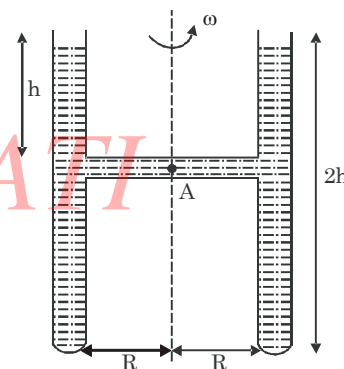
6. A conical flask of mass 10 kg and base area as 10^3 cm^2 is floating in liquid of relative density 1.2, as shown in the figure. The force that liquid exerts on curved surface of conical flask will be (Given $g = 10 \text{ m/s}^2$)



- (A) 20 N downward direction
(B) 40 N in downward direction
(C) can be determined only if θ is given
(D) 20 N in upward direction
7. As shown, a piston chamber of cross section area A is filled with ideal gas. A sealed piston of mass m is right at the middle height of the cylinder at equilibrium. The friction force between the chamber wall and the piston can be ignored. The mass of the rest of the chamber is M . The atmosphere pressure is P_0 . Now slowly pull the piston upwards, find the maximum value of M such that the chamber can be lifted off the ground. The temperature remains unchanged.



- (A) $M = \frac{P_0 A - 2mg}{2g}$ (B) $M = \frac{P_0 A - mg}{2g}$ (C) $M = \frac{P_0 A - mg}{g}$ (D) $M = \frac{P_0 A}{2g}$
8. The shown H shaped apparatus contains an ideal incompressible liquid and has dimension as shown in figure. The diameters of the tubes are small as compared to h and R . The apparatus is rotated with a constant angular velocity ω about a symmetric vertical axis as shown in figure. The pressure at point A is :

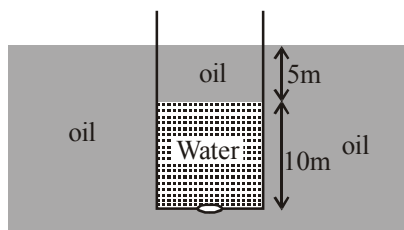


- (A) $P_0 + \rho gh$ (B) $P_0 + \rho gh + \frac{\rho R^2 \omega^2}{2}$
(C) $P_0 + \rho gh - \frac{\rho R^2 \omega^2}{2}$ (D) $P_0 - \rho gh - \frac{\rho R^2 \omega^2}{2}$

Multiple Correct Answer Type

5 Q. [4 M (-1)]

9. A body floats on water and also on an oil of relative density 1.25. Which of the following is/are **TRUE** ?
(A) The body loses more weight in oil than in water
(B) The volume of water displaced is 1.25 times that of oil displaced.
(C) The body experiences equal upthrust from water and oil
(D) To make the body just sink, one will need 1.25 times load in case of oil than in case of water
10. A vessel of uniform cross-section open at the top with an orifice at its bottom contains oil (relative density 0.8) on top of water. It is immersed vertically in a large open tank of same oil as shown in figure. In which of following configuration will liquid level remain same. (Water does not come out or oil does not enter vessel) :- (Take : area of orifice = 1 mm^2 and area of vessel as 100 cm^2)



- (A) (B) (C) (D)

11. A thin walled hemispherical shell of mass m and radius R is pressed against a smooth vertical wall. Through a very small aperture at its top water of density ρ is filled in it completely. Minimum magnitude of force is to be applied to the shell for liquid not to escape from it. Weight of hemispherical

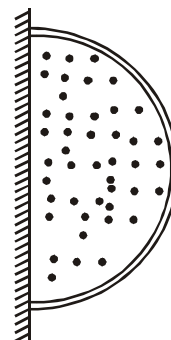
shell is mg and total mass of water is $M = \frac{2}{3}\pi R^3 \rho$:-

(A) Horizontal component of net force is $\frac{1}{2}\pi R^3 \rho g$

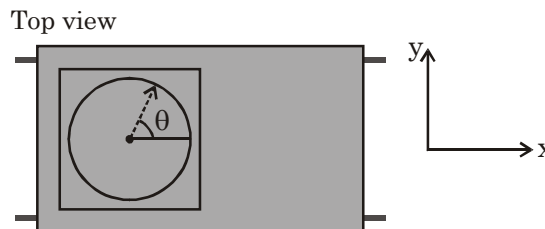
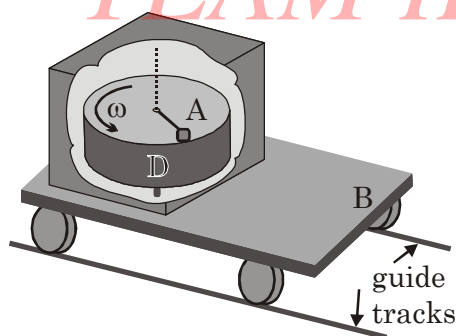
(B) Net external force is $Mg\sqrt{\frac{9}{4} + \left(1 + \frac{m}{M}\right)^2}$

(C) Vertical component of force is $(M + m)g$

(D) Direction of net force is given by $\theta = \arctan \frac{(m+M)}{3M}$, where θ is from horizontal direction.



12. Energy storage devices use spinning flywheels to store energy. Assume that the flywheel is almost perfectly balanced to avoid any vibrations. Let the flywheel D, whose radius is r , rotate with angular speed ω . In addition, assume that the cart B is constrained to move rectilinearly along the smooth guided track along x-axis. Given that the flywheel has a unbalanced weight A of mass m_A attached at the edge. The total mass of the flywheel D, cart B is m_B , determine the following : [Neglect the mass of the wheels, assume that initially (at $t = 0$) everything is at rest, and assume that the unbalanced mass is at the edge of the flywheel along x-axis.] Mark the **CORRECT** statements



(A) When speed of cart is v_B along x axis and angular speed of flywheel is ω momentum of system as function of θ is given by $[m_A(v_B - r\omega \sin \theta) + m_B v_B]\hat{i} + m_A r\omega \cos \theta \hat{j}$

(B) Velocity of A relative to B as a function of θ is given by $-r\omega \sin \theta \hat{i} + r\omega \cos \theta \hat{j}$

(C) Amplitude of the oscillation of B is $\frac{m_A r}{m_A + m_B}$

(D) Maximum velocity of B is $(v_B)_{\max} = \frac{m_A r \omega}{m_A + m_B}$

13. Speed of a particle moving along x-axis is given by $v = \sqrt{16x - x^2 - 48}$ which of the following is/are correct ? (x is in m)

(A) Amplitude of oscillation is 4m

(B) maximum speed of particle is 4m/s

(C) mean position of particle is (8, 0)

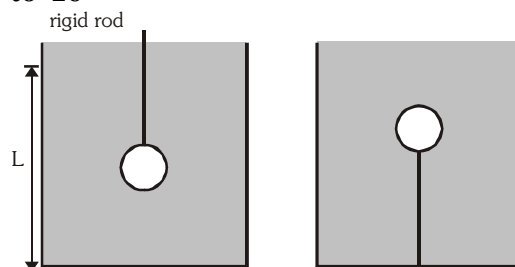
(D) acceleration of particle at $x = 9\text{m}$ is 2m/s^2 .

Linked Comprehension Type (1 Para × 3Q.)
(Single Correct Answer Type)

[3 M (0)]

Paragraph for Question 14 to 16

A cylindrical container of length L is full to the brim with a liquid which has mass density ρ . It is placed on a weight-scale; the scale reading is w . A light ball which would float on the liquid if allowed to do so, of volume V and mass m is pushed gently down and held beneath the surface of the liquid with a rigid rod of negligible volume as shown on the left.



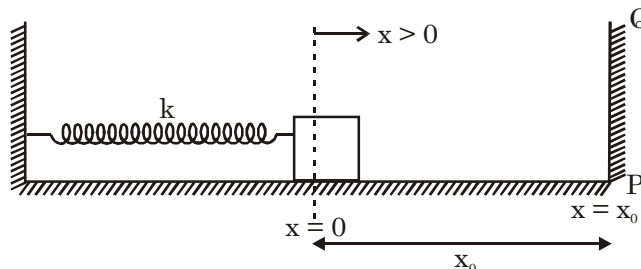
14. What is the mass M of liquid which overflowed while the ball was being pushed into the liquid?
 (A) ρV (B) m (C) $m - \rho V$ (D) none of these
15. What is the reading of the scale when the ball is fully immersed
 (A) $w - \rho Vg$ (B) w (C) $w + mg - \rho Vg$ (D) none of these
16. If instead of being pushed down by a rod, the ball is held in place by a thin string attached to the bottom of the container as shown on the right. What is the tension T in the string?
 (A) $(\rho V - m)g$ (B) ρVg (C) mg (D) none of these

Linked Comprehension Type (1 Para × 2 Q.)
(Multiple Correct Answer Type)

[4 M (0)]

Paragraph for Question no. 17 and 18

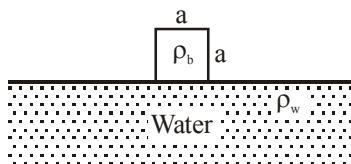
A block is attached to a spring and is placed on a horizontal smooth surface as shown, in which spring is unstretched. Now the spring is given an initial compression $2x_0$ and block is released from rest. Collision with the wall PQ are elastic.



17. Find the time period of motion of the block :-
 (A) $\frac{2\pi}{3} \sqrt{\frac{m}{k}}$ (B) $\frac{4\pi}{3} \sqrt{\frac{m}{k}}$ (C) $\frac{3\pi}{2} \sqrt{\frac{m}{k}}$ (D) $\frac{\pi}{2} \sqrt{\frac{m}{k}}$
18. If coefficient of resitution $e = 1/2$ in the collision between block and the wall then select the **CORRECT** statement(s).
 (A) New maximum compression in the spring is x_0 .
 (B) New maximum compression in the spring is $x_0 \sqrt{\frac{7}{4}}$.
 (C) New angular frequency after collision is $\sqrt{\frac{k}{m}}$
 (D) New angular frequency after collision is $\sqrt{\frac{k}{2m}}$

Matching List Type (4 × 4)**1Q.[3 M (0)]**

19. A uniform cubical block of density ρ_b (in g/cc) & side length a is released from rest from the position where one of its horizontal surface (lower) just touches the water (density $\rho_w = 1\text{gm/cc}$). Various values of ρ_b are given in list-I. Neglect viscosity and surface tension. In list-II, expression for the time period of block are given. Match the correct expression of the time period in list-II with the density given in list-I.

**List-I**

(P) $\frac{1}{5}$

(Q) $\frac{2}{5}$

(R) $\frac{1}{2}$

(S) $\frac{2}{3}$

List-II

(1) $2\pi\sqrt{\frac{\rho_b a}{\rho_w g}}$

(2) $2\pi\sqrt{\frac{\rho_w a}{\rho_b g}}$

(3) $\frac{4\pi}{3}\sqrt{\frac{\rho_b a}{\rho_w g}} + \frac{\rho_b}{\rho_w - \rho_b}\sqrt{\frac{4\rho_w a}{3\rho_b g}}$

(4) $\frac{2\pi}{3}\sqrt{\frac{\rho_b a}{\rho_w g}} + \frac{\rho_b}{\rho_w - \rho_b}\sqrt{\frac{2\rho_w a}{3\rho_b g}}$

Codes :

	P	Q	R	S
(A)	1	3	1	1
(B)	1	2	2	3
(C)	1	1	1	3
(D)	1	4	2	3

SECTION-III**Numerical Grid Type (Single digit Ranging from 0 to 9)****1 Q. [4 M(-1)]**

1. During an earthquake the ground is observed to move horizontally. First it moves suddenly $3\sqrt{2}$ cm to the right, then after 1 second it moves suddenly to the left by $3\sqrt{2}$ cm and returns to its original position. A chandelier hang on a 4m long cord. Find the amplitude (in cm) of the chandelier after the earthquake. Take $\pi^2 = g$

PHYSICS

QUIZ # 28

SECTION-I

Single Correct Answer Type

17 Q. [3 M (-1)]

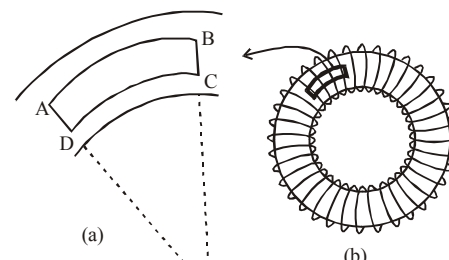
1. Constant current is flowing in toroid & top-view of toroid section in which loop ABCD is present as shown in figure. Then choose the **CORRECT** statement.

(A) $\int \vec{B} \cdot d\vec{\ell}$ is more for AB than DC

(B) $\int \vec{B} \cdot d\vec{\ell}$ is less for AB than DC

(C) $\int \vec{B} \cdot d\vec{\ell}$ is equal for both AB and DC

(D) Can't say anything without knowing about cross section shape of toroid



2. An electric current is flowing through a circular coil of radius R. The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil on its axis is:

(A) $2\sqrt{2}$

(B) 27

(C) 36

(D) 8

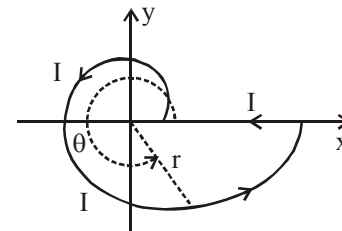
3. A wire carrying a current I is bent into the shape of an exponential spiral, $r = e^\theta$, from $\theta = 0$ to $\theta = 2\pi$ as shown in figure. To complete a loop, the ends of the spiral are connected by a straight wire along the x-axis. Find the magnitude of B at the origin.

(A) $\frac{\mu_0 I}{2\pi\sqrt{2}} [1 - e^{-2\pi}]$

(B) $\frac{\mu_0 I}{4\pi} \sqrt{2} [1 - e^{-2\pi}]$

(C) $\frac{\mu_0 I}{2\pi} [1 - e^{-2\pi}]$

(D) $\frac{\mu_0 I}{4\pi} [1 - e^{-2\pi}]$



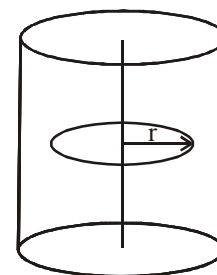
4. The magnetic field inside a solid conducting long wire at distance r from its axis is given as $B = B_0 r^3$ where B_0 is constant. Which of the following relations correctly represents current enclosed in the loop of radius r shown in the figure:

(A) $\frac{2\pi B_0 r^3}{5\mu_0}$

(B) $\frac{\pi B_0 r^4}{2\mu_0}$

(C) $\frac{2\pi B_0 r^2}{\mu_0}$

(D) $\frac{2\pi B_0 r^4}{\mu_0}$



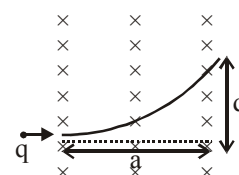
5. In a certain region of space there exists a constant and uniform magnetic field of induction B. The width of the magnetic field is a. A charged particle having charge q, is projected perpendicular to $\vec{B}$ and along the width of the field. If deflection produced by the field perpendicular to the width is d, then the magnitude of the momentum of the particle is :-

(A) $\left(\frac{d^2 + a^2}{2d} \right) qB$

(B) $\frac{a^2}{2d^2} qB$

(C) $\frac{4d^2}{(a+d)} qB$

(D) $\frac{(a^2 - d^2)}{2d} qB$



6. Two long concentric cylindrical conductors of radii a & b ($b < a$) are maintained at a potential difference V & carry equal & opposite currents I . An electron with a particular velocity " U " parallel to the axis will travel undeviated in the evacuated region between the conductors. Then $U =$

(A) $\frac{4\pi V}{\mu_0 I \ln\left(\frac{b}{a}\right)}$ (B) $\frac{2\pi V}{\mu_0 I \ln\left(\frac{a}{b}\right)}$ (C) $\frac{2\pi V}{\mu_0 I \ln\left(\frac{b}{a}\right)}$ (D) $\frac{8\pi V}{\mu_0 I \ln\left(\frac{a}{b}\right)}$

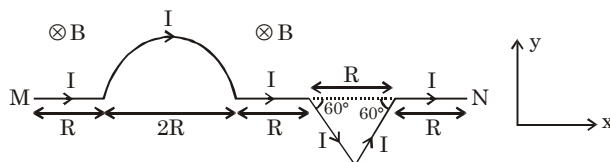
7. A parallel plate capacitor is connected to a battery of emf U . The negative plate of the capacitor is warmed up and it radiates electrons which take off from it practically without any initial speed. Between plates of the capacitor a homogeneous magnetic field directed parallel to plates is created. At what smallest size of the magnetic field will the current in the circuit be zero? Distance between plates is d .

(A) $\sqrt{\frac{mU}{qd^2}}$ (B) $\sqrt{\frac{2mU}{qd^2}}$ (C) $\sqrt{\frac{mU}{2qd^2}}$ (D) $2\sqrt{\frac{mU}{qd^2}}$

8. A current I flows along the length of a thin walled, long metallic hollow cylinder of radius R , distributed uniformly on its surface then pressure on the wall will be :

(A) $\left(\frac{\mu_0 I^2}{8\pi^2 R^2}\right)$ (B) $\left(\frac{\mu_0 I^2}{4\pi^2 R^2}\right)$ (C) $\left(\frac{\mu_0 I^2}{\pi^2 R^2}\right)$ (D) None of these

9. A conducting wire MN carrying a current I is bent into the shape as shown and placed in xy plane. A uniform magnetic field $\vec{B} = -B\hat{k}$ is existing in the region. The net magnetic force experienced by the conducting wire MN is :-



- (A) $2BIR$ (B) $3BIR$ (C) $4BIR$ (D) $6BIR$
10. Two parallel long wires of length 100 m each are separated by a distance of 40 cm from each other. The current in each of them is 200 A. The current flows in the same direction. The force acting between the two wires is :-

(A) 2N repulsive (B) 2N attractive (C) 0.02 N repulsive (D) 0.02 N attractive

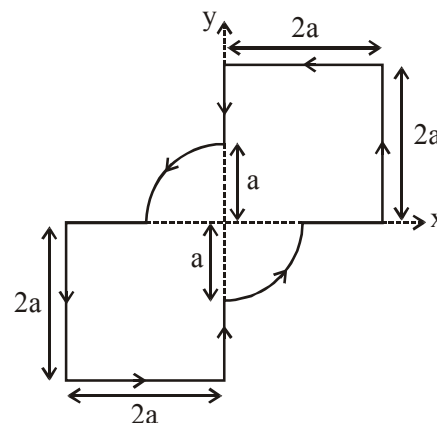
11. A current ' I ' flows through a thin wire as shown in the figure. If there exists an external magnetic field B in the same plane of the wire. The torque acting on the coil is :-

(A) $I\left(\frac{\pi a^2}{2} + 8a^2\right)B$

(B) $I\left(\frac{\pi a^2}{2} + 4a^2\right)B$

(C) $I(\pi a^2 + 8a^2)B$

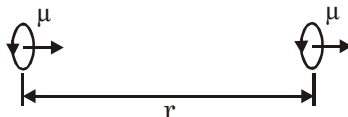
(D) 0



12. Two magnetic dipoles each having parallel magnetic moments M are placed along a common axis separated by a distance d . The magnetic field at a point distant d from each magnet is :

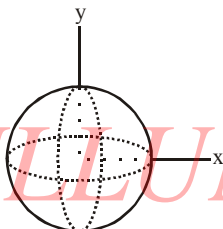
(A) 0 (B) $\frac{\mu_0}{4\pi} \left(\frac{M}{2d^3} \right)$ (C) $\frac{\mu_0}{4\pi} \left(\frac{7M\sqrt{3}}{3d^3} \right)$ (D) $\frac{\mu_0}{4\pi} \left(\frac{5M\sqrt{7}}{6d^3} \right)$

13. Two small magnetic dipole with the same magnetic moments μ are located at a distance r from each other. Determine force of their interaction.



(A) $F = \frac{3\mu_0\mu^2}{2\pi r^4}$ (B) $F = \frac{\mu_0\mu^2}{\pi r^4}$ (C) $F = \frac{1\mu_0\mu^2}{2\pi r^4}$ (D) $F = \frac{\mu_0\mu^2}{2\pi r^4}$

14. Three rings each having equal radius R , are placed mutually perpendicular to each other and each having its centre at the origin of coordinate system. If current I is flowing through each ring then the magnitude of the magnetic field at the common centre is :-

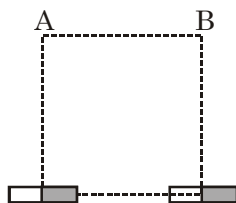


(A) $\frac{3\mu I}{2R}$ (B) $\frac{\sqrt{3}\mu_0 I}{2R}$ (C) 0 (D) $(\sqrt{3}-1)\frac{\mu_0 I}{2R}$

15. The magnetic needle of an oscillation magnetometer, executes 45 oscillations per minute at a place where the dip is 37° . The oscillation frequency at a place where the dip is 60° will be.: (the ratio of the magnetic fields at the two places is 405 : 512).

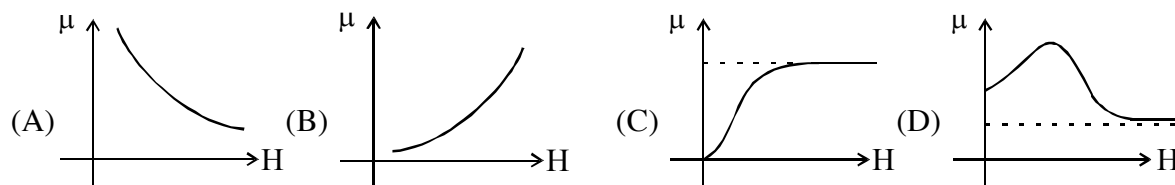
(A) 35 per minute (B) 40 per minute (C) 50 per minute (D) 55 per minute

16. Two small identical magnetic dipoles are placed on the vertices of a square as shown. If magnetic field at A is B_0 , magnitude of magnetic field at B is :



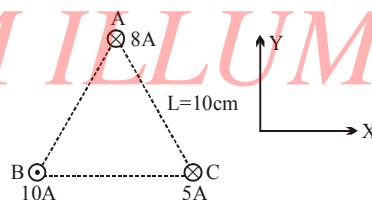
(A) B_0 (B) $B_0 \left(1 - \frac{1}{\sqrt{2}} \right)^2$ (C) $B_0 \left(\frac{3}{2} - \sqrt{2} \right)$ (D) None of these

17. Choose **CORRECT** variation of μ (permeability of material) with magnetising field intensity ($\vec{H}$) for a ferromagnetic material.



Multiple Correct Answer Type**3 Q. [4 M (-1)]**

18. A particle of specific charge ' α ' is projected from origin at $t = 0$ with a velocity $\vec{V} = V_0(\hat{i} + \hat{k})$ in a magnetic field $\vec{B} = -B_0\hat{k}$. Then :
- (A) at $t = \frac{\pi}{\alpha B_0}$, velocity of the particle is $-V_0(\hat{i} - \hat{k})$
- (B) at $t = \frac{\pi}{4\alpha B_0}$, speed of the particle is V_0
- (C) at $t = \frac{2\pi}{\alpha B_0}$, magnitude of displacement of the particle is more than $\frac{2V_0}{\alpha B_0}$
- (D) at $t = \frac{2\pi}{\alpha B_0}$, distance travelled by the particle is less than $\frac{2\sqrt{2}\pi V_0}{\alpha B_0}$.
19. Three long straight wires are at the corners of an equilateral triangle with sides of length $L = 10$ cm. The wires are parallel and carry currents as shown in the figure.

TEAM ILLUMINATI

- (A) Repulsive force per unit length on C is greater than the attractive force on C.
- (B) Repulsive force per unit length on C is less than the attractive force on C.
- (C) Net force per unit length on C is $\sqrt{84} \times 10^{-5} \text{ Nm}^{-1}$, making an angle $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ to the X-axis.
- (D) Net force per unit length on C is $\sqrt{84} \times 10^{-5} \text{ Nm}^{-1}$, making an angle $\frac{\pi}{2} + \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ to the X-axis.
20. A uniform conducting ring of mass $m = 2$ kg, radius $r = 2$ m and resistance 8Ω is kept on smooth, horizontal surface. A time varying magnetic field $\vec{B} = (\hat{i} + t^2\hat{j})$ Tesla is present in the region, where t is time in second and take vertical as y-axis (Take $\pi^2 = 10$). Then :
- (A) Time when ring starts toppling is 1 sec.
- (B) Time when ring starts toppling is $\frac{3}{4}$ sec.
- (C) Heat generated through the ring till the instant when the ring start toppling is $\frac{40}{3}$ Joule.
- (D) Heat generated through the ring till the instant when the ring start toppling is $\frac{80}{3}$ Joule.



QUIZ

**JEE (Main + Advanced) 2020
ENTHUSIAST COURSE
PHASE : TOAS, TEAS & T-AS**

PHYSICS

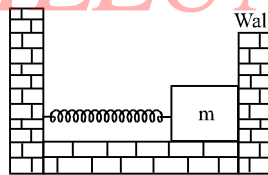
QUIZ # 29

SECTION-I

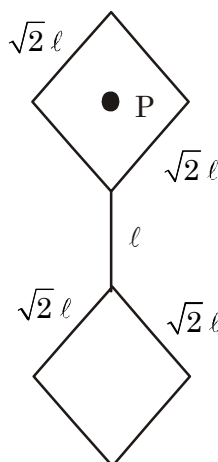
Single Correct Answer Type

5 Q. [3 M (-1)]

- The reading of a spring balance when a mass is weighed on it in air is 40gm. When the mass is immersed in water, its reading is 20gm. The specific gravity of the mass is :
(A) 3 (B) 2 (C) 1.5 (D) 2.5
- The phase difference between two SHM $Y_1 = 10\sin(10\pi t + \pi/3)$ and $Y_2 = 12\sin(8\pi t + \pi/4)$ at $t=0.5s$ is:
(A) $\frac{11\pi}{12}$ (B) $\frac{13\pi}{12}$ (C) π (D) $\frac{17\pi}{12}$
- A bob is attached to a long, light string. The string is deflected by 3° initially with respect to vertical. The length of the string is 1 m. The value of θ at any time t after the bob released can be approximately written as (Use $g = \pi^2$)
(A) $3^\circ \cos \pi t$ (B) $3^\circ \sin \pi t$ (C) $3^\circ \sin(\pi t + \frac{\pi}{6})$ (D) $3^\circ \cos(\pi t + \frac{\pi}{6})$
- In the figure, the block of mass m , attached to the spring of stiffness k is in contact with the completely elastic wall, and the compression in the spring is 'e'. The spring is compressed further by 'e' by displacing the block towards left and is then released. If the collision between the block and the wall is completely elastic then the time period of oscillations of the block will be:



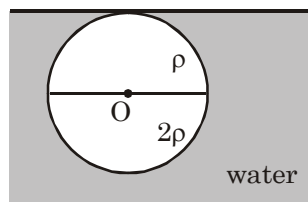
- Two square plates each of mass m are connected by a rod of length ℓ and mass m as shown find time period of SHM in the plane of figure.



-
- (A) $2\pi\sqrt{\frac{8\sqrt{2}\ell}{9g}}$ (B) $2\pi\sqrt{\frac{8\sqrt{2}\ell}{3g}}$ (C) $2\pi\sqrt{\frac{8\ell}{3g}}$ (D) $2\pi\sqrt{\frac{8\ell}{9g}}$

Multiple Correct Answer Type**4 Q. [4 M (-1)]**

6. The potential energy of a particle of mass 0.1 kg moving along the x-axis is given by $U = 5x(x-4)$ J, where x is in meters. It can be concluded that:
- (A) the particle is acted upon by a constant force
 (B) the speed of the particle is maximum at $x = 2$ m
 (C) the particle executes simple harmonic motion
 (D) the period of oscillation of the particle is $\pi/5$ seconds
7. A 20 gm particle is subjected to two simple harmonic motions $x_1 = 2 \sin 10t$, $x_2 = 4 \sin \left(10t + \frac{\pi}{3}\right)$ along same straight line. Where x_1 and x_2 are in metre and t is in sec.
- (A) the displacement of the particle at $t = 0$ will be $2\sqrt{3}$ m
 (B) maximum speed of the particle will be $20\sqrt{7}$ m/s
 (C) magnitude of maximum acceleration of the particle will be $200\sqrt{7}$ m/s².
 (D) Energy of the resultant motion will be 28 J.
8. Total energy of a particle executing oscillating motion is 3 joule and given by $E = x^2 + 2x + v^2 - 2v$ Where x is the displacement from origin and v is velocity of particle at x. Then choose the correct statement(s):
- (A) Amplitude of oscillation is 1m
 (B) Maximum velocity of particle during oscillation is 3 m/sec
 (C) Amplitude of oscillation can be 2m
 (D) Maximum velocity of particle during oscillation is $(\sqrt{5} + 1)$ m/sec
9. A spherical ball of radius R is made by joining two hemispherical parts. The two parts have density ρ and 2ρ . When placed in a water tank, the ball floats while remaining completely submerged. Neglect viscous forces. Mark the correct statement(s) :-



- (A) Density of water is ρ_0 , then $\rho = \frac{2\rho_0}{3}$.
- (B) Position of centre of mass of ball is at distance of $\frac{R}{8}$ from O.
- (C) Moment of inertia of ball about COM of ball perpendicular to the plane of paper is $\frac{123}{160} \rho \pi R^5$
- (D) Time period of small angular oscillations of the ball about its equilibrium position is given by

$$2\pi \sqrt{\frac{123R}{40g}}$$

Linked Comprehension Type
(Single Correct Answer Type)

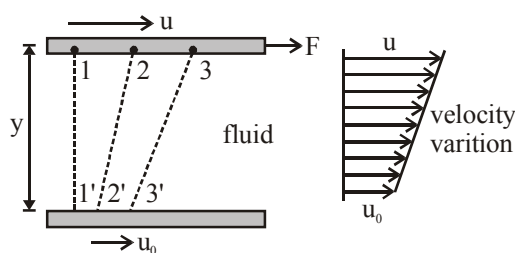
(1 Para × 2Q.) [3 M (0)]

Paragraph for Questions no. 10 and 11

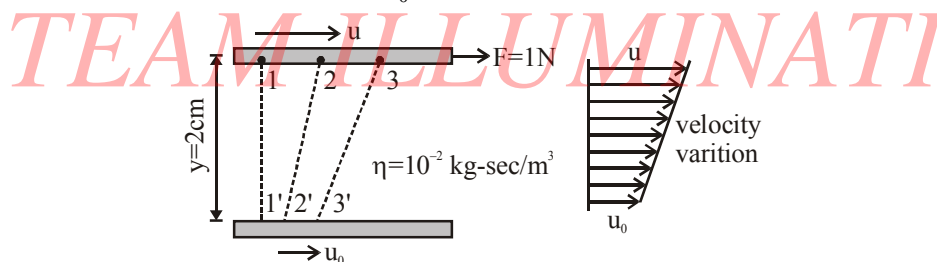
In the shown figure, both the plates are moving. The fluid between the plates moves such that fluid particles in contact of the plates move with same velocity as the plates. The fluid particles in the gap moves in a linear velocity distribution function. As per Newton law of viscosity the tangential force F is

given by $F = \eta A \frac{u - u_0}{y}$, where η = coefficient of viscosity = $10^{-2} \text{ kg -sec/m}^3$, A = area of the plate =

1 m^2 , y = distance between both the plates = 2 cm . The velocity gradient is given by $\frac{du}{dy}$.



10. If the tangential force $F = 1 \text{ Newton}$ and $u_0 = 1 \text{ m/s}$, then the velocity profile for the given figure is :-



- (A) $u = 100 y$ (B) $u = 200 y$ (C) $u = 200 y + 1$ (D) $u = 100 y + 1$
11. In the previous question, the constant velocity of the upper plate would be :
(A) 1 m/s (B) 2 m/s (C) 3 m/s (D) 4 m/s

Matching list based comprehension Type (4 × 4 × 4)

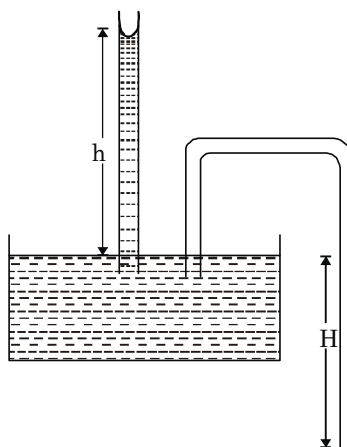
1 Table × 3 Q. [3(-1)]

Single option correct

(Three Columns and Four Rows)

Answer Q.12, 13 & 14 by appropriately matching the information given in the three columns of the following table.

Two capillary tubes of the same cross section are lowered into a vessel with water (figure). The water in the straight tube rises to a height h . The lower end of the bent tube is at a depth H below the level of the water in the vessel.



Answer the questions given below on the basis of given matrix.

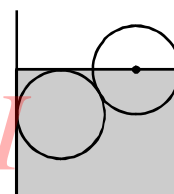
Column - 1	Column - 2	Column - 3
H	h	Shape at Meniscus
(I) $2h_0$	(i) $5h_0$	(P) Convex
(II) 0	(ii) h_0	(Q) Concave
(III) h_0	(iii) $3h_0$	(R) Flat
(IV) $\frac{h_0}{2}$	(iv) $4h_0$	(S) Flow out

12. What will be the shape of meniscus in the bent tube when $H > h$
 (A) (I) (ii) (S) (B) (II) (iii) (P) (C) (III) (i) (R) (D) (IV) (iv) (Q)
13. What will be the shape of meniscus in the bent tube when $H = h$
 (A) (I) (iii) (Q) (B) (II) (iv) (S) (C) (III) (ii) (P) (D) (IV) (i) (R)
14. What will be the shape of meniscus in the bent tube when $H = \frac{h}{2}$
 (A) (I) (iii) (S) (B) (II) (i) (R) (C) (III) (iv) (Q) (D) (IV) (ii) (P)

SECTION-III

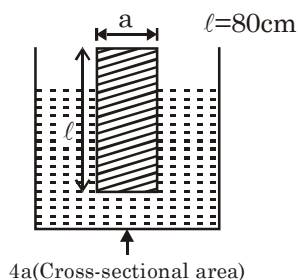
Numerical Grid Type (Single digit Ranging from 0 to 9) 6 Q. [4 M(-1)]

1. There are two cylinder shaped wooden billets, each having mass M , in a vertical wall sewage, which contains water in it. The two billets have same size and same material ; they touch each other and walls of sewage. One of them is totally under water, whilst the other one is immersed only half into water. Friction is negligible



if density of wood is given by $\frac{3}{x}\rho_w$ (ρ_w is density of water) then find x .

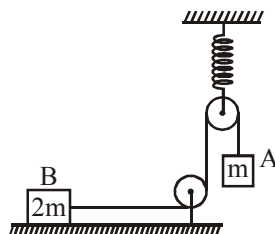
2. A uniform vertical cylinder of cross-sectional area 'a' floats, 90% submerged, in an unknown liquid inside a tank with cross-sectional area four times that of cylinder. When cylinder is pushed down gently and released it performs SHM. The maximum possible amplitude (in cm) for this SHM is



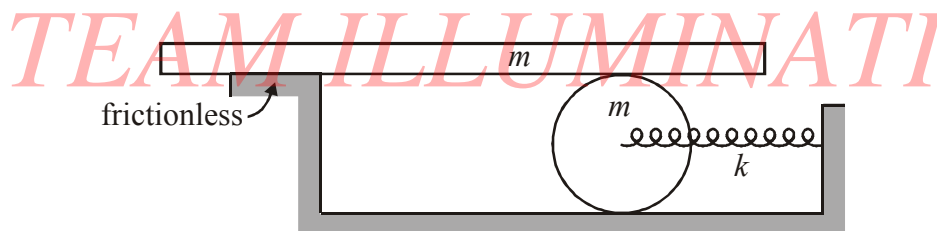
3. A syringe is filled with water upto volume 20 cm^3 . The area of cross-section of the cylinder is 5 cm^2 . The syringe is held vertically and its 90 gm piston is pushed upward by external agent with constant speed. A water beam coming out of the small nozzle (hole area 1 mm^2) has speed 2 m/s . Neglecting friction and viscous nature of water, find the work done by the agent [in 10^{-2} J] in fully emptying the syringe. (Take $g = 10 \text{ m/s}^2$)



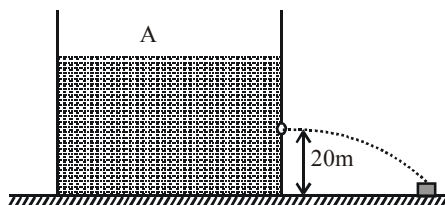
4. In the given figure, spring and pulleys are massless. Block A is performing SHM of amplitude 1m and time period π sec. If block B remains at rest, then minimum value of coefficient of friction between block B and surface will be μ . Fill 10μ in OMR sheet. ($g = 10 \text{ m/s}^2$)



5. The axle of a uniform cylinder with mass m and radius R is connected to a spring with spring constant k , as shown in the Fig. A horizontal board with mass m rests on top of the cylinder, and the board also rests on top of a frictionless support near its left end. The system is slightly displaced from equilibrium. There is no slipping between the cylinder and the board, or between the cylinder and the ground. The angular frequency of the oscillatory motion is p . The value of $\frac{11p^2m}{k}$ is .



6. Water is filled in a uniform container of area of cross section A . A hole of cross section area a ($\ll A$) is made in the container at a height of 20 m above the base. Water streams out and hits a small block on surface at some distance from container. Block is moved on surface in such a way that stream always hits the block. The initial velocity of the block (in m/s) is (Given $\frac{a}{A} = \frac{1}{20}$). (Take $g = 10 \text{ m/s}^2$)



PHYSICS

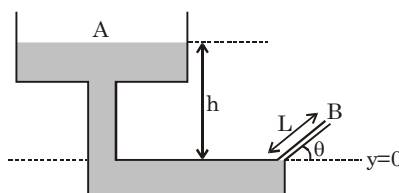
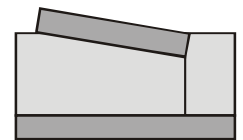
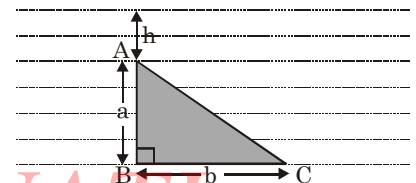
QUIZ # 30

SECTION-I

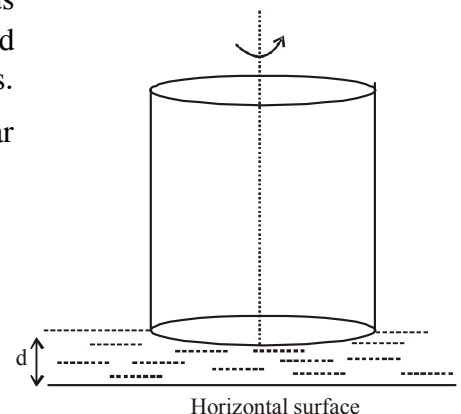
Single Correct Answer Type

9 Q. [3 M (-1)]

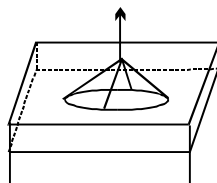
- A rod is oscillating in vertical plane such that it's time period of oscillation is 2 sec. It is hinged at it's end. It has coefficient of linear expansion $2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$. If the temperature increases by 10°C , (Take $g = \pi^2 \text{ m/s}^2$)
 - It's original length is 1.5 m
 - It's time period increases by 10^{-4} sec. (approx.)
 - It's time period decreases by 2×10^{-4} sec. (approx.)
 - It's original length is 3m
- A vertical triangular plate ABC is placed inside water with side BC parallel to water surface as shown. The force on one surface of plate by water is (density of water is ρ and atmospheric pressure P_0) :-
 - $(P_0 + h\rho g) \frac{1}{2} ab$
 - $(P_0 + h\rho g) \frac{1}{2} ab + \frac{a^2 b}{3} \rho g$
 - $[P_0 + \rho g(h + a)] \frac{1}{2} ab$
 - 0
- A wooden plank is floating on the water in a pool. It is tethered to the bottom of the pool by a string attached at mid point of an edge of bottom face, which causes it to float with the diagonal of its vertical cross-section coinciding with the level surface of the water, as shown in the figure. What is the specific gravity of the wood?
 - 1/5
 - 1/4
 - 1/3
 - 1/2
- What is the maximum height above the dashed line attained by the water stream coming out at B from a thin tube of the water tank assembly shown in the figure? Assume $h = 10 \text{ m}$, $L = 2 \text{ m}$, and $\theta = 30^\circ$.



- 10 m
 - 2 m
 - 1.2 m
 - 3.25 m
- In the diagram base of a vertical solid cylinder of mass m and radius r is in contact with a liquid layer. Thickness of liquid layer is d and coefficient of viscosity is η . Cylinder is free to rotate about its axis. If cylinder is given an initial angular velocity ω_0 then angular velocity at time t will be :
 - $\omega_0 e^{-\left(\frac{2\pi\eta r^2 t}{md}\right)}$
 - $\omega_0 e^{-\left(\frac{\pi\eta r^2 t}{2md}\right)}$
 - $\omega_0 e^{-\left(\frac{\pi\eta r^2 t}{md}\right)}$
 - None



6. The surface tension of water is 75 dyne/cm. Find the minimum vertical force required to pull a thin wire ring up (refer figure) if it is initially resting on a horizontal water surface. The circumference of the ring is 20 cm and its weight is 0.1 N



- (A) 0.125 N (B) 0.225 N (C) 0.115 N (D) 0.130 N
7. Figure (a) and (b) show water drop and mercury drop in two identical conical glass pipes, then

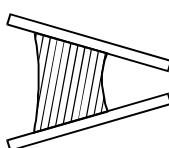


Fig.(a)

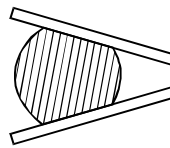
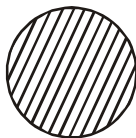
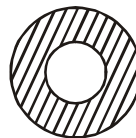


Fig.(b)

- (A) The water drop tends to move towards narrow end and mercury drop towards wide end.
(B) The water drop tends to move towards wide end and mercury drop towards narrow end.
(C) both tend to move towards wide end.
(D) both tend to move towards narrow end.
8. Two balls A and B of same outer radius $2R$ are released from rest in water of sufficient depth. Both balls are made from same material. Ball A is completely solid and ball B is hollow upto radius R . Find ratio of terminal velocity of A and terminal velocity of B (specific gravity of balls is 2)

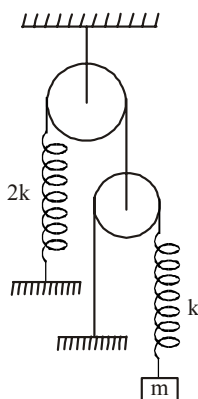


A



B

- (A) 1 (B) $\frac{7}{6}$ (C) $\frac{4}{5}$ (D) $\frac{4}{3}$
9. If the block shown in figure is displaced slightly and released, find the time period of oscillations of the block



- (A) $T = 2\pi\sqrt{\frac{3m}{K}}$ (B) $T = 2\pi\sqrt{\frac{5m}{K}}$ (C) $T = 2\pi\sqrt{\frac{m}{K}}$ (D) $T = 2\pi\sqrt{\frac{2m}{K}}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

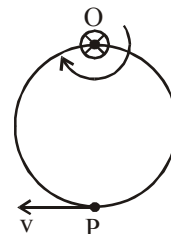
10. A ring of mass m , radius r can oscillate in a vertical plane about its top most point 'O' as shown. Axis is normal to the plane of ring. Maximum speed of lowest point 'P' of ring is v .

(A) Time period of SHM for small amplitudes is $2\pi\sqrt{\frac{2r}{g}}$

(B) Acceleration of centre of mass of ring at lowest position is zero.

(C) Hinge force when centre of mass is at lowest point is $\frac{mv^2}{4r}$

(D) Mechanical energy of oscillation is $\frac{mv^2}{4}$ (when potential energy at mean position is taken as zero.)



11. Mark the **CORRECT** statement about the system. A block is attached to the springs with spring constants k_1 and k_2 and the same unstretched length ℓ as shown in figure :-

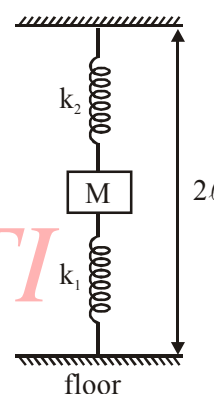
(A) Time period of oscillation is given by $2\pi\sqrt{\frac{x_0}{g}}$, where x_0 is compression of

lower spring in equilibrium position.

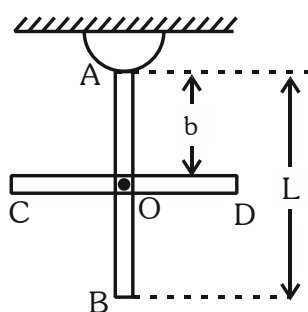
(B) Equivalent force constant of springs is $(k_1 + k_2)$.

(C) Time period of oscillation will not change if system is placed in an upward accelerated elevator.

(D) If system is placed in an upward accelerated elevator the amplitude of oscillation is not same on either side of equilibrium position.



12. Two uniform rods each of mass m and length L are welded together to form the assembly as shown in the figure. Then mark the correct statement(s).



(A) The distance 'b' for which the frequency of small oscillations of the assembly is maximum is $0.316 L$

(B) The maximum frequency of small oscillations of the assembly is nearly $0.2\sqrt{\frac{g}{L}}$

(C) The maximum time period of small oscillations of the assembly is nearly $5\sqrt{\frac{L}{g}}$

(D) The distance 'b' for which the frequency of small oscillations of the assembly is maximum is nearly $0.784 L$

Linked Comprehension Type
(Single Correct Answer Type)
(1 Para × 3Q.) [3 M (-1)]
Paragraph for Question 13 to 15

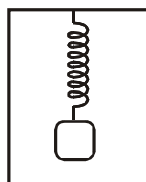
Two particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm & 1Hz respectively. Equilibrium positions of the particles A and B are at the coordinates (3 cm, 0) and (0, 4 cm) respectively. At $t = 0$, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin.

13. Equation of motion of particle A can be written as-
 (A) $x = (2 \text{ cm}) \cos 2\pi t$ (B) $x = (3 \text{ cm}) - (2 \text{ cm}) \cos 2\pi t$
 (C) $x = (2 \text{ cm}) \sin 2\pi t$ (D) $x = (3 \text{ cm}) - (2 \text{ cm}) \sin 2\pi t$
14. Equation of motion of particle B can be written as-
 (A) $y = (2 \text{ cm}) \cos 2\pi t$ (B) $y = (4 \text{ cm}) - (2 \text{ cm}) \cos 2\pi t$
 (C) $y = (2 \text{ cm}) \sin 2\pi t$ (D) $y = (4 \text{ cm}) - (2 \text{ cm}) \sin 2\pi t$
15. Minimum and maximum distance between A and B during the motion is-
 (A) $\sqrt{5}$ cm and $\sqrt{61}$ cm (B) 3 cm and 7 cm
 (C) 1 cm and 5 cm (D) 9 cm and 16 cm

SECTION-III
Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

1. A mass m is resting at equilibrium suspended from a vertical spring of natural length L and spring constant k inside a box as shown :

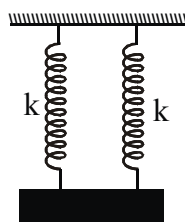
The box begins accelerating upward with acceleration a . Find the maximum speed of the mass (in cm/s) relative to the box in subsequent motion. (Given : $a = 1 \text{ m/s}^2$, $m = 3 \text{ kg}$, $k = 1200 \text{ N/m}$)



2. A small block of mass $m = 0.5 \text{ kg}$ is attached to two springs each of force constant $k = 10 \text{ N/m}$ as shown

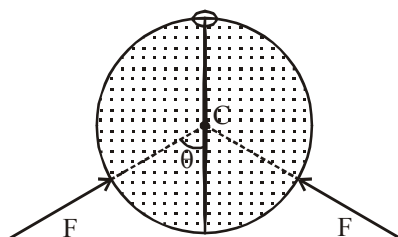
in figure. The block is executing SHM with amplitude $A = \frac{1}{\sqrt{8}} \text{ m}$. When the block is at equilibrium position one of the spring breaks without changing momentum of block. New amplitude of oscillation is

$\frac{e}{16} \text{ m}$. Find the value of e ?



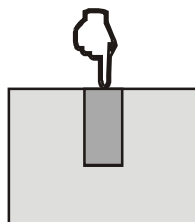
3. Two hemispheres of radius R combined together by two minimum forces ' F ' to form a sphere (completely filled with water) as shown in figure. A small pin hole of radius a at top of the sphere as shown in figure.

If the system is in equilibrium at $\tan \theta = \frac{x}{2}$. Find the value of x . (gravity is present and no other support is given to hemispheres)



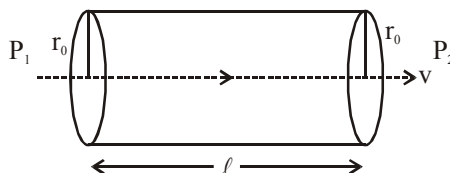
4. A uniform cylinder made of material of density $\rho = 250 \text{ kg/m}^3$ pressed down and held at rest in a pool so that its upper circular surface is in level with the water surface. Length of the cylinder is $l = 20 \text{ cm}$ and radius is $r = 3.0 \text{ cm}$. When released the cylinder jumps out of water moving vertically. Find velocity with which the cylinder leaves the water surface. Density of water is $\rho_0 = 1000 \text{ kg/m}^3$ and acceleration of free fall is $g = 10 \text{ m/s}^2$.

TEAM ILLUMINATI



5. A liquid is flowing through a horizontal uniform cylindrical pipe having constant pressure difference between its ends as shown. Velocity of liquid at the axis of cylinder is v_0 and coefficient of viscosity of

liquid is η . If the velocity of liquid at distance $\frac{r_0}{\sqrt{2}}$ from axis is $\frac{v_0}{n}$. Find the value of n .



PHYSICS

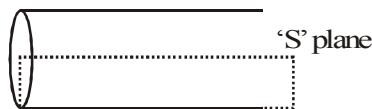
QUIZ # 31

SECTION-I

Single Correct Answer Type

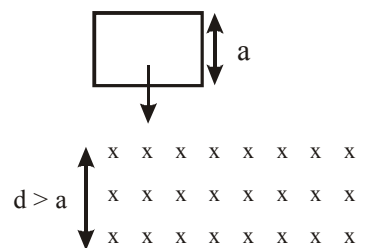
9 Q. [3 M (-1)]

1. A long copper wire of circular cross-section carries a current of 10A. "S" is a plane surface as shown in figure wholly inside the wire passing through axis of cylinder. The magnetic flux per meter of wire, for the plane surface is, in wb/m.



- (A) 1×10^{-5} (B) 1.5×10^{-6} (C) 1×10^{-6} (D) 1.5×10^{-5}

2. Consider a metal wire frame falling freely under gravity in a uniform magnetic field region as shown in the figure. Then mark the correct statement as the loop enters the region ?

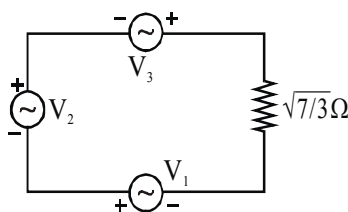


- (A) The speed first decreases then increases
(B) The speed continuously increases
(C) The speed may increase or decrease initially but it will definitely increase afterwards
(D) The speed may decrease continuously

3. The ends of a coil are connected to a galvanometer. When current suddenly I starts flowing in a neighbouring circuit, the instantaneous deflection in the galvanometer is $+7^\circ$. If now the coil be quickly rotated through 180° , what will be the maximum deflection in the galvanometer?

- (A) 7° (B) 14° (C) 21° (D) 28°

4. Three alternating voltage sources $V_1 = 3\sin\omega t$ volt, $V_2 = 5\sin(\omega t + \phi_1)$ volt and $V_3 = 5\sin(\omega t - \phi_2)$ volt connected across a resistance $R = \sqrt{\frac{7}{3}}\Omega$ as shown in the figure (where ϕ_1 and ϕ_2 corresponds to 30° and 127° respectively). Find the peak current (in Amp) through the resistor.



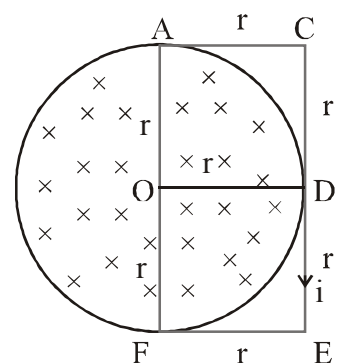
- (A) 3 (B) 4 (C) 5 (D) 6

5. There is a cylindrical region of variable magnetic field $\frac{dB}{dt} = \alpha$ having a wire frame inside as shown each part of the wire frame has resistance r .

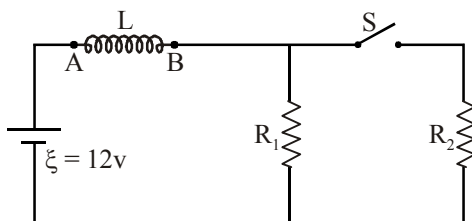
If $i_{DE} = \left\{ \frac{-me}{nr} \right\}$ then find the value of $(m + n)$. (Where e is the e.m.f. developed in the rod AC)

developed in the rod AC)

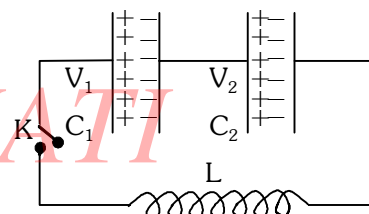
- (A) 3
(B) 4
(C) 5
(D) 7



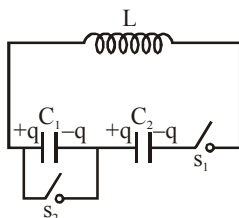
6. The circuit shown has been operating for a long time. The instant after the switch in the circuit labeled S is opened, what is the voltage across the inductor V_L and which labeled point (A or B) of the inductor is at a higher potential ? Take $R_1 = 4.0 \Omega$, $R_2 = 8.0 \Omega$, and $L = 2.5 \text{ H}$.



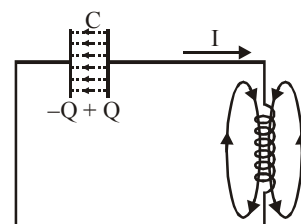
- (A) $V_L = 12 \text{ V}$; Point A is at the higher potential
(B) $V_L = 12 \text{ V}$; Point B is at the higher potential
(C) $V_L = 6 \text{ V}$; Point A is at the higher potential
(D) $V_L = 6 \text{ V}$; Point B is at the higher potential
7. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance L as shown in the figure. Initially key k is open. Now key k is closed and current in the circuit starts increasing. When current in the circuit is maximum



- (A) charge on both the capacitors is same
(B) induced emf in the inductor is zero
(C) potential difference across both the capacitors is half of the induced emf
(D) electrostatic potential energy stored in both the capacitors is same
8. Two capacitors C_1 & C_2 of capacitance $\frac{1}{\pi^2} \times 10^{-2} \text{ F}$ each and an inductor L of inductance $2 \times 10^{-2} \text{ H}$ are connected in series as shown in the figure. Initially charge on each capacitors are $4\sqrt{3} \mu\text{C}$. At $t = 0$ switch S_1 is closed and at $t = \frac{1}{400} \text{ sec}$, switch S_2 is also closed. The maximum charge on capacitor C_2 during LC oscillation is :-



- (A) $3 \mu\text{C}$ (B) $6\sqrt{2} \mu\text{C}$ (C) $3\sqrt{3} \mu\text{C}$ (D) $6\sqrt{3} \mu\text{C}$
9. In the LC circuit shown below, the current is in the direction shown. At this time :-

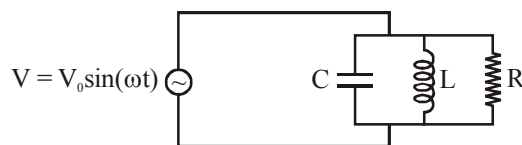


- (A) I is increasing and Q is increasing
(B) I is increasing and Q is decreasing
(C) I is decreasing and Q is increasing
(D) I is decreasing and Q is decreasing

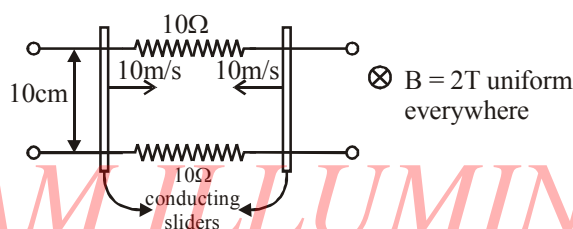
Multiple Correct Answer Type

7 Q. [4 M (-1)]

10. In the following circuit, the AC source is an ideal voltage source. Resonance is defined as a situation where the current through the source and the voltage across the source are in the same phase. At resonance, which of the following will be in the same phase as that of the applied voltage ? (V_R voltage across resistor, I_R current across resistor, I_C current across capacitor, I_L current across inductor)



- (A) V_R (B) I_C (C) I_L (D) I_R
11. The circuit below shows two parallel rails separated by distance of 10 cm. The rails has $10\ \Omega$ resistor each at its middle. The region of space contains magnetic field which is uniform throughout the space. There are two conducting wires on the parallel rails moving towards each other with speed of 10 m/s.



- (A) Current in the circuit is 0.2 A.
(B) Power lost in the circuit is 0.8 W.
(C) The electric field inside the wires is non-conservative in nature.
(D) The electric field inside the wires is conservative in nature.
12. Consider a metal rod of length L that is given a uniform acceleration as shown in figure (a) and an identical rod rotating with constant angular velocity in figure (b).

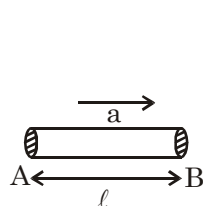


Figure (a)

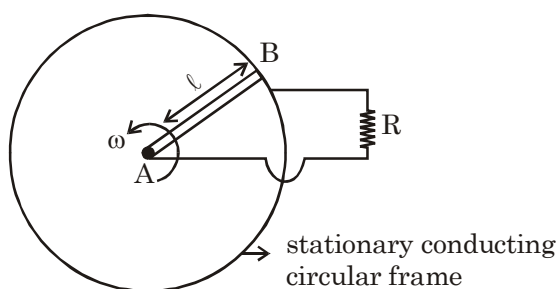


Figure (b)

- (A) If V_A and V_B are potential of end A and B respectively, then $V_A < V_B$ in figure (a)
(B) If V_A and V_B are potential of end A and B respectively, then $V_A > V_B$ in figure (b)
(C) Electric field inside rod has magnitude $\frac{ma}{e}$ in figure (a)
(D) Electric field inside rod has magnitude $\frac{m\omega^2 r}{e}$ in figure (b)

13. A circuit, containing an inductance and a resistance connected in series, has an AC source of 200 V, 50Hz connected across it. An AC current of 10A rms flows through the circuit and the power loss is measured to be 1 kW.

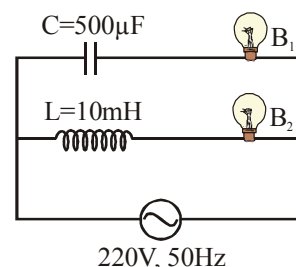
- (A) The inductance of the circuit is $\frac{\sqrt{3}}{10\pi}$ H .
(B) The frequency of the AC when the phase difference between the current and emf becomes $\pi/4$, with the above components is $\frac{50}{\sqrt{3}}$ Hz .
(C) The frequency of the AC when the phase difference between the current and emf becomes $\pi/3$, with the above components is $\frac{25}{\sqrt{3}}$ Hz .
(D) The frequency of the AC when the phase difference between the current and emf becomes $\pi/4$, with the above components is $\frac{25}{\sqrt{3}}$ Hz .

14. In an LCR series AC circuit, voltage across resistance is 2 volt. The inductive and capacitive reactances are 10Ω and 5Ω respectively. The phase difference between voltage across the circuit and current is 30° , then pick up the **CORRECT** statement(s) :

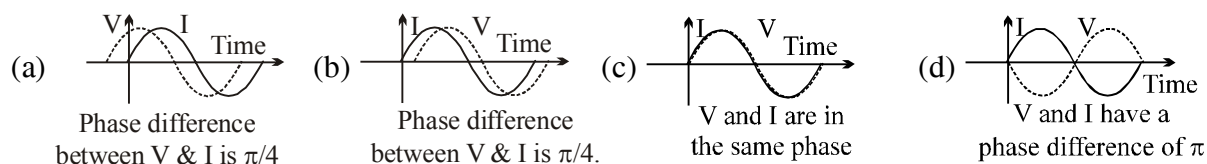
- (A) the resistance of resistor is $5\sqrt{3}\Omega$. (B) the current in the circuit is $\frac{2}{5\sqrt{3}}$ A
(C) the voltage of AC source is $\frac{4}{\sqrt{3}}$ volts (D) the voltage of AC source is $4\sqrt{3}$ volts

15. In the circuit shown in the figure, if both the bulbs B_1 and B_2 are identical :-

- (A) Their brightness will be the same
(B) B_2 will be brighter than B_1
(C) as frequency of supply voltage is increased, brightness of B_1 will increase & that of B_2 will decrease
(D) only B_2 will glow because the capacitor has infinite impedance



16. The diagram shows the variation of V and I in an AC circuit. The circuit only be a series RC or series RL or series LC or series RLC. Consider the four different combinations of V and I graphs. Pick the correct combination/combinations for each graph. Solid curves represent I and broken curves represent V.



Take the angular frequency of the AC voltage source to be 100 rad/s (This is ω)

(I) : $R = 1k\Omega$, $L = 1H$ and $C = 100\mu F$

(II) : $R = 1\Omega$ and $L = 10^{-2}H$

(III) : $R = 100\Omega$ and $C = 10^{-2}\mu F$

(IV) : $L = 1H$, $C = 100\mu F$ and $R = 1\Omega$

(A) $a \rightarrow II$; $d \rightarrow I, IV$

(B) $c \rightarrow I$; $d \rightarrow \text{none}$

(C) $b \rightarrow III$; $c \rightarrow IV$

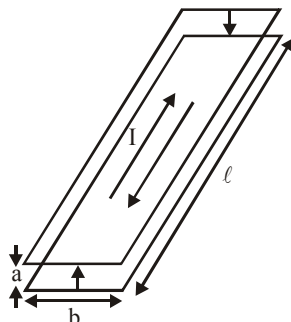
(D) $a \rightarrow II, III$; $b \rightarrow II, III$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Q. No. 17 & 18

Two parallel rectangular superconducting plates of length ℓ , width b and separation a ($\ell \gg b \gg a$) are joined at each end to form a one-turn coil of negligible resistance. The steady state current is I .



17. The energy stored in magnetic field will be :-

- (A) $U = \frac{\mu_0 \ell a I^2}{b}$ (B) $U = \frac{\mu_0 a b I^2}{2\ell}$ (C) $U = \frac{\mu_0 \ell b I^2}{2a}$ (D) $U = \frac{\mu_0 \ell a I^2}{2b}$

18. The magnitude of the force per unit area between the two plates :-

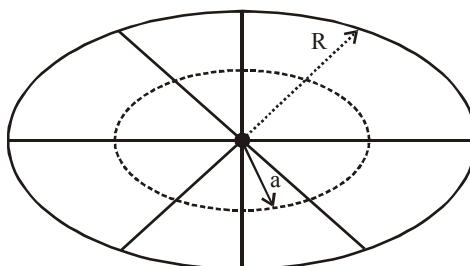
- (A) $\frac{\mu_0 I^2 \ell}{2b}$ (B) $\frac{\mu_0 I^2}{2a^2}$ (C) $\frac{\mu_0 I^2}{2b^2}$ (D) $\frac{\mu_0 I^2}{ab}$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A line charge $\lambda = 10^{-6}$ C/m is fixed on the rim of a wheel of radius 'R' = 1m which is then suspended horizontally, so that it is free to rotate (the spokes are made of wood). In the central region up to radius $a = \frac{1}{2}$ m there is uniform magnetic field, $|\vec{B}_0| = 1$ T pointing up. Now suddenly the field is turned off. If the moment of inertia (I) is = 0.25 kg/m², the final angular velocity ' ω ' of the wheel is $n\pi \times 10^{-6}$ rad/s. Find n.



2. A power line is used to transmit 100 MW power. Power factor of load is 0.866 and resistance of line is 8Ω . the power loss in the wire is 2 %. What is the current (in A) in the wires. fill $\frac{i}{100}$ in OMR sheet.

PHYSICS

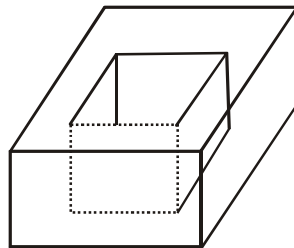
QUIZ # 32

SECTION-I

Single Correct Answer Type

12 Q. [3 M (-1)]

1. From a block of dimensions $(20 \times 20 \times 10) \text{ cm}^3$ a part is removed at the middle as shown of the dimension $(10 \times 10 \times 5) \text{ cm}^3$. This system is kept on the surface of water and water is poured from above. What will be the level of liquid inside the cavity when it sinks ? density of block = 1.1 gm/cc .



- (A) 1 cm (B) 1.5 cm (C) 4 cm (D) never sinks
2. A cube (density 0.5 gm/cc) of side 10 cm is floating in water kept in a cylindrical beaker of base area 1500 cm^2 . When a mass m is kept on wooden block the level of water rises in the beaker by 2 mm . Find the mass m .

(A) 300 gm (B) 100 gm (C) 400 gm (D) 200 gm

3. A container of height ' h ' is filled completely by 2 immiscible liquid of same mass and density ratio $1 : 2$. Cross-section of area is A . A hole of area S is punched in the bottom of the container. ($S \ll A$). Find the time to flow out denser liquid.

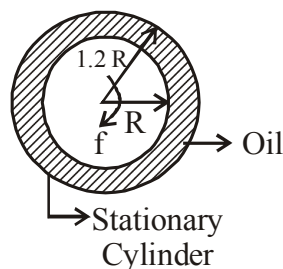
(A) $\frac{A}{5}(4 - \sqrt{2})\sqrt{\frac{h}{g}}$ (B) $\frac{4A}{5}\sqrt{\frac{h}{g}}$ (C) $\frac{A}{5}(3 - \sqrt{2})\sqrt{\frac{h}{3g}}$ (D) $\frac{A}{5}(2 - \sqrt{2})\sqrt{\frac{h}{3g}}$

4. Statement-1 : Coefficient of viscosity of a liquid decreases while that of a gas increase, with increasing temperature.

Statement-2 : Density of a liquid and that of a gas vary in opposite manner with increase in temperature.

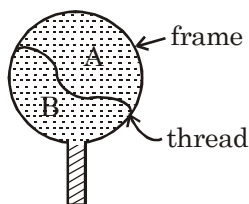
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

5. In a viscometer, two coaxial cylinders of length L are used. The radius of inner solid cylinder is R and outer cylinder is $1.2R$. The gap between the two cylinders is completely filled by a viscous oil. The outer cylinder is kept stationary and the inner cylinder is being rotated with f revolutions per second. Using external constant torque τ . Find the coefficient of viscosity of the oil:

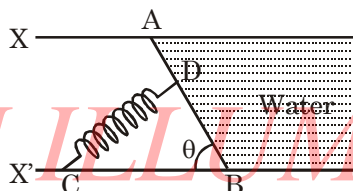


(A) $\frac{\tau}{10\pi^2 R^2 L f}$ (B) $\frac{\tau}{5\pi^2 R^2 L f}$ (C) $\frac{\tau}{20\pi^2 R^2 L f}$ (D) $\frac{\tau}{40\pi^2 R^2 L f}$

6. A thread is tied slightly loose to a wire frame as in figure and the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A is punctured with a pin, the thread :-

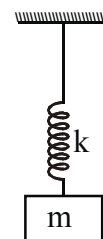


- (A) Becomes concave towards A.
(B) Becomes convex towards A.
(C) Remains in the initial position.
(D) Either (1) or (2) depending on the size of A w.r.t. B
7. AB is a massless frictionless glass rod which can slide on the U-frame XX', with the help of two frictionless rings. A water film is formed between the rod and the U-shaped frame. The rod is held in equilibrium by a hair-spring of stiffness k . Surface tension of water is T and length of rod AB is L . Find the extension in the spring.



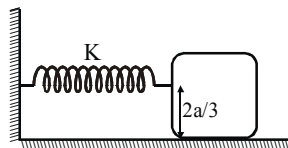
- (A) $\frac{TL}{k} \sin \theta$ (B) $\frac{2TL}{k} \sin \theta$ (C) $\frac{2TL}{k}$ (D) $\frac{2TL}{k \sin \theta}$
8. The displacement of a particle varies according to the relation $x = 3 \sin 100t + 8 \cos^2 50t$. Which of the following is **INCORRECT** about this motion.
- (A) The motion of the particle is S.H.M.
(B) The amplitude of the S.H.M. of the particle is 5 units
(C) The amplitude of the resultant S.H. M. is $\sqrt{73}$ units
(D) The maximum displacement of the particle from the origin is 9 units.
9. Two particles perform simple harmonic motion of same frequency and about same mean position. Their amplitude is same and is equal to A and their time period is T . If at $t = 0$ their separation is maximum and is A , their separation at $t = \frac{T}{12}$ is

- (A) $A \frac{\sqrt{3}}{2}$ (B) A (C) $\frac{A}{\sqrt{2}}$ (D) $\frac{A}{2}$
10. A block of mass 'm' is supported by string spring combination as shown in the diagram. The string is sufficiently long. The system is at rest in equilibrium. The block is pulled by a distance 'x' from equilibrium position and released. It performs oscillatory motion and let $x = x_0$ is maximum possible value for which string is not becoming slacked in subsequent motion. If $x = 2x_0$ then time for which string becomes slacked in one complete oscillation is



- (A) $\sqrt{\frac{3m}{K}}$ (B) $\sqrt{\frac{6m}{K}}$ (C) $3\sqrt{\frac{m}{K}}$ (D) $\sqrt{\frac{12m}{K}}$

11. Find maximum amplitude for safe SHM (block does not topple during SHM) of a cubical block of side 'a' on a smooth horizontal floor as shown in figure (spring is massless)

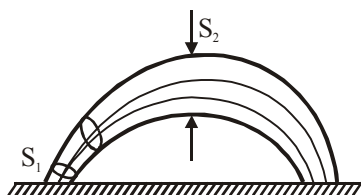


- (A) $\frac{mg}{3K}$ (B) $\frac{3mg}{K}$ (C) $\frac{2mg}{3K}$ (D) None
12. A rod is hinged at some point away from its center and is made to oscillate like a compound pendulum. It oscillates with an angular frequency ω . If the rod is now rotated so that its center of mass is vertically above the point of suspension and released from this position after giving a slight push. What is its angular velocity when it reaches the lowermost position
- (A) ω (B) 2ω (C) $\omega/2$ (D) 4ω

Multiple Correct Answer Type

3 Q. [4 M (-1)]

13. The figure shows a cart of mass 10kg initially kept at rest on a frictionless surface. A jet of liquid of density 1000 kg/m³ is directed on the cart. Velocity of the jet is 10m/s and its cross section area is 2×10^{-4} m². If at $t = 0$, cart is released and allowed to move freely in horizontal direction then choose the CORRECT statement(s) :-
- (A) Force on the cart at $t = 0$ s is 40 N
(B) Velocity of cart at $t = 10$ s is 8 m/s
(C) Acceleration of cart at $t = 10$ s is 1 m/s²
(D) Power supplied to the cart when cart is moving with velocity 5 m/s is 50 W.
14. The potential energy U of a particle is given by $U = \{20 + (x-4)^2\}$ J. Total mechanical energy of the particle is 36 J. Select the correct alternative(s)
- (A) the particle oscillates about point $x = 4$ m
(B) the amplitude of the particle is 4m
(C) the kinetic energy of the particle at $x = 2$ m is 12 J
(D) the motion of the particle is periodic but not simple harmonic
15. Water jet is projected at an angle to the horizontal. At the point of projection, the area of the jet is S_1 and at the highest point, the area of the jet is S_2 . The initial velocity of projection is u .



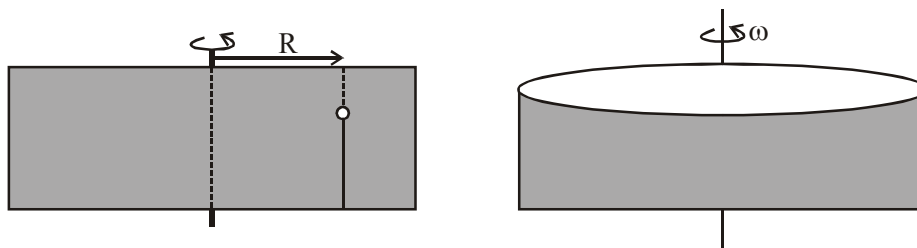
- (A) The angle of projection is $\cos^{-1}\left(\frac{S_1}{S_2}\right)$
- (B) The range on the level ground is $\frac{2u^2}{g} \frac{S_1}{S_2} \sqrt{1 - \frac{S_1^2}{S_2^2}}$
- (C) The maximum height reached from the ground is $\frac{2u^2}{g} \left(1 - \frac{S_1^2}{S_2^2}\right)$
- (D) The rate of volume flow is $S_2 u$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) (1 Para × 3Q.) [3 M (-1)]

Paragraph for Question Nos. 16 & 17

A closed cylindrical container with a vertical axis is completely filled with liquid (ρ_ω). A plastic bead of density ρ_{body} and volume V is placed at distance R from the axis and is anchored to the bottom of the container by a thin thread of length ℓ . If as a result of the containers revolutions, the beads sink by h and final distance of bead from axis is r . (In the final state the total content of the container rotates at the same angular speed.)



16. What is force exerted by liquid on the bead ?

(A) $\rho_\omega V g$

(B) $(\rho_\omega - \rho_{\text{body}}) V g$

(C) $(\rho_\omega - \rho_{\text{body}}) V \sqrt{g^2 + (\omega^2 r)^2}$

(D) $\rho_\omega V \sqrt{g^2 + (\omega^2 r)^2}$

 17. What is ω :-

(A) $\omega = \sqrt{\frac{(R+r)g}{(\ell+h)r}}$

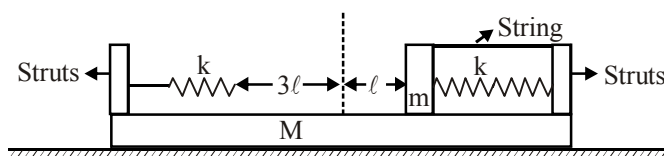
(B) $\omega = \sqrt{\frac{(R-r)g}{(\ell-h)r}}$

(C) $\omega = \sqrt{\frac{(R+r)g}{(\ell-h)r}}$

(D) $\omega = \sqrt{\frac{(R-r)g}{(\ell+h)r}}$

Paragraph for Question Nos. 18 to 20

A plank of mass M is placed on a smooth horizontal surface. Two light identical springs each of stiffness k are rigidly connected to struts at the ends of the plank as shown in figure. When the springs are in their unextended position the distance between their free ends is 3ℓ . A block of mass m is placed on the plank and pressed against one of the springs so that it is compressed by ℓ . To keep the blocks at rest it is connected to the strut by means of a light string, initially the system is at rest. Now the string is burnt.



18. Maximum displacement of plank is :

(A) $\frac{m\ell}{m+M}$

(B) $\frac{5m\ell}{m+M}$

(C) $\frac{3m\ell}{m+M}$

(D) $\frac{2m\ell}{m+M}$

19. What is magnitude of velocity of block relative to plank when block is not in contact with spring.

(A) $\left(\sqrt{\left(\frac{M+m}{Mm} \right) k} \right) \ell$

(B) $\left(\sqrt{\left(\frac{Mm}{M+m} \right) k} \right) \ell$

(C) $\left(2 \sqrt{\left(\frac{M+m}{Mm} \right) k} \right) \ell$

(D) $\left(\frac{1}{2} \sqrt{\left(\frac{Mm}{M+m} \right) k} \right) \ell$

20. Time period of oscillation of block :

(A) $(2\pi + 3) \sqrt{\frac{2Mm}{k(M+m)}}$

(B) $(\pi + 6) \sqrt{\frac{2Mm}{k(M+m)}}$

(C) $(\pi + 3) \sqrt{\frac{Mm}{k(M+m)}}$

(D) $(2\pi + 6) \sqrt{\frac{Mm}{k(M+m)}}$

PHYSICS

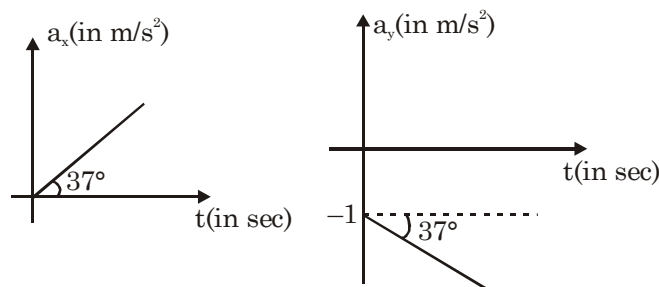
QUIZ # 33

SECTION-I

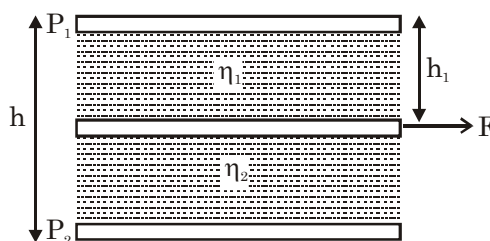
Single Correct Answer Type

7 Q. [3 M (-1)]

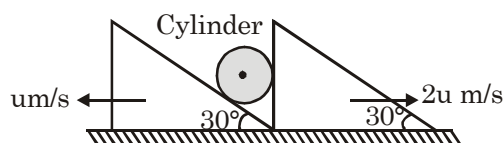
1. In the figure the variation of components of acceleration of a particle of mass 1kg is shown w.r.t. time. The initial velocity of the particle is $\vec{u} = (-3\hat{i} + 4\hat{j}) \text{ m/s}$. The total work done by the resultant force on the particle in time interval from $t = 0$ to $t = 4$ seconds is :-



- (A) 22.5 J (B) 10 J (C) 0 (D) None of these
2. A thin horizontal movable plate is separated from two fixed horizontal plates P_1 and P_2 by two highly viscous liquids of coefficient of viscosity η_1 and η_2 as shown, where $\eta_2 = 4\eta_1$. Area of contact of movable plate with fluid is same. If the distance between two fixed plates is h , then the distance h_1 of movable plate from upper fixed plate such that the movable plate can be moved with a constant velocity by applying a minimum constant horizontal force F on movable plate is (assume velocity gradient to be uniform in each liquid) :-

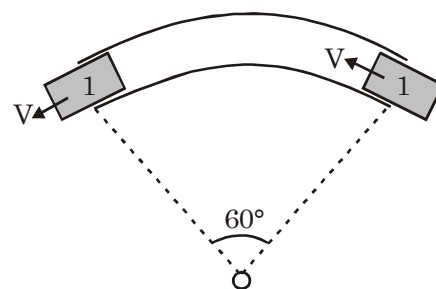


- (A) $\frac{h}{4}$ (B) $\frac{h}{2}$ (C) $\frac{2h}{3}$ (D) $\frac{h}{3}$
3. A uniform ring of mass m and radius a rolls down an inclined plane of inclination θ starting from rest without slipping. When the centre of the ring has travelled a distance ℓ along the inclined plane, its angular momentum about its centre of mass is :-
- (A) $mg\ell \cos \theta$ (B) $\frac{1}{2} ma\sqrt{g\ell \sin \theta}$ (C) $ma\sqrt{g\ell \sin \theta}$ (D) $ma\sqrt{2g\ell \sin \theta}$
4. System is shown in the figure. Assume that cylinder remains in contact with the two wedges. The velocity of cylinder is :-



- (A) $\sqrt{19 - 4\sqrt{3}} \frac{u}{2} \text{ m/s}$ (B) $\frac{\sqrt{13}u}{2} \text{ m/s}$ (C) $\sqrt{3}u \text{ m/s}$ (D) $\sqrt{7}u \text{ m/s}$

5. A car moves around a curve at a constant speed. When the car goes around the arc subtending 60° at the centre, then the ratio of magnitude of instantaneous acceleration to average acceleration over the 60° arc is :-



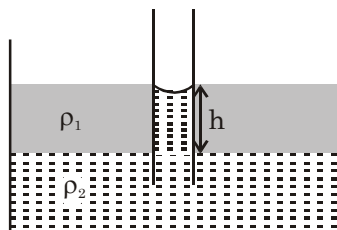
- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$
(C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{3}$

6. A uniform elastic rod of cross-section area A , natural length L and Young's modulus Y is placed on a smooth horizontal surface. Now two horizontal forces (of magnitude F and $3F$) directed along the length of rod and in opposite direction act at two of its ends as shown. After the rod has acquired steady state, the extension of the rod will be :-



- (A) $\frac{2F}{YA}L$ (B) $\frac{4F}{YA}L$ (C) $\frac{F}{YA}L$ (D) $\frac{3F}{2YA}L$

7. A container is partially filled with a liquid of density ρ_2 . A capillary tube of radius r is vertically inserted in this liquid. Now another liquid of density ρ_1 ($\rho_1 < \rho_2$) is slowly poured in the container to a height h as shown. There is only denser liquid in the capillary tube. The rise of denser liquid in the capillary tube is also h . Assuming zero contact angle, the surface tension of heavier liquid is :

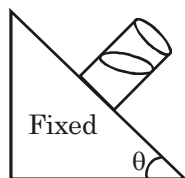


- (A) $r\rho_2gh$ (B) $2\pi r\rho_2gh$ (C) $\frac{r}{2}(\rho_2 - \rho_1)gh$ (D) $2\pi r(\rho_2 - \rho_1)gh$

Multiple Correct Answer Type

7 Q. [4 M (-1)]

8. A cylindrical vessel filled with water is released on a fixed inclined surface of angle θ as shown in figure. The friction coefficient of surface with vessel is μ . Then the constant angle made by the surface of water with the incline will be (Neglect the viscosity of liquid) :-

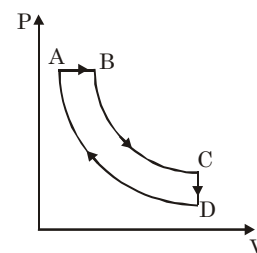


- (A) $\tan^{-1}\mu$ if $\mu < \tan \theta$ (B) 0 if $\mu = 0$ (C) θ if $\mu > \tan \theta$ (D) θ if $\mu < \tan \theta$

9. The water equivalent of a copper calorimeter is 4.5 g. If specific heat of copper is $0.09 \text{ cal/g}^\circ\text{C}$, then choose the correct option from below :-

- (A) Mass of the calorimeter is 0.5 kg
(B) Thermal capacity of the calorimeter is $4.5 \text{ cal}^\circ\text{C}^{-1}$
(C) The heat required to raise the temperature of the calorimeter by 8°C will be 36 cal.
(D) Heat required to melt 15 gm of ice placed in the calorimeter will be 1200 cal

10. n moles of a monoatomic gas undergoes a cyclic process ABCDA as shown. Processes AB $\rightarrow$ Isobaric, BC $\rightarrow$ adiabatic, CD $\rightarrow$ isochoric, DA $\rightarrow$ Isothermal. The maximum temperature and minimum temperature in the cycle are $4T_0$ and T_0 respectively. Then :-



- (A) $T_B > T_C > T_D$
(B) Heat is released by the gas in the process CD.
(C) Heat is supplied to the gas in the process AB.
(D) Total heat supplied to the gas is zero.
11. A closed vessel contains a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A contained in the vessel is 2 times that of B. Choose the correct statement(s) :-
(A) Average kinetic energy per molecule of A is equal to that B.
(B) Root mean square value of translational velocity of B is four times that of A.
(C) Pressure exerted by B is eight times of that exerted by A.
(D) Number of molecules of B in the cylinder is eight times that of A.
12. Two moles of O_2 ($\gamma = \frac{7}{5}$) at temperature T_0 and 3 moles of CO_2 ($\gamma = \frac{4}{3}$) at temperature $2T_0$ are allowed to mix together in a closed adiabatic vessel. The resulting mixture finally comes in thermal equilibrium. Then :-

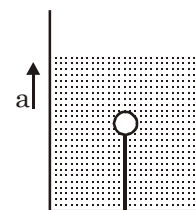
(A) final temperature of the mixture is $\frac{23T_0}{14}$

(B) final temperature of the mixture is $\frac{31T_0}{19}$

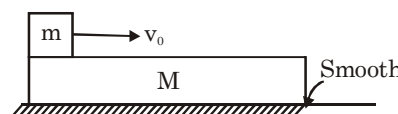
(C) Adiabatic exponent of the mixture formed is $\frac{14}{5}$

(D) Adiabatic exponent of the mixture formed is $\frac{19}{14}$

13. A solid sphere of mass 3kg and specific gravity 0.8 is held stationary relative to a tank filled with water as shown in the figure. The tank accelerates vertically upward with $a = 4 \text{ m/s}^2$. [Take $g = 10 \text{ m/s}^2$]



- (A) The tension in the thread increases due to acceleration of tank.
(B) The tension in the thread decreases due to acceleration of tank.
(C) If the thread snaps, the acceleration of the sphere w.r.t tank is 2.5 m/s^2 .
(D) If the thread snaps the acceleration of the sphere w.r.t. tank is 3.5 m/s^2 .
14. A plank of length L and mass M is placed at rest on a smooth horizontal surface. As shown in figure a small block of mass m is projected with a velocity v_0 . The coefficient of friction between the block and the plank is μ , plank is very long so that block eventually comes to rest on it :-



(A) The final common velocity is $v = \frac{mv_0}{M+m}$

(B) The work done by friction on the plank is $= \frac{1}{2} \frac{m^2 M}{(m+M)^2} v_0^2$

(C) The work done by friction on the block is $= -\frac{1}{2} \frac{mM(M+2m)}{(M+m)^2} v_0^2$

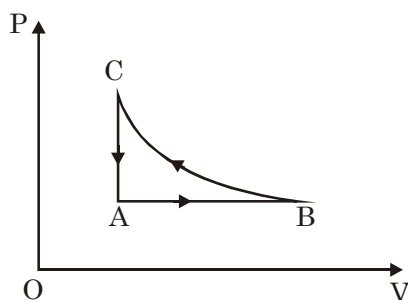
(D) The net work done by friction is $= -\frac{1}{2} \frac{mM}{M+m} v_0^2$

Linked Comprehension Type
(Single Correct Answer Type)

(2 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 15 and 16

A mass of 8g of oxygen gas at the pressure one atmosphere and at temperature 27°C is enclosed in a cylinder fitted with a frictionless piston. Now the gas is heated at constant pressure to 127°C, then it is compressed isothermally to its initial volume and finally it is cooled to its initial temperature at constant volume. ($\ln 2 \approx 0.693$, $\ln 3 \approx 1.099$)



15. Work done by the gas in process A → B is

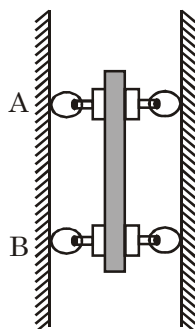
(A) 20.78 J (B) +207.8 J (C) -207.8 J (4) -20.78 J

16. Work done by the gas in process B → C is :-

(A) + 239.1 J (B) 23.91 J (C) -239.1 J (D) -23.91 J

Paragraph for Questions 17 and 18

A bar of mass m is held as shown between 4 uniform discs each of mass m' & radius $r = 75$ mm. the bar has been released from rest, knowing that the normal forces exerted on the discs by vertical wall are sufficient to prevent any slipping. Answer the following questions. (Acceleration due to gravity is g).



17. If $m = 5$ kg and $m' = 2$ kg, then acceleration of the bar is :-

(A) g (B) $\frac{13g}{7}$ (C) $\frac{13g}{17}$ (D) $\frac{g}{17}$

18. If the mass m of the bar is negligible then acceleration of the bar is :-

(A) g (B) $\frac{2g}{3}$ (C) $\frac{13g}{17}$ (D) $\frac{g}{17}$

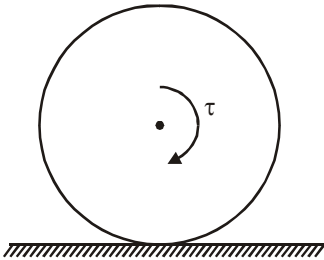
PHYSICS

QUIZ # 34

SECTION-I

Single Correct Answer Type

30 Q. [3 M (-1)]

- A sphere of density ρ , specific heat capacity c and radius r , is hung by a thermally insulated thread in an enclosure which is kept at a temperature slightly lower than that of the sphere. The rate of change of temperature for the sphere depends upon the temperature difference between the sphere and the enclosure, and is proportional to :-
 (1) $\frac{c}{r^3\rho}$ (2) $\frac{r^3\rho}{c}$ (3) $r^3\rho c$ (4) $\frac{1}{r\rho c}$
- A large number of particles are moving, each with speed v having directions of motion randomly distributed. What is the average of magnitude of relative velocity between any two particle (averaged over all the pairs) :-
 (1) v (2) $\left(\frac{\pi}{4}\right)v$ (3) $\left(\frac{4}{\pi}\right)v$ (4) Zero
- If a body (coated black) at 600 K surrounded by atmosphere at 300 K has cooling rate r_0 , the same body at 900 K, surrounded by the same atmosphere, will have the cooling rate close to :-
 (1) $\frac{16r_0}{3}$ (2) $\frac{81r_0}{16}$ (3) $16r_0$ (4) $4r_0$
- A wheel of mass m can be assumed to be a ring. Its radius is R . It is on a level ground. An external couple is applied to it about its axis $\tau = \frac{mgR}{2}$. If it rolls purely, what is the friction force exerted by the ground. (Take mass = 8kg)

 (1) 10N (2) 20N (3) 30N (4) 5N
- A bullet loses 75% of its kinetic energy when passed through a plank. Inside the plank bullet experiences a resistance force proportional to the square of the velocity. Find the total number of planks required so that kinetic energy of the bullet becomes 1.5625% of its initial value :-
 (1) 64 (2) 32 (3) 4 (4) 3
- A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration a_c is varying with time as $a_c = k^2rt^2$, where k is a constant. The power delivered to the particle by the forces acting on it is -
 (1) $2\pi mk^2r^2t$ (2) mk^2r^2t
 (3) $\frac{1}{3}mk^4r^2t^5$ (4) 0
- A uniform rod of mass m and length ℓ is moving with velocity u in a direction perpendicular to its length. A blow of impulse J is given perpendicular to its length at a distance $\frac{\ell}{4}$ from its centre at point P such that instantaneous velocity of point P is $2u$. Then the velocity of its centre of mass will be :-
 (1) $\frac{11u}{7}$ (2) $\frac{12}{7}u$
 (3) $\frac{5}{7}u$ (4) $\frac{17}{7}u$
- Two carnot engines A and B are operated in series. The first one A receives heat at 1200K and rejects to a reservoir at T K. The second engine B receives the heat rejected by the first engine and in turn rejects to a heat reservoir at 300 K. Calculate the value of T , when work outputs of the two engines are equal :-
 (1) 600 K (2) 750 K
 (3) 450 K (4) 900 K

9. In a gravity free space, a man of mass M standing at a height h above the floor, throws a ball of mass m straight down with a speed u . When the ball reaches the floor, the distance of the man above the floor will be:-

(1) $h\left(1 + \frac{m}{M}\right)$ (2) $\left(1 + \frac{M}{m}\right)h$
 (3) h (4) $\frac{m}{M}h$

10. For shown situation find the maximum elongation in the spring. Neglect friction everywhere. Initially, the blocks are at rest and spring is unstretched.



(1) $\frac{4F}{3K}$ (2) $\frac{3F}{4K}$
 (3) $\frac{4F}{K}$ (4) $\frac{2F}{K}$

11. A thin rod AB of length L has linear mass density $\lambda(x) = a + \frac{bx}{L}$, where x is measured from end

A. If the COM of the rod lies at $\frac{3L}{5}$ from end

A then :-

(1) $\frac{a}{b} = \frac{1}{3}$ (2) $\frac{a}{b} = \frac{1}{2}$
 (3) $\frac{a}{b} = \frac{3}{5}$ (4) $\frac{a}{b} = \frac{2}{5}$

12. A solid floats with $\frac{2}{3}$ of its volume immersed in a liquid and with $\frac{3}{4}$ of its volume immersed in another liquid. What fraction of its volume will be immersed if it floats in a homogenous mixture form of equal volumes of the liquids ?

(1) $\frac{6}{7}$ (2) $\frac{8}{11}$
 (3) $\frac{11}{16}$ (4) $\frac{12}{17}$

13. The Coriolis force is a pseudo force that occurs in rotating reference frame. The force is given by $\vec{F} = -2m(\vec{\omega} \times \vec{v})$ where $\vec{\omega}$ is the angular velocity of frame and $\vec{v}$ is the velocity of a particle as measured from the frame. Earth rotates from west to east. A projectile is launched from the equator due north. The direction of the Coriolis force on this projectile is :-

(1) east (2) west
 (3) up (4) the force is zero

14. Water in an electric kettle is boiling. The power absorbed by water is 3935 W. Assuming that pressure of vapor in kettle is equal to atmospheric pressure, determine the speed at which vapor comes out of the opening if opening has a cross sectional area of 2 cm^2 . Take latent heat of water as $2.26 \times 10^6 \text{ J/kg}$. Assume that steam behaves like an ideal gas.

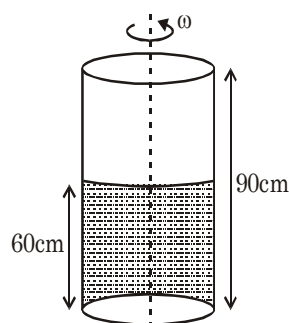
$R = \frac{25}{3} \text{ J/mol.k}$

(1) 17 m/s (2) 13 m/s
 (3) 15 m/s (4) 5 m/s

15. The molar heat capacity for a process undergone by monoatomic gas is given by $\frac{R}{2}$. Which of the following represents the process undergoes by monoatomic gas :-

(1) $\frac{V}{T} = \text{constant}$ (2) $PV^{3/2} = \text{constant}$
 (3) $PV^{-2} = \text{constant}$ (4) $TV = \text{constant}$

16. Cylinder of radius 20 cm and height 90 cm is filled with water up to height 60 cm. Find maximum angular velocity so that water does not spill out

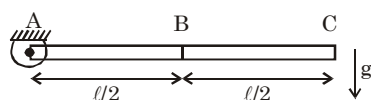


(1) $5\sqrt{3} \text{ rad/s}$ (2) $10\sqrt{3} \text{ rad/s}$
 (3) 5 rad/s (4) 10 rad/s

17. A particle is performing motion in the xy plane such that its x and y coordinates at any time are given by $x = a \sin \omega t$ and $y = b \cos \omega t$. The radius of curvature of its trajectory when it crosses x axis is given by

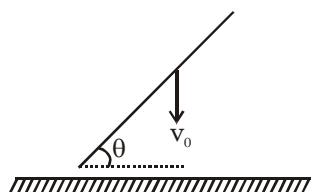
- (1) $\frac{a^2}{b}$ (2) $\frac{b^2}{a}$
(3) a (4) ∞

18. A uniform rod of mass m and length ℓ is released from the horizontal position as shown. Find the force applied by one half (AB) of the rod on the other half (BC) just after the rod is released :-



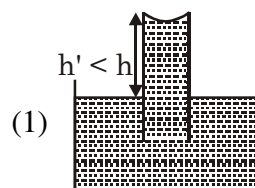
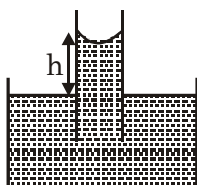
- (1) $\frac{mg}{8}$ (2) $\frac{mg\sqrt{3}}{16}$
(3) $\frac{mg}{16}$ (4) zero

19. A uniform rod of mass m and length ℓ strikes a horizontal surface with speed v_0 elastically. The initial angular velocity of rod is zero. Find angle θ for which velocity of COM becomes zero just after the collision :-

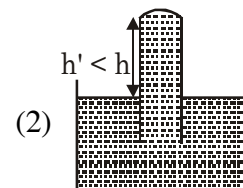


- (1) 45° (2) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(3) $\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ (4) None

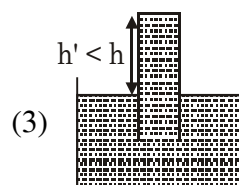
20. Water rises to a height h in the capillary tube. When the length of capillary tube is reduced to h' ($h' < h$) then which of the following representation is correct ?



(1)



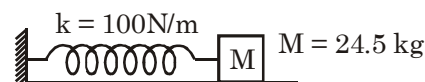
(2)



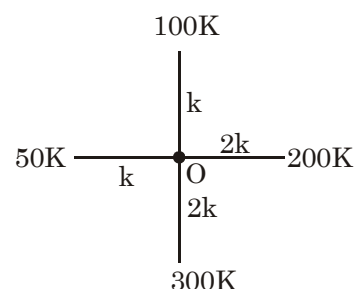
(3)

(4) Liquid will overflow

21. Time period of the system is π sec. Find the mass of spring :-

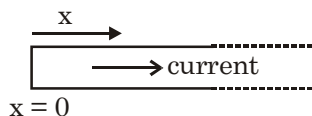


- (1) 1 kg (2) 2 kg
(3) 1.5 kg (4) 4.5 kg
22. Two spheres A & B are identical except that sphere A is white whereas sphere B is black. After they have been in thermal contact for long time with each other and their surroundings, in the visible range :-
- (1) A radiates less than B
(2) Both emit the same amount of radiation
(3) A radiates more than B
(4) A radiates more than B only if its temperature is high enough
23. Four rods having identical dimensions are connected at point O as shown. Find the temperature of the junction O :-

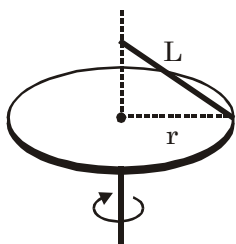


- (1) 200 K (2) 191.67 K
(3) 187.67K (4) 202.67 K

24. Temperature difference of ΔT is maintained across a very long rod having cross-sectional area A . Thermal conductivity varies as $k_0 e^{+x/a}$ (when k_0 and a are constant) Find thermal current flowing through the rod

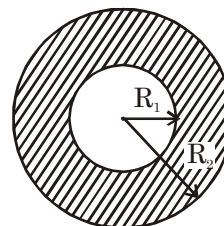


- (1) $\frac{k_0 A \Delta T}{2a}$ (2) $\frac{k_0 A \Delta T}{a}$
(3) $\frac{k_0 A \Delta T}{\ell n 2}$ (4) None
25. A disk rotates at constant angular velocity around its vertical axis of symmetry. A rod of length $L = 1\text{m}$ is placed onto the disk in a way that its one end touches the disk at a distance of $r = 0.8\text{m}$ from the centre, while its other end is above the centre as shown. The rod is then released and rotates together with the disk in this position. Find the angular velocity of the disk. Use $g = 9.8\text{ m/s}^2$:-



- (1) 3.3 rad/s (2) 5.71 rad/s
(3) 7 rad/s (4) 4.05 rad/s
26. A container of water equivalent 10 gm contains 10 gm ice at -10°C some of water at 20°C is added to it so that ice just melts. Find the mass of water required :-
- (1) 22.5 gm (2) 47.5 gm
(3) 31.5 gm (4) 65 gm

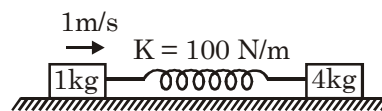
27. The space between two concentric sphere of radius $R_1 = 3\text{m}$ and $R_2 = 5\text{m}$ is filled with a material of thermal conductivity k ($k = 400 \frac{\text{J}}{\text{smK}}$). Inner and outer surfaces are maintained at 200 K and 500 K respectively. find distance from the centre where the temperature is 350 K :-



- (1) 4.5 m (2) 4m
(3) 3.75 m (4) 4.33 m
28. Block A is projected with speed v on a smooth horizontal surface towards block B as shown. Assume all the collisions to be elastic. Find the total no. of collision that will take place :-



- (1) 2 (2) 3
(3) 4 (4) Infinite
29. 2 blocks of mass 1kg and 4 kg are attached to the two ends of a spring. Block of mass 1kg is given velocity 1 m/s towards right when the spring was in natural length. The minimum speed of 1 kg block in the subsequent motion will be:-



- (1) 0.8 m/s (2) 0.6 m/s
(3) 0.2 m/s (4) None
30. A particle mass m is subjected to a force given by $F = F_0 \cos \omega t$, where F_0 and ω are constants. Find the average velocity of the particle till it stops for the first time :-

- (1) $\frac{F_0}{m\omega}$ (2) $\frac{2F_0}{m\pi\omega}$
(3) $\frac{4F_0}{m\pi\omega}$ (4) $\frac{2F_0}{m\omega}$

QUIZ # 34 (TOAS, TEAS & T-AS)				ANSWER KEY	
SECTION-I					
Single Correct Answer Type				30 Q. [3 M (–1)]	
1. Ans. (4)	2. Ans. (3)	3. Ans. (1)	4. Ans. (2)	5. Ans. (4)	6. Ans. (2)
7. Ans. (1)	8. Ans. (2)	9. Ans. (1)	10. Ans. (1)	11. Ans. (1)	12. Ans. (4)
13. Ans. (4)	14. Ans. (3)	15. Ans. (4)	16. Ans. (2)	17. Ans. (2)	18. Ans. (3)
19. Ans. (2)	20. Ans. (1)	21. Ans. (3)	22. Ans. (1)	23. Ans. (2)	24. Ans. (2)
25. Ans. (3)	26. Ans. (2)	27. Ans. (3)	28. Ans. (2)	29. Ans. (4)	30. Ans. (2)

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PHYSICS

QUIZ # 35

SECTION-I

Single Correct Answer Type

14 Q. [3 M (-1)]

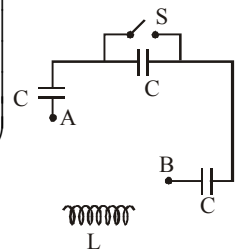
1. Three capacitor of capacitance C each are connected in series as shown in the figure. Initially switch S is open. Now capacitors are charged by a battery of emf V by connecting between terminal A and B . After long time battery is disconnected and inductor of inductance L is connected between A and B at time $t = 0$ so that an oscillatory circuit is formed. Now at an instant t_0 switch S is closed, then find the amplitude of charge oscillations of the remaining capacitors.

(A) $CV \sqrt{\frac{1}{6} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$

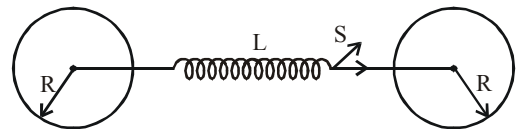
(B) $\frac{CV}{3} \sqrt{\frac{1}{5} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$

(C) $CV \sqrt{\frac{1}{4} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$

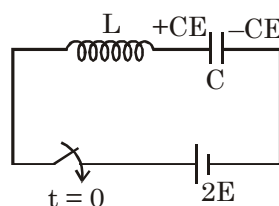
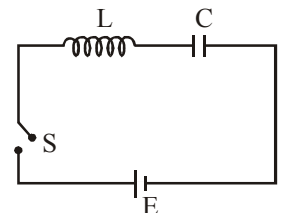
(D) $\frac{CV}{3}$



2. Two conducting spheres of equal radii R are placed far apart and are connected by an ideal inductor as shown. Initially one of the spheres has a charge Q while other is uncharged. Switch is closed at time, $t = 0$. Then :-



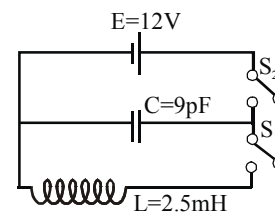
- (A) Minimum time after which all charge can be transferred to other sphere is $\sqrt{2\pi^3 \epsilon_0 LR}$
- (B) Minimum time after which all charge can be transferred to other sphere is $\sqrt{8\pi^3 \epsilon_0 LR}$
- (C) First sphere will acquire a negative charge at some instant of time
- (D) At some instant of time both spheres will have zero charge
3. An ideal DC source of emf E is connected with an uncharged capacitor and inductor. If switch is closed at $t = 0$, what is the maximum charge on the capacitor in subsequent flow of charge ?
- (A) $2CE$ (B) CE
- (C) $1.5 CE$ (D) $4CE$
4. Maximum charge on capacitor after switch is closed :-



- (A) $2CE$ (B) $4CE$ (C) $3CE$ (D) None of these

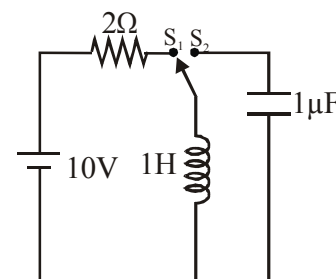
5. In the circuit shown, the capacitor is initially charged with a 12 V battery, when switch S_1 is open and switch S_2 is closed. S_1 is then closed and, at the same time, S_2 is opened. The maximum value of current in the circuit is

(A) 0.38 mA (B) 0.84 mA
(C) 0.72 mA (D) 0.1 mA



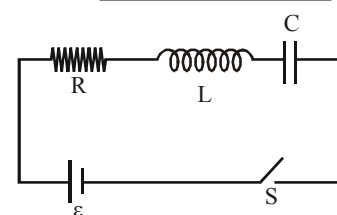
6. For the circuit shown, the switch was closed at switch S_1 for a long time till steady state condition reached. At time $t = 0$, the switch S_1 is opened and S_2 is closed. The current through inductor as a function of time will be

(A) $5 \cos 100 t$
(B) $10 \cos 100 t$
(C) $5 \cos 1000 t$
(D) $10 \cos 1000 t$



7. A resistor of resistance R , an inductor of inductance L and a capacitor of capacitance C are connected in series and the combination is connected to a battery of emf ε as shown. At $t = 0$, the switch S is closed. Let, after a very long time, the thermal energy developed in circuit be Q , energy stored in circuit be U and energy supplied by the battery be E . Then :-

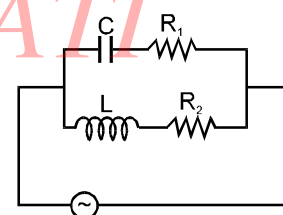
(A) $Q : U : E = 1 : 1 : 2$ (B) $Q : U : E = 1 : 3 : 4$ (C) $Q : U : E = 3 : 1 : 4$ (D) $Q : U : E = 1 : 2 : 3$



8. In the circuit the rms value of voltage across the capacitor C , inductor L and resistor R_1 are respectively $5\sqrt{5}$, 10V and 5V.

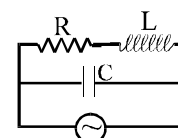
Then the peak voltage across R_2 is

(A) 7 V (B) 10 V
(C) 15 V (D) $5\sqrt{2}$ V



9. The A.C. circuit shown in figure. Find the frequency (ω_0) of the AC voltage source so that current through the source will be in same phase as of voltage of source.

(A) $\omega_0 = \sqrt{\frac{1}{LC} + \frac{R^2}{L^2}}$ (B) $\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$ (C) $\omega_0 = \frac{1}{\sqrt{LC}}$ (D) $\omega_0 = \sqrt{\frac{1}{LC} + \frac{R^2}{4L} + \frac{R}{2L}}$



10. A coil (which can be modelled as a series circuit) has been designed for high Q performance at a rated voltage and a specified frequency. If the frequency of operation is doubled and the coil is operated at the same rated voltage, then the Q factor and the active power P consumed by the coil will be affected as below

(A) P is doubled, Q is halved (B) P is halved, Q is doubled
(C) P remains constant but Q increases 4 times (D) P decreases 4 times but Q is doubled

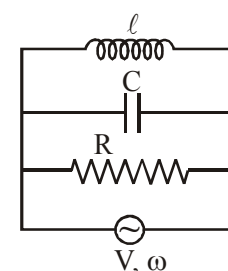
11. Consider L, C, R circuit as shown in figure, with a.c. source of peak value V and angular frequency ω . Then the peak value of current through the ac source

(A) $\frac{V}{\sqrt{\frac{1}{R^2} + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

(B) $V \sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2}$

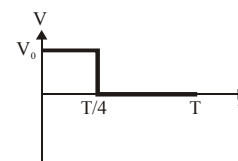
(C) $\frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

(D) $\frac{VR\omega C}{\sqrt{\omega^2 C^2 + R(\omega^2 C^2 - 1)^2}}$



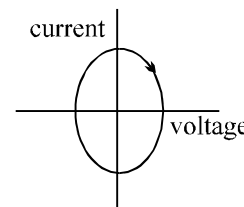
12. A periodic voltage V varies with time t as shown in the figure. T is the time period. The r.m.s. value of the voltage is :-

(A) $\frac{V_0}{8}$ (B) $\frac{V_0}{2}$ (C) V_0 (D) $\frac{V_0}{4}$



13. The graph shows the current versus the voltage in a driven RLC circuit at a fixed frequency. The arrow indicates the direction that this curve is drawn as time progresses. In this plot, the

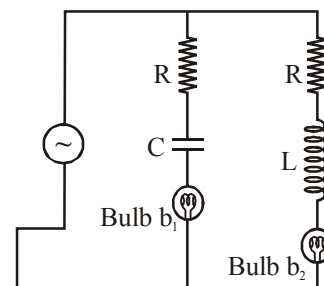
(A) Current lags the voltage by about 90 degrees
(B) Current leads the voltage by about 90 degrees
(C) Current and voltage are in phase
(D) Current and voltage are 180 degrees out of phase



14. Two identical incandescent light bulbs are connected as shown in the figure. When an AC voltage source of frequency f is applied in circuit, which of the following observations will be **CORRECT**?

(A) Maximum brightness of both bulbs does not depend on

frequency f , if $f = \frac{1}{2\pi}(\sqrt{1/LC})$



(B) Both bulbs will glow with same brightness provided frequency $f = \frac{1}{2\pi}(\sqrt{1/LC})$
(C) Bulb b_1 will light up initially and goes off, bulb b_2 will be ON constantly
(D) Bulb b_1 will blink and bulb b_2 will be ON constantly

Multiple Correct Answer Type

6 Q. [4 M (-1)]

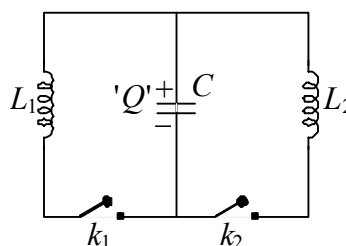
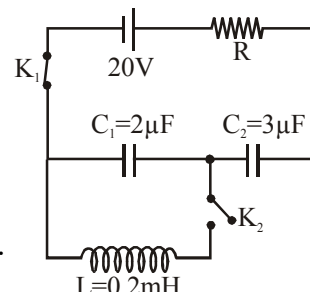
15. A circuit containing capacitors C_1 and C_2 , shown in the figure is in the steady state with key K_1 closed. At the instant $t = 0$, K_1 is opened and K_2 is closed. The ratio of energy in the inductor and that of capacitor C_1 becomes one third at time 't' for the first time. If charge on capacitor C_1 at that instant is q_1 and that on capacitor C_2 is q_2 then choose the **CORRECT** statement(s) :-

(A) Angular frequency of oscillations of the L.C. circuit will be 5×10^4 rad/s.
(B) Time 't' will be $21 \mu\text{s}$.

(C) Charge on capacitor C_1 at that instant will be, $q_1 = 12\sqrt{3}\mu\text{C}$.

(D) Charge on capacitor C_2 at that instant will be, $q_2 = 24\mu\text{C}$.

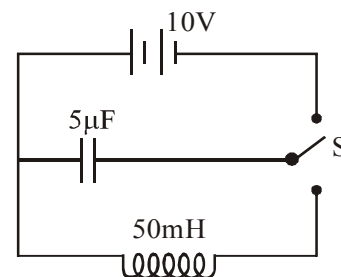
16. The given arrangement carries a capacitor with capacitance 40mF and two inductors $L_1 = 25\text{ H}$ and $L_2 = 100\text{ H}$. If the capacitor initially carries a charge of 10 mC , then :-



(A) the maximum current through the inductor L_1 when key k_1 is closed is 20 mA
(B) the maximum current through the inductor L_2 when key k_2 is closed is 5mA
(C) the maximum current through inductor L_2 when both the keys are closed is $\sqrt{5}\text{ mA}$
(D) time period of oscillation of charge is minimum when both the keys are closed

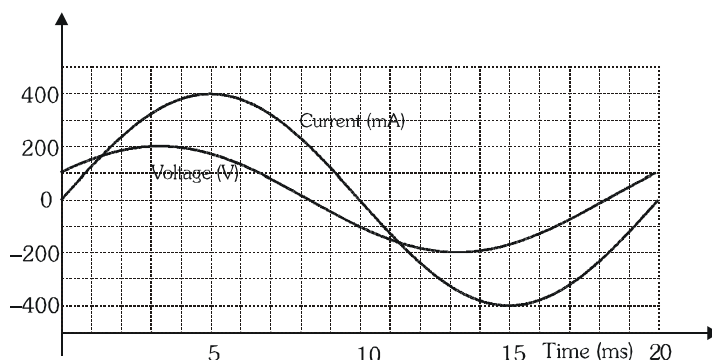
17. In the circuit shown, switch S is first pushed up to charge the capacitor. Switch S is then pushed down.

- (A) Maximum value of energy stored in the inductor is $250 \times 10^{-6} \text{ J}$
(B) Maximum value of energy stored in the capacitor is $250 \times 10^{-6} \text{ J}$
(C) Maximum current in the circuit is 100 mA
(D) Maximum charge on the capacitor is $5 \mu\text{C}$



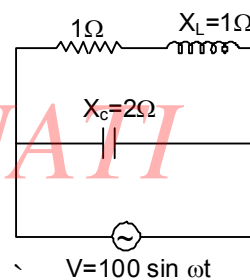
18. The given graph shows variation with time in the source voltage and steady state current drawn by a series RLC circuit. Which of the following statements is/are correct?

- (A) Current lags the voltage.
(B) Resistance in the circuit is $250\sqrt{3} \Omega$.
(C) Reactance in the circuit is 250Ω .
(D) Average power dissipation in the circuit is $20\sqrt{3} \text{ W}$.



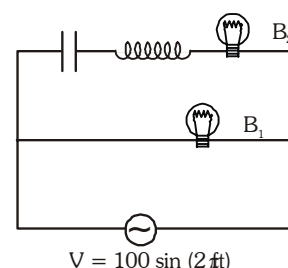
19. In the given a.c. circuit, choose the correct statement(s).

- (A) Impedance of circuit is 2Ω
(B) Power factor of circuit is $\frac{1}{\sqrt{2}}$
(C) Peak value of current through resistance is $50\sqrt{2} \text{ A}$
(D) Average power supplied by source is 2500 W.



20. Two similar bulbs of rating (100W, $50\sqrt{2} \text{ V}$) are connected in circuit as shown when frequency of source is varied, brightness of B_2 changes. It is observed that at a 50 Hz frequency both the bulbs have same brightness. Select correct alternative/s.

- (A) Current in both the bulbs will be in same phase when both have same brightness.
(B) Current through source will be 2A when both the bulbs have same brightness.
(C) Current through source will be $2\sqrt{2} \text{ A}$ when both the bulbs have same brightness.
(D) B_1 will be more brighter for all frequencies except 50 Hz



PHYSICS

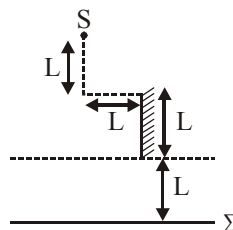
QUIZ # 36

SECTION-I

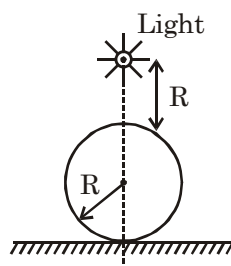
Single Correct Answer Type

17 Q. [3 M (-1)]

1. A plano-convex lens, when silvered at its plane surface is equivalent to a concave mirror of focal length 28 cm. When its curved surface is silvered and the plane surface not silvered, it is equivalent to a concave mirror of focal length 10 cm, then the refractive index of the material of the lens is:-
 (A) 9/14 (B) 14/9 (C) 17/9 (D) none
2. An astronomical telescope has an objective lens of focal length 100 cm and for relax eye adjustment magnification is 10. The separation between objective and eye piece lens is :
 (A) 110 cm (B) 90 cm (C) 10 cm (D) 125 cm
3. A lensmaker has to design an achromatic combination of effective power $-0.5D$. The two types of glasses to be used (say A and B) has dispersive powers in the ratio $\omega_A : \omega_B = 3 : 5$. Then the lens made from glass A will have focal length: -
 (A) +80cm (B) -80 cm (C) $\frac{+400}{3}$ cm (D) $\frac{-400}{3}$ cm
4. An object in a lift is dropped freely and simultaneously, the lift having a plane mirror at its floor, is operated. The lift starts ascending with an acceleration a . If the velocity of the image relative to the object be 40 ms^{-1} upwards, after 1 second. The value of a in ms^{-2} will be: ($g = 10 \text{ ms}^{-2}$)
 (A) 2 (B) 10 (C) 5 (D) None of these
5. A point source of light is placed in front of a plane mirror as shown in the figure. Determine the length of reflected patch of light on the screen Σ :-

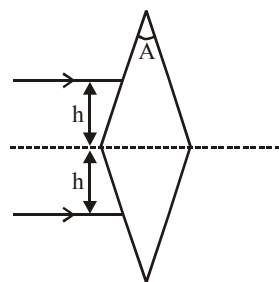


- (A) L (B) $2L$ (C) $\frac{3L}{2}$ (D) $\frac{L}{2}$
6. An opaque sphere of radius R lies on a horizontal plane. A light source is placed above sphere as shown. Then
 - (i) area of shadow on the plane is $2\pi R^2$
 - (ii) area of shadow on the plane is $3\pi R^2$
 - (iii) if the sphere is just submerged in some liquid, area of the shadow on plane decreases
 - (iv) if the sphere is just submerged in some liquid, area of the shadow on plane increases

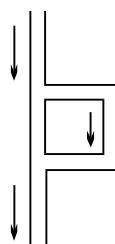


- (A) (i), (ii) (B) (ii), (iii) (C) (i), (iii) (D) (ii), (iv)

7. Two identical isosceles thin prisms of prism angle A and refractive index μ are placed with their bases touching each other. This system acts as a converging lens. What is the focal length of this system for parallel rays at a distance h from the base of the prism ?



- (A) $\frac{2h}{(\mu-1)A}$ (B) $\frac{h}{(\mu-1)A}$
 (C) $\frac{h}{2(\mu-1)A}$ (D) None of these
8. A sound wave is travelling in a uniform pipe with gas of adiabatic exponent γ . If u is the particle velocity at any point in medium and c is the wave velocity, then relative change in pressure $\frac{dP}{P}$ while wave passes through this point is :-
- (A) $\frac{u}{\gamma c}$ (B) $\gamma \sqrt{\frac{u}{c}}$ (C) $\gamma \frac{u}{c}$ (D) $\frac{u^2}{\gamma c^2}$
9. A girl stands 130 m away from a high wall. She claps her hands together at a steady rate such that 37 claps are made in 30 seconds ; starting at time $t = 0$ and ending at $t = 30$ s. Each clap, except the first, coincides with the echo of the one before. Calculate the speed of sound.
 (A) 320 m/s (B) 327 m/s (C) 340 m/s (D) 313 m/s
10. The average signal to noise ratio for a particular tape recorder is 50 db. This means that the intensity of an average reproduced signal is about how many times more than the intensity of the tape noise?
 (A) 7.07 times (B) 50 times (C) 625 times (D) 100,000 times
11. A sound consists of four frequencies $\rightarrow$ 300 Hz, 600 Hz, 1200 Hz and 2400 Hz. A sound 'filter' is made by passing this sound through a bifurcated pipe as shown. The sound waves have to travel a distance of 50 cm more in the right branch-pipe than in the straight pipe. The speed of sound in air is 300 m/s. Then, which of the following frequencies will be almost completely muffled or "silenced" at the outlet?

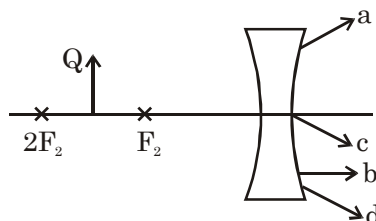


- (A) 300 Hz (B) 600 Hz (C) 1200 Hz (D) 2400 Hz
12. A resonant pipe closed at one end is marked 'C' and another similar pipe open at both ends is marked 'O'. Then,
 (A) the resonant frequencies in the fundamental mode are equal when pipe C is half the length of pipe O.
 (B) the resonant frequencies in the fundamental mode are equal when pipe O is half the length of pipe C.
 (C) when the pipes are of equal length, some of the resonant frequencies from pipe C are equal to the resonant frequencies from the pipe O.
 (D) when the pipes are of equal length, some of the resonant frequencies from pipe O are equal to the resonant frequencies from the pipe C.

13. A pipe open at the top end is held vertically with some of its lower portion dipped in water. At a certain depth of immersion, the air column of length $\frac{3}{8}$ m in the pipe resonates with a tuning fork of frequency 680 Hz. The speed of sound in air is 340 m/s. The pipe is now raised up by a distance 'x' until it resonates in the 'next overtone' with the same tuning fork. The value of 'x' is
(A) 20 cm (B) 40 cm (C) 50 cm (D) 25 cm
14. A source of frequency f is rotating in a circular path of radius R . An observer in the same plane is also rotating in a circular path of radius $2R$. The centers of both circular paths are separated by $6R$. Both these are rotating with same angular velocity ω in same direction. Find the difference between maximum and minimum frequency received by observer. 'C' is speed of sound.
(A) $\frac{2C\omega Rf}{C^2 - (\omega R)^2}$ (B) $\frac{6C\omega Rf}{C^2 - (\omega R)^2}$ (C) $\frac{3C\omega Rf}{C^2 - (\omega R)^2}$ (D) $\frac{4C\omega Rf}{C^2 - (\omega R)^2}$
15. Three musicians experiment with the Doppler effect. Musician A rides in a car at a speed u directly away from musician B who is stationary. Musician C rides in a car directly toward B and travels at the same speed as A. Musician A plays a note at frequency f_A on his trumpet. B hears the note, adjusts his trumpet, and plays the same note he heard. Choose the incorrect statement :



- (A) The note heard by C coming from B will be the same pitch as f_A .
(B) The note heard by A coming from B will be higher in pitch than f_A .
(C) The note heard by A coming from B will be lower in pitch than f_A .
(D) The note heard by B coming from A will be lower in pitch than f_A .
16. A diverging lens and an object are positioned as shown in figure at left. Which of the rays a, b, c and d could emanate from point Q at the top of the object ?



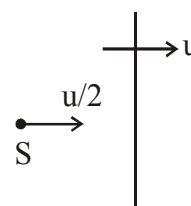
- (A) a, b and c (B) a and c (C) a and b (D) all
17. A large convex spherical mirror in an amusement park is facing a plane mirror 10 m away. A boy of height 1m standing midway between the two sees himself twice as tall as in plane mirror as in spherical one. In other words, the angle subtended at the observer by the image in plane mirror is twice the angle subtended by the image in the spherical mirror. What is the focal length of the convex mirror ?
(A) 5m (B) 2.5 (C) 1.8 m (D) 3.6 m

Multiple Correct Answer Type

3 Q. [4 M (–1)]

18. Sound waves from different source are interfering and pressure at same point in space is given as –
 $P = P_0 \sin (200 \pi t + \pi/2) \cos (3 \pi t + \pi/3)$
(A) The frequency of one source is above 100Hz and other source is below 100Hz.
(B) Beat frequency is 3 Hz
(C) Beat frequency is 200 Hz
(D) Beat frequency can be 3 Hz as well as 200 Hz

19. Choose the **CORRECT** statement(s).
- (A) Concave mirror can not form erect magnified image of an object placed in front of it.
 (B) Concave mirror can form inverted magnified image of an object placed in front of it.
 (C) Convex mirror can not form real image of an object placed in front of it.
 (D) Convex mirror can not form magnified image of an object placed in front of it.
20. A wall is moving with velocity u and a source of sound (frequency f) moves with velocity $\frac{u}{2}$ in the same direction as shown. Assuming that the sound travels with a speed of $10u$, pick the right option or options.
- (A) wavelength of incident sound is $\frac{19u}{2f}$
 (B) frequency of the incident sound is $\frac{18}{19}f$
 (C) wavelength of reflected sound is $\frac{u}{f}$
 (D) frequency of the reflected sound is $\frac{12f}{19}$.



TEAM ILLUMINATI

PHYSICS

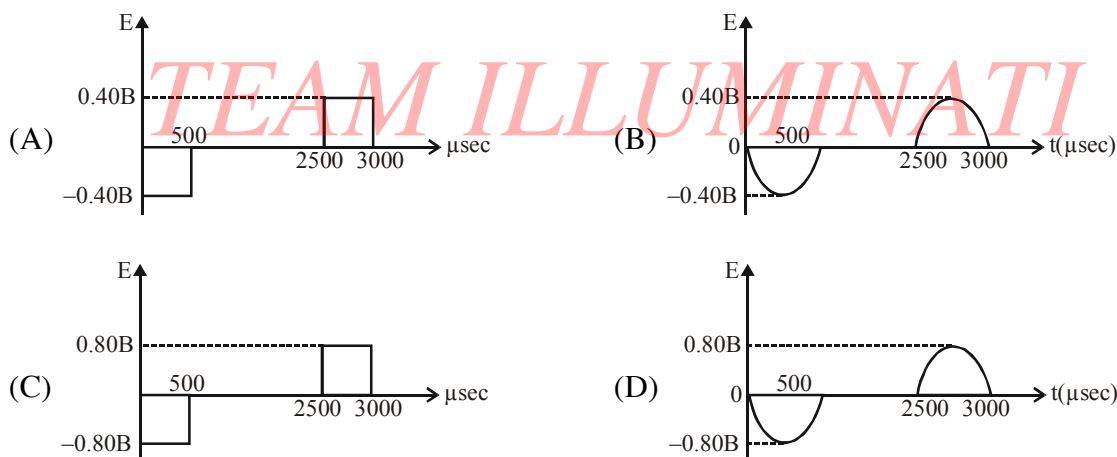
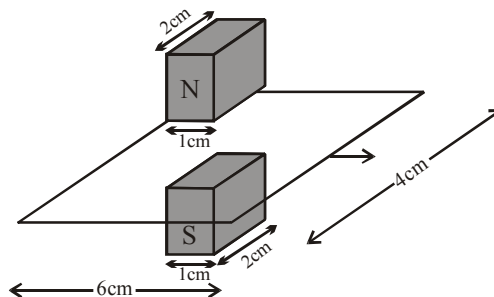
QUIZ # 37

SECTION-I

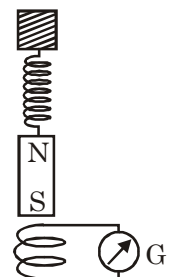
Single Correct Answer Type

6 Q. [3 M (-1)]

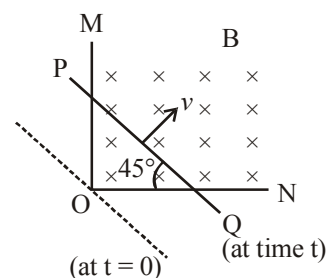
1. A magnetic field (B), uniform between two magnets can be determined by measuring the induced voltage in the loop as it is pulled through the gap at uniform speed 20 m/sec. Size of magnet is $2\text{ cm} \times 1\text{ cm} \times 2\text{ cm}$ and size of coil $4\text{ cm} \times 6\text{ cm}$ as shown in figure. The correct variation of induced emf with time is : (Assume at $t = 0$, the coil enters the magnetic field) :-



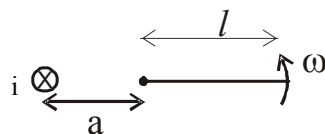
2. A magnet NS is suspended from a spring and while it oscillates, the magnet moves in and out of the coil C, having very small resistance. The coil is connected to a galvanometer G. Then, as the magnet oscillates,
- (A) G shows deflection to the left and right with constant amplitude.
(B) G shows deflection on one side.
(C) G shows no deflection.
(D) G shows deflection to the left and right but the amplitude steadily decreases.



3. Electric circuit is composed of three conducting rods MO, ON & PQ as shown in figure. The resistance of the rods per unit length is known to be λ . The rod PQ slides as shown in figure at a constant velocity v keeping it tilt angle relative to ON & MO fixed at 45° . The whole system is embedded in a uniform magnetic field B , which is directed perpendicularly into the page. Then induced electric current is :
- (A) Proportional to time t
(B) Inversely proportional to time t
(C) Proportional to square of time t
(D) Independent of time t

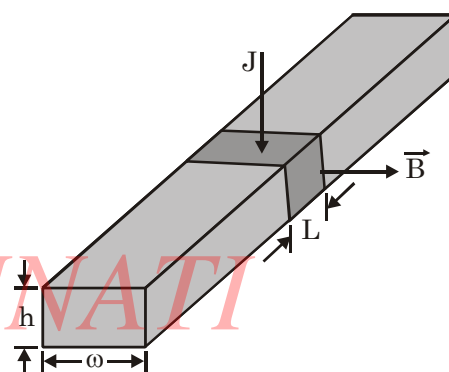


4. A rod of length ℓ is rotating in the plane of paper with an angular speed ω about one of its ends which is at a distance a from an infinitely long conducting wire carrying current i inwards the plane of paper. Find induced emf in the rod at the instant as shown in figure.

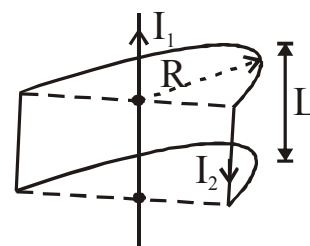


- (A) $\frac{\mu_0 i \omega}{4\pi} \left[\ell - a \ln \left(\frac{\ell + a}{a} \right) \right]$ (B) $\frac{\mu_0 i \omega}{2\pi} \left[\ell - a \ln \left(\frac{\ell + a}{a} \right) \right]$
(C) $\frac{\mu_0 i \omega}{8\pi} \left[\ell - a \ln \left(\frac{\ell + a}{a} \right) \right]$ (D) 0

5. Heart-lung machines and artificial kidney machines employ blood pumps. A mechanical pump can mangle blood cells. Figure represents an electromagnetic pump. The blood is confined to an electrically insulating tube, represented as a rectangle of width ω and height h . Two electrodes fit into the top and the bottom of the tube. The potential difference between them establishes an electric current through the blood, with current density J over a section of length L . A perpendicular magnetic field exists in the same region. The section of liquid in the magnetic field experiences a pressure increase given by :-



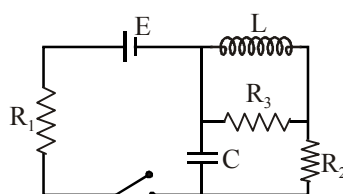
- (A) JLB (B) JhB
(C) $\frac{J\omega hB}{L}$ (D) No pressure increase occurs.
6. An infinitely long straight wire carrying a current I_1 is partially surrounded by a loop, as shown in figure. The loop has a length L and radius R , current I_2 . The axis of loop coincides with the wire. Find force between two system



- (A) zero (B) $\frac{\mu_0 I_1 I_2 L}{2\pi R}$
(C) $\frac{\mu_0 I_1 I_2 L}{\pi R}$ (D) $\frac{\mu_0 I_1 I_2 L}{R}$

Multiple Correct Answer Type

7. Current in R_3 in the shown circuit

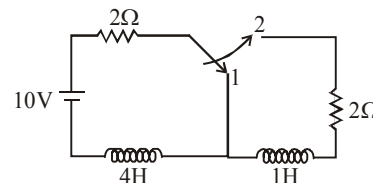


- (A) just after closing the switch is zero (B) long after closing the switch is zero
(C) just after closing the switch is $\frac{E}{R_3}$ (D) long after closing the switch is $\frac{E}{R_3}$

7 Q. [4 M (-1)]

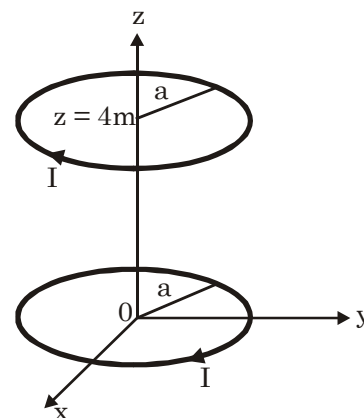
8. A network is given as shown. Switch was at position 1 for long time. at $t = 0$ it is shifted to position 2, then

- (A) Current just after switching is $\sqrt{20}$ Amp
(B) Current just after switching is 4 Amp
(C) Current in the circuit varies $i = \frac{5}{2} + \frac{3}{2}e^{-4t/5}$
(D) Current just after switching is 0



9. Two parallel, circular loops carrying a current of 40 A each are arranged as shown in figure. The first loop is situated in the x - y plane with its center at the origin, and the second loop's center is at $z = 4$ m. If the two loops have the same radius $a = 3$ m, determine the magnetic field at:

- (A) $z = 0, \vec{B} = \frac{-2432\pi \times 10^{-7}}{75} \hat{k}T$
(B) $z = 4, \vec{B} = \frac{-2432\pi \times 10^{-7}}{75} \hat{k}T$
(C) $z = 0, \vec{B} = \frac{-1685\pi \times 10^{-7}}{24} \hat{k}T$
(D) $z = 4, \vec{B} = \frac{-1685\pi \times 10^{-7}}{24} \hat{k}T$



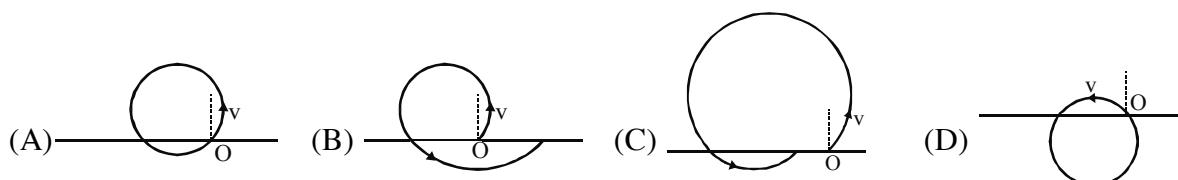
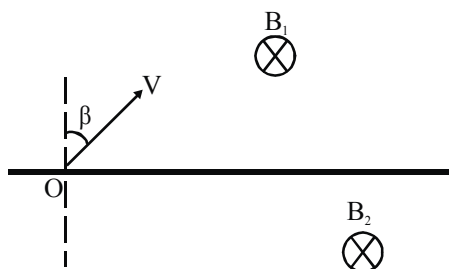
10. A wire of 3 mm radius is made up of an inner material ($0 < r < 2$ mm) for which conductivity $\sigma = 10^7$ SI units, and an outer material ($2 \text{ mm} < r < 3$ mm) for which conductivity $\sigma = 4 \times 10^7$ SI units. If the wire carries a total current of 100 mA dc,

- (A) The current in both the material is the same
(B) The current in the inner material is less than that in the outer material.

(C) The magnetic field in the inner material as function of r is $\frac{r}{1200} T$

(D) The magnetic field in the outer wire as a function of r is $3 \times 10^{-6} \left(\frac{10^5 r^2}{6} - \frac{1}{20} \right) T$

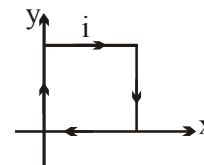
11. In the diagram shown, a particle of charge $+Q$ and mass M is projected making an angle β with the vertical line. Draw the possible path on which the charge will move. Above the dark line magnetic field is B_1 and below it is B_2 . (Consider all possible cases for values of B_1 and B_2)



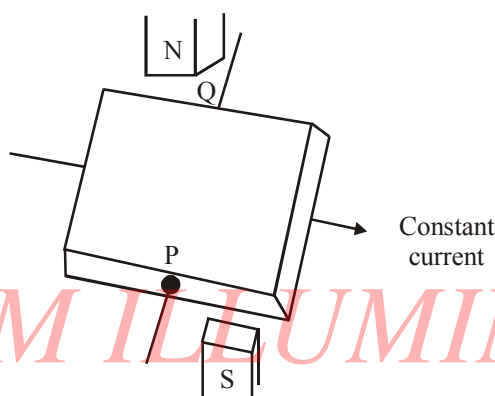
12. Figure shows a square current carrying coil of edge length L . The magnetic field on the coil is given by

$$\vec{B} = \frac{B_0 y}{L} \hat{i} + \frac{B_0 x}{L} \hat{j} \text{ where } B_0 \text{ is a positive constant. (A is area of coil)}$$

- (A) If coil is free to rotate about x axis torque on the coil is given by $\frac{1}{2} iAB_0 \hat{i}$.
 (B) If coil is free to rotate about y - axis torque on coil is given by $-\frac{1}{2} iAB_0 \hat{j}$.
 (C) Resultant force on coil is zero.
 (D) Equation for the torque $\vec{\mu} \times \vec{B}$ where μ is magnetic moment of coil is not valid on the coil if any of the side is fixed as axis.



13. Figure shows the essential parts of an apparatus to demonstrate the Hall effect. Which of the following statement (s) is/are correct ?



- (A) In the arrangement above, the Hall potential difference is developed across PQ
 (B) The magnitude of the Hall potential is greater if the applied magnetic flux density is increased
 (C) The magnitude of the Hall potential is less if the width PQ of the specimen is decreased
 (D) Hall potential is independent of material used for conductor

Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question nos. 14 to 16

If the velocity of a charged particle moving through a uniform magnetic field has a component parallel to that field the particle will move in a helical path as shown in figure 1. The net force F on the particle can be described by the equation.

$$F = qvB \sin \theta$$

where q is the charge on the particle, v is the velocity of the charge, B is the magnetic field strength, and θ is the angle between the velocity and the magnetic field. The pitch p of the helix is the distance between adjacent turns. The radius r of the helix is determined by the component of velocity of the charge perpendicular to the magnetic field and is independent of the component of the velocity parallel to the magnetic field.

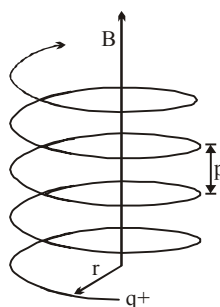


Figure 1

A nonuniform magnetic field that is stronger at its ends than at its middle can trap a charged particle by reflecting it back and forth between ends. Such a situation is shown in figure 2.

In a similar phenomenon electrons and protons above the atmosphere are reflected back and forth between the north and south poles of the earth's magnetic field forming the two Van Allen belts high above the atmosphere. Occasionally a solar flare shoots additional electrons and protons into the Van Allen belts. This pushes the electrons to a lesser height. The repulsion due to additional electrons drives the charged particles along the earth's magnetic field lines into the atmosphere where they collide with the air molecules forcing them to a higher energy level emitting photons. Oxygen atoms emit green light and nitrogen atoms emit pink. This light forms the curtain of lights in the sky known in the northern hemisphere as the aurora borealis

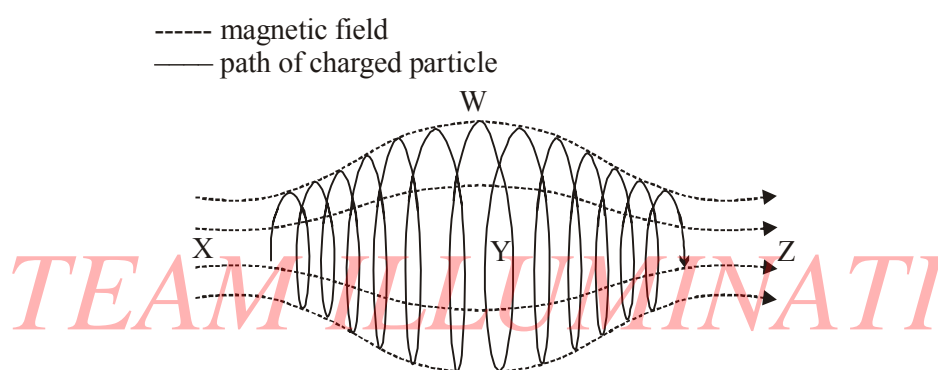


Figure 2 : Path of a charged particle trapped in a changing magnetic field

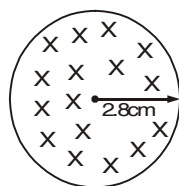
14. If the magnetic field in figure-2 represents a portion of the Earth's magnetic field, at which labelled point would you expect to find the aurora borealis?
 (A) W (B) X (C) Y (D) Z
15. Which of the following most accurately describes the change in the pitch and period of the helical path traveled by a negatively charged particle moving through a magnetic field that is gradually increasing?
 (A) The pitch increases and the period decreases
 (B) The pitch increases and the period increases
 (C) The pitch decreases and the period remains the same
 (D) Both the pitch and the period decrease.
16. The electrons and protons caught in the Van Allen belt reflect from one pole to the other. Looking up from the North pole, if electrons rotate counterclockwise then protons rotate
 (A) counterclockwise both when looking up from the north pole or when looking up from the south pole.
 (B) clockwise both when looking up from the north pole of when looking up from the south pole.
 (C) clockwise when looking up from the north pole but counterclockwise when looking up from the south pole.
 (D) counterclockwise when looking up from the north pole but clockwise when looking up from the south pole.

SECTION-III

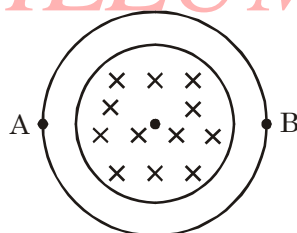
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

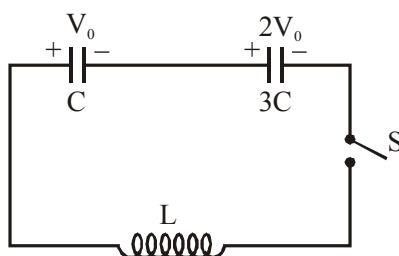
1. The magnetic field of a cylindrical magnet that has a pole-face radius 2.8 cm can be varied sinusoidally between minimum value 16.8 T and maximum value 17.2 T at a frequency of $\frac{60}{\pi}$ Hz. Cross section of the magnetic field created by the magnet is shown. At a radial distance of 2cm from the axis. If the amplitude of the electric field (in mN/C) induced by the magnetic field variation is 40 xmN/C. Then find the value of x.



2. A resistance less ring has 2 bulbs A & B rated at 2 V, 10 W and 2 V, 20W respectively. The ring encloses an ideal solenoid whose magnetic field is as shown. The radius of solenoid is 1m and the number of turns/length = 1000/m. The current changes at rate of 9 A/sec. Find the power dissipated P in bulb B. Fill $P \times 10^4$ in nearest integer.



3. A ring made of a superconductor is placed into a homogeneous magnetic field whose intensity grows from zero to $B_0 = 2T$. The plane of the ring is perpendicular to the force lines of the field. The induced current appearing in the ring is $1/x$ (amp). The radius of the ring $r = \frac{1}{\sqrt{\pi}}$ cm and its inductance $L = 1mH$. Find x.
4. Two capacitors of capacitance C and 3C are charged to potential difference V_0 and $2V_0$ respectively and connected to an inductor of inductance L as shown in the figure. Initially the current in the inductor is zero. Now the switch S is closed. Find the maximum current (in mA) in the inductor. (Take $C = 12 \mu F$, $L = 1H$ & $V_0 = 1V$)



PHYSICS

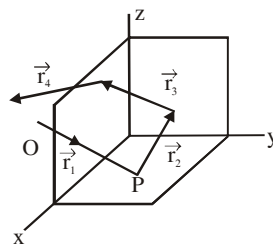
QUIZ # 38

SECTION-I

Single Correct Answer Type

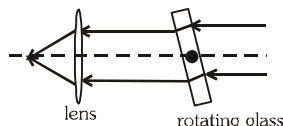
13 Q. [3 M (-1)]

1. A plane mirror is placed in y - z plane facing towards negative x -axis. The mirror is moving parallel to y -axis with a speed of 2 cm/s. A point object P is moving in front of the mirror with a velocity $(1\text{ cm/s})\hat{i} + (1\text{ cm/s})\hat{j}$. Then which of the following statements is **INCORRECT**?
- (A) The velocity of image is $(-\hat{i} + \hat{j})$ cm/s.
- (B) The velocity of image with respect to mirror is $-(\hat{i} + \hat{j})$ cm/s.
- (C) The velocity of image with respect to object is $-(2\hat{i})$ cm/s.
- (D) The unit vector in the direction of reflected ray is $\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$.
2. Three plane mirrors are arranged so that they are mutually perpendicular as shown in figure. Let the line of intersection of the mirrors be the coordinate axes. A ray OP represented by vector $\vec{r}_1 = (x_1, y_1, z_1)$ is reflected from the mirror. Which of the following statement is **INCORRECT**?

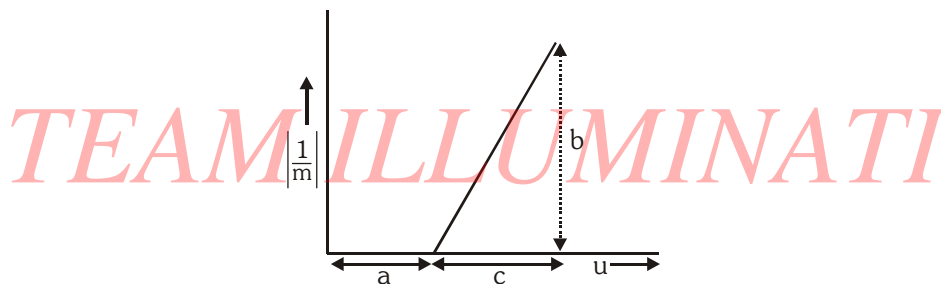


- (A) The components of the reflected vector from mirror in x - y plane is $\vec{r}_2 = (x_1, y_1, -z_1)$
- (B) If $\vec{r}_2$ is now reflected at the y - z plane, the components of the reflected vector is $\vec{r}_3 = (-x_1, y_1, -z_1)$
- (C) If $\vec{r}_3$ is reflected at the z - x plane, the components of the reflected vector is $\vec{r}_4 = (-x_1, -y_1, -z_1)$
- (D) For an incident ray that is reflected from the mirrors in some other order, the components of final ray $\vec{r}_4$ will change.
3. A man of height 2 m stands on a straight road on a hot day. The vertical temperature in the air results in a variation of refractive index with height y as $\mu = \mu_0 \sqrt{1+ay}$ where μ_0 is the refractive index of air near the road and $a = 2 \times 10^{-6}/\text{m}$. What is the actual length of the road, man is able to see
- (A) 2000 m (B) 390 m (C) infinite distance (D) None of these
4. A light ray incident along vector $2\hat{i} + 4\hat{j} + \sqrt{5}\hat{k}$ strikes on the x - z plane from medium I of refractive index $\sqrt{3}$ and enters into medium II of refractive index μ_2 . The value of μ_2 for which angle of refraction becomes 90° , is :-
- (A) $\frac{4\sqrt{3}}{5}$ (B) $\frac{3\sqrt{3}}{5}$ (C) $\frac{2\sqrt{3}}{5}$ (D) $\frac{\sqrt{3}}{5}$

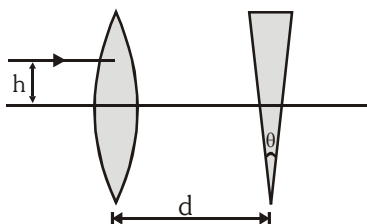
5. A slab of high quality flat glass, with parallel faces, is placed in the path of a parallel light beam before it is focussed to a spot by a lens. The glass is rotated slightly back and forth from the vertical orientation, about an axis coming out of the page, as shown in the diagram. According to ray optics the effect on the focused spot is:



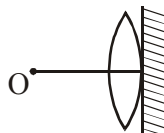
- (A) There is no effect on the spot
(B) The spot moves towards then away from the lens.
(C) The spot moves up and down parallel to the lens.
(D) The spot blurs out of focus.
6. The graph shows how the reciprocal of magnification $\left(\frac{1}{\mu} = \frac{4}{3}\right)$ produced by a converging thin lens varied with object distance u in displacement method. What was the focal length of the lens used?



- (A) a/b (B) bc/a (C) c/b (D) ab/c
7. A ray of light parallel to the axis of a converging lens (having focal length f) strikes it at a small distance 'h' from its optical centre. A thin prism having angle θ and refractive index μ is placed normal to the axis of lens at a distance 'd' from it. What should be the value of μ so that the ray emerges parallel to the lens axis

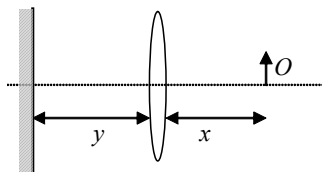


- (A) $\frac{h}{f\theta}$ (B) $\frac{h}{f\theta} + 1$ (C) $\frac{h}{(d+f)\theta}$ (D) $\frac{h}{(d+f)\theta} + 1$
8. Behind a thin converging lens having both the surfaces of the same radius 10 cm, a plane mirror has been placed. The image of an object at a distance 40 cm from the lens is formed at the same position. What is the refractive index of the lens ?

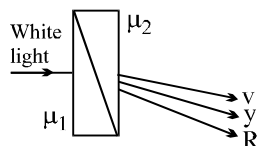


- (A) 1.5 (B) 5/3 (C) 9/8 (D) None

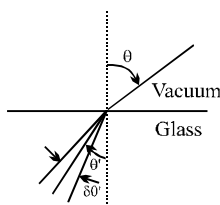
9. A convex lens of focal length f and a plane mirror are y distance apart. An object O is kept on the principal axis of the lens at a distance x from the lens. The values of x and y for the image of O to not coincide with O is



- (A) $x = f, y = f$ (B) $x = f, y = 2f$ (C) $x = 2f, y = f$ (D) $x = 2f, y = 2f$
10. In determining the focal length of a convex lens by two needle method which of the following is not possible :-
- (A) Positions of object needle and image needle can be interchanged for their distances from convex lens
- (B) For same distance between object needle and image needle there are two positions of lens at which parallax can be removed
- (C) Minimum distance between object needle and image needle must be equal to four times the focal length of lens
- (D) Positions of object and image needles can not be interchanged in the experimental setup
11. A parallel beam of white light falls on a convex lens. Images of blue, red and green light are formed on other side of the lens at distances x, y and z respectively from the pole of the lens. Then :-
- (A) $x > y > z$ (B) $x > z > y$ (C) $y > z > x$ (D) None
12. A beam of white light is incident on a combinations of their prisms with equal angles. In the out going beam the rainbow colours are observed as shown. Then



- (A) $\mu_1 < \mu_2$ (B) $\mu_2 = \mu_1$
- (C) $\omega_1 < \omega_2$ (D) the obtained colour pattern is not possible
13. A beam of light has a small wavelength spread $\delta\lambda$ about a central wavelength λ . The beam travels in vacuum until it enters a glass plate at an angle θ relative to the normal to the plate, as shown in figure. The index of refraction of the glass is given by $n(\lambda)$. The angular spread $\delta\theta'$ of the refracted beam is given by

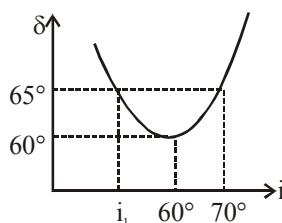


- (A) $\delta\theta' = \left| \frac{1}{n} \delta\lambda \right|$ (B) $\delta\theta' = \left| \frac{dn(\lambda)}{d\lambda} \delta\lambda \right|$
- (C) $\delta\theta' = \left| \frac{\tan \theta'}{n} \frac{dn(\lambda)}{d\lambda} \delta\lambda \right|$ (D) $\delta\theta' = \left| \frac{\sin \theta}{\sin \theta'} \frac{\delta\lambda}{\lambda} \right|$

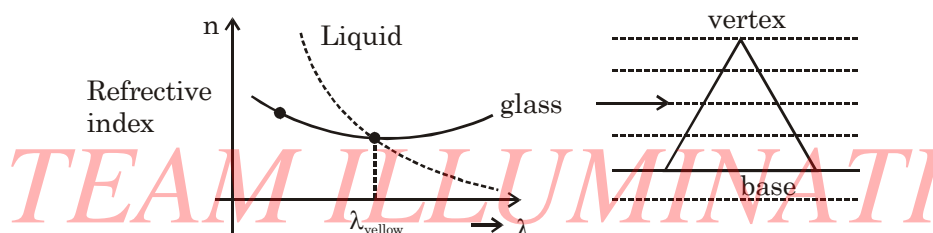
Multiple Correct Answer Type

2 Q. [4 M (-1)]

14. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Choose the **CORRECT** statement(s).



- (A) The angle of prism is 60°
 (B) The refractive index of the prism is $n = \sqrt{3}$
 (C) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
 (D) for minimum deviation, the angle of emergence $e = 60^\circ$.
15. When a ray of white light is incident on the prism parallel to the base. The variation in refractive index Vs wavelength (λ) graph given as shown.

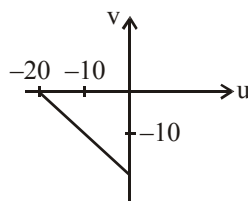


- (A) Yellow ray travels without deviation. (B) Blue ray is deviated towards the vertex.
 (C) Red ray is deviated towards the base. (D) there is no dispersion.

Linked Comprehension Type (1 Para $\times$ 2Q. & 1 Para $\times$ 3Q.) [3 M (-1)] (Single Correct Answer Type)

Paragraph for Question no. 16 and 17

There is a very simple but very useful way to find the location of image, its magnification and its properties in case of lens or a mirror. Let us consider the case of a concave mirror. Take a graph on xy-plane as shown. Assume that x-axis represents object distance (u) and y-axis represents image distance (v).



Let us consider a concave mirror with focal length -10 cm. Mark a point on plane as $(-10, -10)$ [in other words (f, f)]. Suppose we want to find the location of a real object located at 20 cm from the mirror. By cartesian sign convention, $u = -20$ cm. Draw a straight line from $x = -20$ to y-axis such that it passes through point $(-10, -10)$. Then the point at which it cuts the y-axis is the image. It can be easily seen from mirror formula as well as the geometry of graph that $v = -20$ cm. Note that this is not a graph of u vs v .

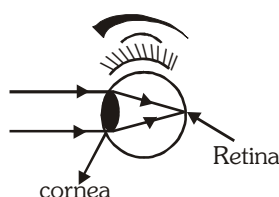
16. If the slope of the line is m , the magnification of the mirror is given by :

- (A) m (B) $-m$ (C) $\frac{1}{m}$ (D) $-\frac{1}{m}$

17. For a particular mirror and a particular object image combination, the x intercept is positive but y intercept is negative. x intercept is smaller in modulus than y intercept.
- (A) the mirror is convex (B) the mirror is concave
 (C) the mirror can be either convex or concave (D) this combination is not possible.

Paragraph for Question 18 to 20

A simple model of the human eye is shown in the figure. It is based on the assumption that all of the incident light gets refracted at the cornea. At the outer surface the eye, the cornea act as a converging lens with a fixed focal length of 2 cm approximately. For very distant objects parallel light rays incident on the cornea get refracted by the cornea to produce a focused image on the retina, which then transmits an electric impulse to the brain brought the optic nerve.



Vision defects come into play if the image of a distant object is not formed on the retina. The two most common vision defects are Myopia and Hypermetropia. Myopia or near sightedness occurs when the cornea focuses the image of distant object in front of the retina whereas Hypermetropia or far sightedness occurs when the cornea focuses the image behind the retina. Vision defects can be removed by introducing another lens in front of the eye such that the lens system produces a focused image on the retina. For a very distant object the object distance may be taken as infinity and the following relation holds for the

$$\text{combination : } \left(\frac{1}{v} \right) = \left(\frac{1}{f_e} \right) + \left(\frac{1}{f_l - x} \right)$$

where $f_e \rightarrow$ focal length of the cornea, $f_l \rightarrow$ focal length of the correcting lens, 'x' $\rightarrow$ distance between correcting lens and cornea. 'v' $\rightarrow$ image distance as measured from the cornea.

18. To remove the vision defects myopia and hypermetropia, the lens used respectively are
- (A) Converging, Diverging (B) Diverging, Converging
 (C) Diverging, Diverging (D) Converging, Converging
19. The focal length of a woman's cornea is found to be 1.6 cm. If she is wearing a converging lens of focal length – 7.5 cm at a distance of $x = 0.5$ cm from her cornea then the image distance 'v' measured from the cornea for a distant object is
- (A) 0.5 cm (B) 1.0 cm (C) 2.0 cm (D) 2.5 cm
20. The cornea and the correcting lens touch each other can act together as a single lens in calls of a contact lens. If the focal length of both the cornea and the contact lens is halved, then the image distance 'v' for a distant object is
- (A) same as the old value (B) is $(1/4)^{\text{th}}$ of the old value
 (C) is twice of the old value (D) is $(1/2)$ of the old value

PHYSICS

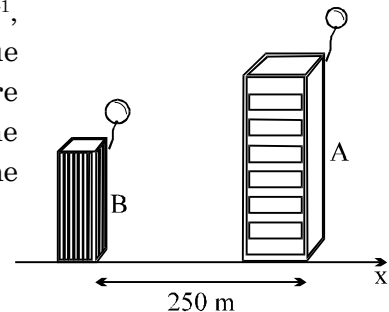
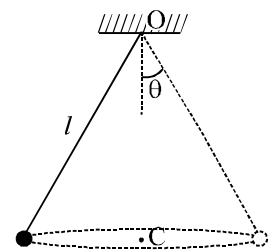
QUIZ # 39

SECTION-I

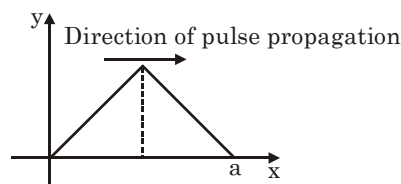
One or more than one Correct Answer Type

6 Q. [4M(-2,0,+1)]

- Three simple harmonic motions in the same direction having the same amplitude a and same period are superimposed. If each differs in phase from the next by 45° , then :-
 - the resultant amplitude $(1 + \sqrt{2})a$
 - the phase of the resultant motion relative to the first is 90°
 - the energy associated with the resulting motion is $(3 + 2\sqrt{2})$ times the energy associated with any single motion.
 - the resulting motion is not simple harmonic
- A particle of mass m is suspended from point O and undergoes circular motion in horizontal plane as conical pendulum as shown in figure.
 - Angular momentum of particle about point of suspension does not remains constant.
 - Angular momentum of particle about centre of circle remains constant.
 - Average of total force during half rotation is $\frac{2mg \tan \theta}{\pi}$.
 - Average of total torque about axis OC during half rotation is zero
- Two balloons are simultaneously released from two buildings A and B. Balloon from A rises with constant velocity 10 ms^{-1} , While the other one rises with constant velocity of 20 ms^{-1} . Due to wind the balloons gather horizontal velocity $V_x = 0.5y$, where 'y' is the height from their respective point of release. The buildings are at a distance of 250 m & after some time 't' the balloons collide.
 - $t = 5 \text{ sec}$.
 - Distance between balloon initially is $50\sqrt{29} \text{ m}$.
 - Distance between balloon initially is 100m.
 - $t = 10 \text{ sec}$
- The wave function of a triangular wave pulse is defined by the relation below at time $t = 0 \text{ sec}$.



$$y = \begin{cases} mx & \text{for } 0 \leq x \leq \frac{a}{2} \\ -m(x-a) & \text{for } \frac{a}{2} \leq x \leq a \\ 0 & \text{every where else, where } m \ll 1 \end{cases}$$



The wave pulse is moving in the $+X$ direction in a string having tension T and mass per unit length μ . The total energy present with the wave pulse is :-

- $\frac{m^2 Ta}{2}$
- $m^2 Ta$
- mTa
- $\frac{mTa}{2}$

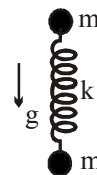
5. A mass m is hung on an ideal massless spring. Another equal mass is connected to the other end of the spring. The whole system is at rest. At $t = 0$, m is released and the system falls freely under gravity. Assume that natural length of the spring is L_0 , its initial stretched length is L and the acceleration due to gravity is g . What is distance between masses as function of time.

(A) $L_0 + (L - L_0) \cos \sqrt{\frac{2k}{m}} t$

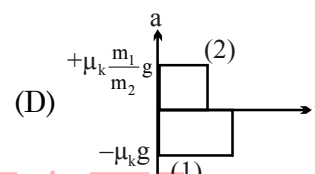
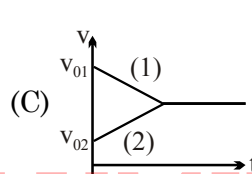
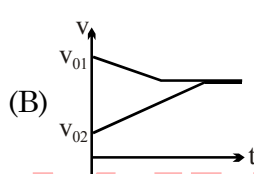
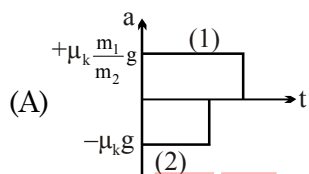
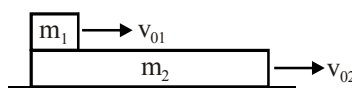
(B) $L_0 \cos \sqrt{\frac{2k}{m}} t$

(C) $L_0 \sin \sqrt{\frac{2k}{m}} t$

(D) $L_0 + (L - L_0) \sin \sqrt{\frac{2k}{m}} t$



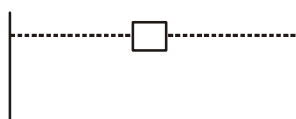
6. Block m_1 is projected on a long plank of mass m_2 . Plank is placed on a smooth horizontal surface. There is friction between block and plank, coefficient of friction is μ . Block m_1 has initial velocity v_{01} and plank has initial velocity v_{02} with $(v_{01} > v_{02})$. Which of following graphs is/are **CORRECT**.



Linked Comprehension Type (2 Para $\times$ 2 Q.) [3M(-1)] (Single Correct Answer Type)

Paragraph for Questions 7 and 8

A cube made of wood having specific gravity 0.4 and side length a is floated in a large tank full of water.



7. Which action would change the depth to which block is submerged?
 (A) more water is added in the tank (B) atmospheric pressure increases
 (C) the tank is accelerated upwards (D) None of these
8. What can be the maximum amplitude of it's vertical simple harmonic motion?
 (A) $\frac{a}{2}$ (B) $0.4 a$ (C) $0.6 a$ (D) $0.2 a$

Paragraph for Questions 9 and 10

Consider a gas at temperature T occupying a volume V consisting of a mixture of two gases having N_a & N_b atoms of masses m_a & m_b respectively.

9. Give an expression for the total pressure exerted by the gas

(A) $\frac{(N_a + N_b)kT}{V}$

(B) $\frac{1.5(N_a + N_b)kT}{V}$

(C) $\frac{(N_a + N_b)RT}{V}$

(D) none

10. Suppose now that $N_a = N_b$ and that the different atoms combine at constant volume to form molecules of mass $m_a + m_b$. Once the temperature returns to its original value, what would be the ratio of the pressure after combination to the pressure before?

(A) 1

 (B) $1/3$

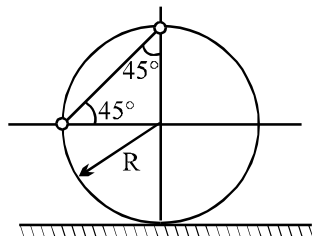
 (C) $1/2$

(D) none

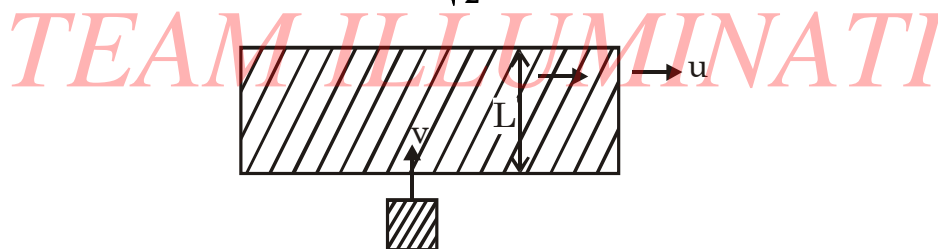
SECTION-II**Numerical Answer type Questions****8 Q. [3(-0)]**

(Upto Second Decimal Place)

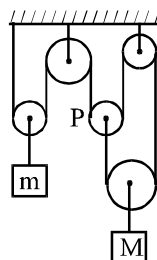
1. Two beads connected by a light inextensible string are placed over the fixed vertical ring as shown in figure. If mass of each bead is 100 gms and all surfaces are frictionless. The tension in the string just after the beads are released from the shown position is T N then find T^2 .



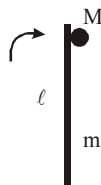
2. A long piece of paper is being pulled on a horizontal surface with a constant velocity u along its length. Width of the paper is L . A small block moving horizontally, perpendicular to the direction of motion of the paper, with velocity v ; slides onto the paper. The coefficient of friction between the block and the paper is μ . Find maximum value of v (in m/s) such that the block does not cross the opposite edge of the paper. $\mu = 0.1$, $L = \sqrt{\frac{5}{2}}$ m, $u = 3$ m/s



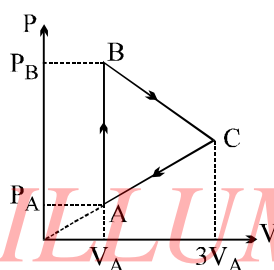
3. A point source of sound is located somewhere along the x-axis. Experiments show that the same wave front simultaneously reaches listeners at $x = -8$ m and $x = +2.0$ m. A third listener is positioned along the positive y-axis. What is her y-coordinate (in m) if the same wave front reaches her at the same instant as it does the first two listeners?
4. A point source of sound is located inside sea water having bulk modulus $B_w = 2.0 \times 10^9$ N/m². A driver located at a distance of 10m from the source registers a pressure amplitude of $\Delta P_0 = 3000 \pi$ N/m² and gives equation of wave as $y = A \sin(15\pi x - 21000\pi t)$. Where y and x are in meter and t in second. If displacement amplitude of sound wave at the location of driver is $x \times 10^{-7}$ m then value of x will be
5. A source of sound starts receding away from a stationary listener at $t = 0$, such that its distance from the listener at any later time t is $x = 10t(t + 5)$, where x is in meters and t in seconds. Further the frequency of the source is decreasing with time and is given by : $\nu = (1000/t)$ Hz, where t is in seconds. Find the frequency (in Hz) of the sound heard by the listener when the source is 187.5 m away from him. Speed of sound = 340 m/s.
6. All pulleys are massless. The string is light and inextensible. $m = 1$ kg and $M = 2$ kg. Find acceleration of centre of pulley 'P' in m/s². ($g = 10$ m/s²)



7. A uniform stick of mass 18 kg and length 3 m spins around on a frictionless horizontal plane, with its Centre of Mass stationary. A mass M is placed on the plane, and the stick collides elastically with it, as shown (with the contact point being the end of the stick). What should M (in kg) be so that after the collision the stick has translational motion, but no rotational motion?



8. A sample of ideal gas is taken through the cyclic process shown in the figure. The temperature of the gas at state A is $T_A = 200$ K. At states B and C the temperature of the gas is the same. If the greatest temperature of the gas in kelvins during the cyclic process is $T_{\max}$ then write the value of $\frac{T_{\max}}{1000}$.



TEAM ILLUMINATI

PHYSICS

QUIZ # 40

SECTION-I

One or More than one correct answer type

6 Q. [4(-2,0+1)]

1. A block of mass m is released from height $4h$ from rest onto a spring

A having spring constant $K_A = \frac{mg}{2h}$ as shown in figure. The block

reaches the minimum value of $y = h$ during its downward motion.

Then y_{eq} represents equilibrium position of block & a_{max} represents maximum acceleration of block

(A) $K_B = \frac{4mg}{h}$, $y_{eq} = h$

(B) $K_B = \frac{4mg}{h}$, $y_{eq} = \frac{17h}{9}$

(C) $a_{max} = 5g$ upwards

(D) $a_{max} = 4g$ upwards

2. A non-uniform ball of radius R and radius of gyration about geometric center $= R/2$, is kept on a smooth surface. The geometric center coincides with the center of mass. The ball is struck horizontally with a sharp impulse $= J$ at a height ' h ' above the surface. The ball will :-

(A) slip on surface for all cases.

(B) roll purely if $h = 5R/4$.

(C) roll purely if $h = 3R/2$.

(D) not rotate if $h = R$.

3. A man with a gun is standing on Equator. He keeps two targets (one eastward and other westward of him); along the equator 1 km away from him. Bullet is fired towards each target.

(A) The bullet that is fired eastward will hit below the target

(B) The bullet that is fired eastward will hit above the target

(C) The bullet that is fired westward will hit below the target

(D) The bullet that is fired westward will hit above the target

4. Which among the following forces is conservative?

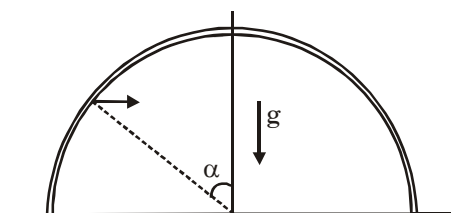
(A) $\vec{F} = xy^3\hat{i} + yx^3\hat{j}$

(B) $\vec{F} = y^3\hat{i} + xy^2\hat{j}$

(C) $\vec{F} = \frac{y^2}{x}\hat{i} + 2y\ln\left(\frac{x}{b}\right)\hat{j}$

(D) $\vec{F} = 2xy^2\hat{i} - x^2y^2\hat{j}$

5. In a circus ride, a motorcyclist moves along the inner surface of a sphere of radius R . Racing, he begins to describe a horizontal circle in the upper hemisphere. After that, for greater effect, the lower hemisphere is removed. The coefficient of friction of the tires on the surface of the sphere is equal to μ , and the angle between the vertical and the direction to the motorcyclist from the centre of the sphere is equal to α .



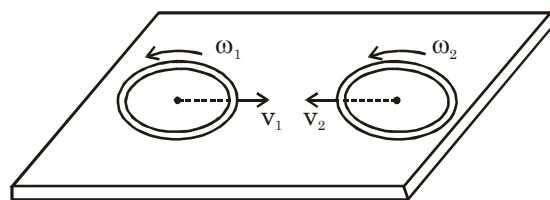
(A) The minimum speed of motorcyclist to maneuver the curve is $\sqrt{\frac{gR \sin \alpha (\tan \alpha + \mu)}{\mu \tan \alpha - 1}}$

(B) The minimum speed of motorcyclist to maneuver the curve is $\sqrt{\frac{gR \sin \alpha (\tan \alpha - \mu)}{\mu \tan \alpha + 1}}$

(C) Normal force acting on the motorcyclist is $\frac{mgR \cos \alpha (\tan \alpha + 1)}{\mu \tan \alpha - 1}$

(D) Normal force acting on the motorcyclist is $\frac{mgR \cos \alpha (\tan \alpha - 1)}{\mu \tan \alpha + 1}$

6. Two identical thin rotating rings move along a smooth horizontal plane towards each other. Their velocities v_1 and v_2 are directed along a straight line connecting the centers of the rings. Angular velocities of the rings ω_1 and ω_2 . Determine the angular velocity of the rings after the collision, if the slipping relative to each other disappears at the last moment a blow.



(A) $\omega'_1 = \frac{3\omega_1 - \omega_2}{4}$ (B) $\omega'_1 = \frac{3\omega_2 - \omega_1}{4}$ (C) $\omega'_2 = \frac{3\omega_1 - \omega_2}{4}$ (D) $\omega'_2 = \frac{3\omega_2 - \omega_1}{4}$

Matching list type (4×4 or 4×5 or 4×6) with one or more options correct (Two Columns with List I having 4 options & List II contains 4 or 5 or 6 options) 4Q.[3M(-1,0)]

7. A standing wave, formed in an organ pipe aligned along the x-axis, is given by $S = 5 \times 10^{-6} \sin(4\pi x) \cos(1320\pi t)$ Where x is in m and t is in seconds.

List-I

List-II

(P) Magnitude of change in pressure is maximum

(1) $x = \frac{1}{4} \text{ m}$

(Q) Magnitude of change in pressure is minimum

(2) $x = \frac{1}{2} \text{ m}$

(R) Maximum amplitude of vibration

(3) $x = 1 \text{ m}$

(S) Minimum amplitude of vibration

(4) $x = \frac{1}{8} \text{ m}$

(5) $x = \frac{3}{8} \text{ m}$

(A) $P \rightarrow 1,2,3; Q \rightarrow 4,5; R \rightarrow 4,5; S \rightarrow 1,2,3$

(B) $P \rightarrow 1,3; Q \rightarrow 3,5; R \rightarrow 2,3,4; S \rightarrow 1,2,3,4,5$

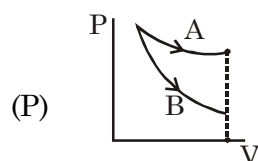
(C) $P \rightarrow 2,3; Q \rightarrow 2,4; R \rightarrow 2,3,5; S \rightarrow 3,4,5$

(D) $P \rightarrow 1,3,4; Q \rightarrow 1,3; R \rightarrow 2,3; S \rightarrow 2,3,4,5$

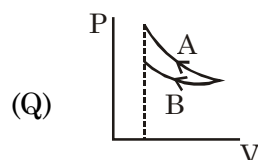
8. An ideal gas undergoes two processes A and B. One of these is isothermal and the other is adiabatic.

List-I

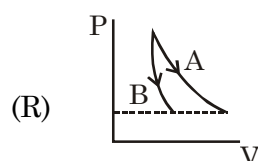
List-II



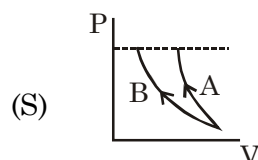
(1) Heat supplied during curve A is positive



(2) Work done by gas in both processes positive



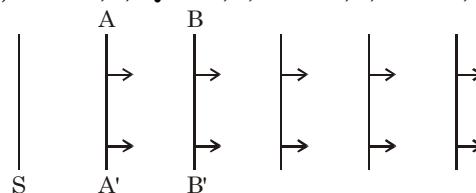
(3) Internal energy increases in adiabatic process



(4) Temperature of gas in process B is constant

- (A) $P \rightarrow 1,2; Q \rightarrow 2,3; R \rightarrow 3,4; S \rightarrow 1,2$ (B) $P \rightarrow 2,3; Q \rightarrow 1,4; R \rightarrow 3,4; S \rightarrow 3,4$
(C) $P \rightarrow 1,2; Q \rightarrow 3,4; R \rightarrow 1,2; S \rightarrow 3,4$ (D) $P \rightarrow 3,4; Q \rightarrow 1,3; R \rightarrow 2,4; S \rightarrow 1,4$

9. The diagram shows plane wavefronts for sound wave travelling in air towards right. Each of these wavefronts represent successive pressure maxima for the pressure wave. Initially the source S, observer O and medium are all at rest. The source is a large plane diaphragm and observer is an detector. Wave fronts being considered in column-II have been emitted after the action in column-I has taken place.



List-I

- (P) Source starts moving towards right
(Q) Air starts moving towards right
(R) Observer and source both move towards left with same speed.
(S) Source and medium (air) both move towards right with same speed

List-II

- (1) distance between any two wavefronts will increase.
(2) distance between any two wavefronts will decrease
(3) the time needed by sound to move from point A to B in space will increase.
(4) time needed by sound to move from point A to B in space will decrease.
(5) frequency received by observer is increase.

- (A) $P \rightarrow 1,2; Q \rightarrow 2,5; R \rightarrow 4,5; S \rightarrow 3,4$ (B) $P \rightarrow 1; Q \rightarrow 4,5; R \rightarrow 2,3; S \rightarrow 1,4$
(C) $P \rightarrow 3,5; Q \rightarrow 2,3; R \rightarrow 1,4; S \rightarrow 1,5$ (D) $P \rightarrow 2,5; Q \rightarrow 1,4; R \rightarrow 1; S \rightarrow 4,5$

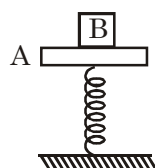
10. Match the column I with column II appropriately.

List-I

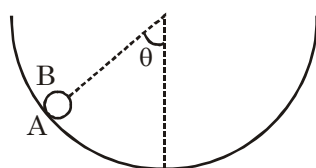
- (P) A bob B hanging from a string A of length 3m is projected to left with a speed of 10 m/s



- (Q) The plat form is pushed down by a distance $2x_0$ below mean and released x_0 is compression in mean position



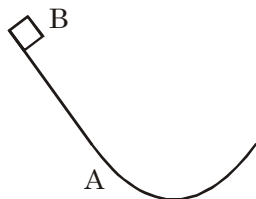
- (R) A spherical solid ball B is released on a perfectly rough spherical surface A as shown $\theta = \pi/3$



List-II

- (1) The acceleration of centre of mass of B at some time can be equal to g
(2) The force exerted by A on B can be zero at some point
(3) The speed of the body B varies sinusoidally with time when force exerted by A on B is not zero

- (S) Block B is released on a smooth track A as shown



- (4) The motion of B is oscillatory

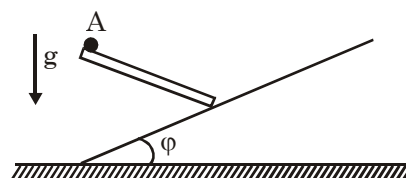
- (5) The motion of B is periodic
(A) $P \rightarrow 2,3$; $Q \rightarrow 4,5$; $R \rightarrow 2,3$; $S \rightarrow 4,5$ (B) $P \rightarrow 1,2$; $Q \rightarrow 1,2,3$; $R \rightarrow 1,4,5$; $S \rightarrow 1,2$
(C) $P \rightarrow 1,4$; $Q \rightarrow 1,2$; $R \rightarrow 1,2,3$; $S \rightarrow 1,4,5$ (D) $P \rightarrow 3,4$; $Q \rightarrow 1,2,5$; $R \rightarrow 2,3$; $S \rightarrow 1,3,4$

SECTION-II

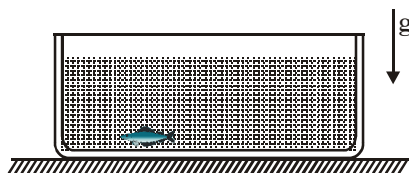
Numerical-Answer Type (Upto Second Decimal Place)

8 Q. [3(-0)]

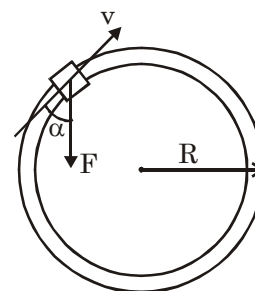
1. A ball is released on a smooth plane 'A' of constant length as shown. After that it moves smoothly to another smooth incline of inclination $\phi = 60^\circ$. Find the angle of plane 'A' with vertical (in degrees) so that the ball reaches the incline in the shortest time ?



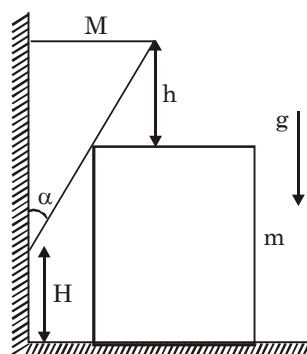
2. On a smooth floor is a massless vessel filled with water of density ρ_0 ; water volume V_0 . A beetle of volume V and density ρ , which appeared at the bottom of the vessel, after some time begins to crawl along the bottom of the vessel with a velocity $u = 2.5$ m/s relative to it. How fast (in cm/s) will the vessel move along the floor ? Assume the water level to remain horizontal all the time. Take specific gravity of beetle to be 1.5 and $V_0/V = 100$.



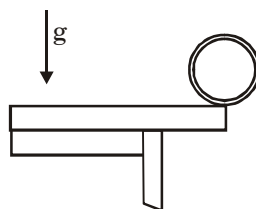
3. On rails forming a horizontal circular path of radius 10 m, a trolley of mass 20π kg rolls with speed of 5 m/s. The worker runs after it and begins to stop it, pulling the cable tied to the trolley with a force $F = 20$ N of constant magnitude, making a constant angle 120° to the direction of the speed of the trolley. How many turns will the trolley make in a circle before stopping? Neglect friction.



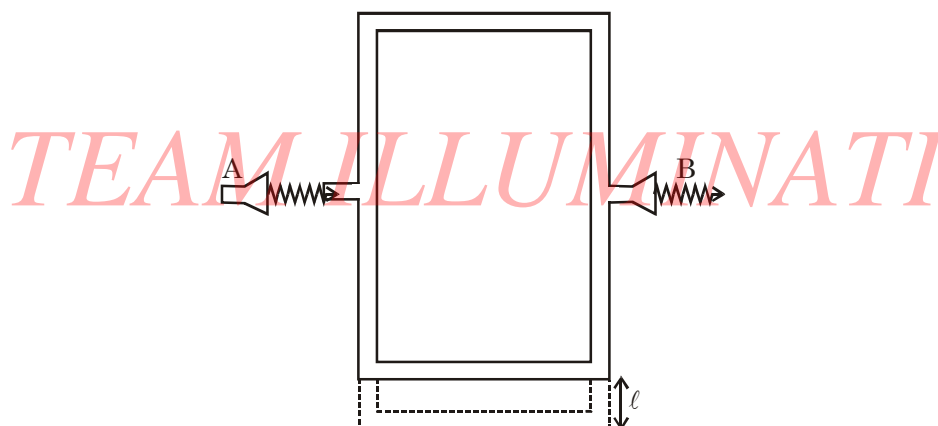
4. The wedge of mass M with angle α at the apex fits snugly to the vertical wall and rests on a bar of mass m located on the horizontal plane. The top of the wedge is at a height H above this plane, and the end of the wedge is at a height $h < H$ above the top surface of the bar. The bar is first held in this position, and then released. Find the speed of bar (in m/s) at the time of separation from the wedge. Neglect friction. [Take $\alpha = 30^\circ$, $m/M = 2$, $h = 2$ m, $H = 3$ m]



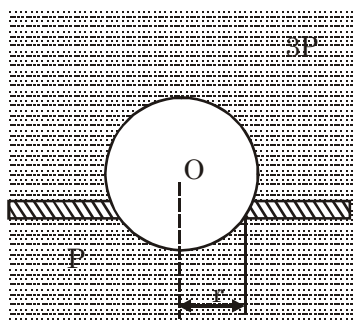
5. The center of the thin ring is just above the edge of the table. The ring begins to roll off the table without slipping from a state of rest. At what angle (in degrees) will the ring turn until it is separated from the edge of the table?



6. A disc of mass ' $m = 2 \text{ kg}$ ' and radius ' $R = 3 \text{ meter}$ ' is rotating with angular speed $\omega = 1 \text{ rad/s}$ on a frictionless horizontal surface about an axis perpendicular to and passing through the centre of disc. Find the force (in Newton) applied by one half of the disc on other.
7. The device for the demonstration of sound interference has first two identical - upper and lower - acoustic ducts. At what minimum distance ℓ (in meters) is it necessary to lower the lower acoustic circuit in order to maximally weaken the sound of the horn B at the frequency $\nu = 100 \text{ Hz}$. Take speed of sound to be 340 m/s inside the ducts.



8. The ball closes a hole of radius $r = 1 \text{ cm}$ in a flat wall separating the liquid, whose pressure is $3P$ and P . How hard (in Newton) is the ball pressed against the hole? [Take $P = 10^5 \text{ N/m}^2$]



PHYSICS

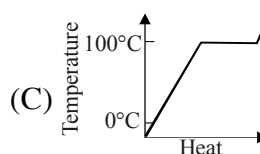
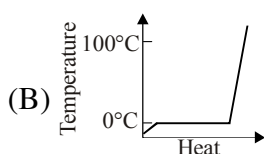
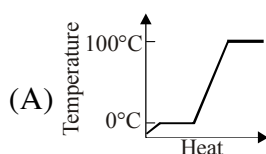
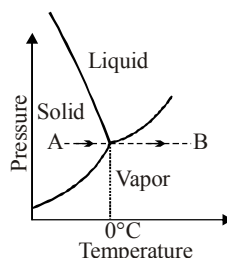
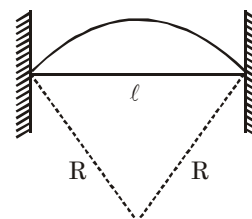
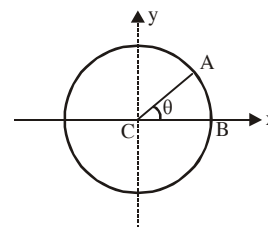
QUIZ # 41

SECTION-I

Single Correct Answer Type

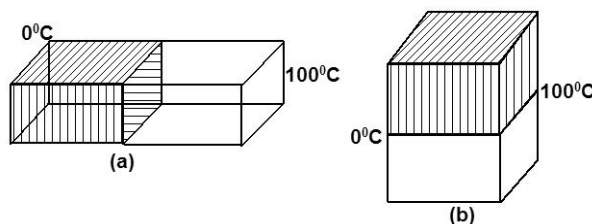
8 Q. [3 M (-1)]

- Two lines AC and BC are drawn on a circular disc as shown in figure. The disc is made of a material with linear expansion coefficients $\alpha_x > \alpha_y$. Here α_x and α_y are linear expansion coefficient for expansion in x-direction and y-direction respectively. C is the centre of disc. If the disc is heated uniformly. Angle ACB ($=\theta$) will
 (A) increase
 (B) decrease
 (C) remains constant
 (D) may increase or decreases depending on value of temperature increment
- A liquid of volumetric thermal expansion coefficient $= \gamma$ and bulk modulus B is filled in a spherical tank of negligible heat expansion coefficient. Its radius is R and wall thickness is 't' ($t \ll R$). When the temperature of the liquid is raised by θ , the tensile stress developed in the walls of the tank is :
 (A) $B\gamma\theta R/2t$ (B) $B\gamma\theta R/t$ (C) $2B\gamma\theta R/t$ (D) $B\gamma\theta R/4t$
- A liquid whose coefficient of volume expansion is γ_r completely fills a sealed metal tank, at negligible pressure. The coefficient of linear expansion of the metal is α , and the compressibility of the liquid is k. If the temperature of the system is increased by t, the pressure of the liquid will be
 (A) $\gamma_r \frac{t}{k}$ (B) $\frac{t}{k} (\gamma_r - 3\alpha)$ (C) $tk(\gamma_r - 3\alpha)$ (D) $\frac{t}{k} (\gamma_r - \alpha)$
- A rod of length ℓ , coefficient of thermal expansion α is heated by a temperature difference ΔT . It was constrained between two walls at a distance ℓ . As a result, it bends to form a circular arc. The radius of arc is :-
 (A) $\frac{\ell}{2\alpha\Delta T}$ (B) $\frac{\ell}{6\alpha\Delta T}$
 (C) $\frac{\ell}{2\sqrt{6\alpha\Delta T}}$ (D) $\frac{\ell}{3\sqrt{3\alpha\Delta T}}$
- The phase diagram for water is shown in figure. If the temperature of a certain amount of ice is increased by following the path represented by the dashed line from A to B in the phase diagram, which of the graphs of temperature as a function of heat added is correct ? Treat water vapour as an ideal gas.

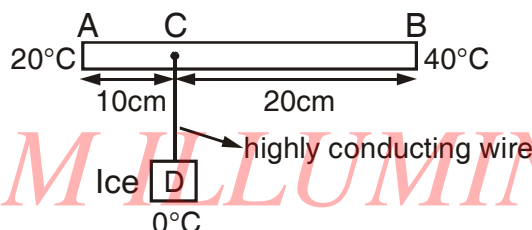


(D) None of these

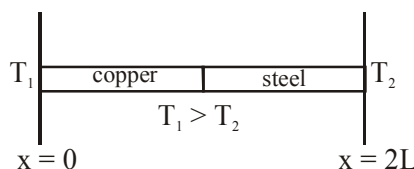
6. Two identical square rods of metal are welded end to end as shown in the Fig (a). Assume that 10cal of heat flows through the rods in 2min. Now the rods are welded as shown in the Fig.(b). The time it would take for 10cal to flow through the rods now, is



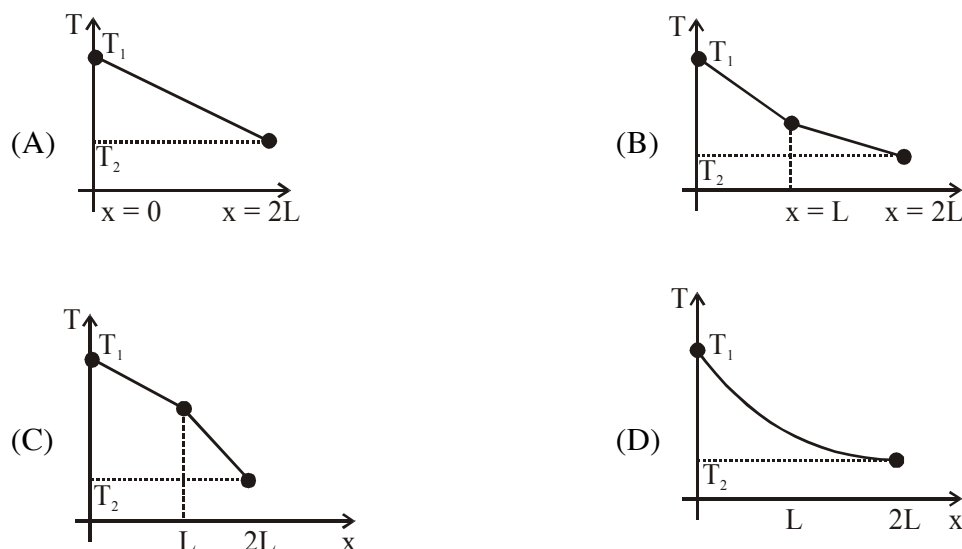
- (A) 0.75 min (B) 0.5 min (C) 1.5 min (D) 1 min
7. In the figure shown, AB is a rod of length 30 cm and area of cross-section 1.0 cm^2 and thermal conductivity 336 S.I. units. The ends A & B are maintained at temperatures 20°C and 40°C respectively. A point C of this rod is connected to a box D, containing ice at 0°C , through a highly conducting wire of negligible heat capacity. The rate at which ice melts in the box is : [Latent heat of fusion for ice $L_f = 80 \text{ cal/gm}$]



- (A) 84 mg/s (B) 84 g/s (C) 20 mg/s (D) 40 mg/s
8. A copper rod and a steel rod of equal cross-sections and lengths (L) are joined side by side and connected between two heat baths as shown in the figure.



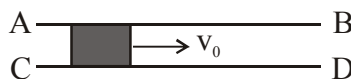
If heat flows through them from $x = 0$ to $x = 2L$ at a steady rate and conductivities of the metals are K_{cu} & K_{steel} ($K_{\text{cu}} > K_{\text{steel}}$), then the temperature varies as (convection and radiation are negligible)



Multiple Correct Answer Type

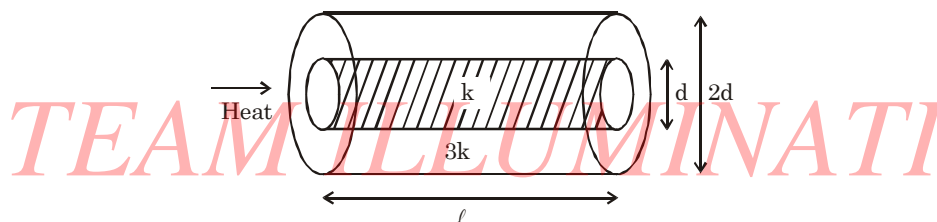
6 Q. [4 M (-1)]

9. In the figure, AB and CD are two vertical walls of a horizontal rectangular groove. A rectangular block of mass m just fits in the groove. Base of the block is smooth and coefficient of friction between walls and block is μ . Contact area of block with each wall is A . Linear coefficient of thermal expansion and young's modulus of material of block are α and Y respectively. Temperature θ of block increases uniformly as $\Delta\theta = t$ (t = time). Initial velocity of block is v_0 then

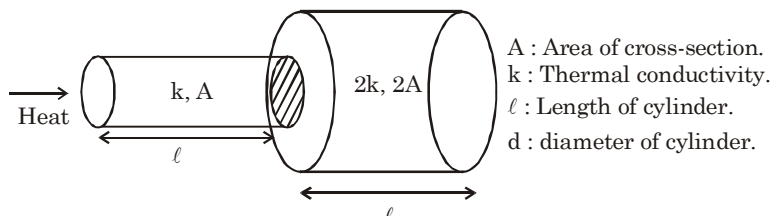


- (A) Retardation of block at time t is $\frac{\mu YA \alpha t}{m}$ (B) Velocity at time t is $\left(v_0 - \frac{\mu YA \alpha t^2}{m} \right)$
(C) Time taken by block to stop is $\sqrt{\frac{mv_0}{\mu YA \alpha}}$ (D) work done by friction is $\frac{1}{2}mv_0^2$

10. Different combination of cylinder are shown in the figure. Select the correct alternatives for the thermal resistance of the following combinations :



A thin cylinder of diameter 'd' is surrounded by another cylinder.



- (A) Thermal resistance of '1' combination is $\frac{2\ell}{5k\pi d^2}$.
(B) Thermal resistance of '2' combination is $\frac{4\ell}{5kA}$
(C) Thermal resistance of '1' combination is $\frac{5\ell}{2k\pi d^2}$
(D) Thermal resistance of '2' combination is $\frac{5\ell}{4kA}$

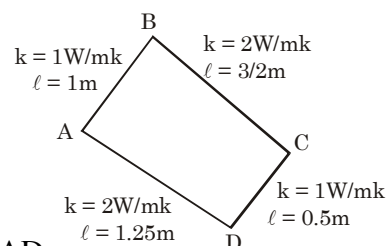
11. Consider 4 well lagged rods as shown. Point A is maintained at 100°C and C at 0°C . All rods have same cross sectional area :-

(A) Temperature of junction D = $\frac{400}{7}^\circ\text{C}$

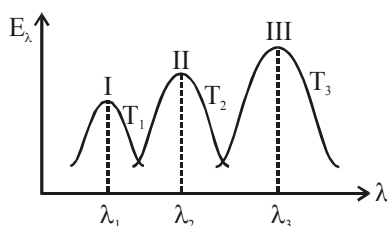
(B) Temperature of junction B = $\frac{300}{7}^\circ\text{C}$.

(C) Rate of heat flowing through AB is less than heat flowing through AD.

(D) Rate of heat flowing through AB is more than rate of heat flowing through AD.



12. A solid sphere and a hollow sphere of the same material and of equal radii are initially at the same temperature and are kept in surrounding which is at lower temperature.
(A) Both the spheres will emit thermal radiation at the same rate in the beginning
(B) Both the spheres will emit thermal radiation at equal rates at all instants of time
(C) The initial rate of cooling dT/dt will be the same for the two spheres
(D) Both the spheres will absorb thermal radiation from the surrounding at the same rates at all instants of time
13. Three identical bodies are at temperature T_1 , T_2 and T_3 having e_1 , e_2 and e_3 as their respective emissivities. The thermal spectrum obtained for them is as shown in the diagram. Choose the correct order of temperatures and emissivities:-



- (A) $T_1 > T_2 > T_3$ (B) $T_1 < T_2 < T_3$ (C) $e_1 < e_2 < e_3$ (D) $e_1 > e_2 > e_3$
14. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C . The cylinder temperature rises uniformly to 25°C in 5 min and finally becomes constant at 45°C . Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings,
(A) the rate of loss of heat of the cylinder to surrounding at 20°C is 2W
(B) the rate of loss of heat of the cylinder to surrounding at 45°C is 12W
(C) the rate of loss of heat of the cylinder to surrounding at 20°C is 5W.
(D) the rate of loss of heat of the cylinder to surrounding at 45°C is 30W.

Linked Comprehension Type (1 Para $\times$ 3Q.) [3 M (-1)]
(Single Correct Answer Type)

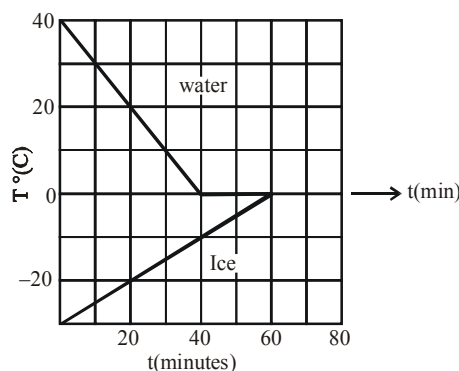
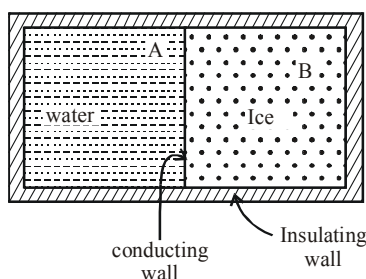
Paragraph for Questions Nos. 15 to 17

A 0.60 kg sample of water and a sample of ice are placed in two compartments A and B that are separated by a conducting wall, in a thermally insulated container. The rate of heat transfer from the water to the ice through the conducting wall is constant P , until thermal equilibrium is reached. The temperature T of the liquid water and the ice are given in graph as functions of time t . Temperature of the each compartment remain homogeneous during whole heat transfer process.

Given specific heat of ice = 2100 J/kg-K

Given specific heat of water = 4200 J/kg-K

Latent heat of fusion of ice = $3.3 \times 10^5 \text{ J/kg}$



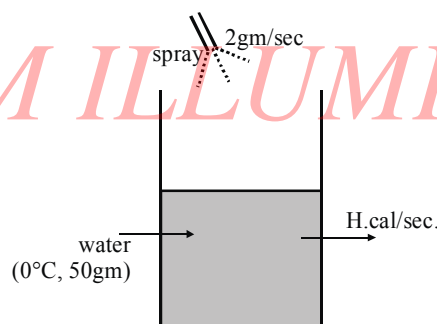
15. The value of rate P is :-
 (A) 42.0 W (B) 36.0 W (C) 21.0 W (D) 18.0 W
16. The initial mass of the ice in the container is equal to :-
 (A) 0.36 kg (B) 1.2 kg (C) 2.4 kg (D) 3.6 kg
17. The mass of the ice formed due to conversion from the water till thermal equilibrium is reached, is equal to :-
 (A) 0.12 kg (B) 0.15 kg (C) 0.25 kg (D) 0.40 kg

SECTION-II

Numerical Answer Type Question

1 Q.[3(0)]
(upto second decimal place)

1. A container (water equivalent of container is negligible) contains 50 gm of water at 0°C . If H amount of heat is extracted per second from water and keeping the spray closed, then whole water freezes in 10 sec. If spray is also working and sprays the water with 2 gm/sec at 0°C , then water freezes in t_1 second keeping the rate of extraction of heat from water same. Then calculate $6t_1$. (Latent heat capacity of ice = 80 cal/gm)



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. Emissivity e is a property of surface. Suppose for a surface emissivity e varies with kelvin temp. T as $e = CT$ (C is constant). If energy emission rate at temp. 600 K from the surface is 160 W, what will be the energy emission rate (in watt) at 300 K.
2. A lizard of mass 4g is warming it self in the bright sunlight. It's surface area is 4.2 cm^2 , but its body casts a shadow of area 1.4 cm^2 on a piece of paper held perpendicular to sun rays. The intensity of sun rays at earth is 1000 W/m^2 . Assume that specific heat capacity of lizard is same as that of water and its body behaves like a black body, also neglect the radiation from the lizard's body. How long (in minutes) must lizard lie in sun in order to raise it's temperature by 5°C . Specific heat capacity of water is $4200\text{ J/kg}^{\circ}\text{C}$.

If your answer is n fill value $\frac{n}{5}$.

PHYSICS

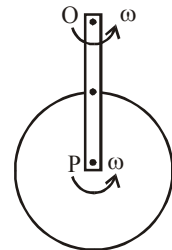
QUIZ # 42

SECTION-I

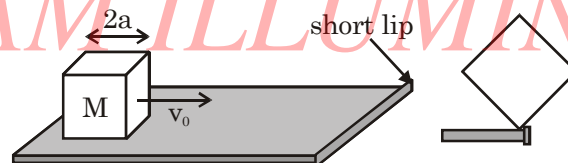
Single Correct Answer Type

11 Q. [3 M (-1)]

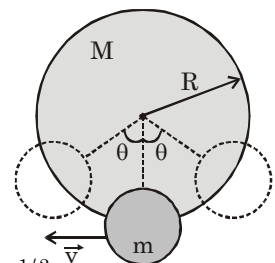
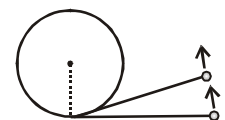
1. A rod of mass m and length $2R$ can rotate about an axis passing through O in vertical plane. A disc of mass m and radius R is hinged to the other end P of the rod and can freely rotate about P . When disc is at lowest point both rod and disc has angular velocity ω . If rod rotates by maximum angle $\theta = 60^\circ$ with downward vertical, the ω in terms of R and g will be (all hinges are smooth) :-



- (A) $\sqrt{\frac{9g}{16R}}$ (B) $\sqrt{\frac{3g}{23R}}$ (C) $\frac{1}{3}\sqrt{\frac{g}{R}}$ (D) None of these
2. If the law were changed so that traffic in India travelled on the right hand side of the road (instead of on the left), what would happen to the length of the day.
 (A) Slightly increase (B) Slightly decrease
 (C) Will remain unaltered (D) Can't be said
3. A cube with sides of length $2a$ and mass M is moving with an initial speed v_0 along a frictionless table. When the cube reaches the end of the table it is caught abruptly by a short lip and begins to rotate. The minimum speed v_0 such that the cube falls off the table is :-

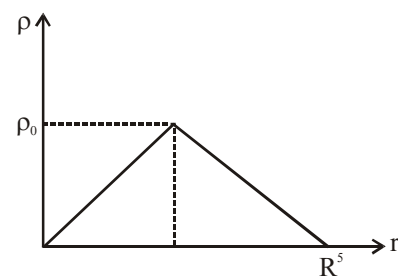


- (A) $\sqrt{\frac{16ag(\sqrt{2}-1)}{3}}$ (B) $\sqrt{16ag(\sqrt{2}-1)}$ (C) $\sqrt{ag(\sqrt{2}-1)}$ (D) $\sqrt{\frac{ag}{3}(\sqrt{2}-1)}$
4. A ball is attached to a string that is attached to a pole. When the ball is hit, the string wraps around the pole and the ball spiral inwards sliding on the Motionless smooth surface. Neglecting air resistance, what happens as the ball swings around the pole? Consider angular momentum about centre of pole.
 (A) The mechanical energy and angular momentum are conserved.
 (B) The angular momentum of ball is conserved but the mechanical energy of the ball increases.
 (C) The angular momentum of the ball is conserved and the mechanical energy of the ball decreases.
 (D) The mechanical energy of the ball is conserved and angular momentum of ball decreases
5. A smaller disk of radius r and mass m is attached rigidly to the face of a second large disk of radius R and mass M as shown in Figure. The center of the small disk is located at the edge of the large disk. The large disk is mounted at its center on a frictionless axle. The assembly is rotated through a small angle θ from its equilibrium position and released. The speed of the center of the small disk as it passes through the equilibrium position is



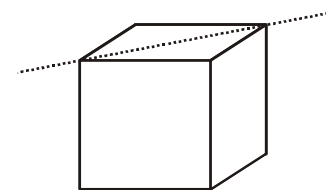
- (A) $v = 2 \left[\frac{Rg(1 - \cos \theta)}{(m/M) + (r/R)^2 + 1} \right]^{1/2}$ (B) $v = 2 \left[\frac{Rg(1 - \cos \theta)}{(m/M) + (R/r)^2 + 2} \right]^{1/2}$
- (C) $v = 2 \left[\frac{Rg(1 - \cos \theta)}{(M/m) + (r/R)^2 + 2} \right]^{1/2}$ (D) $v = 2 \left[\frac{Rg(1 - \cos \theta)}{(M/m) + (r/R)^2 + 1} \right]^{1/2}$

6. Variation of density of solid sphere of radius R with $(r)^5$ is shown in the diagram, where r is the distance from centre of the sphere. Moment of inertia of sphere about its symmetric axis is :-



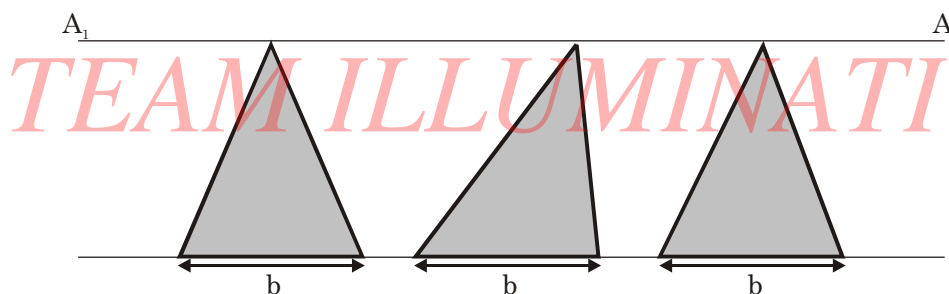
- (A) $\frac{1}{2} \pi \rho_0 R^5$ (B) $\frac{4}{15} \pi \rho_0 R^5$
(C) $\frac{4\pi}{3} \rho_0 R^5$ (D) $\frac{2\pi}{5} \rho_0 R^5$

7. The moment of inertia of a cube of mass m and side a about the diagonal of one of its face is :-

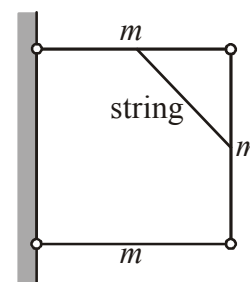


- (A) $\frac{ma^2}{12}$ (B) $\frac{ma^2}{6}$
(C) $\frac{5ma^2}{12}$ (D) $\frac{2ma^2}{3}$

8. Three plates of same thickness (negligible) and made of same material are kept as shown. Choose the correct option :-

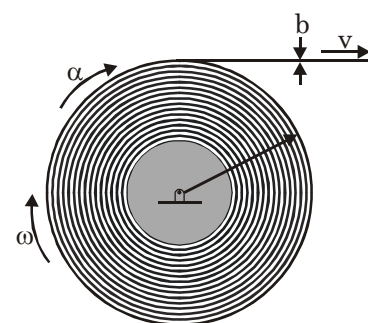


- (A) All must have same mass and moment of inertia about $A_1 A_2$
(B) All must have same mass and different moment of inertia about $A_1 A_2$
(C) All may have same mass and may have different moment of inertia about $A_1 A_2$
(D) all will have different mass and different moment of inertia.
9. Three sticks with mass m form three sides of a square, as shown in figure. They are connected with pivots to each other and to a wall. The midpoints of the top and right sticks are connected by a massless string. What is the tension in the string?



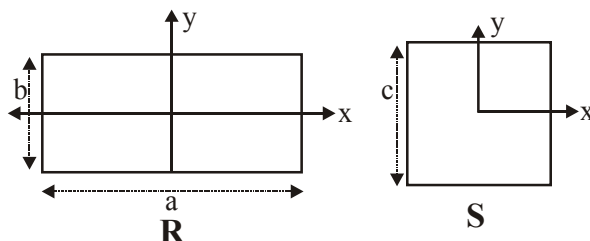
- (A) $4mg$ (B) $\sqrt{2}mg$
(C) $2\sqrt{2}mg$ (D) $4\sqrt{2}mg$

10. In a continuous printing process, paper is drawn into the press at a constant speed v . Denoting by ' r ' the radius of the paper roll at any given time and by ' b ' the thickness of the paper, what is the angular acceleration of the paper roll.



- (A) $\frac{bv^2}{2\pi r^3}$ (B) $\frac{2bv^2}{\pi r^3}$
(C) $\frac{bv^2}{4\pi r^3}$ (D) $\frac{4bv^2}{\pi r^3}$

11. A uniform square plate S (side c) and a uniform rectangular plate R (sides b, a) ($a > b$) have identical areas and masses figure. given (a) $I_{xR}/I_{xS} < 1$; (b) $I_{yR}/I_{yS} > 1$; (c) $I_{zR}/I_{zS} > 1$ which of the following is **CORRECT** :-



- (A) a, b (B) b, c (C) a, b, c (D) None

Multiple Correct Answer Type

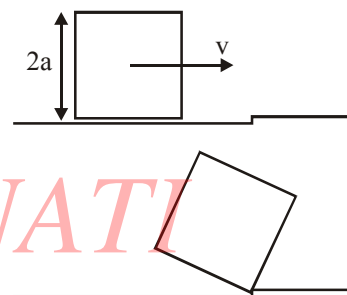
5 Q. [4 M (-1)]

12. A cube of mass m and side 2a is sliding with velocity v along a frictionless, horizontal floor. The cube hits a very low step head-on and the leading edge comes abruptly to rest such that the cube then rotates about the step. See the figure below.

- (A) Momentum, angular momentum, energy are conserved during the process

- (B) Angular velocity of cube after collision is $\frac{2v}{7a}$.

- (C) Angular velocity of cube after collision is $\frac{3v}{8a}$.



- (D) Minimum velocity required for cube to roll over in same direction of initial velocity is $\sqrt{\frac{16ag}{3}(\sqrt{2}-1)}$

13. A uniform disc of mass 'm' and radius 'R' is free to rotate in horizontal plane about a vertical smooth fixed axis passing through its centre. There is a smooth groove of small width along the radius of the disc and one small ball of same mass is attached to a spring as shown in figure. Initially spring is in its natural length of $\frac{R}{2}$. The disc is given initial angular velocity ω_0 and released (spring constant $k =$

$\frac{m\omega_0^2}{2}$). Then select correct statement(s) :

- (A) Angular velocity of disc when elongation in spring becomes $\frac{R}{2}$ will be $\frac{\omega_0}{2}$

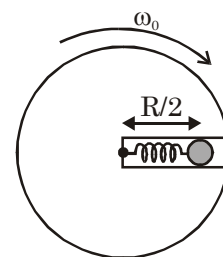
- (B) Radial velocity of ball when elongation in spring becomes $\frac{R}{2}$ will be $\frac{\omega_0 R}{2}$

- (C) Rate of change of energy of spring with respect to time when elongation in spring becomes $\frac{R}{2}$

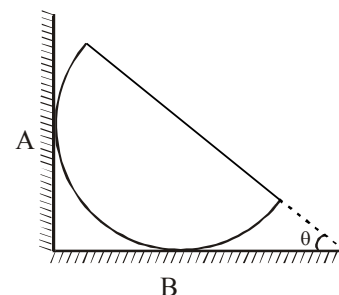
will be $\frac{m\omega_0^3 R^2}{8}$

- (D) Rate of change of energy of spring with respect to time when elongation in spring becomes

$\frac{R}{2}$ will be $\frac{m\omega_0^3 R^2}{2}$



14. A semi-circular disc rests in a vertical plane with its curved edge on rough horizontal and equally rough vertical surface. The coefficient of friction is μ for both surfaces, then
(A) friction acts upwards at A and towards left at B
(B) for greatest value of θ (for no slipping) normal contact force

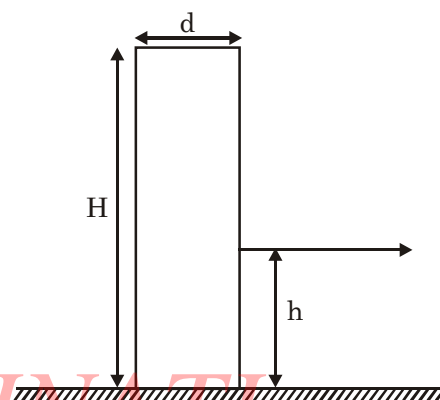


at A is $\mu \frac{w}{1+\mu^2}$, where w = weight of disc

(C) greatest value of θ will be $\sin^{-1} \left[\frac{3\pi \left(\frac{\mu + \mu^2}{1 + \mu^2} \right)}{4} \right]$

(D) None of the above

15. A curved wooden sculpture that can be taken as cylinder of height H and diameter d . A thread is attached at a point that is h height above the base, and the sculpture is placed in the middle of a rough horizontal table-top. The coefficient of friction between table top and the sculpture is $\mu_{\text{kinetic}} = \mu_{\text{static}} = \mu$. sculpture is to be pulled horizontally by the thread and drag it in one continuous movement to the edge of the table without toppling. sculpture acceleration must satisfy :-



(A) $a \geq \left(\frac{2\mu h - d}{H - 2h} \right) g$ (B) $a \leq \left(\frac{2\mu h - d}{H - 2h} \right) g$ (C) $a \leq \left(\frac{2\mu h + d}{H - 2h} \right) g$ (D) $a \geq \left(\frac{2\mu h + d}{H - 2h} \right) g$

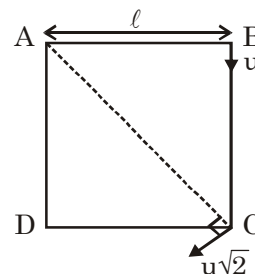
16. Velocities of some points of a square plate are shown, mass of square plate is ' m '. Choose **CORRECT** option(s).

(A) KE of plate is $\frac{mu^2}{3}$

(B) KE of plate is $\frac{mu^2}{6}$

(C) Plate is in state of pure rotation about point A

(D) Velocity of centre of plate is $\frac{u}{\sqrt{2}}$



Linked Comprehension Type (1 Para x 2Q.)
(Single Correct Answer Type)

[3 M (-1)]

Paragraph for Questions 17 & 18

Two cylinders with the same mass, the same length and the same external radius are initially resting on an inclined plane. The coefficient of sliding friction on the inclined plane, μ is known and has the same value for both the cylinders. The first cylinder is empty (tube) and the second is homogeneous solid cylinder. Both of them are released from the same height on the inclined plane.

17. What is the condition for angle θ of the inclined plane so that both cylinders will perform pure rolling:-

(A) $\mu \geq \tan \theta$

(B) $\mu \geq \frac{\tan \theta}{2}$

(C) $\mu \geq \frac{\tan \theta}{3}$

(D) $\mu \geq \frac{2 \tan \theta}{3}$

18. Choose the correct statement for the situation when the cylinders have reached the bottom of the inclined plane :- (Assume pure rolling)
- (A) The rotational kinetic energy of the first cylinder is more than that of the second cylinder but its translational kinetic energy is lesser.
- (B) The rotational kinetic energy of the first cylinder is less than that of the second cylinder but its translational kinetic energy is more.
- (C) The rotational kinetic energy as well as translational kinetic energy of the first cylinder is more than that of the second cylinder.
- (D) The rotational kinetic energy as well as translational kinetic energy of the first cylinder is less than that of the second cylinder.

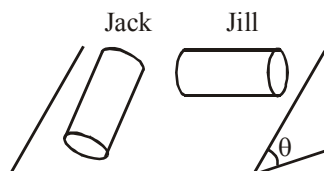
Linked Comprehension Type (1 Para × 2 Q.)
(Multiple Correct Answer Type)

[4 M (–1)]

Paragraph for Question no. 19 and 20

Jack and Jill,
 went up the hill,
 to fetch a pail of water.
 Jack fell down,
 and broke his crown,
 and Jill came tumbling after.

We have all heard this famous nursery rhyme in our school days. But does it give correct physics. Let us examine. We can assume the hill to be like an inclined plane of constant inclination θ . Assume that friction coefficient is μ . As Jack & Jill climbed it up, friction must be sufficient so that they do not slide down. At the top, some how jack fell down & started sliding down the hill like a solid cylinder and Jill come rolling after him with lesser acceleration. Both are assumed to be like solid cylinder.



19. The mistake(s) in the rhyme was :
- (A) If Jack has climbed up the hill, he can't slide down.
- (B) The acceleration of Jack should be less than acceleration of Jill.
- (C) The acceleration of Jack should be equal to the acceleration of Jill.
- (D) There are no mistakes.
20. Initially Jill climbed up the hill without slipping and after that she came rolling down the hill without sliding. What is/are possible value(s) of μ .
- (A) $\frac{\tan \theta}{2}$ (B) $2 \tan \theta$ (C) $\frac{4}{3} \tan \theta$ (D) $\frac{3}{4} \tan \theta$

PHYSICS

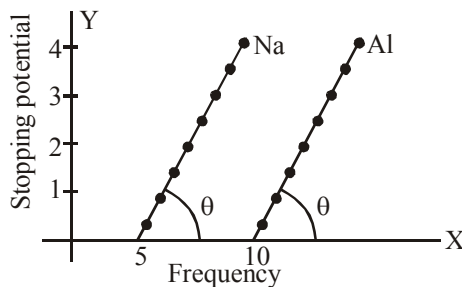
QUIZ # 43

SECTION-I

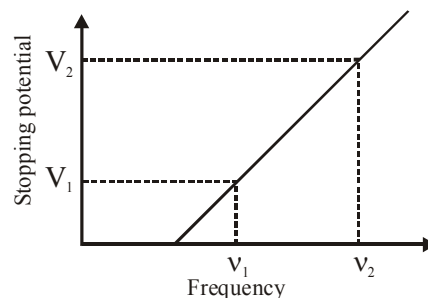
Single Correct Answer Type

18 Q. [3 M (-1)]

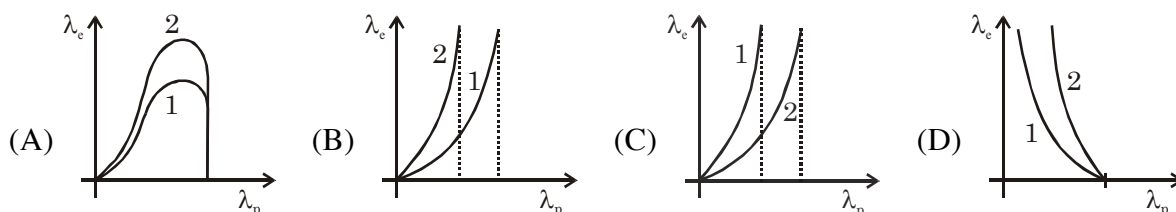
1. From the figure describing photoelectric effect we may infer correctly that



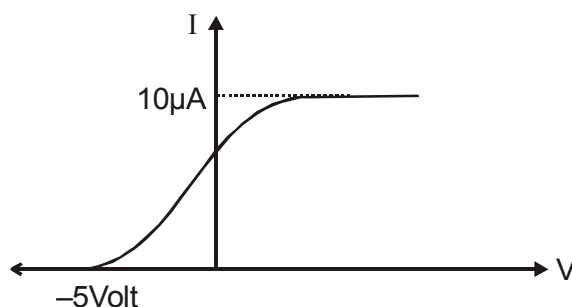
- (A) Na and Al both have the same threshold frequency
(B) Maximum kinetic energy for both the metals depends linearly on the frequency
(C) The change in stopping potentials are different for Na and Al for the same change in frequency
(D) Al is a better photo sensitive material than Na
2. An electromagnetic radiation which has an electric field given by $E = E_0 \sin \omega t + E_0 \sin (\omega t + \theta) + E_0 \sin (\omega t + 2\theta) + \dots + E_0 \sin (\omega t + N\theta)$ is incident on a metal plate having a constant work function. If photo electric emission takes place then
- (A) Maximum kinetic energy of emitted photoelectron depends on N
(B) Maximum kinetic energy of emitted photoelectron does not depend on N
(C) Number of photoelectrons emitted does not depend on N
(D) Both number of photoelectrons as well as maximum KE of emitted photoelectrons depend on N
3. Below figure shows the graph of stopping potential versus the frequency of a photosensitive metal. The work function of the metal is :



- (A) $\frac{(V_2 v_2 + V_1 v_1)e}{v_2 + v_1}$ (B) $\frac{(V_2 v_1 + V_1 v_2)e}{v_2 + v_1}$ (C) $\frac{(V_2 v_2 - V_1 v_1)e}{v_2 - v_1}$ (D) $\frac{(V_2 v_1 - V_1 v_2)e}{v_2 - v_1}$
4. In a photoelectric effect experiment, a graph was plotted between de broglie wavelength of fastest emitted photoelectron (λ_c) and wavelength of incident photons (λ_p). Two different emitters were used such that work function $\phi_1 > \phi_2$. Select the correct plot.



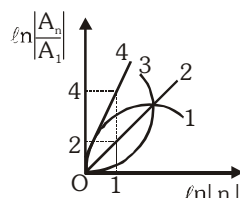
5. In the photoelectric experiment, if we use a monochromatic light, the $I - V$ curve is as shown. If work function of the metal is 2eV , estimate the power of light used. (Assume efficiency of photo emission = $10^{-3}\%$, i.e. number of photoelectrons emitted are $10^{-3}\%$ of number of photons incident on metal.)



- (A) 2 W (B) 5 W (C) 7 W (D) 10 W
6. In the photoelectric experiment, fastest electron has speed v if wavelength used is λ . If wavelength used is $\frac{3\lambda}{4}$, the speed of fastest electron will be :-

- (A) $\left(\frac{3}{4}\right)^{\frac{1}{2}} v$ (B) $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$ (C) Less than $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$ (D) Greater than $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$

7. The figure shows a graph between $\ell n \left| \frac{A_n}{A_1} \right|$ and $\ell n |n|$, where A_n is the area enclosed by the n^{th} orbit in a hydrogen like atom. The correct curve is-



- (A) 4 (B) 3 (C) 2 (D) 1
8. In the Bohr's model of hydrogen-like atom the force between the nucleus and the electron is modified

as $F = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right)$, where β is a constant. For this atom, the radius of the n^{th} orbit in terms of the

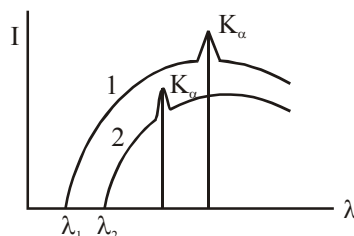
Bohr radius $\left(a_0 = \frac{\epsilon_0 h^2}{m\pi e^2} \right)$ is :

- (A) $r_n = a_0 n - \beta$ (B) $r_n = a_0 n^2 + \beta$ (C) $r_n = a_0 n^2 - \beta$ (D) $r_n = a_0 n + \beta$
9. If a physical quantity A associated with a hydrogen atom satisfies:

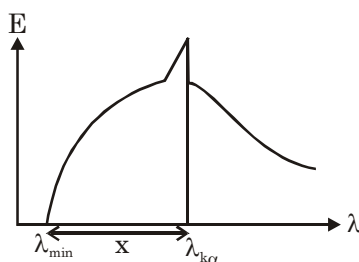
$\frac{A}{\sqrt{r}} = \text{constant}$, where r is the radius of the electronic orbit of the hydrogen atom (not necessarily in the ground state), then A could be :

- (A) wavelength of emitted radiation from the state
(B) frequency emitted radiation from the state
(C) Angular momentum of a revolving electron in the state
(D) Energy of the emitted radiation from the state

10. An electron is in second bohr orbit of hydrogen atom, acceleration of the electron is :-
 (A) $\frac{h^2}{\pi^2 m^2 r^3}$ (B) $\frac{h^2}{mr^2}$ (C) $\frac{h^2}{m^2 r^2}$ (D) $\frac{m^2 h^2}{\pi^2 r^2}$
11. When an electron in a hydrogen atom makes a transition from 2nd excited state to ground state, it emits a photon of frequency f . The frequency of photon emitted when an electron of Li^{++} makes a transition from 1st excited state to ground state is :-
 (A) $\frac{243}{32}f$ (B) $\frac{81}{32}f$ (C) $\frac{243}{8}f$ (D) $\frac{27}{8}f$
12. Magnetic field at the centre of hydrogen like atom (atomic number = Z) due to motion of electron in n^{th} orbit is proportional to
 (A) $\frac{n^3}{Z^5}$ (B) $\frac{n^4}{Z}$ (C) $\frac{Z^2}{n^3}$ (D) $\frac{Z^3}{n^5}$
13. A diatomic molecule is made of two masses m_1 and m_2 which are separated by a distance r . If we calculate its rotational energy by applying Bohr's rule of angular momentum quantization, its rotational kinetic energy will be given by (n is an integer).
 (A) $\frac{(m_1 + m_2)h^2 n^2}{4\pi^2 m_1 m_2 r^3}$ (B) $\frac{n^2 h^2}{2(m_1 + m_2)r^2}$ (C) $\frac{2n^2 h^2}{(m_1 + m_2)r^2}$ (D) $\frac{(m_1 + m_2)n^2 h^2}{8\pi^2 m_1 m_2 r^2}$
14. When two different materials A and B having atomic number Z_1 and Z_2 are used as the target in Coolidge X-ray tube at different operating voltage V_1 and V_2 respectively their spectrums are found as below. The correct relation is :



- (A) $V_1 > V_2$ and $Z_1 > Z_2$ (B) $V_1 < V_2$ and $Z_1 < Z_2$
 (C) $V_1 < V_2$ and $Z_1 > Z_2$ (D) $V_1 > V_2$ and $Z_1 < Z_2$
15. The k_α wavelength of an element with atomic number z is λ_z . The k_α wavelength of an element with atomic number $2z$ is λ_{2z} . How are λ_z and λ_{2z} related ?
 (A) $\lambda_z > 4\lambda_{2z}$ (B) $\lambda_z = 4\lambda_{2z}$
 (C) $\lambda_z < 4\lambda_{2z}$ (D) Depending on z , λ_z could be greater than or less than $4\lambda_{2z}$
16. Difference between λ_{k_α} and $\lambda_{\min}$ become twice if accelerating potential changes from 3100 volt to 12400 volt. Find λ_{k_α} (Given $hc = 12400 \text{ eV}\text{\AA}$) :-



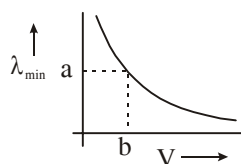
- (A) $6 \text{\AA}$ (B) $5 \text{\AA}$ (C) $7 \text{\AA}$ (D) $5.5 \text{\AA}$

17. From Coolidge tube experiment, X-rays are used to detect fracture present in bones. The wavelength range required to detect fracture is $1.24 \text{ \AA}$ to $2.48 \text{ \AA}$. Find minimum accelerating voltage required to operate given X-ray machine : (Given : $hc = 12400 \text{ eV \AA}$)
(A) 10000 Volt (B) 5000 Volt (C) 10520 Volt (D) 7440 Volt
18. The constant 'a' in Mosely's law $\sqrt{\nu} = a(Z - b)$ for K_{α} X-ray is related to Rydberg constant R as ('c' is the speed of light)
(A) $R = \frac{4a^2}{3c}$ (B) $R = \frac{3c}{4a}$ (C) $R = \frac{3c^2}{4}$ (D) $R = \frac{2c}{a}$

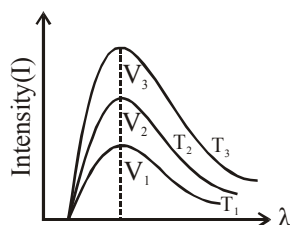
Multiple Correct Answer Type

2 Q. [4 M (-1)]

19. $\lambda_{\min}$ versus accelerating voltage V graph is shown for a Coolidge tube.



- (A) graph is parabolic (B) graph is hyperbolic
(C) $hc = aeb$ (D) $\lambda_{\min} = \frac{hc}{eV}$
20. The continuous X-ray spectrum for the three potential differences between target and cathode, V_1 , V_2 and V_3 is shown. Temperatures of filament in these cases are T_1 , T_2 and T_3 respectively, then :



- (A) $V_1 > V_2 > V_3$ (B) $T_1 > T_2 > T_3$ (C) $V_1 = V_2 = V_3$ (D) $T_1 < T_2 < T_3$

PHYSICS

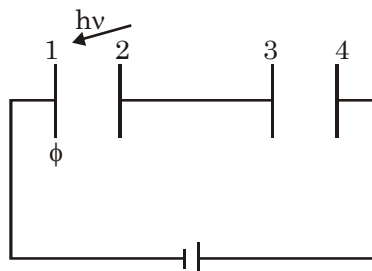
QUIZ # 44

SECTION-I

Single Correct Answer Type

11 Q. [3 M (-1)]

- A xenon arc lamp is covered with an interference filter that only transmits light of 400 nm wavelength. The transmitted light strikes a metal surface, a stream of electrons emerges from the metal. The filter is then replaced with one transmitting at 300 nm and the lamp adjusted so that the intensity of striking the surface is the same as it was for the 400 nm light. With the 300 nm light :-
 (A) More electrons are emitted in a given time interval
 (B) The electrons which are emitted are more energetic
 (C) Both are true
 (D) Both are false
- An experiment is carried out in which light is shone on a photocell whose plate (cathode) has a work function of 1.5 eV. Two series of measurements were performed. For the first measurement, light has a wavelength 414 nm and intensity I and for the second measurement light has a wavelength of 517.5 nm and intensity 3I. The ratio of stopping potential for the wavelength is : (Take : $h = 4.14 \times 10^{-15} \text{ eVs}$)
 (A) $\frac{4}{3}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) $\frac{3}{2}$
- Two identical capacitors are arranged as shown. Work function of plate 1 is ϕ eV. The emf of battery is $\frac{\phi}{e}$. If energy of incident photon is $h\nu$ then maximum kinetic energy of e^- reaching plate 2 is :-

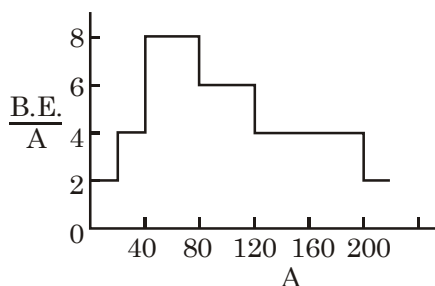


- (A) $h\nu - \phi/2$ (B) $h\nu - \phi$ (C) $h\nu - 2\phi$ (D) $h\nu$
- The work functions of metals A and B are in the ratio 1 : 2. If light of frequencies f and $2f$ are incident on the surfaces of A and B respectively, the ratio of the maximum kinetic energies of photoelectrons emitted is (f is greater than threshold frequency of A, $2f$ is greater than threshold frequency of B) :-
 (A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 1 : 4
- In a photo-electric experiment light of wavelength 400 nm was used. Work function of the metal is 2.1 eV. The ejected electrons are accelerated through potential of 5V and then enters a transverse magnetic field. The ratio of maximum to minimum possible radius of electrons in magnetic field is :
 (A) $\sqrt{\frac{3.1}{2}}$ (B) $\sqrt{\frac{5}{3.1}}$ (C) $\sqrt{\frac{6}{5}}$ (D) $\sqrt{\frac{31}{21}}$
- A radioactive nucleus (initial mass number A and atomic number Z) emits 2 α -particles and 3 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be:-
 (A) $\frac{A-Z-1}{Z-7}$ (B) $\frac{A-Z-1}{Z-4}$ (C) $\frac{A-Z-4}{Z-8}$ (D) $\frac{A-Z-12}{Z-4}$
- A nucleus (of nuclear density ρ) disintegrates into two daughter nuclei with masses in the ratio 8 : 27. Density of the smaller nucleus is :-
 (A) $\frac{2}{3}\rho$ (B) $\frac{2}{5}\rho$ (C) $\frac{8}{27}\rho$ (D) ρ

8. The equation $x \rightarrow y + \alpha$ represents a decay process. If x is at rest just before decay and k_1 and k_2 represent the kinetic energy of y and α having mass m_2 and m_1 respectively, then

(A) $\frac{k_1}{k_2} = \frac{m_1}{m_2}$ (B) $\frac{k_1}{k_2} = \frac{m_1^2}{m_2^2}$ (C) $\frac{k_1}{k_2} = \sqrt{\frac{m_2}{m_1}}$ (D) None of these

9. The following histogram represents the binding energy per particle (B.E./A) in MeV as a function of the mass number A of a nucleus.



A nucleus with mass number $A = 180$ fissions into two nuclei of equal masses. In the process.

- (A) 180 MeV of energy is released (B) 180 MeV of energy is absorbed
 (C) 360 MeV of energy is released (D) 360 MeV of energy is absorbed
10. When the nucleus of a radium-226, which is at rest, decays, an α particle and the nucleus of radon are created. The released energy during the decay is 4.87 MeV, which appears as the kinetic energy of the two resulted particles. Calculate the kinetic energy of α particle and radon nucleus.
- (A) 4.78 MeV and 0.09 MeV (B) 4.67 MeV and 0.2 MeV
 (C) 4.84 MeV and 0.03 MeV (D) 4.81 MeV and 0.06 MeV
11. The mass of certain nuclei is given as follows:

$${}_{84}^{210}\text{Po} : 209.982857 \text{ amu}$$

$${}_{82}^{206}\text{Pb} : 205.974449 \text{ amu}$$

$${}_{83}^{209}\text{Bi} : 208.980383 \text{ amu}$$

$${}_{83}^{210}\text{Bi} : 209.984105 \text{ amu}$$

$${}_{84}^{209}\text{Po} : 208.982416 \text{ amu}$$

$${}_{85}^{210}\text{At} : 209.987131 \text{ amu}$$

$$m_n = 1.007276 \text{ amu}$$

$$m_p = 1.008665 \text{ amu}$$

$$m_e = 0.000549 \text{ amu}$$

$$m_\alpha = 4.002603 \text{ amu}$$

Which nuclear reaction can ${}_{84}^{210}\text{Po}$ undergo ?

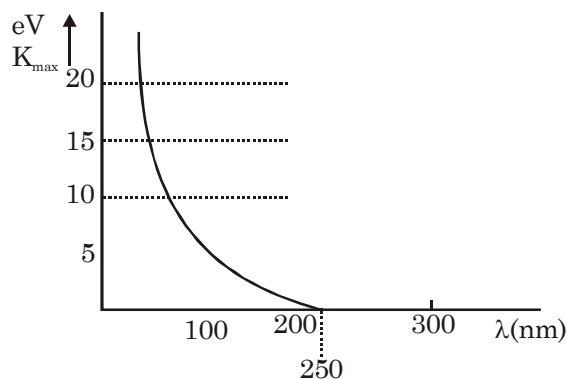
- (1) α decay (2) β^- decay (3) β^+ decay (4) k capture
 (A) all 4 (B) 1 only (C) 1, 2 & 4 only (D) 1 & 3 only

Multiple Correct Answer Type

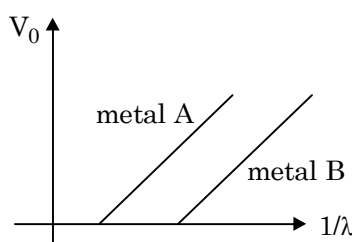
5 Q. [4 M (-1)]

12. In a photoelectric-effect experiment, if ' f ' is the frequency of incident radiations on the metal surface and I is the intensity of the incident radiations, then :-
- (A) If ' f ' is increases, keeping I and work function constant, then stopping potential and maximum kinetic energy of photoelectron increases.
 (B) If distance between cathode and anode increases, stopping potential remains same.
 (C) If ' I ' is increased keeping ' f ' and work function constant, saturation current increases but stopping potential remains same.
 (D) Work function is decreased keeping f and I constant. Then, stopping potential and maximum kinetic energy of photoelectrons increases.

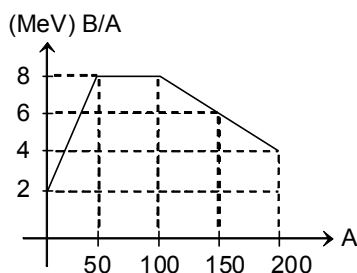
13. In a photoelectric effect experiment, the maximum kinetic energy of the ejected photoelectrons is measured for various wavelengths of the incident light. Figure shows a graph of this maximum kinetic energy $K_{\max}$ as a function of the wavelength λ of the light falling on the surface of the metal. Which of the following statement/s is/are **CORRECT** ?



- (A) Threshold frequency for the metal is 1.2×10^{15} m
(B) Work function of the metal is 4.968 eV
(C) Maximum kinetic energy of photoelectrons corresponding to light of wavelength 100 nm is nearly 7.4 eV
(D) Photoelectric effect takes place with red light ($\lambda \approx 700$ nm)
14. In an experiment on photoelectric effect, graph of stopping potential V_0 against reciprocal of wavelength $1/\lambda$ of the incident light for two different metals A and B are as shown. This shows that :-



- (A) work function of metal A is smaller than that of metal B
(B) threshold frequency for A is smaller than that of metal B
(C) when light of a certain wavelength is incident on both metals maximum kinetic energy of photoelectrons emitted from A will be greater than that of photoelectrons emitted from B
(D) metals A and B have the same threshold frequency
15. If a graph of Binding energy (B/A) per nucleon versus mass number (A) looks like as shown in figure. Then using the curve choose **CORRECT** option(s).



- (A) Fusion of two nuclei with mass numbers 30 and 45 will have Q value 99 MeV
(B) Fission of nuclei with mass number 80 into two nuclei of equal mass number will have Q value 256 MeV
(C) Fission of nuclei with mass number 150 into two nuclei of equal mass number will release energy
(D) Fission of nuclei with mass number 80 into two nuclei of equal mass number will release energy

16. Choose the **CORRECT** option(s) :

- (A) Strong nuclear force is a spin dependent force.
- (B) Strong nuclear force between an electron and proton is same as between a proton and neutron.
- (C) Strong nuclear force is a short range force.
- (D) Strong nuclear force is always attractive in nature.

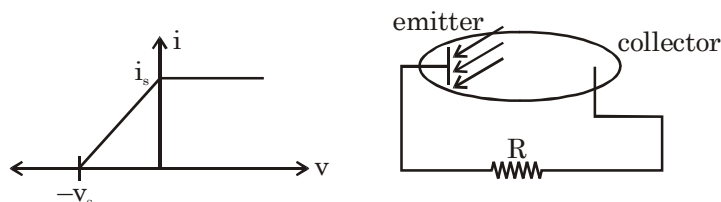
Linked Comprehension Type

(2 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

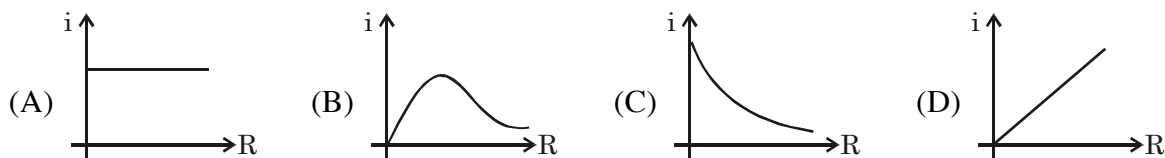
Paragraph for Question no. 17 and 18

Let us assume that the current voltage graph for a photoelectric tube consists of two straight lines as shown.



On such a photoelectric tube light of wavelength 310 nm is incident. The work function of the emitter plate is 2eV. The tube effectively acts as a solar cell to supply current to an external resistor R. Saturation current is 2A.

17. How does the current depend on R :-



18. At what value of R will be the power dissipated in R maximum :-

- (A) 1Ω
- (B) 2Ω
- (C) $\frac{1}{2}\Omega$
- (D) $\frac{1}{4}\Omega$

Paragraph for question nos. 19 and 20

The first nuclear reaction ever observed was by Ernest Rutherford in 1919. It was triggered by α -particles incident on an isotope of nitrogen $^{14}_7\text{N}$. He observed a proton was emitted along with another element x. Let us assume that $^{14}_7\text{N}$ nucleus was initially stationary. For this reaction to occur, α -particle must touch the nitrogen nucleus. The distance between their centres at this moment is d. For this problem, we will neglect the effect of outer electrons in $^{14}_7\text{N}$. Symbols have their usual meanings.

19. Value of d is

- (A) $R_0(2^{1/3} + 7^{1/3})$
- (B) $R_0(2^{2/3} + 7^{2/3})$
- (C) $R_0(2^{2/3} + 14^{2/3})$
- (D) $R_0(2^{2/3} + 14^{1/3})$

20. The minimum initial kinetic energy of α -particle so that reaction can occur is :-

- (A) $\frac{18ke^2}{R_0(2^{2/3} + 14^{2/3})}$
- (B) $\frac{14ke^2}{R_0(2^{2/3} + 14^{1/3})}$
- (C) $\frac{18ke^2}{R_0(2^{2/3} + 14^{1/3})}$
- (D) $\frac{14ke^2}{R_0(2^{2/3} + 7^{2/3})}$

PHYSICS

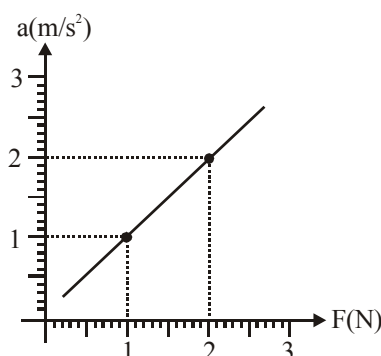
QUIZ # 45

SECTION-I

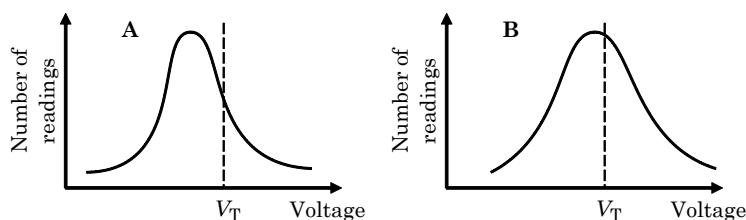
Single Correct Answer Type

9 Q. [3 M (-1)]

1. In an experiment to determine the inertial mass of an object using Newton's second law, following graph is obtained between net force on the object and the acceleration produced in it. The mass of the object within error limits is :-



- (A) 1.0 kg (B) 1 kg (C) (1.0 ± 0.1) kg (D) (1.0 ± 0.15) kg
2. In an experiment to determine the radius of a chalk by screw gauge, the diameter is measured and readings are $d_1 = 1.002$ cm, $d_2 = 1.004$ cm and $d_3 = 1.006$ cm. Select the correct alternatives:-
- (A) Mean absolute error in radius is 0.0013 cm
 (B) Mean absolute error in diameter is 0.0013 cm
 (C) Error is 0 cm
 (D) % age error in the measurement of diameter is 1.3%
3. Two voltmeters A and B are used to measure a voltage across a resistance. Several readings are taken and represented in the following graphs. Here V_T is the true value of the voltage being measured. Which of the following statements is/are correct?

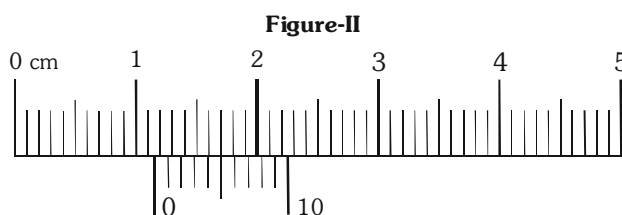
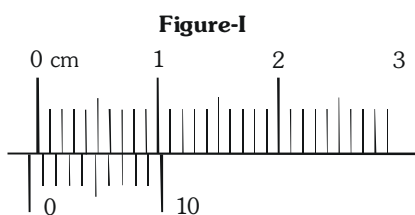


- (A) Voltmeter A is more precise but less accurate than voltmeter B.
 (B) Voltmeter A is less precise but more accurate than voltmeter B.
 (C) Voltmeter A is more precise as well as more accurate than voltmeter B.
 (D) Voltmeter A is less precise as well as less accurate than voltmeter B.
4. A screw gauge having 5 main scale division = 0.1 cm is designed to have a least count of 0.0002 cm. If n be the number of divisions on circular scale and m be the pitch, then
- (A) $n = 10$, $m = 0.02$ cm (B) $n = 90$, $m = 0.1$ cm
 (C) $n = 100$, $m = 0.04$ cm (D) $n = 100$, $m = 0.02$ cm

5. The length of a cylinder is measured with a meter rod having least count 0.1 cm . Its diameter is measured with vernier calipers having least count 0.01 cm . Given that length is 5.0 cm . and radius is 2.0 cm . The percentage error in the calculated value of the volume will be
 (A) 1% (B) 2% (C) 2.5% (D) 4%
6. A physical parameter a can be determined by measuring the parameters b, c, d and e using the relation $a = \frac{b^\alpha c^\beta}{d^\gamma e^\delta}$. If the maximum errors in the measurement of b, c, d and e are $b_1\%, c_1\%, d_1\%$ and $e_1\%$, then the maximum error in the value of a determined by the experiment is :-
 (A) $(b_1 + c_1 - d_1 - e_1)\%$ (B) $(b_1 + c_1 + d_1 + e_1)\%$
 (C) $(\alpha b_1 + \beta c_1 + \gamma d_1 + \delta e_1)\%$ (D) $(\alpha b_1 + \beta c_1 - \gamma d_1 - \delta e_1)\%$
7. A student performs an experiment to determine the Young's modulus of a wire, exactly 2m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of $\pm 0.04 \text{ mm}$ at a load of exactly 1.0 kg . The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of $\pm 0.01 \text{ mm}$. Take $g = 10 \text{ m/s}^2$ (exact). The Young's modulus obtained from the reading is
 (A) $(2.0 \pm 0.3) \times 10^{11} \text{ N/m}^2$ (B) $(2.0 \pm 0.2) \times 10^{11} \text{ N/m}^2$
 (C) $(2.0 \pm 0.1) \times 10^{11} \text{ N/m}^2$ (D) $(2.0 \pm 0.05) \times 10^{11} \text{ N/m}^2$
8. A wire has a mass $(0.3 \pm 0.003) \text{ g}$, radius $(0.5 \pm 0.005) \text{ mm}$ and length $(6 \pm 0.06) \text{ cm}$. Then maximum % error in measurement of density is :
 (A) 1 (B) 2 (C) 3 (D) 4
9. Intensity observed in an interference pattern $I = I_0 \sin^2 \theta$. At $\theta = 30^\circ$, intensity $I = 5 \pm 0.002$. The percentage error in angle if $I_0 = 20 \text{ W/m}^2$ is
 (A) $4\sqrt{3} \times 10^{-2}\%$ (B) $\frac{4}{\pi} \times 10^{-2}\%$ (C) $\frac{4\sqrt{3}}{\pi} \times 10^{-2}\%$ (D) $\sqrt{3} \times 10^{-2}\%$

Multiple Correct Answer Type**6 Q. [4 M (-1)]**

10. In ordinary Vernier calipers, 10^{th} division of the Vernier scale coincides with 9^{th} division of the main scale. In a specially designed Vernier calipers the Vernier scale is so constructed that 10^{th} division on it coincides with 11^{th} division on the main scale. Each division on the main scale equals to 1 mm . The calipers have a zero error as shown in the figure-I. When the Vernier caliper is used to measure a length, the concerned portion of its scale is shown in figure-II.



- (A) Zero error in the calipers has magnitude 0.7 mm .
 (B) The length being measured is 1.08 cm .
 (C) The length being measured is 1.22 cm .
 (D) Though the given Vernier scale does not follow the principle of Vernier scale, yet can be used satisfactorily.

11. Select the **CORRECT** statement(s) :
- (A) Accuracy of measurement by screw gauge can be increased by making more divisions on circular scale
 (B) If N division on main scale is equal to (N + m) divisions on vernier scale, then accuracy of measurement can be increased by taking m smallest
 (C) Zero errors can be minimised by taking large number of readings.
 (D) Zero error in a given instrument always have a fixed value.
12. While measuring length of small iron rod, following five readings were obtained.
 12.66 mm, 12.58 mm, 12.62 mm, 12.80 mm and 12.72 mm
 The experimenter reported the result of the experiment based on the above readings.
 (A) Number of significant figures in the length reported will be 3.
 (B) Number of significant figures in the length reported will be 4.
 (C) Number of significant figures in the error reported will be 1.
 (D) Number of significant figures in the error reported will be 2.
13. The resistance of a copper wire is given by $R = R_0 [1 + \alpha(\theta - 20)]$ where $R_0 = (4.00 \pm 0.01)\Omega$ is the resistance of wire at $\theta = 20^\circ\text{C}$. The value of resistance coefficient of copper is $(4 \times 10^{-3} \pm 1 \times 10^{-4})/^\circ\text{C}$. It is connected with a battery of emf $(10.0 \pm 0.1)\text{ V}$.
 (A) The resistance R at $\theta = 25^\circ\text{C}$ is $(4.08 \pm 0.02)\Omega$
 (B) The resistance R at $\theta = 50^\circ\text{C}$ is $(4.48 \pm 0.03)\Omega$
 (C) The percentage error in calculation of power loss in resistance at $\theta = 25^\circ\text{C}$ is 3%
 (D) The percentage error in calculation of power loss in resistance at $\theta = 50^\circ\text{C}$ is 4%
14. The length of a strip measured with a meter rod is 10.0 cm. Its width measured with a vernier callipers is 1.00 cm. The least count of the meter rod is 0.1 cm and that of vernier callipers is 0.01 cm. What can be the error in its area ?
 (A) $+0.01\text{ cm}^2$ (B) -0.1 cm^2 (C) $+0.2\text{ cm}^2$ (D) -0.2 cm^2
15. A projectile is thrown with velocity $U = 20\text{m/s} \pm 5\%$ at an angle 60° . If the projectile falls back on the ground in the same level then find out the range of the projectile as
 $R = \dots\dots X \dots\dots \text{m} \pm \dots\dots Y \dots\dots \%$. Consider $g = 10\text{m/s}^2$.
 (A) $X = 34.6$ (B) $X = 25.3$ (C) $Y = 5$ (D) $Y = 10$

SECTION-II
Numerical Answer Type Question
(upto second decimal place)
5Q.[3(0)]

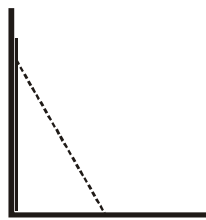
1. In a searle's experiment, the diameter of the wire as measured by a screw gauge is 0.050 cm. Pitch of screw gauge is 2 mm and number of divisions on circular scale are 200. The length, measured by a scale of least count 0.1 cm is 100 cm. When a weight of 50 N is suspended from the wire elongation is measured to be 0.125 cm by a micrometer of least count 0.001 cm. Find the maximum percentage error is the measurement of young's modulus of the material of the wire from these data.
2. A screw gauge has a screw having 2 threads in 1 mm. The circular scale has 50 divisions. Find the diameter (in mm) of wire, if the main scale shows 6th division and the vernier reads 46.
3. Consider the following data:
 10 main scale division = 1 cm, 10 vernier division = 9 main scale divisions. Zero of vernier scale is to the right of the zero marking of the main scale with 6 vernier divisions coinciding with main scale divisions. The measured reading for length measurement is 4.3 cm on main scale with 2 vernier divisions coinciding with main scale graduations. Estimate the actual length (in cm).
4. The length of a rectangular sheet is 1.5 cm and breadth is 1.203 cm. The area (in cm^2) of the face of rectangular sheet to the correct number of significant figures is :
5. A physical quantity $y = \frac{a^4 b^2}{(cd)^{1/3}}$ has four variables a, b, c and d. The percentage error in a, b, c and d are 2%, 3%, 4% and 5% respectively. The error (in %) in y will be

SECTION-I

Multiple Correct Answer Type

10 Q. [4 M (–1)]

1. A uniform rod of length ℓ leans against a frictionless wall. The ground is also frictionless. The rod is initially held motionless, with its bottom end an infinitesimal distance from the wall. The rod is then released, whereupon the bottom end begins to slide away from the wall, and the top end begins to slide down the wall.



(A) Rod loses contact with the wall when it makes an angle $\sin^{-1}\left(\frac{2}{3}\right)$ with the vertical.

(B) Rod loses contact with the wall when it makes an angle $\cos^{-1}\left(\frac{2}{3}\right)$ with the vertical.

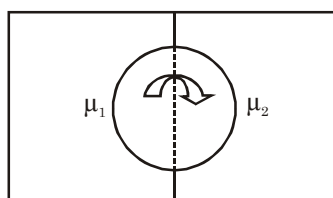
(C) Horizontal component of velocity of centre of mass after a long time is $\frac{\sqrt{g\ell}}{3}$

(D) Horizontal component of velocity of centre of mass after a long time is $\sqrt{g\ell}$

2. A body is projected with speed v in a medium where it experiences drag force $F = bv^a$ ($a > 0$). For what value(s) of a will it move infinitely large distance ?

(A) 1 (B) 2 (C) 3 (D) 4

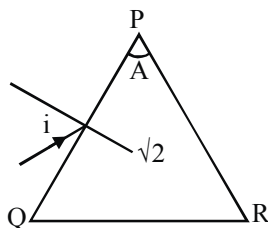
3. A horizontal tabletop is divided into two parts that have different surface texture. A thin uniform disk spinning clockwise about its axis is placed flat on the tabletop. The coefficients of sliding friction between the disk and the two sides of the table are μ_1 and μ_2 as shown ($\mu_1 > \mu_2$). Initially, the center of the disk is located right over the line dividing the tabletop.



- (A) Centre of mass of disk will remain at rest.
 (B) Initial acceleration of centre of mass of disk is towards left.
 (C) Initial acceleration of centre of mass of disk is downwards.

(D) Magnitude of acceleration of centre of mass is $\frac{g}{\pi}(\mu_1 - \mu_2)$

4. A prism of refractive index $\sqrt{2}$ and apex angle A is shown. Light is incident from PQ side at angle of incidence i ($0 < i \leq 90^\circ$) :-



- (A) if $A = 40^\circ$ then light incident at all angles will emerge from surface PR .
(B) if $A = 80^\circ$ then light incident at some angles will emerge from surface PR and at some other angles light will suffer TIR at surface PR .
(C) if $A = 100^\circ$ then light incident at all angles will be reflected back from surface PR .
(D) whatever is the value of A , light will emerge from the surface PR for some value of i .

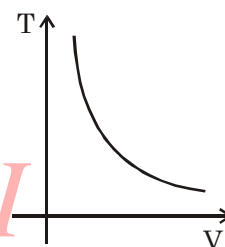
5. An ideal monoatomic gas undergoes a process given by a rectangular hyperbola on T - V graph

- (A) Bulk modulus for this process will be $2P$ (where P is pressure)

- (B) Molar specific heat for the process is $\frac{R}{2}$.

- (C) Bulk modulus for this process will be $\frac{P}{2}$ (where P is pressure)

- (D) Molar specific heat for the process is R .



6. From a height of $H = 70$ m we dropped a stone. The stone is connected to another identical stone by an inextensible and massless string which we held in our hand. The length of the string is 40 m. After 2 second of releasing the first stone, we release the second stone also. Choose the **CORRECT** statement(s) from the following statements. Neglect air resistance.

- (A) The first stone strikes the ground 4 sec after it is released.
(B) The velocity of the first stone at the time of striking is 30 m/s
(C) The total mechanical energy of the stones has decreased during the motion.
(D) The velocity of the second when the first is about to hit the ground is $10\sqrt{6}$ m/s.

7. A spherical ball of mass M falls in a dust cloud. The dust cloud is made up of stationary dust particles of mass m each. They are uniformly present in space with number density n (number of dust particles per unit volume). As the ball falls, it collides elastically with dust particles coming its way. This produces a resistive force on the falling ball.

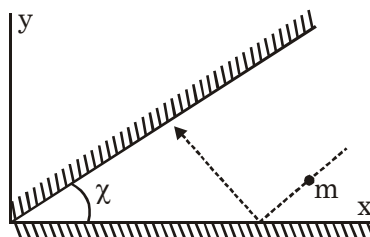
- (A) The resistive force is proportional to velocity of the ball v .
(B) The resistive force is proportional to square of the velocity of the ball v .
(C) The resistive force is doubled if the radius of the ball was double

- (D) When gravitational force balances the resistive force, the velocity of the ball is $\sqrt{\frac{Mg}{\pi mnR^2}}$

8. Choose the **CORRECT** statement(s) :-

- (A) If the temperature of air in an organ pipe increases, its resonance frequency also increases
(B) The difference in lateral pressure in the air-flow causes a spinning ball to move in a curved path, its path bending towards that side where air pressure is smaller
(C) Effective acceleration due to gravity at the equator is smaller than that at the poles.
(D) When a body is weighed on a weighing machine in an elevator descending down with constant velocity, the reading on the weighing machine shows the correct weight

9. Consider a particle of mass m moving in two dimensions between two perfectly reflecting walls which intersect at an angle χ at the origin (see figure). Assume that when the particle is reflected, its speed is unchanged and its angle of incidence equals its angle of reflection. The particle is attracted to the origin by a potential $U(r) = -\frac{C}{r^3}$, where C is some constant.



Now start the particle at a distance R from the origin on the x -axis with a velocity vector $V = (V_x, V_y)$. Assume $V_y \neq 0$, $V_x < 0$.

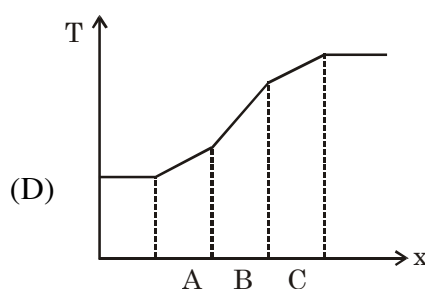
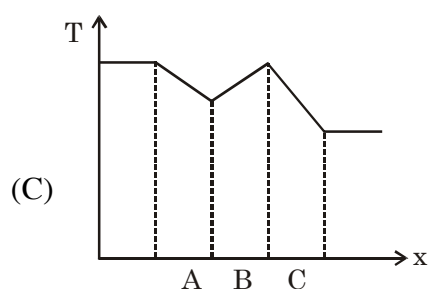
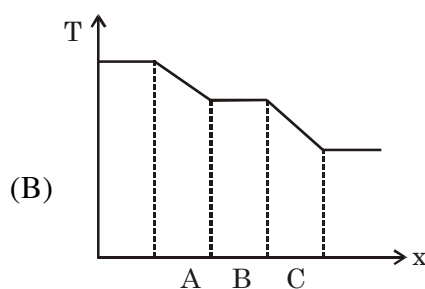
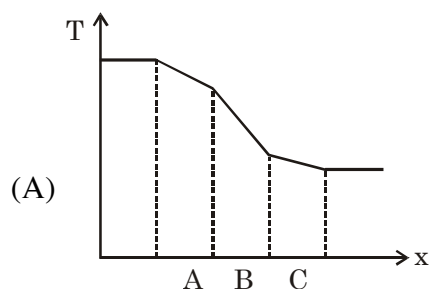
- (A) The equation for distance of closest approach to the origin is $\frac{1}{2}mv^2 - \frac{C}{R^3} = \frac{m^2v_y^2R^2}{2mr_{ca}^2} - \frac{C}{r_{ca}^3}$ where r_{ca} is distance of closest approach.

- (B) The particle will reach the origin if $U > \frac{m^3R^6v_y^6}{54C^2}$ (Here U is total energy)

- (C) The particle will reach the origin if $R < \left(\frac{3C}{mv_y^2}\right)^{1/3}$

- (D) The particle will never reach the origin for any value of R or U . (Here U is total energy)

10. Graphs give the temperature along an x -axis that extends directly through a wall consisting of three layer with different non-zero and finite coefficient of thermal conductivities. The air temperature on both sides of wall are different and uniform. Out of the situations as shown by the graph, which is/are impossible?

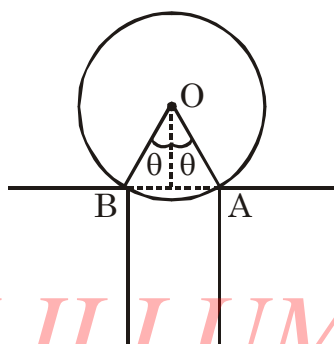


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

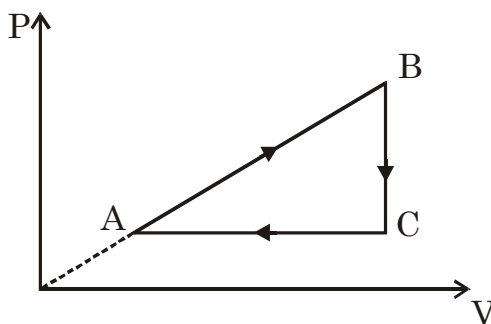
8 Q. [4 M (0)]

1. A uniform solid sphere with mass M , radius R is initially rolling without slipping towards left with speed v on a horizontal surface. It then encounters a ditch of width d as shown in the figure. The initial speed v is smaller than a value $v_{\max}$ such that when the sphere arrives at the near edge of the ditch at point A, it falls off while keeping in touch with point A without slipping, until it hits the other edge at point B. The angular speed of the sphere right before it hits point B is $\sqrt{\frac{\alpha g (1 - \cos \theta)}{\beta R} + \frac{v^2}{R^2}}$. Fill the value of $\alpha - \beta$ in OMR sheet.

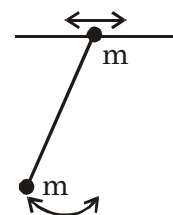


TEAM ILLUMINATI

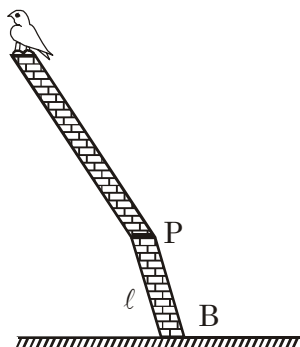
2. One mole of a mono atomic ideal gas is taken around a cyclic process shown in the figure. It is given $T_B = 4T_A$. The efficiency of the cyclic process in % will be (round off to nearest integer):-



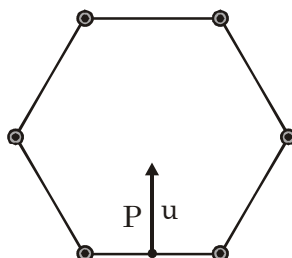
3. A car moving towards a stationary observer from a large distance with a speed of 72 km/hr blows horn for 5 sec. Find the time interval (in sec) for which sound will be heard by the observer. [Speed of sound is 300 m/s] (Round off to nearest integer)
4. Two equal masses m are fixed on the two ends of a uniform rigid rod of length L having same mass m . The upper mass is restricted to move on a smooth horizontal rail and the rod is free to swing along the rail. The angular frequency for small oscillation is given by $n\sqrt{\frac{2g}{7L}}$, here n will be.
5. Two identical wires, one made of iron and the other of aluminum are stretched along - side on a sonometer board by equal stretching forces. [Density of iron = 7.5 gm/cc, density of aluminum = 2.7 g/cc]. The frequency of lowest harmonic for which both wires vibrate in unison, given that the length of the wires is 1 m, their diameters 1 mm and tension 75π is given as β kHz. Find the value of 12β .



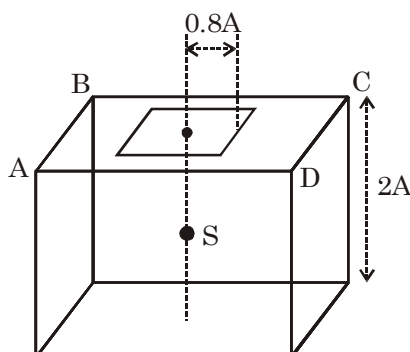
6. A tall, slender, cylindrical brick chimney of height L is slightly perturbed from its vertical equilibrium position so that it topples over, rotating rigidly around its base B until it breaks at a point P . The most likely value for the distance ℓ of P from B is $\frac{2L}{\alpha}$. Fill the value of α in OMR sheet. (Assume that the chimney breaks because the torque is too great and the chimney bends and snaps as shown in figure).



7. Six equal uniform rods, fastened at their ends by frictionless pivots, form a regular hexagon and lie on a frictionless surface. A blow is given at a right angle to the midpoint of one of them at point P in Figure so that it begins to slide with velocity u . The opposite rod begins to move with velocity $v = \frac{u}{5\beta}$. Fill the value of β in OMR sheet.



8. A cube of side $2A$ has a point source S at its centre. All the faces of the cube are blackened except the top surface which has got a square sheet of black paper of side $1.6A$. If the source starts oscillating inside a negligibly thin tunnel perpendicular to the direction of face $ABCD$, and having the mean position of oscillation at the centre of the cube, then the maximum amplitude of oscillation so that the source may not be seen from the top surface is found to be βA . Then find 10β . (Refractive index of material of cube $\mu = \sqrt{3.25}$. Assume black surface absorbs all light incident on it.)



SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

(One or more options correct)

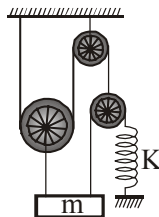
1. Match the column 1 with column 2. Spring constant is k and x is maximum change in length of the spring. If $T_0 = 2\pi\sqrt{\frac{m}{k}}$ and T is time period of the body having mass m .

Column-I

Column-II

(A) $T < T_0$

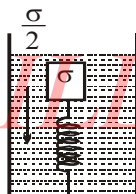
(P)



Block is displaced through distance x_0 and released. pulleys are massless and frictionless.

(B) $T = T_0$

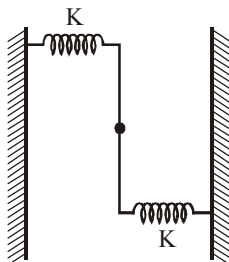
(Q)



Fluid is non viscous of density $\frac{\sigma}{2}$, mass of block m , density σ and it is displaced x_0 downward and released

(C) Change in length of spring $x > x_0$

(R)



Rod of length ℓ mass m is free to oscillate in horizontal plane. Rod is displaced through small angle θ and released such that each spring is compressed through x_0 . Whole system is placed on a smooth horizontal table.

(D) Linear S.H.M.

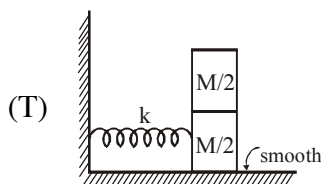
(S)



Potential energy of diatomic molecules when separation is r , is

$$U = - \left\{ a + \frac{k(r - r_0)^2}{b} \right\}$$

where r_0 equilibrium distance and $b = 1$ the atoms are displaced x_0 each in opposite direction and released

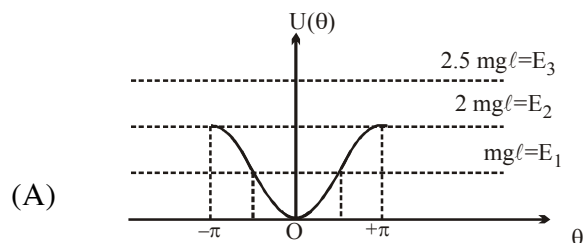


Both blocks do not slide over each other spring is compressed slightly by x_0 and released.

2. Column-I represents potential energy graph for certain system. Column-II gives statements related to graphs.

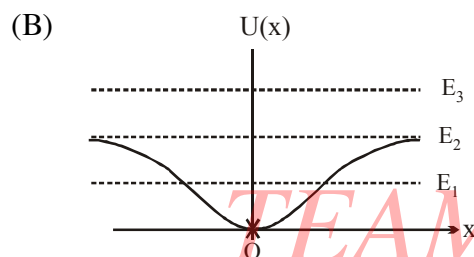
Column I

Column II



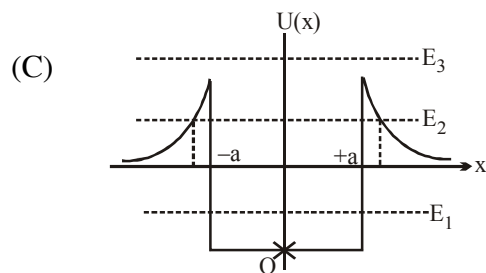
U vs θ graph for a bob hanging vertically from a string with its lowest position as reference level and θ is angle of string from vertical line

- (P) If total energy is E_3 , it is not possible for the body to have any turning point in its motion.



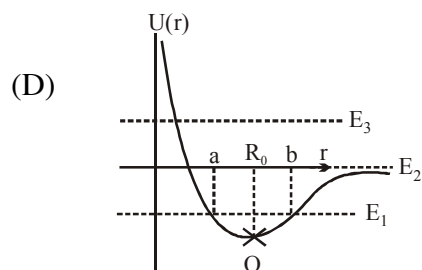
A particle moving along x axis with potential energy function as $U(x) = [1 - e^{-x^2}]$

- (Q) For a small displacement about point O, potential energy function is quadratic in variable plotted on x-axis.



Potential energy function of a particle in an arbitrary force field

- (R) For a small displacement about position O, motion is simple harmonic.



Graph represents potential energy for a particle

- (S) If total energy is $E_{\text{total}} < E_2$, particle executes periodic and oscillatory motion for all energy values greater than energy at O.

- (T) Point O is position of stable equilibrium

PHYSICS

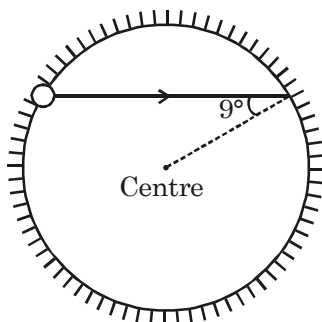
QUIZ # 47

SECTION-I

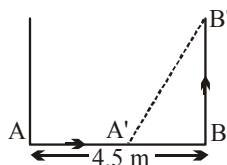
Single Correct Answer Type

20 Q. [4 M (–1)]

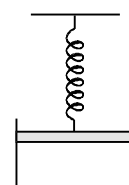
1. A small hole is made in a sphere as shown whose inner surface is made reflecting. Find number of reflections before the ray comes out of the sphere.



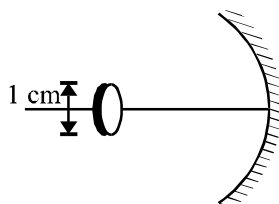
- (A) 20 (B) 19 (C) 9 (D) 11
2. A particle is projected with speed 6 m/s as shown. It is tied to a string having breaking strength 80 N. (Length of string is 1m and mass of particle is 1kg)
- (A) String will not break (B) Particle will oscillate
(C) String will break in the subsequent motion (D) Particle will complete the circle.
3. Two particles instantaneously at A & B respectively 4.5 meters apart are moving with uniform velocities as shown in the figure. The former towards B at 1.5 m/sec and the latter perpendicular to AB at 1.125 m/sec. The instant when they are nearest is:



- (A) 2 sec (B) 3 sec (C) 4 sec (D) $1\frac{23}{25}$ sec
4. A steel rod rigidly clamped at one end and free at the other has longitudinal stationary waves set up in it and it vibrates in a mode in which there are eight displacement nodes other than the rigid end. The young's modulus for steel is 2.0×10^{11} Pa and its density is 7.8×10^3 kg m⁻³. The frequency of the source that sets the longitudinal waves in the rod is 10 kHz. The length of the rod is
- (A) 1.2 m (B) 2.1 m (C) 0.6 m (D) 3.2 m
5. A vessel containing a liquid is heated with its contents. The pressure at the bottom of vessel due to the liquid should. (no evaporation take place and there is no spilling)
- (A) increase (B) decrease
(C) increases if $\gamma_{\text{liq}} > 3 \alpha_{\text{vessel}}$ (D) decreases if $\gamma_{\text{liq}} > 3 \alpha_{\text{vessel}}$
6. One mole of an ideal gas is kept enclosed under a light piston (area = 10^{-2} m²) connected by a compressed spring (spring constant 100 N/m). The volume of gas is 0.83 m³ and its temperature is 100 K. The gas is heated so that it compresses the spring further by 0.1 m. The work done by the gas in the process is (Take R = 8.3 J/mole and suppose there is no atmosphere):
- (A) 3 J (B) 6 J (C) 9 J (D) 1.5 J



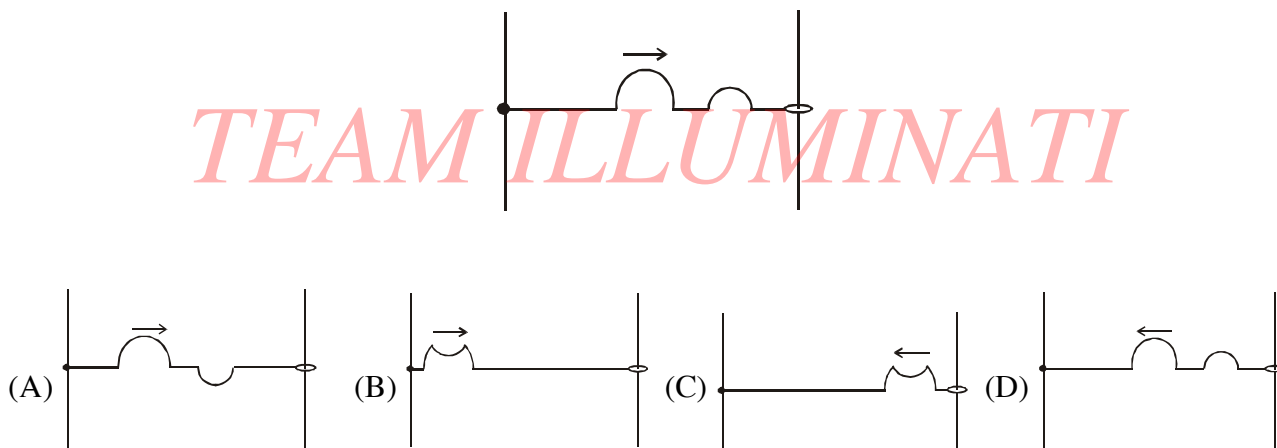
7. A concave mirror of radius of curvature 20 cm is shown in the figure. A circular disc of diameter 1 cm is placed on the principal axis of mirror with its plane perpendicular to the principal axis at a distance 15 cm from the pole of the mirror. The radius of disc increasing according to the law $r = (0.5 + 0.1t)$ cm/sec.



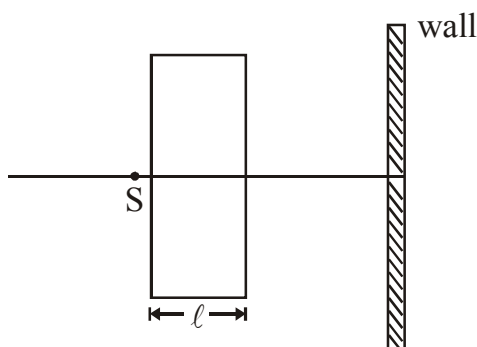
The area of image of the disc at $t = 1$ second is:

- (A) $1.2 \pi \text{ cm}^2$ (B) $1.44 \pi \text{ cm}^2$ (C) $1.52 \pi \text{ cm}^2$ (D) none of these
8. The right end of a string is tied to a ring that can slide without friction up and down the rod, left end of the string is tied, rigidly to another rod. Two pulses are sent along the string toward the right rod, the pulses reflect from right rod and then from left rod and so on. At a given point of time the string's appearance is illustrated in the figure below. The string's appearance at a later point after complete or partial reflection may be

TEAM ILLUMINATI



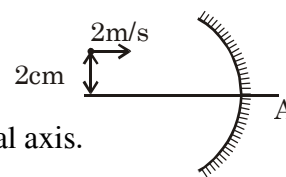
9. A steel rope of length L , area of cross-section A , Young's modulus Y , is hanging from a ceiling in equilibrium. The elastic potential energy per unit volume at a point $L/3$ from ceiling is [Density = d]
(A) $2(dgL)^2/9Y$ (B) $(dgL)^2/9Y$ (C) $2(dgL)/9AY$ (D) $2(dgL)^2/3Y$
10. A beam of light from a point source S passes the motionless glass slab of thickness ℓ and hits the wall in time t_1 . When glass slab moves towards wall with speed $v \ll C$ then light beam hits the wall in time t_2 . Then ($C \rightarrow$ velocity of light in free space, $n \rightarrow$ refractive index of glass slab)



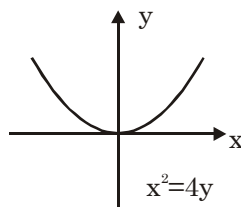
- (A) $t_1 = t_2$ (B) $t_1 < t_2$ (C) $|t_1 - t_2| = \frac{1}{C} \left(\frac{nv\ell}{C} \right)$ (D) None

11. An object is moving parallel to the principal axis of a mirror with a speed of 2m/s. Find the speed of image when it is at a distance of 60 cm in front of the concave mirror of radius of curvature 80 cm.

(A) 4m/s parallel to principal axis. (B) 8 m/s parallel to principal axis.
 (C) 16 m/s parallel to principal axis. (D) None



12. A particle is moving with constant speed of 2 m/s along the curve $x^2 = 4y$. Find the acceleration of particle at the instant when it crosses the origin.



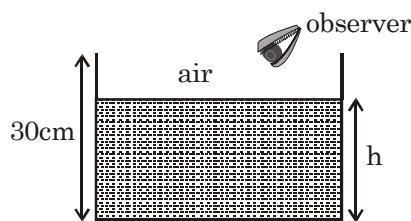
(A) 2m/s^2 (B) 1 m/s^2 (C) 4 m/s^2 (D) 10 m/s^2

13. Two masses 3kg and 6 kg are joined together by a string of length $\sqrt{2}\text{m}$. The initial separation between them is 1m and particle of mass 6 kg is projected perpendicular to the line joining between them with speed 2 m/s. Find tension in the string when it becomes taut (Ignore gravity).



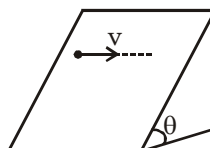
(A) 1N (B) 2N (C) $2\sqrt{2}\text{ N}$ (D) None

14. To what height h should water be filled so that container appears to be filled by 75%
 $\left(\mu_w = \frac{4}{3}\right)$:-



(A) 20 cm (B) 27 cm (C) 24 cm (D) 26 cm

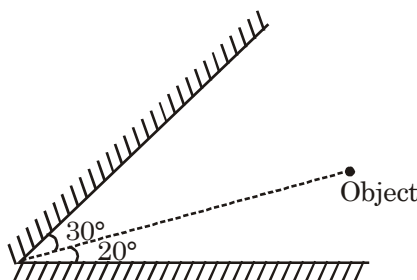
15. A particle is projected perpendicular to the line of greatest slope as shown on an inclined plane having $\mu = \tan\theta$. Find the speed of particle after a long time.



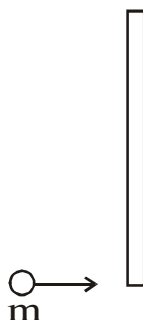
(A) zero (B) $\frac{v}{2}$ (C) $\frac{v}{3}$ (D) v

16. A road passes at some distance from a standing man. A truck is coming on the road with some constant acceleration. The truck driver blows a whistle of frequency 500 Hz when the line joining the truck and the man makes an angle of θ with the road. The man hears the note at a frequency of 600 Hz when the truck is closest to him. Also the speed of truck has got doubled during this time. The value of θ is :-
 (A) 30° (B) 45° (C) 60° (D) 53°

17. Number of images formed by the two plane mirrors will be :-



- (A) 5 (B) 6 (C) 7 (D) 8
18. An object is kept on the principal axis of a concave mirror of focal length 10 cm, at distance 15 cm from the mirror. Mirror is cut in two equal pieces along principal axis and the piece below principal axis is removed. If remaining piece is shifted up by 2 cm perpendicular to principal axis then find by what distance (in cm) image gets shifted perpendicular to principal axis ?
(A) 6 (B) 4 (C) 3 (D) 5
19. A body of mass 1 kg is thrown vertically up in air from earth's surface. A constant air resistive force acts on the body opposite to its velocity. Body returns to earth's surface with speed equal to half of initial speed. Find resistive force (in N) acting on the body.
(A) 10 (B) 8 (C) 4 (D) 6
20. A uniform rod of mass 6m is kept on smooth horizontal surface. A small ball of mass m strikes the rod at one end with certain velocity and comes to rest just after collision. Coefficient of restitution between ball and rod is



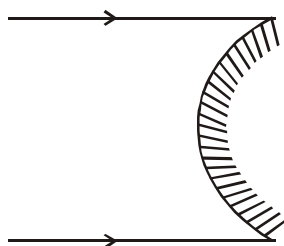
- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{2}{5}$

SECTION-II

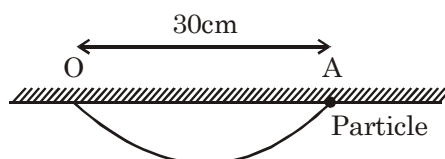
Numerical Answer Type Question (upto second decimal place)

5Q.[3(0)]

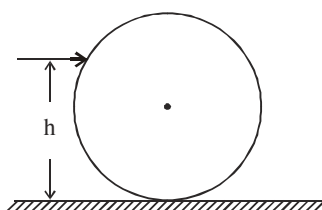
- A wave pulse starts propagating in + x direction along a non - uniform wire of length 10 m with mass per unit length given by $\mu = \mu_0 + \alpha x$ and under a tension of 100 N. The time taken by the pulse (in sec) to travel from the lighter end ($x = 0$) to the heavier end is ($\mu_0 = 10^{-2}$ kg/m and $\alpha = 9 \times 10^{-3}$ kg/m²).
- A parallel beam strikes the convex mirror (as shown) of radius of curvature 10cm and aperture diameter 12cm. the focal length of the mirror for given beam is $1.25 f_0$ cm, find f_0 .



3. A small ball of mass 100 g is attached to a light and inextensible string of length 50 cm. The string is tied to a support O and the mass m released from point A which is at a horizontal distance of 30 cm from the support. Calculate the speed (in m/s) of the ball at its lowest point of the trajectory.



4. A uniform disc of radius 6 cm is kept on smooth horizontal surface with its plane vertical as shown in figure. At height h from horizontal surface when a horizontal force is exerted on the disc, it starts rolling without sliding. Find h (in cm).



5. A heavy disc with radius $r = 0.5$ m is rolling down hanging on two string wound around the disc very tightly. The free ends of the string are attached to a fixed horizontal support. The string are always under tension during the motion. At some instant, the angular velocity of the disc is $\omega = 2$ rad/sec and angle between string is $\alpha = 60^\circ$. Find the velocity of centre of mass of the disc (in m/s) at this moment.

